

Inverse Fabius Theory

Analyticity, asymptotics, computability, and dyadic sampling

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Abstract

This volume replaces five overlapping reports about the inverse Fabius function and the Rvachev law. It develops their common analytic and probabilistic platform once, then treats dense-open analyticity of a precisely specified totalized elementary-expression grammar, its nonrepresentability results, positive compositional inverse iterates, exact inverse-dyadic germs, Bell and Bernoulli inversion, Barnes–Rvachev deconvolution, all-orders Lambert–W endpoint asymptotics, dyadic self-sampling and Richardson filters, exact inverse moduli, and certified computation.

Every nonconjectural retained assertion has a complete human-readable proof. Exact Lean counterparts are identified by module and declaration; manuscript results with no exact formal counterpart are described as human-proved frontier results. Numerical experiments are used only as reproducible checks. Open claims occur only as explicitly named conjectures or research problems, and false, vacuous, or duplicated former conjectures are quarantined rather than silently recycled.

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Part I

Scope, Status, and the Common Fabius–Rvachev Platform

1 The shared analytic and probabilistic platform

The five source reports use the same bounded Fabius function, Rvachev density, random series, Fourier product, and dyadic spline approximants. They are fixed once in this part; later parts introduce only the notation peculiar to their question.

1.1 Fabius, Rvachev, and the random series

Let $V_1, V_2, \dots$ be independent uniform random variables on $[0, 1]$, put $U_k := 2V_k - 1$, and define the uncentered and centered laws

$$X := \sum_{k=1}^{\infty} 2^{-k} V_k, \quad Y := \sum_{k=1}^{\infty} 2^{-k} U_k, \quad S_n := \sum_{k=1}^n 2^{-k} U_k. \quad (1.1)$$

Thus $Y = 2X - 1$. Throughout the volume, X and V_k refer to the uncentered $[0, 1]$ model, while Y and U_k refer to its centered $[-1, 1]$ model. The density of Y is Rvachev's even up-function up, supported on $[-1, 1]$ [2, 3, 7]. The bounded Fabius function, originating in Fabius's probabilistic construction [1], satisfies

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1, \quad (1.2)$$

and is a strictly increasing C^∞ bijection from $[0, 1]$ to itself. We write

$$Q_F := F^{-1} : [0, 1] \longrightarrow [0, 1]. \quad (1.3)$$

The reflection laws are

$$F(1 - x) = 1 - F(x), \quad Q_F(1 - y) = 1 - Q_F(y). \quad (1.4)$$

Let p_n be the density of S_n and set

$$F_n^{\text{spl}}(x) := p_n(x - 1), \quad 0 \leq x \leq 1. \quad (1.5)$$

On every compact subinterval of $(0, 1)$, F_n^{spl} is strictly increasing for all sufficiently large n . Its increasing inverse branch will be denoted

$$G_n^{\text{spl}} := (F_n^{\text{spl}})^{-1}. \quad (1.6)$$

The omitted random tail is a scaled copy of Y :

$$Y - S_n \stackrel{d}{=} 2^{-n} Y. \quad (1.7)$$

1.2 The finite Thue–Morse spline

Put $\varepsilon_j = (-1)^{\text{wt}_2(j)}$, the signed Thue–Morse sequence [4]. The unequal-box inclusion–exclusion formula for the corresponding box spline [6] gives

$$F_n^{\text{spl}}(x) = \frac{2^{\binom{n}{2}}}{(n-1)!} \sum_{j=0}^{2^n-1} \varepsilon_j (x - 2^{-n} - j2^{1-n})_+^{n-1}. \quad (1.8)$$

Thus every $G_n^{\text{spl}}(y)$ is the inverse of an explicitly known piecewise polynomial. Formula (1.8) alone does not reveal which polynomial piece controls a prescribed quantile or why its coefficients should simplify. The local theorem in [appendix C](#) answers both questions at the inverse-dyadic outputs $y = F(2^{-r})$.

1.3 Fourier product and reciprocal-sinc coefficients

With the angular-frequency convention $\widehat{f}_{\text{ang}}(\xi) = \int f(x)e^{-ix\xi} dx$,

$$\begin{aligned}\Phi_{\text{ang}}(\xi) &:= \widehat{\text{up}}_{\text{ang}}(\xi) = \prod_{k=1}^{\infty} \text{sinc}_{\text{rad}}(2^{-k}\xi), \\ \widehat{p}_n_{\text{ang}}(\xi) &= \prod_{k=1}^n \text{sinc}_{\text{rad}}(2^{-k}\xi), \quad \text{sinc}_{\text{rad}} t := \frac{\sin t}{t}, \quad \text{sinc}_{\text{rad}} 0 := 1.\end{aligned}\tag{1.9}$$

Near the origin define rational coefficients b_j by

$$\frac{1}{\Phi_{\text{ang}}(\xi)} = \sum_{j=0}^{\infty} b_j \xi^{2j}, \quad b_0 = 1.\tag{1.10}$$

Euler's product gives

$$\log \frac{1}{\Phi_{\text{ang}}(\xi)} = \sum_{k=1}^{\infty} \gamma_k \xi^{2k}, \quad \gamma_k = \frac{\zeta(2k)}{k\pi^{2k}(4^k - 1)}.\tag{1.11}$$

Equivalently, in Bernoulli form,

$$\gamma_k = \frac{(-1)^{k+1} 2^{2k-1} B_{2k}}{k(2k)!(4^k - 1)}.\tag{1.12}$$

The coefficient array is generated by complete Bell polynomials [5]:

$$b_j = \frac{1}{j!} \mathcal{B}_j(1!\gamma_1, 2!\gamma_2, \dots, j!\gamma_j),\tag{1.13}$$

or by the recurrence

$$b_j = \frac{1}{j} \sum_{k=1}^j k \gamma_k b_{j-k}.\tag{1.14}$$

The first values are

$$b_1 = \frac{1}{18}, \quad b_2 = \frac{31}{16200}, \quad b_3 = \frac{2347}{42865200}, \quad b_4 = \frac{1904369}{1311675120000}.\tag{1.15}$$

The merged volume keeps the four polynomial/function families visually distinct. The notation used below is summarized here once:

Object	Reserved notation
finite-prefix spline and its local inverse	F_n^{spl} and G_n^{spl}
forward and inverse compositional iterates	$G_n = F^{o n}$ and $K_n = (F^{-1})^{o n}$
uncentered and centered Appell families	P_n and A_n
angular and 2π Fourier products	$\Phi_{\text{ang}}(\xi)$ and $\Phi_{2\pi}(z) = \Phi_{\text{ang}}(2\pi z)$
endpoint reversion kernel	$\mathcal{K}(z, w)$

Thus a superscript “spl” always signals a finite convolution prefix; bare G_n, K_n are reserved for composition iterates; and the two Fourier coordinates are never silently interchanged.

The corpus proves the uniform derivative expansion

$$(F_n^{\text{spl}})^{(m)}(x) = \sum_{j=0}^{J-1} (-1)^j b_j 4^{-jn} F^{(m+2j)}(x) + \mathcal{O}_{m,J}(4^{-Jn})\tag{1.16}$$

uniformly on $[0, 1]$ for every fixed m, J . We shall invert this expansion.

1.4 Terminating jets at inverse powers of two

Let

$$x_r := 2^{-r}, \quad y_r := F(x_r). \quad (1.17)$$

The global dilation identity implies

$$F^{(k)}(x_r) = 2^{\binom{k+1}{2}} F(2^{k-r}), \quad 0 \leq k \leq r, \quad F^{(k)}(x_r) = 0 \quad (k > r). \quad (1.18)$$

Write $d_0 = 1$ and

$$d_r = r! 2^{\binom{r}{2}} F(2^{-r}). \quad (1.19)$$

Then the nonconstant part of the terminating Taylor jet is

$$J_r(z) := \sum_{k=1}^r \alpha_{r,k} z^k, \quad \alpha_{r,k} = 2^{\binom{k+1}{2} - \binom{r-k}{2}} \frac{d_{r-k}}{k!(r-k)!}. \quad (1.20)$$

Thus the Taylor series of F at x_r is the polynomial $y_r + J_r(z)$, although $F(x_r + z)$ is not equal to that polynomial on any neighborhood. The discrepancy is flat at $z = 0$.

2 Fabius, Rvachev, Thue–Morse, and the derivative atlas

2.1 The Rvachev bump and the signed global extension

Let up denote Rvachev's compactly supported C^∞ function, normalized by

$$\text{supp up} = [-1, 1], \quad \text{up}(0) = 1,$$

and related to the bounded Fabius function by

$$\text{up}(t) = F(1 - |t|) \quad (-1 \leq t \leq 1). \quad (2.1)$$

In particular, up is even, strictly decreasing on $[0, 1]$, and

$$\text{up}(\tfrac{1}{2}) = F(\tfrac{1}{2}) = \tfrac{1}{2}. \quad (2.2)$$

For $k \geq 0$, put

$$\varepsilon_k = (-1)^{\text{wt}_2(k)}, \quad (2.3)$$

where $\text{wt}_2(k)$ is the sum of the binary digits of k . The signed global extension Θ can be written blockwise as

$$\Theta(t) = \varepsilon_k \text{up}(t - (2k + 1)), \quad 2k \leq t \leq 2k + 2. \quad (2.4)$$

It agrees with F on $[0, 1]$, satisfies $\Theta'(t) = 2\Theta(2t)$ globally, and obeys

$$|\Theta(t)| \leq 1. \quad (2.5)$$

The translate signs in (2.4) are precisely the Thue–Morse signs.

2.2 Exact all-order derivatives

Define the superfactorial dyadic scale

$$Q_m := 2^{\binom{m+1}{2}} = 2^{m(m+1)/2}. \quad (2.6)$$

Repeated differentiation of the dilation equation gives, for every $m \geq 1$ and $x \in (0, 1)$,

$$F^{(m)}(x) = Q_m \Theta(2^m x). \quad (2.7)$$

Consequently

$$\left| F^{(m)}(x) \right| \leq Q_m. \quad (2.8)$$

At the matched dyadic grid this becomes the exact Thue–Morse jet law

$$F^{(m)}\left(\frac{a}{2^m}\right) = \begin{cases} 0, & a \text{ even,} \\ Q_m \varepsilon_{\lfloor a/2 \rfloor}, & a \text{ odd.} \end{cases} \quad (2.9)$$

The factor Q_m is the source of nonanalytic growth. Since

$$Q_m^{1/m} = 2^{(m+1)/2}, \quad (2.10)$$

no fixed ordinary exponential can neutralize it after division by $m!$. The only serious issue is cancellation among the many chain-rule terms created by composition.

2.3 Shape and orbit dynamics

The repository proves that F is a strictly increasing bijection of $[0, 1]$, that F' is strictly increasing on $[0, \frac{1}{2}]$ and strictly decreasing on $[\frac{1}{2}, 1]$, and that

$$F'(0) = 0, \quad F'(\tfrac{1}{2}) = 2, \quad F'(1) = 0. \quad (2.11)$$

Moreover F is strictly convex on $[0, \frac{1}{2}]$ and strictly concave on $[\frac{1}{2}, 1]$. Comparing with the chord joining $(0, 0)$ and $(\frac{1}{2}, \frac{1}{2})$ gives

$$F(x) < x \quad (0 < x < \tfrac{1}{2}), \quad F(x) > x \quad (\tfrac{1}{2} < x < 1). \quad (2.12)$$

Thus every noncentral interior orbit moves monotonically toward an endpoint:

$$0 < x < \tfrac{1}{2} \implies x > F(x) > F^{\circ 2}(x) > \cdots > 0, \quad (2.13)$$

$$\tfrac{1}{2} < x < 1 \implies x < F(x) < F^{\circ 2}(x) < \cdots < 1. \quad (2.14)$$

This elementary dynamical fact will force strict unimodality of the exponential weights in the derivative spine sum.

Part II

Analyticity and Obstructions to Totalized Elementary Expressions

3 Statement

3.1 The two inputs

Write $\mathcal{A}(f) = \{x \in \mathbb{R} : f \text{ is real analytic at } x\}$ for the *analytic locus* of $f: \mathbb{R} \rightarrow \mathbb{R}$. The proof combines one negative fact about the Fabius function with one positive fact about the explicitly defined totalized elementary-expression class introduced below. Throughout this chapter, “elementary” is shorthand for that class; it is not a claim about an unspecified larger differential-field convention.

Theorem 3.1 (Nowhere analyticity; already formalized). *Let F be the Fabius function, extended by 0 to the left of 0 and by 1 to the right of 1. Then $\mathcal{A}(F) = \mathbb{R} \setminus [0, 1]$. In particular F is real analytic at no point of $[0, 1]$, endpoints included.*

Proof. On $(0, 1)$ this is the case $n = 1$ of the forward-engine theorem [Theorem 9.4](#); its complete derivative-spine proof is given in [Sections 13](#) and [16](#). It remains only to account for the totalization and the endpoints. The function is constant, hence analytic, on each component of $\mathbb{R} \setminus [0, 1]$. If it were analytic at 0, its analytic germ would agree with the zero function on a left-hand interval. The identity theorem for real power series would then make it zero on a two-sided neighbourhood of 0, contrary to strict increase on $[0, 1]$. At 1 the same argument, applied to $1 - F$, uses the constant right-hand continuation. Thus precisely the points outside $[0, 1]$ are analytic points. $\square$

This is the unnumbered corollary to equation (24) of Arias de Reyna, *Arithmetic of the Fabius function* (arXiv:1702.06487v3), transferred from Rvachev’s up to F and completed at the right endpoint. The same statement is machine checked as `Fabius.canonical_fabius_analyticAt_iff` in the module `FabiusFunction.NowhereAnalytic`; the proof above records the human argument and points to the full spine estimate used in its interior step.

Theorem 3.2 (Density of the analytic locus). *If $f \in \mathcal{E}$, then $\mathcal{A}(f)$ is a dense open subset of $\mathbb{R}$.*

Proof. By [Lemma 6.1](#), the analytic locus of every real function is open, so only density requires proof. Induct on the derivation of f in [Definition 4.1](#). A constant and the identity are analytic everywhere.

Suppose first that the final constructor is binary. By the induction hypothesis, the analytic loci of its two arguments are dense and open; their intersection is therefore dense. On that intersection both arguments are analytic, hence so are their sum and product. Thus the analytic locus of the result contains a dense set. Negation is even more immediate: it preserves every analytic point.

It remains to treat the unary constructors. For the reciprocal, logarithm, and a fixed real power, the outer function is analytic away from the single value 0; for the arcsine it is analytic away from $\{-1, 1\}$; and for the exponential, sine, cosine, and arctangent there are no exceptional values. These assertions are [Lemma 5.1](#). In every case the exceptional set is finite, so [Lemma 6.2](#), applied to the inner function supplied by the induction hypothesis, makes the analytic locus of the composite dense. These are all the constructors in [Definition 4.1](#), and the induction is complete. $\square$

Corollary 3.3 (Main result). *Let $U \subseteq [0, 1]$ have nonempty interior and let $g \in \mathcal{E}$. Then g and F do not agree on U . In particular no member of the totalized expression class $\mathcal{E}$ agrees with F on $(0, 1)$.*

Proof. By [Theorem 3.2](#) the set $\mathcal{A}(g)$ is dense, so it meets the nonempty open set $\text{int } U$; pick $x \in \text{int } U \cap \mathcal{A}(g)$. If g and F agreed on U they would have the same germ at x , because $\text{int } U$ is a neighbourhood of x contained in U ; analyticity at a point depends only on the germ, so F would be analytic at $x \in [0, 1]$, contradicting [Theorem 3.1](#). $\square$

Remark 3.4. Hypothesising nonempty *interior* rather than openness costs nothing in the proof and buys a little: U may be a closed interval, or an interval with a nowhere dense closed set removed, or any set that is merely fat somewhere. (Removing *all* the rationals would not do: the irrationals in an interval have empty interior, and the hypothesis genuinely fails.) What cannot be weakened is fatness itself: on a finite set an elementary function certainly can agree with F , since a polynomial interpolates any finite table of values.

Corollary 3.3 is the strengthened form used here of the assertion “the Fabius function is not elementary”, and it is strictly stronger than the bare non-elementarity of F as a function on $\mathbb{R}$. The bare statement leaves open the possibility that some elementary function agrees with F on all of $(0, 1)$ and differs from it outside; Corollary 3.3 rules that out, and rules it out on every subinterval.

Remark 3.5. Only a very weak consequence of [Theorem 3.2](#) is actually used: that an elementary function is analytic at *at least one* point of every nonempty open set. Density is nevertheless the statement one must prove, because it is the statement that survives the induction: “analytic somewhere” is not preserved by intersection, whereas “dense open” is.

4 The totalized elementary-expression class

4.1 Definition

For the theorem to have an exact scope, we fix a concrete grammar of total real functions. It contains the usual real expressions built from constants, the identity, field operations, fixed powers, exponentials, logarithms, and the listed trigonometric functions wherever those expressions are classically defined. It does not silently include arbitrary algebraic branches or arbitrary interleaved algebraic–exponential–logarithmic towers; the precise extensions proved later are stated separately.

Definition 4.1 (Totalized elementary-expression class). The class $\mathcal{E}$ of *totalized elementary expressions* $\mathbb{R} \rightarrow \mathbb{R}$ is the smallest class of functions such that

1. every constant function $x \mapsto c$ lies in $\mathcal{E}$, and the identity $x \mapsto x$ lies in $\mathcal{E}$;
2. if $f, g \in \mathcal{E}$ then $x \mapsto f(x) + g(x)$, $x \mapsto f(x)g(x)$ and $x \mapsto -f(x)$ lie in $\mathcal{E}$;
3. if $f \in \mathcal{E}$ then $x \mapsto f(x)^{-1}$ lies in $\mathcal{E}$;
4. if $f \in \mathcal{E}$ and $r \in \mathbb{R}$ then $x \mapsto f(x)^r$ lies in $\mathcal{E}$;
5. if $f \in \mathcal{E}$ then each of $\exp \circ f$, $\log \circ f$, $\sin \circ f$, $\cos \circ f$, $\arcsin \circ f$, $\arctan \circ f$ lies in $\mathcal{E}$.

Two features of Definition 4.1 deserve comment.

Composition is a theorem, not an axiom. Definition 4.1 contains no clause “ $f, g \in \mathcal{E} \Rightarrow f \circ g \in \mathcal{E}$ ”. It does not need one.

Proposition 4.2. $\mathcal{E}$ is closed under composition.

Proof. Induct on the derivation of the outer function f . Every clause of Definition 4.1 commutes with precomposition by a fixed g : the constant $x \mapsto c$ composed with g is again that constant; the identity composed with g is g itself, which is elementary by hypothesis; and $(f_1 + f_2) \circ g = (f_1 \circ g) + (f_2 \circ g)$, $(\exp \circ f) \circ g = \exp \circ (f \circ g)$, and likewise for every remaining clause. $\square$

Consequently $\mathcal{E}$ contains every function one expects: differences $f - g = f + (-g)$, quotients $f/g = f \cdot g^{-1}$, natural powers, all polynomial and rational functions, $\sqrt{\cdot}$ and the n -th roots, the absolute value $|x| = \sqrt{x^2}$, the tangent $\sin / \cos$, the hyperbolic functions $\sinh x = (e^x - e^{-x})/2$, $\cosh$, $\tanh$, the arccosine $\pi/2 - \arcsin$, and the inverse hyperbolic sine $\operatorname{arsinh} x = \log(x + \sqrt{1 + x^2})$.

Total functions and junk values. Every member of $\mathcal{E}$ is a *total* function $\mathbb{R} \rightarrow \mathbb{R}$. This is a deliberate choice, and it is the choice that makes [Corollary 3.3](#) strongest. The formalization inherits `Mathlib`’s conventions

$$\begin{aligned} 0^{-1} &= 0, & \log x &= \log |x|, & \log 0 &= 0, & 0^r &= 0 \ (r \neq 0), & 0^0 &= 1, \\ x^r &= |x|^r \cos(\pi r) \ (x < 0), & \arcsin x &= \pm \frac{\pi}{2} \ (\pm x \geq 1), \end{aligned}$$

so that clauses (3)–(5) of Definition 4.1 never fail to produce a function.

One might worry that junk values weaken the theorem, by admitting “elementary” functions that are not classically elementary. They do the opposite. Admitting more functions into $\mathcal{E}$ makes the assertion “ $g \in \mathcal{E}$ and $g = F$ on U ” easier to satisfy, hence its refutation stronger. And nothing is lost in the other direction: if a classical elementary expression is well formed at every point of an open set U — no vanishing denominator, no nonpositive logarithm, no real power of a nonpositive base, no arcsine of a value outside $[-1, 1]$ — then on U it agrees pointwise with the total function produced by the same derivation, since on U the junk values are never consulted. So every such classical elementary function on U is the restriction to U of a member of $\mathcal{E}$, and Corollary 3.3 applies to it.

Warning 4.3 (Odd roots need a different derivation). The condition “no real power of a nonpositive base” in the box above cannot be dropped, and it is the one place where the total-function convention bites. `Mathlib`’s x^r at a negative base is $|x|^r \cos(\pi r)$, so $(-8)^{1/3} = 2 \cos(\pi/3) = 1$, whereas the classical real cube root of -8 is -2 . The clause $x \mapsto f(x)^r$ of Definition 4.1 therefore supplies the n -th roots only on $[0, \infty)$.

Nothing is lost, because the classical real odd root is in $\mathcal{E}$ by a *different* derivation,

$$x \longmapsto \frac{x}{|x|} \cdot |x|^{1/n},$$

which uses only division, the absolute value and a real power, and which evaluates to 0 at $x = 0$ under the same conventions. This is `IsElementary.signedRpow`, and `Fabius.signedRoot_pow` verifies formally that for odd n its n -th power is the identity, so it really is the real n -th root and not merely a plausible substitute.

4.2 What is not claimed

$\mathcal{E}$ contains `Mathlib`’s total real-power representatives of the principal n -th roots on $[0, \infty)$ for positive n and, by Warning 4.3, the classical odd-root functions on all of $\mathbb{R}$. It is not, as defined, closed under passage to an arbitrary algebraic function: Definition 4.1 has no clause producing a continuous branch of $P(x, y) = 0$ for a polynomial P with elementary coefficients whose Galois group is not solvable, and Liouville’s differential-field definition of “elementary” does admit such branches. Section 7 closes that gap, not by enlarging $\mathcal{E}$ but by proving the conclusion of Corollary 3.3 directly for such branches. What remains genuinely outside the result is stated there.

Two smaller gaps are worth naming here.

The first is that $\mathcal{E}$ is a class of *total* functions, so a classical elementary expression enters it only through a derivation whose junk values are never consulted on the region of interest; Warning 4.3 records the one place where the obvious derivation is the wrong one.

The second concerns variable exponents. Definition 4.1 provides $x \mapsto f(x)^r$ for a *fixed* real r , and `IsElementary.rpow_of_pos` supplies f^g whenever the base f is positive *at every real* — which covers $(1 + x^2)^{\sin x}$ and any base that is everywhere positive (no uniform lower bound is needed), but not $x \mapsto x^x$, whose base is positive only on $(0, \infty)$. That case is handled instead by the region statement below.

The sign of an everywhere nonvanishing base is not an obstruction. `Mathlib`’s x^y is sign-dependent at a negative base — there $x^y = |x|^y \cos(\pi y)$ — but a single elementary formula covers both signs at once,

$$f^g = \exp(\log f \cdot g) \cdot \cos\left(g\pi \cdot \frac{1-f/|f|}{2}\right),$$

since $(1 - f/|f|)/2$ is the indicator of $f < 0$, making the cosine $\cos 0 = 1$ where $f > 0$ and $\cos(g\pi)$ where $f < 0$; no absolute value is needed inside the logarithm because `Mathlib` already sets $\log x = \log |x|$. This is `IsElementary.rpow_of_ne_zero`, and it needs only that the base never vanishes. This is a sufficient uniform closure rule, not a claim that every power whose base has a zero is non-elementary: fixed natural powers are already covered by other constructors. At

$f(x) = 0$ the conventions $0^r = 0$ for $r \neq 0$ and $0^0 = 1$ are discontinuous in the exponent, so the nonvanishing formula does not extend uniformly there. Nothing is lost for Corollary 3.3, and this is formalized rather than asserted: `Fabius.eqOn_rpow_of_pos` states that on any set where the base is positive, f^g agrees pointwise with $\exp(\log f \cdot g)$, which is a member of $\mathcal{E}$ whenever f and g are. Since Corollary 3.3 already rules out agreement on *every* nonempty open subset of $[0, 1]$, and the base of a classical variable-exponent expression is positive wherever the expression is well formed, such an expression is excluded too.

We also make no claim about elementary functions of a complex variable, and none about the closely related but distinct question of whether F satisfies an algebraic differential equation.

5 Analyticity of the generators

Each generator of Definition 4.1 is real analytic away from a finite set of arguments. All of these facts are in `Mathlib`, most of them in the form “ C^n for every n in $\mathbb{N}_0 \cup \{\infty, \omega\}$ ”; specializing to the analytic exponent ω and applying `ContDiffAt.analyticAt` converts them to the statements below.

Lemma 5.1.

1. $\exp, \sin, \cos$ and $\arctan$ are real analytic at every point of $\mathbb{R}$.
2. $\log$ is real analytic at every $x \neq 0$ — at negative arguments as well, because $\log$ is even under the convention $\log x = \log |x|$.
3. For each fixed $r \in \mathbb{R}$, the function $x \mapsto x^r$ is real analytic at every $x \neq 0$; at negative x this uses the identity $x^r = |x|^r \cos(\pi r)$.
4. $\arcsin$ is real analytic at every $x \notin \{-1, 1\}$; outside $[-1, 1]$ it is locally constant.
5. $x \mapsto x^{-1}$ is real analytic at every $x \neq 0$.

Proof. The four entire generators in (1) have convergent Taylor series at every real centre. On either component of $\mathbb{R} \setminus \{0\}$ the totalized logarithm is $\log |x|$, hence is analytic, proving (2). For fixed r , on $(0, \infty)$ the power is $\exp(r \log x)$, while on $(-\infty, 0)$ it is the constant multiple $\cos(\pi r) \exp(r \log(-x))$; this proves (3), including the cases where the constant multiple vanishes. On $(-1, 1)$, $\arcsin$ is analytic because its derivative $(1 - x^2)^{-1/2}$ is analytic and it has the usual convergent local primitive; on each exterior component the chosen totalization is constant. Only ± 1 remain exceptional, proving (4). Finally $x \mapsto x^{-1}$ is represented near every nonzero centre by the convergent geometric series

$$\frac{1}{x} = \frac{1}{a} \sum_{m \geq 0} \left(-\frac{x-a}{a} \right)^m \quad (|x-a| < |a|),$$

which proves (5). □

Remark 5.2. The bad sets are sets of *values* of the inner function, not sets of points of the domain — $\{0\}$ for the reciprocal, the logarithm and a real power, $\{-1, 1\}$ for the arcsine, and $\emptyset$ for the rest. This is the shape the induction consumes, and it is why Lemma 6.2 below is stated with a finite set $B \subseteq \mathbb{R}$ of forbidden values.

6 Density of the analytic locus

6.1 Openness

Lemma 6.1. *For every $f: \mathbb{R} \rightarrow \mathbb{R}$ the set $\mathcal{A}(f)$ is open.*

Proof. Analyticity at x means that f agrees near x with the sum of a convergent power series; that series then converges on a whole ball around x and, after a change of origin, represents f near every point of that ball. So $\mathcal{A}(f)$ contains a neighbourhood of each of its points. $\square$

6.2 The unary step

The following lemma is where the argument is actually made. Recall the shape of the difficulty. If f is analytic on a dense open set and g is analytic everywhere except at finitely many values, one would like to say that $g \circ f$ is analytic wherever f is analytic and $f(x)$ avoids the bad values. That set need not be dense: consider $f \equiv 0$ and $g(y) = y^{-1}$. What saves the statement is that in that degenerate case $g \circ f$ is constant, hence analytic anyway. Separating the two cases is exactly the isolated-zeros theorem.

Lemma 6.2 (Key lemma). *Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ and let $B \subseteq \mathbb{R}$ be finite. Suppose $\mathcal{A}(f)$ is dense and g is analytic at every $y \notin B$. Then $\mathcal{A}(g \circ f)$ is dense.*

Proof. Let U be a nonempty open set; we must produce a point of $U \cap \mathcal{A}(g \circ f)$. Since $\mathcal{A}(f)$ is dense there is $x_0 \in U \cap \mathcal{A}(f)$, and by Lemma 6.1 the set $V = U \cap \mathcal{A}(f)$ is an open neighbourhood of x_0 . Distinguish two cases.

Case 1: f is constant equal to some $c \in B$ on a neighbourhood of x_0 . Then $g \circ f$ equals the constant $g(c)$ on that neighbourhood, so $g \circ f$ is analytic at x_0 itself, and $x_0 \in U$.

Case 2: for no $c \in B$ is f eventually equal to c near x_0 . Fix $c \in B$. The function $f - c$ is analytic at x_0 , so by the isolated-zeros theorem for analytic functions of one real variable, either $f - c$ vanishes identically near x_0 — excluded in this case — or $f - c$ is nonvanishing on some punctured neighbourhood of x_0 . Thus for each of the finitely many $c \in B$ there is a punctured neighbourhood of x_0 on which $f \neq c$. Intersecting these finitely many punctured neighbourhoods with the punctured neighbourhood $V \setminus \{x_0\}$ leaves a punctured neighbourhood of x_0 , which is nonempty because $\mathbb{R}$ has no isolated points. Any point t of it satisfies $t \in U$, f analytic at t , and $f(t) \notin B$; hence g is analytic at $f(t)$ and $g \circ f$ is analytic at t . $\square$

Remark 6.3. The case distinction is not an artefact of the write-up. Without it the lemma is false as stated, for the naive reason above. What makes the two cases exhaustive is the isolated-zeros theorem, applied to $f - c$: an analytic germ either vanishes identically or vanishes nowhere on some punctured neighbourhood. A merely C^∞ inner function admits neither alternative — it can equal c on a sequence accumulating at x_0 without being constant there — so analyticity of f , and not just smoothness, is what the step consumes. Within the induction of this section it is the only place where a theorem of substance about real analytic functions is used; Sections 7 and 8 call on the implicit function theorem as well.

6.3 Assembly

The complete structural induction proving Theorem 3.2 is placed beside the theorem. The present section supplies precisely its two non-formal ingredients: Lemma 6.1 makes the invariant open, while Lemma 6.2 shows that a unary constructor with finitely many exceptional input values preserves density. Together with the exceptional sets computed in Lemma 5.1, this exhausts the grammar of Definition 4.1; there is no second proof to add here.

Remark 6.4 (Sharpness). Density cannot be strengthened to “ $\mathcal{A}(f) = \mathbb{R}$ ”: the elementary functions $x \mapsto x^{-1}$, $x \mapsto |x|$ and $x \mapsto \arcsin x$ have analytic loci $\mathbb{R} \setminus \{0\}$, $\mathbb{R} \setminus \{0\}$ and $\mathbb{R} \setminus \{-1, 1\}$. Nor can “dense open” be strengthened to “open with finite complement”: the elementary function $x \mapsto (\sin x)^{-1}$ omits the infinite discrete set $\pi\mathbb{Z}$. What density does give for free is that the complement is closed and nowhere dense, being the complement of a dense open set. It need not be *discrete*, and one more reciprocal settles that: $x \mapsto (\sin(1/x))^{-1}$ is elementary — it is `IsElementary.id.inv.sin.inv`

– and its analytic locus omits every $1/(k\pi)$ with $k \in \mathbb{Z} \setminus \{0\}$, together with 0, where the function is not even continuous. Those points accumulate at 0. So “nowhere dense” is the correct description of the exceptional set and cannot be improved to “discrete”.

7 Algebraic functions over the elementary functions

Definition 4.1 reaches principal n -th roots on nonnegative inputs for positive n , and classical odd roots on all of $\mathbb{R}$, but not a general algebraic function. This section removes that restriction. It does so without enlarging $\mathcal{E}$, for a reason worth stating: an algebraic clause inside the inductive definition would cost Proposition 4.2. Composing a branch y with an elementary h produces a branch over the coefficients $a_i \circ h$, but it is a *continuous* branch only if h is continuous, and elementary functions need not be — $x \mapsto x^{-1}$ is not. Rather than weaken the closure theorem, we prove the conclusion directly.

Theorem 7.1 (Branches of polynomial equations). *Let $U \subseteq \mathbb{R}$ be open, let $a_0, \dots, a_n$ be real analytic on U with a_n nowhere zero on U , and let y be continuous on U with*

$$\sum_{i=0}^n a_i(x) y(x)^i = 0 \quad (x \in U).$$

Then y is real analytic at some point of every nonempty open subset of U .

Density is again the right conclusion, and for the same kind of reason as before: $y = |\cdot|$ is a continuous branch of $y^2 - x^2 = 0$ on all of $\mathbb{R}$, with $a_0 = -x^2$, $a_1 = 0$, $a_2 = 1$, and it is not analytic at the origin.

Proof. The implicit function theorem applies only where $\partial P/\partial y \neq 0$, and the classical way to control the locus where it vanishes is the discriminant. `Mathlib` has no usable discriminant theory — `Polynomial.discr` carries no bridge to separability, and there is no continuity of roots in the coefficients — so we induct on the degree instead, which needs neither.

Write g for $\partial P/\partial y$ evaluated along the branch and split U into $V = \{x \in U : g(x) \neq 0\}$, which is open, and $W = U \setminus \overline{V}$, which is also open. On V the implicit function theorem applies directly. On W we have $g \equiv 0$, so y satisfies the *derivative* equation, of degree $n - 1$, whose leading coefficient $n a_n$ is still nowhere zero; the inductive hypothesis applies to it. Finally V together with a dense subset of W is dense in U , because an open set disjoint from V is disjoint from $\overline{V}$ and hence contained in W . The set W absorbs the *interior* of the degenerate locus $\{g = 0\}$; what the argument gives up is the boundary $\partial V \cap U$, and that is exactly why the conclusion is density rather than analyticity everywhere. For $y = |\cdot|$ the degenerate locus is $\{0\}$, so $V = \mathbb{R} \setminus \{0\}$, $\overline{V} = \mathbb{R}$, $W = \emptyset$, and the origin is conceded to the splitting step — correctly, since $|\cdot|$ is not analytic there.

The induction terminates because the degree drops. Its base case $n = 0$ is vacuous rather than positive: the equation reads $a_0(x) = 0$ while the leading coefficient hypothesis says $a_0(x) \neq 0$, so the region is empty.

The implicit function step is `Mathlib's ContDiffAt.implicitFunction`, which is stated for every smoothness exponent in $\mathbb{N}_0 \cup \{\infty, \omega\}$; instantiating at ω and converting with `ContDiffAt.analyticAt` gives an analytic implicit function theorem. Its one nontrivial hypothesis, invertibility of the partial derivative as a continuous linear map, is discharged in one variable by `HasDerivAt.hasFDerivAt_equiv` together with uniqueness of the Fréchet derivative. $\square$

Corollary 7.2. *Let $U \subseteq [0, 1]$ be nonempty and open, let $a_0, \dots, a_n \in \mathcal{E}$, let a_n vanish nowhere on U , and let y be continuous on U with $\sum_{i=0}^n a_i(x) y(x)^i = 0$ for every $x \in U$. Then y does not agree with the Fabius function on U ; in particular not on $(0, 1)$.*

Proof. Each a_i has dense open analytic locus by Theorem 3.2. The intersection W of the $n + 1$ of them is a finite intersection of dense open sets, hence dense and open, so $U \cap W$ is nonempty and open, and Theorem 7.1 applies to it: y is analytic at some $x \in U \cap W \subseteq U$. Now argue as in Corollary 3.3. $\square$

Every hypothesis on the branch in Corollary 7.2 is confined to U , and that is what makes the corollary say what one wants it to say. A version with global hypotheses would be much weaker than it looks. For the standard Bring equation $y^5 - y - x = 0$, a global continuous branch would be a continuous right inverse of $h(w) = w^5 - w$. Such a right inverse is injective; its image is an interval, and $h(y(x)) = x$ forces that image to be unbounded in both directions, hence all of $\mathbb{R}$. It would therefore force h itself to be injective, a contradiction. The localized statement nevertheless reaches the positive real branch over $(0, 1)$, an algebraic function not expressible by radicals.

The corollary is genuinely stronger than Corollary 3.3 in that direction, and independent of it in the other: it says nothing about $\exp$ or $\log$ of an algebraic function, because it is a single algebraic step over elementary coefficients rather than a closure. A tower that interleaves algebraic extensions with exponentials and logarithms at several levels is covered only when it can be presented in one of the two forms.

8 The inverse, and why inverting cannot help

F is a homeomorphism of $[0, 1]$ onto itself, so it has a continuous inverse. We write $F^{-1}: \mathbb{R} \rightarrow \mathbb{R}$ for the formal development's clamped totalization: the inverse on $[0, 1]$, the constant 0 on $(-\infty, 0]$, and the constant 1 on $[1, \infty)$. Everything above concerned F ; this section treats that totalization, and then closes the class under the admissible inverse-branch rule.

In the formalization this function is `Fabius.fabiusInv`, defined by `Set.IccExtend` of the order isomorphism. The extension is continuous, so nothing is hidden in the gluing; its constant tails are what the endpoint half of Theorem 8.2 exploits.

8.1 A right inverse is an implicit branch

The observation that organizes this section is that no new analysis is needed. The equation

$$h(g(x)) = x$$

says exactly that g is a continuous branch of $P(x, z) = h(z) - x = 0$, an implicit equation whose partial derivative in z is h' . So the analytic implicit function theorem quoted in Section 7 — already proved there, for an unrelated purpose — delivers the analytic inverse function theorem with nothing further to check.

Theorem 8.1 (Analytic inverse function theorem). *If h is analytic at $g(x_0)$ with $h'(g(x_0)) \neq 0$, g is continuous at x_0 , and $h \circ g$ is the identity near x_0 , then g is analytic at x_0 .*

Proof. Set $H(x, z) = h(z) - x$. It is analytic near $(x_0, g(x_0))$, satisfies $H(x_0, g(x_0)) = 0$, and has $\partial_z H(x_0, g(x_0)) = h'(g(x_0)) \neq 0$. The analytic implicit-function theorem therefore gives an analytic function $\tilde{g}$ near x_0 with $h(\tilde{g}(x)) = x$ and $\tilde{g}(x_0) = g(x_0)$. After shrinking the neighbourhood, h is one-to-one near $g(x_0)$ by the ordinary inverse-function theorem; continuity of g keeps $g(x)$ in that same neighbourhood. The two right inverses must then agree, $g = \tilde{g}$, so g is analytic at x_0 . $\square$

8.2 The inverse of the Fabius function

Theorem 8.2. *F^{-1} is real analytic at no point of $[0, 1]$; its analytic locus is exactly $\mathbb{R} \setminus [0, 1]$.*

Proof. In the interior, run Theorem 8.1 backwards. If F^{-1} is analytic at $y_0 \in (0, 1)$ then F , being its inverse near that point, is analytic at $F^{-1}(y_0) \in (0, 1)$ — provided $(F^{-1})'(y_0) \neq 0$. It has no critical point anywhere in the interior. This is the one place where F' enters at all — everywhere else F is used through its continuity and through its failure to be analytic — and what is needed of it is

only that $F' > 0$ on $(0, 1)$. Given that, the easy half of the inverse function theorem – the half that assumes the inverse already exists – yields

$$(F^{-1})'(y) = \frac{1}{F'(F^{-1}(y))} \quad (0 < y < 1),$$

a nonzero real number. The core inverse module formalizes this reciprocal derivative, its strict positivity, and the stronger fact that F^{-1} is C^∞ throughout $(0, 1)$. [Theorem 3.1](#) then supplies the contradiction.

The direction deserves to be stated, because the shorter route is circular. Differentiating $F^{-1} \circ F = \text{id}$ to obtain $(F^{-1})'(F(x)) \cdot F'(x) = 1$ presupposes that F^{-1} is differentiable, which is exactly what is at issue; strict positivity of F' is what supplies it.

At the endpoints the argument is different, and simpler. F^{-1} is identically 0 to the left of 0; an analytic germ agreeing with 0 on a set with an accumulation point vanishes identically, so F^{-1} would vanish just to the right of 0 as well, contradicting strict monotonicity. Symmetrically at 1. $\square$

Corollary 8.3. *No member of $\mathcal{E}$ agrees with F^{-1} on any subset of $[0, 1]$ with nonempty interior; in particular the restriction of F^{-1} to $(0, 1)$ has no representative in this totalized expression class.*

Proof. If a member g of $\mathcal{E}$ agreed with F^{-1} on a set $U \subseteq [0, 1]$ having nonempty interior, density in [Theorem 3.2](#) would provide $y \in \text{int } U$ at which g is analytic. Equality on the neighbourhood $\text{int } U$ gives g and F^{-1} the same germ at y , making F^{-1} analytic there and contradicting [Theorem 8.2](#). $\square$

One might have hoped that inverting improves matters. F is flat to infinite order at the endpoints, so F^{-1} is steep there, and steepness is not obviously an obstruction to being elementary – the signed real cube root $x \mapsto (x/|x|)|x|^{1/3}$ is steep at the origin and perfectly elementary. [Theorem 8.2](#) says the hope is misplaced, and the reason is structural rather than quantitative: analytic functions are closed under inversion wherever the derivative survives, so an analytic inverse would hand analyticity straight back.

8.3 Closing the class under inversion

Nothing about the argument was special to F . The same reasoning shows that the invariant of [Theorem 3.2](#) survives the admissible inverse-branch rule below.

Theorem 8.4 (Dense analytic locus of a right inverse). *Let f be analytic on a dense subset of $\mathbb{R}$, and let g be continuous on a nonempty open set U with $f(g(y)) = y$ there. Then $\mathcal{A}(g) \cap U$ is relatively open and dense in U .*

Proof. Relative openness follows from [Lemma 6.1](#); we prove density. Let $I \subseteq U$ be any nonempty open interval. The right-inverse identity makes g injective on I . A continuous injective real function on an interval is strictly monotone, so $J = g(I)$ is a nonempty open interval and $g : I \rightarrow J$ is a homeomorphism. Moreover $f|_J$ is its inverse and is therefore injective.

The open set J meets the dense analytic locus of f . Choose a nonempty open interval $V \subseteq J$ on which f is analytic. Since f is injective on J , it is not constant on V . Hence the analytic function f' is not identically zero on V ; by the isolated-zero theorem, some $z \in V$ satisfies $f'(z) \neq 0$. Put $y = f(z) \in I$. Near y we have $f \circ g = \text{id}$, and $g(y) = z$; [Theorem 8.1](#) shows that g is analytic at y . Thus every nonempty open interval in U meets $\mathcal{A}(g)$, as required. $\square$

Applied to every nonempty open subset at once, this says that a continuous inverse branch is analytic on a relatively dense subset of its domain, and the domain need not be open: `Fabius.analyticDenseOn_of_rightInverse`. For a domain with empty interior the statement is vacuous, which is the correct reading – there is no open set on which to test.

The conclusion is “analytic somewhere in U ”, not “analytic on U ”, and that is not slack in the argument: the real cube root is a continuous right inverse of the entire function $x \mapsto x^3$ on all of $\mathbb{R}$ and is not analytic at the origin. In the formalization the right inverse has to be written $t \mapsto (t/|t|) |t|^{1/3}$, since `Real.rpow` makes $(-8)^{1/3} = 1$ (Warning 4.3); that this signed form really does cube back to t is `Fabius.signedRoot_pow`. Critical points of f do produce singularities of the inverse. It is the same phenomenon that made density, rather than analyticity everywhere, the right invariant in Theorem 3.2.

Definition 8.5. $\mathcal{E}^{-1}$ is the smallest class containing $\mathcal{E}$, closed under the operations of Definition 4.1, and closed under the following rule: if $f \in \mathcal{E}^{-1}$, U is open, g is continuous on U with $f(g(y)) = y$ for $y \in U$, and g is analytic on the interior of the complement of U , then $g \in \mathcal{E}^{-1}$.

Two features of the rule are deliberate, and the Lambert W function shows why each is needed. It is local in U : W satisfies its defining identity $W(y)e^{W(y)} = y$ only on $[-1/e, \infty)$, so a rule demanding a global identity would not reach it. And it asks for analyticity on the *interior* of the complement rather than at every point outside U : the branch point $-1/e$ sits on the boundary of the natural $U = (-1/e, \infty)$, and W is certainly not analytic there. Dropping that boundary costs nothing, since the boundary of an open set is nowhere dense, and for W the constant extension then satisfies the analyticity clause on all of $(-\infty, -1/e)$.

The Lean result is a conditional criterion for any supplied total function satisfying these hypotheses. Since `Mathlib` does not define W , the formalization neither constructs a standard real branch nor verifies a particular totalization of it.

That clause is a condition on the chosen extension of the branch past U , not a fact that comes for free. Extension by a constant always meets it, and that is the honest reading of the rule: a branch is admitted together with a declaration of how it is continued. Other conventions also meet it without being constant — `Mathlib` sets $\log y = \log |y|$, which is analytic off the origin. At the other extreme $U = \emptyset$ reduces the clause to “ g is entire”, so the rule also adjoins every everywhere-analytic function; harmlessly, since those already have dense analytic locus, which is all the invariant asks. Inversion may be applied at any depth: to members of $\mathcal{E}$, to branches of those, and so on without limit.

One asymmetry with $\mathcal{E}$ deserves stating, since Section 4 made a point of it. Composition is a *theorem* for $\mathcal{E}$ (Proposition 4.2), proved by inducting on the derivation of the outer function, because every clause of Definition 4.1 commutes with precomposition. That induction does not replay for $\mathcal{E}^{-1}$: the inverse clause does not commute with precomposition, since g being a branch of $f(z) = x$ says nothing about $g \circ h$ being a branch of anything. So $\mathcal{E}^{-1}$ is not claimed to be closed under composition, and Theorem 8.6 does not need it — the induction there is over the derivation, and every clause of Definition 8.5 preserves dense analyticity on its own.

Theorem 8.6. *Every member of $\mathcal{E}^{-1}$ is analytic on a dense set.*

Proof. Induction. The elementary generators are Theorem 3.2; the operations are as in Section 6; and the inverse rule splits on whether the test set meets U . Given a nonempty open B : if $B \cap U \neq \emptyset$ then Theorem 8.4 applies to the nonempty open set $B \cap U$ and produces a point of B at which g is analytic; and if $B \cap U = \emptyset$ then $B \subseteq U^c$, so $B \subseteq \text{int}(U^c)$ because B is open, and every point of B will do by the rule’s analyticity clause. $\square$

Corollary 8.7. *Neither F nor F^{-1} agrees with any member of $\mathcal{E}^{-1}$ on any subset of $[0, 1]$ with nonempty interior. In particular neither is representable on $(0, 1)$ by elementary functions, elementary operations, and any finite tower of totalized inverse branches satisfying Definition 8.5.*

Proof. Let $h \in \mathcal{E}^{-1}$ and suppose it agrees with either F or F^{-1} on a set $U \subseteq [0, 1]$ with nonempty interior. By Theorem 8.6, $\mathcal{A}(h)$ is dense, hence meets $\text{int } U$. At such a point the agreement gives equal germs, so the corresponding Fabius function would be analytic there, contradicting Theorem 3.1 or Theorem 8.2. $\square$

It is worth being explicit about what Corollary 8.7 does and does not rest on. The inverse need not be two-sided: a continuous right inverse on an open set is enough. On the other hand, the conditional Lambert- W constructor also requires the chosen totalization to be analytic on the interior of that open set's complement. The branch-domain boundary is deliberately excluded from this side condition. Because `Mathlib` does not define W , the formalization neither constructs a standard real branch nor verifies any particular totalization.

Part III

Inverse Iterates and Their Taylor Obstructions

9 Statement of the problem and strengthened results

The bounded Fabius function is the unique smooth increasing bijection $F: [0, 1] \rightarrow [0, 1]$ satisfying

$$F(0) = 0, \quad F(1 - x) = 1 - F(x), \quad F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (9.1)$$

with the usual constant continuation outside $[0, 1]$ when the bounded totalization is used. Put

$$I := F^{-1}, \quad G_n := F^{\circ n}, \quad K_n := I^{\circ n}, \quad G_0 = K_0 = \text{id}. \quad (9.2)$$

Associativity of composition gives the exact inverse identity

$$K_n = G_n^{-1}, \quad G_n = K_n^{-1}. \quad (9.3)$$

The inverse is C^∞ on $(0, 1)$, though it has singular endpoint slopes.

Theorem 9.1 (Main theorem: inverse iterates are nowhere analytic). *For every integer $n \geq 1$, the n -fold inverse iterate*

$$K_n = I^{\circ n}$$

is not real analytic at any point of $[0, 1]$.

Proof. First let $y \in (0, 1)$ and put $x = K_n(y)$. The derivative formula (17.1) gives $K'_n(y) > 0$. If K_n were analytic at y , the analytic inverse-function theorem would make its local inverse G_n analytic at x , contradicting Theorem 9.4. At 0, the quotient argument (17.8) shows that K_n has no finite right derivative, so it cannot be analytic even in the one-sided endpoint sense. Reflection, $K_n(1 - y) = 1 - K_n(y)$, gives the same contradiction at 1. Under the bounded totalization convention there is also a shorter endpoint obstruction: K_n is constant on the exterior side but strictly increasing on the interior side, which is incompatible with an analytic germ. $\square$

The proof yields a stronger statement about formal Taylor series. For a smooth function h near x , write

$$R_h(x) := \text{rad} \sum_{m \geq 0} \frac{h^{(m)}(x)}{m!} z^m$$

for the radius of convergence of its formal Taylor series; the series need not represent h even when this radius is positive.

Theorem 9.2 (Co-countable zero-radius set for inverse iterates). *For each fixed $n \geq 1$ there exists a co-countable dense set $Y_n \subset (0, 1)$ such that*

$$R_{K_n}(y) = 0, \quad \limsup_{m \rightarrow \infty} \left(\frac{|K_n^{(m)}(y)|}{m!} \right)^{1/m} = +\infty \quad (y \in Y_n). \quad (9.4)$$

More precisely, one may take $Y_n = G_n(E_n)$, where the explicit co-countable dense set E_n is defined in (16.1) from forward orbit arithmetic.

Proof. Take the co-countable dense set E_n of (16.1) and put

$$Y_n := G_n(E_n). \quad (9.5)$$

The homeomorphism $G_n : (0, 1) \rightarrow (0, 1)$ preserves density and sends the countable complement of E_n to the countable complement of Y_n . For $y = G_n(x)$ with $x \in E_n$, Theorem 9.5 gives $R_{G_n}(x) = 0$, and the formal-reversion theorem Theorem 17.1 gives $R_{K_n}(y) = 0$. The displayed limsup is then exactly the Cauchy–Hadamard formula. The spine estimate establishing the premise is proved in Section 16. $\square$

The exceptional points are genuine and instructive.

Theorem 9.3 (Affine but nonrepresenting center jet). *For every $n \geq 1$,*

$$K_n(\tfrac{1}{2}) = \tfrac{1}{2}, \quad K'_n(\tfrac{1}{2}) = 2^{-n}, \quad K_n^{(m)}(\tfrac{1}{2}) = 0 \quad (m \geq 2). \quad (9.6)$$

Thus the Taylor series of K_n at $1/2$ is an affine polynomial of infinite radius, but it does not agree with K_n on any neighborhood.

Proof. The exact centre derivatives are $F(\tfrac{1}{2}) = \tfrac{1}{2}$, $F'(\tfrac{1}{2}) = 2$, and $F^{(m)}(\tfrac{1}{2}) = 0$ for $m \geq 2$. Writing $T_{\frac{1}{2}}[G_n]$ for the formal Taylor series at the centre, formal composition gives $T_{\frac{1}{2}}[G_n](h) = \tfrac{1}{2} + 2^n h$. Its unique formal compositional inverse is $\tfrac{1}{2} + 2^{-n} h$, which proves all three derivative identities for K_n . If that affine Taylor series represented K_n on a neighbourhood, then K_n would be analytic there, contrary to Theorem 9.1. $\square$

The forward theorem needed for the proof is itself stronger than nowhere analyticity.

Theorem 9.4 (Forward engine theorem). *For every integer $n \geq 1$, $G_n = F^{\circ n}$ is not real analytic at any point of $[0, 1]$.*

Proof. For the co-countable dense set E_n in (16.1), the finite spine expansion Theorem 13.1, strict ratio unimodality Lemma 14.2, the binary-transition lemma Lemma 15.1, and the defect bound Proposition 11.2 give the dominant-spine estimate Proposition 16.1. Hence Theorem 9.5 gives zero Taylor radius at every point of E_n . Were G_n analytic at any point of $[0, 1]$, its power series would represent it throughout a neighbourhood; that neighbourhood meets the dense set E_n , a contradiction. This also covers the endpoints because every endpoint neighbourhood meets $(0, 1)$. $\square$

Theorem 9.5 (Forward co-countable zero-radius set). *Fix $n \geq 1$. There exists a co-countable dense set $E_n \subset (0, 1)$ such that, for every $x \in E_n$,*

$$\limsup_{m \rightarrow \infty} \left(\frac{|G_n^{(m)}(x)|}{m!} \right)^{1/m} = +\infty. \quad (9.7)$$

Consequently $R_{G_n}(x) = 0$ for every $x \in E_n$.

Proof. The set E_n in (16.1) is co-countable and dense because it removes only one point, a finite tie set, and finitely many homeomorphic preimages of the countable dyadic set. At $x \in E_n$, the dominant-spine lower bound Proposition 16.1 holds for infinitely many m :

$$|G_n^{(m)}(x)| \geq c(x, n) Q_m H_n(x)^m, \quad Q_m = 2^{m(m+1)/2}.$$

Since $m! \leq m^m$, taking m -th roots gives

$$\left(\frac{|G_n^{(m)}(x)|}{m!} \right)^{1/m} \geq c(x, n)^{1/m} H_n(x) \frac{2^{(m+1)/2}}{m} \rightarrow \infty \quad (9.8)$$

along that subsequence. Cauchy–Hadamard now gives $R_{G_n}(x) = 0$. $\square$

The logical order of the report is deliberate. We first establish the forward engine by a derivative-spine argument. We then transfer both nonanalyticity and zero Taylor radius through local inversion. The endpoint and center analyses are treated separately because the inverse is not a local diffeomorphism at the endpoints and because the center has a convergent but nonrepresenting Taylor series.

10 Why a direct dyadic-jet induction fails

It is useful to see the obstruction before introducing the cure. For two smooth functions f and h ,

$$(f \circ h)^{(m)}(x) = \sum_{\pi \in \Pi_m} f^{(|\pi|)}(h(x)) \prod_{B \in \pi} h^{(|B|)}(x), \quad (10.1)$$

where Π_m is the set of partitions of $\{1, \dots, m\}$. Even if $h^{(r)}(x) = 0$ for all large r , partitions with many small blocks can survive. Similarly, even if the high derivatives of f vanish at $h(x)$, the term $f'(h(x))h^{(m)}(x)$ survives. There is no pointwise “eventually zero jet” induction for F^{on} .

At the opposite extreme, a crude absolute-value bound on (10.1) introduces Bell numbers. Since the number of partitions is superexponential on an ordinary exponential scale, such a bound loses the delicate comparison with Q_m . The correct normalization must exploit the quadratic exponent in Q_m , not merely count partitions. This leads to the defect below.

11 The dyadic defect of a Faà di Bruno partition

Let $\pi \in \Pi_m$ have $k = |\pi|$ blocks of sizes $r_1, \dots, r_k$, so $r_1 + \dots + r_k = m$. Define

$$D(\pi) := \binom{m}{2} - \binom{k}{2} - \sum_{i=1}^k \binom{r_i}{2}. \quad (11.1)$$

An equivalent and more transparent formula is

$$D(\pi) = \sum_{1 \leq i < j \leq k} (r_i r_j - 1). \quad (11.2)$$

Every summand is nonnegative. Therefore $D(\pi) = 0$ only in the two extremal cases: there is one block, or every block is a singleton. These are exactly the two unbranched chain-rule spines.

Lean crosswalk: the finite positive-list defect is complete. The module `PartitionDefect.lean` has exactly three public definitions and thirty-three public theorems; the following inventory is exhaustive. Its list $r = [r_1, \dots, r_k]$ records positive block sizes, but it does not construct a set partition of $\{1, \dots, m\}$.

- The three definitions are `pairSum`, `blockPairDefect`, and `partitionDefect`.
- The unordered-pair and triangular-number engine consists of `pairSum_nil`, `pairSum_cons`, `pairSum_one`, `pairSum_add`, `pairSum_congr`, `pairSum_map`, `choose_add_two`, and `choose_list_sum_two`.
- The conventional defect formulas, excess decomposition, and fixed-block lower bound are `blockPairDefect_eq_mul_sub_one`, `partitionDefect_eq_pairSum_mul_sub_one`, `partitionDefect_nonneg`, `choose_sum_two_eq_choose_length_add_sum_add_partitionDefect`, `partitionDefect_eq_choose_sum_sub_choose_length_sub_sum`, `length_le_sum_of_pos`, `sum_map_sub_one`, `pairSum_add_eq`, `pairSum_map_add_eq`, `partitionDefect_eq_linear_add_pairwise_excess`, `partitionDefect_lower_bound`, and `partitionDefect_fixed_block_bound`.
- The zero and sharp-equality classifiers are `pairSum_eq_zero_iff_pairwise`, `blockPairDefect_eq_zero_iff`, `partitionDefect_eq_zero_iff`, `sub_one_mul_sub_one_eq_zero_iff`, `partitionDefect_eq_lower_bound_iff`, and `partitionDefect_fixed_block_eq_iff`.

- The first-positive-shell closure is `add_sub_one_le_mul_of_pos`, `mul_eq_add_sub_one_iff`, `firstShell_le_fixedBlockProduct`, `fixedBlockProduct_eq_firstShell_iff`, `firstShell_le_partitionDefect`, `partitionDefect_eq_firstShell_iff`, and `partitionDefect_twoBlock_firstShell`.

For positive entries the module proves

$$D(r) = (k-1) \sum_i (r_i - 1) + \sum_{i < j} (r_i - 1)(r_j - 1) \geq (k-1)(m-k).$$

It proves $D(r) = 0$ exactly when $k \leq 1$ or every entry is one; equality in the fixed- m , fixed- k bound exactly when every pair contains a singleton (equivalently at most one entry exceeds one); and, under precisely $2 \leq k < m$, the first-shell bound $D(r) \geq m-2$, with equality exactly at $k=2$ or $k=m-1$ together with that same one-nonsingleton condition. The explicit profile $[m-1, 1]$ attains the shell for $m \geq 3$.

This closure does *not* formalize the set-partition family Π_m , the quadratic-scale factorization (11.4), the weighted Bell sum (11.5), its asymptotics, the two-spine reduction, the finite spine expansion, or either the forward- or inverse-iterate theorem. Those steps remain paper-only.

Let $u(\pi)$ be the number of singleton blocks and let

$$s(\pi) := k - u(\pi) \tag{11.3}$$

be the number of non-singleton blocks.

Lemma 11.1 (Quadratic-scale factorization). *For every partition $\pi \in \Pi_m$,*

$$\frac{Q_k \prod_{r_i \geq 2} Q_{r_i}}{Q_m} = 2^{s(\pi) - D(\pi)}. \tag{11.4}$$

Proof. Using $\log_2 Q_r = r(r+1)/2$ and omitting singleton blocks from the product,

$$\begin{aligned} 2 \log_2 \frac{Q_k \prod_{r_i \geq 2} Q_{r_i}}{Q_m} &= k(k+1) + \sum_{r_i \geq 2} r_i(r_i+1) - m(m+1) \\ &= k^2 + s(\pi) + \sum_{r_i \geq 2} r_i^2 - m^2. \end{aligned}$$

On the other hand, expanding (11.1) gives

$$2D(\pi) = m^2 - k^2 + s(\pi) - \sum_{r_i \geq 2} r_i^2.$$

Subtracting proves (11.4). □

The next proposition is the combinatorial engine. Its role is stronger than saying that each nonextremal partition has positive defect: it shows that the sum over all of them is still exponentially small.

Proposition 11.2 (Weighted defect decay). *For each fixed $K > 0$ there exist constants $C_K > 0$ and $\rho_K \in (0, 1)$ such that*

$$\sum_{\substack{\pi \in \Pi_m \\ 1 < |\pi| < m}} (2K)^{s(\pi)} 2^{-D(\pi)} \leq C_K \rho_K^m \quad (m \geq 2). \tag{11.5}$$

Proof. We give the full counting argument because this is the point at which a naive Bell number estimate fails.

Write $k = |\pi|$, let

$$e := m - k \tag{11.6}$$

be the total block excess over singletons, and let $r_{\max} = m - t$ be the largest block size. Convexity of $r \mapsto \binom{r}{2}$ shows that, for fixed m and k , the defect is minimized by one block of size $m - k + 1$ and $k - 1$ singleton blocks. Hence

$$D(\pi) \geq e(k - 1) = e(m - e - 1). \quad (11.7)$$

The contribution between a largest block and all other blocks gives the second bound

$$D(\pi) \geq (m - t)t - (k - 1) \geq t(m - t - 1). \quad (11.8)$$

First consider partitions with $1 \leq e \leq m/4$. Their non-singleton blocks contain at most $2e$ elements, because the number of such blocks is at most e . Choosing those elements and partitioning them gives at most

$$(2e + 1)m^{2e}(2e)^{2e} \quad (11.9)$$

possibilities. Also $s(\pi) \leq e$, while (11.7) gives $D(\pi) \geq e(3m/4 - 1)$. Uniformly for $1 \leq e \leq m/4$, the base-two logarithm of the counting and weight factor is at most

$$\log_2(2e + 1) + 4e \log_2 m + e \log_2 \max(1, 2K).$$

For fixed K this is at most $em/4$ once m is large enough, whereas the negative defect exponent is at most $-e(3m/4 - 1)$. Thus the e -th contribution is at most $2^{-em/3}$ after enlarging the threshold once more, uniformly in e . The resulting geometric sum is $\mathcal{O}(2^{-m/3})$.

Next consider partitions with $1 \leq t \leq m/4$. Choose the t elements outside a largest block and partition them. Up to harmless overcounting there are at most m^{tt} choices, and $s(\pi) \leq t + 1$. By (11.8), $D(\pi) \geq t(3m/4 - 1)$. The same explicit estimate as above, now with t in place of e (and one harmless extra factor $\max(1, 2K)$), bounds the t -th contribution by $2^{-tm/3}$ for all sufficiently large m , uniformly for $1 \leq t \leq m/4$. Its sum is again exponentially small.

It remains to treat the middle region $e > m/4$ and $t > m/4$. If $t \leq 3m/4$, then (11.8) gives $D(\pi) \geq c_1 m^2$ for all sufficiently large m and an absolute $c_1 > 0$. Suppose instead that $t > 3m/4$, so every block has size less than $m/4$. If $k > m/4$, then (11.7) gives $D(\pi) \geq c_2 m^2$. If $k \leq m/4$, then

$$\begin{aligned} D(\pi) &= \sum_{i < j} r_i r_j - \binom{k}{2} \\ &= \frac{m^2 - \sum_i r_i^2}{2} - \binom{k}{2} > \frac{3m^2}{8} - \frac{m^2}{32} = \frac{11m^2}{32}. \end{aligned}$$

In the middle region, the total number of partitions is at most m^m and the weight $(2K)^{s(\pi)}$ is at most $\max(1, 2K)^m$. Both are overwhelmed by 2^{-cm^2} . Combining the three regions proves (11.5), after increasing C_K to cover the finitely many small values of m . $\square$

12 A two-spine composition lemma

The weighted defect estimate converts Faà di Bruno's formula into an asymptotically binary chain rule.

Lemma 12.1 (Two-spine reduction). *Let h be smooth near x , let f be smooth near $y = h(x)$, and put $B = |h'(x)|$. Suppose there are constants $H > 0$ and $K \geq 1$ such that*

$$|f^{(r)}(y)| \leq Q_r \quad (r \geq 1), \quad (12.1)$$

$$|h^{(r)}(x)| \leq K Q_r H^r \quad (r \geq 2). \quad (12.2)$$

Then, with $L = \max(B, H)$,

$$(f \circ h)^{(m)}(x) = f'(y)h^{(m)}(x) + f^{(m)}(y)h'(x)^m + o(Q_m L^m) \quad (m \rightarrow \infty). \quad (12.3)$$

Proof. In (10.1), remove the one-block partition and the all-singleton partition. They give exactly the two displayed terms. For any remaining partition π with k blocks, u singleton blocks, and $s = k - u$ non-singleton blocks, the corresponding term has absolute value at most

$$Q_k B^u \prod_{r_i \geq 2} (K Q_{r_i} H^{r_i}) \leq Q_m L^m (2K)^s 2^{-D(\pi)},$$

where lemma 11.1 was used in the last step. Summing and applying proposition 11.2 bounds the entire remainder by $C_K Q_m L^m \rho_K^m = o(Q_m L^m)$. $\square$

Remark 12.2 (Why this lemma is composition-stable). The estimate does not require analyticity, Gevrey bounds of factorial type, or a closed formula for every Bell polynomial. It uses only the exact quadratic dyadic scale Q_r and an ordinary exponential envelope H^r for the inner derivatives. The conclusion has the same form, so it can be iterated a fixed number of times.

13 The orbit-spine asymptotic for $F^{\circ n}$

Fix $x \in (0, 1)$ and $n \geq 1$. Define the finite orbit

$$x_j := G_j(x) = F^{\circ j}(x), \quad 0 \leq j \leq n-1, \quad (13.1)$$

and the local slopes

$$a_j := F'(x_j) > 0. \quad (13.2)$$

The prefix and suffix weights are

$$B_0 := 1, \quad B_j := \prod_{r=0}^{j-1} a_r = G'_j(x) \quad (j \geq 1), \quad (13.3)$$

$$A_{j,n} := \prod_{r=j+1}^{n-1} a_r, \quad 0 \leq j < n, \quad (13.4)$$

where empty products equal 1. Finally set

$$H_n(x) := \max_{0 \leq j < n} B_j. \quad (13.5)$$

Theorem 13.1 (Finite spine expansion). *For each fixed $n \geq 1$ and $x \in (0, 1)$,*

$$G_n^{(m)}(x) = Q_m \left[\sum_{j=0}^{n-1} A_{j,n} B_j^m \Theta(2^m x_j) + o(H_n(x)^m) \right] \quad (m \rightarrow \infty). \quad (13.6)$$

Proof. We induct on n . For $n = 1$, $x_0 = x$, $A_{0,1} = B_0 = H_1 = 1$, and (13.6) is exactly (2.7).

Assume the formula for G_j . It immediately supplies a global-in-order envelope at the fixed point x : there is $K \geq 1$ such that

$$\left| G_j^{(r)}(x) \right| \leq K Q_r H_j(x)^r \quad (r \geq 2). \quad (13.7)$$

Indeed the asymptotic gives the estimate for all sufficiently large r , and K can be enlarged to absorb the finitely many remaining orders.

Apply lemma 12.1 with $f = F$ and $h = G_j$. Since $h(x) = x_j$, $h'(x) = B_j$, $f'(x_j) = a_j$, and $f^{(m)}(x_j) = Q_m \Theta(2^m x_j)$, we obtain

$$G_{j+1}^{(m)}(x) = a_j G_j^{(m)}(x) + Q_m B_j^m \Theta(2^m x_j) + o(Q_m H_{j+1}(x)^m), \quad (13.8)$$

where $H_{j+1} = \max(H_j, B_j)$. Multiplication by a_j changes every old suffix factor $A_{i,j}$ into $A_{i,j+1}$, while the new term has suffix factor $A_{j,j+1} = 1$. This is precisely (13.6) for $j + 1$. $\square$

Remark 13.2 (Combinatorial interpretation). Each summand in (13.6) corresponds to a chain-rule tree with one high-valence vertex, located at orbit level j , and unary vertices everywhere else. All trees with two or more genuinely branching vertices are swallowed by the $o(H_n^m)$ remainder. This is why “spine” is an apt name.

14 Strict unimodality of the spine weights

The finite sum in (13.6) could still cancel if several B_j have the same maximal exponential size. The orbit dynamics show that this is exceptional.

Lemma 14.1 (Decreasing slope sequence). *Let $x \in (0, 1) \setminus \{\frac{1}{2}\}$. Then the sequence*

$$a_0, a_1, a_2, \dots \quad (14.1)$$

is strictly decreasing.

Proof. If $x < \frac{1}{2}$, then (2.13) says $x_{j+1} < x_j < \frac{1}{2}$. Since F' is strictly increasing on the left half, $a_{j+1} = F'(x_{j+1}) < F'(x_j) = a_j$. If $x > \frac{1}{2}$, then $x_{j+1} > x_j > \frac{1}{2}$ and F' is strictly decreasing on the right half, giving the same conclusion. $\square$

Lemma 14.2 (Unique or adjacent double maximum). *For $x \in (0, 1) \setminus \{\frac{1}{2}\}$, the finite positive sequence $B_0, \dots, B_{n-1}$ is strictly log-concave in the ratio sense:*

$$\frac{B_{j+1}}{B_j} = a_j \quad \text{is strictly decreasing in } j. \quad (14.2)$$

Hence its maximum is unique unless $a_k = 1$ for some $0 \leq k < n - 1$; in that case the only maximizers are the adjacent pair $B_k = B_{k+1}$.

Proof. Equation (14.2) follows from (13.3) and lemma 14.1. A positive finite sequence whose successive ratios strictly decrease first rises while the ratio exceeds 1, possibly has one flat step when the ratio equals 1, and then falls. The conclusion follows. $\square$

Because F' rises strictly from 0 to 2 on $[0, \frac{1}{2}]$, there is a unique $u \in (0, \frac{1}{2})$ with $F'(u) = 1$. Symmetry gives the only other solution $1 - u$. For a fixed n , the tie set is therefore

$$T_n := \bigcup_{\substack{k \geq 0 \\ k+1 < n}} G_k^{-1}(\{u, 1 - u\}). \quad (14.3)$$

Thus $T_1 = \emptyset$. Each G_k is a homeomorphism of $[0, 1]$, so in general T_n is finite, with at most $2(n - 1)$ points.

15 Binary transitions force recurrent large amplitudes

Let $\mathcal{D}_2$ denote the dyadic rationals in $(0, 1)$.

Lemma 15.1 (Binary-transition lemma). *If $y \in (0, 1) \setminus \mathcal{D}_2$, then there are infinitely many $m \geq 1$ such that*

$$|\Theta(2^m y)| \geq \frac{1}{2}. \quad (15.1)$$

Proof. Choose the nonterminating binary expansion

$$y = 0.b_1 b_2 b_3 \dots \quad (b_j \in \{0, 1\}).$$

Since y is not dyadic, its digits are not eventually constant: eventual zeros give a terminating expansion, while eventual ones give the alternative expansion of a dyadic rational. Hence there are infinitely many transitions $b_m \neq b_{m+1}$.

Modulo 2, the number $2^m y$ is

$$b_m + 0.b_{m+1}b_{m+2} \cdots$$

At a transition this lies in $[\frac{1}{2}, \frac{3}{2}]$. By the block formula (2.4), the centered argument of up then lies in $[-\frac{1}{2}, \frac{1}{2}]$. Since up is even and decreasing on $[0, 1]$, (2.2) gives

$$|\Theta(2^m y)| = \text{up}(\text{centered argument}) \geq \text{up}(\frac{1}{2}) = \frac{1}{2}.$$

There are infinitely many such m . □

Remark 15.2. The lemma uses only adjacent binary digit changes, not equidistribution or normality. It therefore applies to every non-dyadic point, including highly non-generic automatic, Sturmian, or sparse-digit numbers.

16 Proof of the zero-radius theorem

For fixed n , define

$$E_n := (0, 1) \setminus \left(\left\{ \frac{1}{2} \right\} \cup T_n \cup \bigcup_{j=0}^{n-1} G_j^{-1}(\mathcal{D}_2) \right). \quad (16.1)$$

Every G_j is a homeomorphism, so each preimage $G_j^{-1}(\mathcal{D}_2)$ is countable. The tie set T_n is finite. Thus E_n is co-countable and dense in $(0, 1)$.

Proposition 16.1 (Dominant-spine lower bound). *Let $x \in E_n$, and let j_* be the unique index for which $B_{j_*} = H_n(x)$. There are infinitely many m such that*

$$\left| G_n^{(m)}(x) \right| \geq c(x, n) Q_m H_n(x)^m \quad (16.2)$$

with a constant $c(x, n) > 0$ independent of m .

Proof. Because $x \in E_n$, the orbit point x_{j_*} is not dyadic. By lemma 15.1, there is an infinite set $M \subset \mathbb{N}_0$ such that

$$|\Theta(2^m x_{j_*})| \geq \frac{1}{2} \quad (m \in M). \quad (16.3)$$

For every $j \neq j_*$, put $r_j = B_j/H_n(x) < 1$. Divide (13.6) by $Q_m H_n(x)^m$. Along M , the dominant summand has absolute value at least $A_{j_*,n}/2$, while the sum of all other spines is bounded by

$$\sum_{j \neq j_*} A_{j,n} r_j^m \rightarrow 0.$$

The normalized remainder also tends to zero. Therefore, for all sufficiently large $m \in M$, the normalized absolute value is at least $A_{j_*,n}/4$. We may take $c(x, n) = A_{j_*,n}/4 > 0$. □

This proposition is the last engine needed by the unique proofs of theorems 9.4 and 9.5 given with their statements. Namely, taking m th roots yields (9.8); Cauchy–Hadamard gives zero Taylor radius on the dense set E_n , and an analytic neighbourhood cannot meet such a point.

17 The inverse-iterate theorem

17.1 Interior diffeomorphism structure

Let $I = F^{-1}$. Since $F'(x) > 0$ for $0 < x < 1$, every G_n is a strictly increasing C^∞ diffeomorphism of $(0, 1)$ onto itself, and its inverse is K_n . For $y \in (0, 1)$ and $x = K_n(y)$,

$$K'_n(y) = \frac{1}{G'_n(x)} = \prod_{j=0}^{n-1} \frac{1}{F'(F^{\circ j}(x))} > 0. \quad (17.1)$$

Equivalently, writing $y_j = I^{\circ j}(y)$,

$$K'_n(y) = \prod_{j=0}^{n-1} I'(y_j). \quad (17.2)$$

The nonzero derivative in (17.1) is the interior mechanism used in the unique proof of [theorem 9.1](#): an analytic germ for K_n would have an analytic local inverse, contrary to [theorem 9.4](#). That argument proves local nonrepresentability, but it does not say whether the formal Taylor series of K_n converges to some unrelated analytic germ. Formal reversion settles that stronger question.

17.2 Formal-radius invariance under local inversion

Theorem 17.1 (Positive convergence radius is invariant under formal reversion). *Let g and k be C^∞ local inverses near x_0 and $y_0 = g(x_0)$, with $g'(x_0) \neq 0$. Then*

$$R_g(x_0) > 0 \iff R_k(y_0) > 0. \quad (17.3)$$

Consequently $R_g(x_0) = 0$ if and only if $R_k(y_0) = 0$.

Proof. Translate the centers and write the formal Taylor series as

$$\mathsf{T}_g(z) = a_1 z + a_2 z^2 + \cdots, \quad \mathsf{T}_k(w) = b_1 w + b_2 w^2 + \cdots, \quad a_1 b_1 = 1.$$

Differentiating the smooth identities $g \circ k = \text{id}$ and $k \circ g = \text{id}$ to every finite order gives the formal composition identities

$$\mathsf{T}_g \circ \mathsf{T}_k = w, \quad \mathsf{T}_k \circ \mathsf{T}_g = z. \quad (17.4)$$

Thus the two formal Taylor series are the unique formal compositional inverses of one another.

Assume $R_k(y_0) > 0$. Then T_k defines an analytic germ with nonzero linear coefficient. By the analytic inverse-function theorem it has a convergent analytic inverse germ. The Taylor series of that inverse is the unique formal inverse of T_k , hence equals T_g . Therefore $R_g(x_0) > 0$. Interchanging g and k proves the converse. $\square$

Corollary 17.2 (Transport of the zero-versus-positive Taylor-radius locus). *For every $n \geq 1$ and $x \in (0, 1)$,*

$$R_{K_n}(G_n(x)) = 0 \iff R_{G_n}(x) = 0. \quad (17.5)$$

The same equivalence holds with “ $= 0$ ” replaced by “ > 0 ”.

Proof. At x and $G_n(x)$, the smooth germs of G_n and K_n are mutually inverse and $G'_n(x) > 0$. Applying [theorem 17.1](#) gives the equivalence for positive radius. A power-series radius is nonnegative, so negating that equivalence gives the displayed zero-radius equivalence. $\square$

Applied to the set Y_n of (9.5), the radius-transport corollary is precisely the formal-series step in the unique proof of [theorem 9.2](#). That theorem also gives an independent route to interior nowhere analyticity: an analytic point has an open analytic neighborhood, but every nonempty open interval meets the dense set Y_n on which the Taylor radius is zero.

17.3 Endpoint obstruction and iterated inverse asymptotics

There is an elementary endpoint obstruction even before using the sharp asymptotic. Since $F'(0) = 0$,

$$\frac{F(x)}{x} \longrightarrow 0 \quad (x \downarrow 0). \quad (17.6)$$

Writing $y = F(x)$ gives

$$\frac{I(y)}{y} = \frac{x}{F(x)} \longrightarrow +\infty. \quad (17.7)$$

Moreover $F(x) < x$ on $(0, 1/2)$, so $I(y) > y$ there and therefore $K_n(y) \geq I(y)$. Hence

$$\frac{K_n(y)}{y} \longrightarrow +\infty. \quad (17.8)$$

Thus K_n has no finite right derivative at 0 and cannot be analytic there. The symmetry $I(1 - y) = 1 - I(y)$, inherited by every iterate, gives the same conclusion at 1. This is the endpoint ingredient invoked in the unique proof of [theorem 9.1](#); it works under either the one-sided or totalized endpoint convention.

The repository's formally backed endpoint theorem yields substantially more. Put $L = \log 2$. Its quadratic logarithmic law is

$$-\log F(x) \sim \frac{(\log(1/x))^2}{2L} \quad (x \downarrow 0). \quad (17.9)$$

Theorem 17.3 (Iterated inverse endpoint scale). *For each fixed $n \geq 1$, as $y \downarrow 0$,*

$$-\log K_n(y) \sim c_n (\log(1/y))^{2^{-n}}, \quad c_n = (2 \log 2)^{1-2^{-n}}. \quad (17.10)$$

Consequently, for every $\alpha > 0$,

$$\frac{K_n(y)}{y^\alpha} \longrightarrow +\infty. \quad (17.11)$$

Thus K_n is not Hölder continuous of any positive order at either endpoint.

Proof. For $y \downarrow 0$, let $T_0 = \log(1/y)$ and recursively put $T_j = \log(1/K_j(y))$. If $u = I(v)$, then $v = F(u)$, so (17.9) gives

$$\log(1/u) \sim \sqrt{2L \log(1/v)}. \quad (17.12)$$

Hence $T_{j+1} \sim \sqrt{2LT_j}$. Induction gives

$$T_n \sim c_n T_0^{2^{-n}}, \quad c_{n+1} = \sqrt{2Lc_n}, \quad c_0 = 1.$$

Solving the constant recursion yields $c_n = (2L)^{1/2+1/4+\dots+1/2^n} = (2L)^{1-2^{-n}}$, proving (17.10). Since $T_n = o(T_0)$,

$$\log \frac{K_n(y)}{y^\alpha} = \alpha T_0 - T_n \longrightarrow +\infty,$$

which proves (17.11). Reflection gives the endpoint 1. $\square$

The sharper repository expansion inserts a lower-branch Lambert- W phase and a tiny period-one ripple. Iterating that full expansion should produce a hierarchy of nested Lambert phases; the coarse law above is the exact leading scale and is all that is needed for the regularity theorem.

17.4 The fixed center: a convergent Taylor series that fails to represent

The proof of [Theorem 9.3](#) was placed with its statement. Its conclusion is worth emphasizing: the entire nonlinear remainder is flat at the centre but nonzero on every punctured neighbourhood.

17.5 No local elementary representation

The repository's non-elementarity development proves that every function in its formal elementary real-variable class has a dense open analytic locus.

Corollary 17.4 (Local non-elementarity). *Fix $n \geq 1$. No elementary function in the repository's class can agree with $K_n = I^{\circ n}$ on a subset of $(0, 1)$ having nonempty interior. The same statement holds for $G_n = F^{\circ n}$.*

Proof. Agreement on an open interval with a function having a dense analytic locus would make the corresponding iterate analytic at some point of that interval, contrary to [theorems 9.1 and 9.4](#). $\square$

17.6 What formal reversion does and does not preserve

The forward lower bound on E_n has the explicit superfactorial scale

$$|G_n^{(m)}(x)| \geq cQ_m H_n(x)^m \quad \text{for infinitely many } m, \quad Q_m = 2^{m(m+1)/2}. \quad (17.13)$$

Formal reversion proves zero radius for the inverse Taylor series, but by itself it does not identify a comparable pointwise leading term for $K_n^{(m)}$. The inverse Bell polynomials contain additional cancellation. A direct inverse-spine asymptotic is therefore a genuine next problem, not an automatic corollary. The numerical section suggests that the inverse Taylor roots grow even faster at the chosen diagnostic point, but no asymptotic claim is based on those finite-order data.

18 Conjectures and research directions

The main theorem is complete, but it leaves a structured exceptional countable set on which the formal Taylor series may converge without representing the function. The formal-radius transport theorem shows that the inverse classification is exactly the homeomorphic image of the forward positive-radius locus. Of the five submitted conjecture environments, only the direct inverse-spine statement remains a manuscript conjecture. Three are retained as explicit warning records: one has a false inverse clause, one is already proved from exact quarter-point facts, and one duplicates the corrected forward report's sole live conjecture. The former nested-Lambert statement is demoted below to a precise research problem because its proposed coefficient scale, phase recursion, and uniform remainder had not been specified. Neither the surviving conjecture nor that problem has an exact proved Lean counterpart.

Warning 18.1 (Former Conjecture 14.1 has a false inverse clause). The submitted statement first proposed that the positive-radius locus of G_n is precisely its eventually-zero-jet locus, then claimed that [theorem 17.1](#) gives the same eventual-zero classification for K_n . The second clause is false: formal reversion preserves convergence, not polynomiality. At $x = 1/4$ the exact forward Taylor germ is

$$T_{1/4}[F](z) = \frac{5}{72} + z + 4z^2,$$

because $F'(1/4) = 1$, $F''(1/4) = 8$, and all higher derivatives vanish. At $y = F(1/4) = 5/72$, the inverse Taylor germ therefore solves $w = z + 4z^2$ and is

$$z = \frac{\sqrt{1 + 16w} - 1}{8},$$

an infinite convergent series. Thus $K_1 = I$ has positive Taylor radius at $5/72$ without an eventually-zero jet. The forward-only assertion remains a plausible open classification problem, especially for $n \geq 2$, but no inverse eventual-zero equivalence is asserted here.

The center is exceptional in both directions because its forward and inverse formal series are affine. At noncentral dyadic points for $n = 1$, however, finite nonlinear forward Taylor polynomials revert to infinite convergent inverse series, exactly as the quarter-point example shows.

Warning 18.2 (Former Conjecture 14.2 is already discharged). The submitted tie-cancellation statement is not a live conjecture. If $x \in T_n$, the tied adjacent weights satisfy $a_k = 1$. Strict monotonicity of F' , together with $F'(1/4) = F'(3/4) = 1$, forces $x_k \in \{1/4, 3/4\}$. Hence $x_{k+1} = F(x_k) \in \{5/72, 67/72\}$. The first orbit point is dyadic, so $\Theta(2^m x_k) = 0$ for every sufficiently large m by (2.9); the second is non-dyadic, so lemma 15.1 supplies infinitely many m with $|\Theta(2^m x_{k+1})| \geq 1/2$. Since the fixed suffix coefficient $A_{k+1,n}$ is positive, along that subsequence the surviving tied spine term has absolute value at least $(A_{k+1,n}/2)H_n(x)^m$. Thus persistent cancellation is already excluded by exact quarter-point facts and the binary-transition lemma; no additional Thue–Morse correlation conjecture remains.

Conjecture 18.3 (Direct inverse-spine asymptotic). *For $y = G_n(x)$ away from an explicit countable exceptional set, the derivatives of K_n admit an asymptotic expansion on the same universal quadratic scale Q_m ,*

$$K_n^{(m)}(y) = Q_m \left[\sum_{j=0}^{n-1} \tilde{A}_{j,n}(x) \tilde{B}_j(x)^m \Theta(2^m x_j) + o(\tilde{H}_n(x)^m) \right],$$

with computable inverse prefix weights $\tilde{B}_j$ obtained from the forward slopes and with a unique dominant term off finite tie surfaces.

The implicit identity $G_n \circ K_n = \text{id}$ and the same partition defect provide a plausible route, but one must control the recursive inverse Bell polynomials without assuming the desired envelope. A proof would upgrade abstract zero-radius transport to explicit derivative lower bounds for the inverse iterates. The finite-order replay does not establish the displayed scale, weights, exceptional set, or remainder.

Warning 18.4 (Former Conjecture 14.4 duplicates the forward report). The submitted statement that, for every fixed $z > 0$, the weighted partition sums

$$\sum_{\substack{\pi \in \Pi_m \\ 1 < |\pi| < m}} z^{s(\pi)} 2^{-D(\pi)}$$

admit a full asymptotic expansion whose leading term comes from the two first-defect shells: one block of size 2 with the rest singleton, and one block of size $m - 1$ with one singleton is exactly Conjecture 14.3 of the corrected forward-iterate report. It remains open manuscript mathematics there, with no exact Lean counterpart, but it is not a second independent inverse-iterate conjecture. A sharp version would give an explicit two-spine rate and the first correction term for both forward and prospective inverse spines.

Research problem 18.5 (Nested Lambert phase closure). *Put $y_0 = y$, $y_{r+1} = I(y_r)$ and, for $0 \leq r < n$,*

$$T_r := \log(1/y_r), \quad \rho_r := \sqrt{2T_r/L}, \quad \vartheta_r := \rho_r + \beta \pmod{1}.$$

For fixed n set $q_n := T_0^{-2^{-n}}$ and $\ell := \log T_0$. Determine whether, for every truncation order N , there are recursively computable functions $C_{n,j}$, polynomial in ℓ and real analytic and one-periodic in each of $\vartheta_0, \dots, \vartheta_{n-1}$, such that

$$T_n = c_n q_n^{-1} + \sum_{j=0}^{N-1} C_{n,j}(\ell, \vartheta_0, \dots, \vartheta_{n-1}) q_n^j + \mathcal{O}_{n,N}((1 + \ell)^{D_{n,N}} q_n^N), \quad (18.1)$$

where $c_n = (2L)^{1-2^{-n}}$, $D_{n,N}$ is finite, and the remainder constant is uniform in all phase vectors attained as $y \downarrow 0$. Here “recursively computable” is to mean the following explicit composition-and-truncation procedure. With the one-step coefficients h_j of [theorem K.1](#), form

$$\mathcal{I}_N(T) := L\rho + L\beta - \log \rho - \sum_{j=1}^N \frac{h_j(\log \rho, \rho + \beta)}{\rho^j}, \quad \rho = \sqrt{2T/L}.$$

Starting from $\widehat{T}_0 = T_0$, expand $\mathcal{I}_N(\widehat{T}_r)$ in q_n , retain all powers below q_n^N , and call the result $\widehat{T}_{r+1}$. Prove, uniformly in the attained phases, that

$$T_r - \widehat{T}_r = \mathcal{O}_{n,N} \left((1 + \ell)^{D'_{n,N}} q_n^N \right) \quad (0 \leq r \leq n),$$

including the phase-error estimates needed before each periodic h_j is composed with $\sqrt{2\widehat{T}_r/L} + \beta$. Finally, determine the first coefficient in which each ϑ_r occurs and prove that its dependence is nonconstant; this is the precise “periodic ripple is transported” claim. If the single lattice $q_n^{\mathbb{Z}}$ is not closed under the recursion, identify and prove minimality of the larger exponent semigroup.

The one-step theorem [theorem K.1](#) supplies the input to this problem, and (17.10) supplies its leading term. Neither result by itself justifies the multiscale closure, the phase substitutions, or the uniform remainder in (18.1); the interior Taylor-reversion experiment tests none of them.

18.1 Promising formalization route

A Lean formalization can be modular.

1. Define Q_m and formalize the partition defect and the exact factorization ([11.4](#)).
2. Prove a sufficiently explicit weighted defect bound for finite set partitions.
3. Use the existing derivative atlas and bounded-derivative API to formalize the two-spine reduction and the fixed- n orbit-spine induction.
4. Reuse strict monotonicity, convexity, and order-isomorphism modules to prove the finite tie-set and dense dominant-amplitude statements.
5. Finish the forward theorem with the existing power-series radius and analytic-at infrastructure.
6. Formalize the purely algebraic uniqueness of the compositional inverse of a formal power series with invertible linear coefficient.
7. Combine the analytic inverse-function theorem with formal-series convergence to prove [theorem 17.1](#) and transport E_n to Y_n .
8. Import the formal quadratic endpoint law and prove the elementary induction for [theorem 17.3](#).

The formal-reversion layer should be reusable well beyond the Fabius development.

18.2 Generalization to other atomic functions

The proof abstracts beyond the classical dyadic case. Suppose a smooth increasing map f has derivatives

$$f^{(m)}(x) = q^{\binom{m+1}{2}} \Psi(q^m x), \quad (18.2)$$

for some $q > 1$, with bounded Ψ , recurrent nonvanishing along a dense set of q -adic orbits, and monotone orbit slopes yielding a unique dominant prefix. The same partition calculation replaces $2^{-D(\pi)}$ by $q^{-D(\pi)}$. Once forward iterates are controlled, [theorem 17.1](#) automatically transfers every zero-radius statement to inverse iterates. This suggests a general theorem for invertible Rvachev-type distribution functions and multiscale refinable maps.

19 Summary of the proof mechanism

The complete argument can be compressed into the following chain.

1. Exact differentiation gives $F^{(m)} = Q_m \Theta \circ (2^m \cdot)$ with $Q_m = 2^{m(m+1)/2}$ and $|\Theta| \leq 1$.
2. In Faà di Bruno's set-partition formula, every non-spine partition has a positive quadratic defect $D(\pi)$.
3. The weighted sum of all non-spine partitions is exponentially small despite Bell-number multiplicity.
4. Hence, after division by Q_m , the derivative $G_n^{(m)}$ is asymptotic to a finite sum of orbit spines:

$$A_{j,m} B_j^m \Theta(2^m x_j), \quad 0 \leq j < n.$$

5. Away from a finite tie set, monotone orbit dynamics makes one B_j uniquely maximal; away from countably many dyadic orbit preimages, binary transitions keep its amplitude at least $1/2$ infinitely often.
6. Therefore $R_{G_n}(x) = 0$ on a co-countable dense set E_n , and G_n is nowhere analytic.
7. Smooth local inverse Taylor series are unique formal compositional inverses. Positive convergence radius is invariant under reversion.
8. Thus $R_{K_n}(y) = 0$ on the co-countable dense set $Y_n = G_n(E_n)$, and no interior point of K_n can be analytic.
9. At the endpoints, $K_n(y)/y^\alpha \rightarrow \infty$ for every $\alpha > 0$; hence endpoint analyticity is impossible. At the center, the Taylor series is affine but does not represent the function.

This proves [theorems 9.1](#) and [9.2](#) and explains why finite self-composition cannot regularize either the Fabius function or its inverse.

A Auxiliary details on the partition defect

This appendix records two identities useful for sharpening [proposition 11.2](#).

Proposition A.1 (Exact minimum at fixed block count). *Let $m \geq 1$ and $1 \leq k \leq m$. Among partitions of m with k blocks,*

$$\min D(\pi) = (k-1)(m-k). \quad (\text{A.1})$$

Equality holds exactly when one block has size $m - k + 1$ and every other block is a singleton.

Proof. For fixed m, k , formula (11.1) shows that minimizing D is equivalent to maximizing $\sum_i \binom{r_i}{2}$. Strict convexity of $r \mapsto \binom{r}{2}$ transfers one unit from a smaller nonsingleton block to a larger block and increases the sum. Iteration leaves one large block and $k-1$ singletons. Direct substitution gives (A.1). $\square$

Proposition A.2 (First defect shell). *For $m \geq 3$, the least positive defect is $m-2$. It occurs for the two endpoint forms*

$$(2, 1, \dots, 1) \quad \text{and} \quad (m-1, 1). \quad (\text{A.2})$$

which coincide in the boundary case $m=3$ and are distinct for $m \geq 4$.

Proof. From (A.1), the least positive value over $2 \leq k \leq m-1$ is

$$\min_{2 \leq k \leq m-1} (k-1)(m-k) = m-2,$$

attained only at the endpoint values $k=2$ and $k=m-1$; these are the same value when $m=3$. The equality classification in proposition A.1 gives the displayed forms and the stated distinctness. $\square$

This identifies the candidate leading correction to the spine sum. At the near-singleton end, choose one pair and leave all other elements singleton; at the near-giant end, choose the singleton outside a block of size $m-1$. Both carry the same dyadic penalty $2^{-(m-2)}$, but their derivative coefficients and orbit factors are different. A refined expansion should sum these shells before estimating the higher-defect remainder.

Part IV

Inverse-Dyadic Germs and Rvachev–Appell Deconvolution

B Bell-polynomial transfer from forward splines to quantiles

B.1 A general inverse perturbation theorem

The first new step is abstract. It is useful well beyond the Fabius function. Let $q = 1/4$ and abbreviate $\varepsilon_n = q^n$. From (1.16),

$$F_n^{\text{spl}}(x) = F(x) + \sum_{j=1}^{J-1} a_j(x) \varepsilon_n^j + \mathcal{O}(\varepsilon_n^J), \quad a_j(x) = (-1)^j b_j F^{(2j)}(x). \quad (\text{B.1})$$

Theorem B.1 (Uniform inverse-transfer expansion). *Fix $0 < \eta < 1/2$ and an integer $J \geq 1$. There exist smooth functions $c_1, \dots, c_{J-1}$ on $[\eta, 1-\eta]$ such that*

$$\boxed{G_n^{\text{spl}}(y) = Q_F(y) + \sum_{j=1}^{J-1} c_j(y) \varepsilon_n^j + \mathcal{O}_{\eta,J}(\varepsilon_n^J)} \quad (\text{B.2})$$

uniformly for $y \in [\eta, 1-\eta]$. Put $x = Q_F(y)$ and $\widehat{c}_m := m! c_m(y)$. With $\mathcal{B}_{m,k}$ denoting the partial exponential Bell polynomial and $\mathcal{B}_{0,0} = 1$, the coefficients obey

$$c_m(y) = -\frac{1}{F'(x)} \left[\frac{1}{m!} \sum_{k=2}^m F^{(k)}(x) \mathcal{B}_{m,k}(\widehat{c}_1, \dots, \widehat{c}_{m-k+1}) + \sum_{j=1}^m \frac{1}{(m-j)!} \sum_{k=0}^{m-j} a_j^{(k)}(x) \mathcal{B}_{m-j,k}(\widehat{c}_1, \dots, \widehat{c}_{m-j-k+1}) \right]. \quad (\text{B.3})$$

In the second line, the term with $m=j$ is interpreted as $a_m(x)$.

Proof. On the compact interval $K = Q_F([\eta, 1-\eta]) \Subset (0,1)$, F' has a positive minimum. The derivative version of (1.16) therefore implies $(F_n^{\text{spl}})' > 0$ on a fixed neighborhood of K for all large n , so the inverse branch G_n^{spl} exists there.

Write $x = Q_F(y)$ and use the finite ansatz

$$\delta_n^{[J]}(y) := \sum_{m=1}^{J-1} c_m(y) \varepsilon_n^m.$$

Substitute $x + \delta_n^{[J]}$ into the finite expansion (B.1). The coefficient of ε_n^m in $F(x + \delta_n^{[J]}) - F(x)$ is

$$F'(x)c_m + \frac{1}{m!} \sum_{k=2}^m F^{(k)}(x) \mathcal{B}_{m,k}(1!c_1, 2!c_2, \dots).$$

The coefficient of ε_n^{m-j} in $a_j(x + \delta_n)$ is the corresponding second Bell sum in (B.3). Solving the resulting triangular equation for c_m gives the formula. Taylor's theorem with a uniform remainder shows that $F_n^{\text{spl}}(x + \delta_n^{[J]}) = y + \mathcal{O}_{\eta,J}(\varepsilon_n^J)$. The uniform positive lower bound for $(F_n^{\text{spl}})'$ then converts this residual in the function value into $G_n^{\text{spl}}(y) = x + \delta_n^{[J]}(y) + \mathcal{O}_{\eta,J}(\varepsilon_n^J)$, proving the claimed finite-order expansion without assuming convergence of an infinite series in ε_n . $\square$

B.2 The first two quantile corrections

The first coefficient already has a simple geometric meaning: it is the logarithmic derivative of the density scale.

Corollary B.2 (Explicit coefficients through 16^{-n}). At $x = Q_F(y)$,

$$c_1(y) = \frac{1}{18} \frac{F''(x)}{F'(x)} \quad (\text{B.4})$$

and

$$c_2(y) = -\frac{31}{16200} \frac{F^{(4)}(x)}{F'(x)} + \frac{1}{324} \frac{F''(x)F'''(x)}{F'(x)^2} - \frac{1}{648} \frac{F''(x)^3}{F'(x)^3}. \quad (\text{B.5})$$

Proof. For $m = 1$, $c_1 = -a_1/F'$ and $a_1 = -b_1F''$. For $m = 2$,

$$c_2 = -\frac{a_2}{F'} + \frac{a_1 a_1'}{F'^2} - \frac{F'' a_1^2}{2F'^3}.$$

Insert $a_1 = -F''/18$ and $a_2 = 31F^{(4)}/16200$. $\square$

Remark B.3. The forward expansion is linear in the derivative tower of F , but the inverse coefficients are necessarily nonlinear rational functions of that tower. The cubic term in (B.5) is not a higher Fourier correction; it is created solely by series reversion. This distinction is why simply applying the forward Richardson formula to a numerically inverted value does not reveal the correct leading constant.

B.3 Gaussian q -binomial Richardson filters for quantiles

For $s \geq 1$ and $0 \leq j \leq s-1$, define the Lagrange weight for evaluation at zero on the geometric nodes $1, q, \dots, q^{s-1}$ by

$$\lambda_{s,j} := \prod_{\substack{0 \leq \ell \leq s-1 \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell}, \quad q = \frac{1}{4}. \quad (\text{B.6})$$

Factoring the products gives

$$\lambda_{s,j} = \frac{(-1)^{s-1-j} q^{\binom{s-j}{2}}}{(q; q)_j (q; q)_{s-1-j}} = \frac{(-1)^{s-1-j} q^{\binom{s-j}{2}}}{(q; q)_{s-1}} \begin{bmatrix} s-1 \\ j \end{bmatrix}_q. \quad (\text{B.7})$$

In addition to the usual cancellations, the complete residual moments have a closed form.

Corollary B.4 (Quarter-scale geometric moments). *For every integer $k \geq 1$,*

$$\sum_{j=0}^{s-1} \lambda_{s,j} q^{kj} = (-1)^{s-1} q^{\binom{s}{2}} \left[\begin{matrix} k-1 \\ s-1 \end{matrix} \right]_q, \quad (\text{B.8})$$

where the Gaussian binomial is zero when $k < s$. Also $\sum_j \lambda_{s,j} = 1$.

Proof. Take $p = s - 1$ in the general geometric Lagrange theorem [Theorem H.4](#). Then $\lambda_{s,j} = w_{p,j}(q)$, and [equations \(H.7\) and \(H.10\)](#) become exactly (B.8). The case $k = 0$ is the constant interpolation identity $\sum_j \lambda_{s,j} = 1$. $\square$

Define the extrapolated quantile

$$\mathcal{G}_{n,s}(y) := \sum_{j=0}^{s-1} \lambda_{s,j} G_{n+j}^{\text{spl}}(y). \quad (\text{B.9})$$

Theorem B.5 (q -binomial acceleration of finite-prefix quantiles). *Uniformly on every $[\eta, 1 - \eta] \in (0, 1)$,*

$$\mathcal{G}_{n,s}(y) = Q_F(y) + (-1)^{s-1} c_s(y) q^{\binom{s}{2}} q^{sn} + \mathcal{O}_{\eta,s}(q^{(s+1)n}). \quad (\text{B.10})$$

More generally, if the full inverse germ is inserted, the coefficient of $c_k(y) q^{kn}$ for $k \geq s$ is the Gaussian-binomial moment in (B.8).

Proof. Insert (B.2) for each depth $n + j$ and interchange the finite sums. Formula (B.8) cancels every $k < s$ term and gives the displayed leading coefficient at $k = s$. $\square$

The first filters are

$$\mathcal{G}_{n,2} = \frac{4G_{n+1}^{\text{spl}} - G_n^{\text{spl}}}{3}, \quad (\text{B.11})$$

$$\mathcal{G}_{n,3} = \frac{G_n^{\text{spl}} - 20G_{n+1}^{\text{spl}} + 64G_{n+2}^{\text{spl}}}{45}, \quad (\text{B.12})$$

$$\mathcal{G}_{n,4} = \frac{-G_n^{\text{spl}} + 84G_{n+1}^{\text{spl}} - 1344G_{n+2}^{\text{spl}} + 4096G_{n+3}^{\text{spl}}}{2835}. \quad (\text{B.13})$$

They have orders 16^{-n} , 64^{-n} , and 256^{-n} respectively.

C Exact inverse-dyadic local polynomials and algebraic quantile germs

C.1 The plateau mechanism

Put

$$\xi_r := -1 + 2^{-r} = x_r - 1, \quad a := 2^{-n}, \quad Q := a^2 = 4^{-n}. \quad (\text{C.1})$$

The corpus proves the exact Thue–Morse derivative plateau

$$p_n^{(r)}(x) = 2^{\binom{r+1}{2}} \quad \text{for almost every } x \in [\xi_r - a, \xi_r + a], \quad n \geq r + 1. \quad (\text{C.2})$$

Therefore $z \mapsto p_n(\xi_r + z)$ is an ordinary polynomial of degree at most r on $[-a, a]$. The next theorem identifies that polynomial exactly.

Let $M(t) = \mathbb{E} e^{tY}$ be the moment generating function of Y . Since $M(t) = \Phi_{\text{ang}}(it)$, (1.10) gives

$$\frac{1}{M(t)} = \sum_{j=0}^{\infty} (-1)^j b_j t^{2j}. \quad (\text{C.3})$$

Theorem C.1 (Exact reciprocal-sinc local polynomial). *For $r \geq 1$, $n \geq r + 1$, and $|z| \leq 2^{-n}$,*

$$\boxed{F_n^{\text{spl}}(x_r + z) = y_r + \mathcal{A}_r(z, Q),} \quad (\text{C.4})$$

where

$$\boxed{\mathcal{A}_r(z, Q) := J_r(z) + \sum_{j=1}^{\lfloor r/2 \rfloor} (-1)^j b_j Q^j J_r^{(2j)}(z).} \quad (\text{C.5})$$

Thus a formally infinite reciprocal-sinc operator becomes an exact finite operator on the terminating inverse-dyadic jet.

Proof. Set $q_n(z) := p_n(\xi_r + z)$ on $[-a, a]$. By (C.2), q_n is a polynomial of degree at most r . The tail factorization $Y = S_n + aY'$ with $Y' \stackrel{d}{=} Y$ gives

$$\text{up}(x) = \mathbb{E} p_n(x - aY). \quad (\text{C.6})$$

Differentiating at $x = \xi_r$ shows, for $0 \leq k \leq r$,

$$\text{up}^{(k)}(\xi_r) = \mathbb{E} q_n^{(k)}(-aY). \quad (\text{C.7})$$

The polynomial $M(-aD)q_n = M(aD)q_n$ has degree at most r ; its k th derivative at zero is the right side of (C.7). Hence, as polynomials,

$$M(aD)q_n(z) = \sum_{k=0}^r \frac{\text{up}^{(k)}(\xi_r)}{k!} z^k = y_r + J_r(z). \quad (\text{C.8})$$

Apply the inverse operator (C.3). Derivatives of order greater than r vanish, so the series truncates exactly:

$$q_n(z) = \sum_{j=0}^{\lfloor r/2 \rfloor} (-1)^j b_j a^{2j} D^{2j}(y_r + J_r(z)).$$

Finally use $F_n^{\text{spl}}(x_r + z) = p_n(\xi_r + z)$. □

Remark C.2 (Why this is stronger than the global asymptotic). The uniform expansion (1.16) has an arbitrarily high but nonzero remainder. At the plateau center, the exact polynomial argument bypasses that remainder completely. The identity is valid at finite n throughout a shrinking interval of radius 2^{-n} , not merely coefficientwise as $n \rightarrow \infty$.

C.2 The algebraic inverse branch

Theorem C.3 (Exact algebraic quantile germ). *For each fixed $r \geq 1$, there is a unique real-analytic function $\Delta_r(Q)$ near $Q = 0$ satisfying*

$$\Delta_r(0) = 0, \quad \mathcal{A}_r(\Delta_r(Q), Q) = 0. \quad (\text{C.9})$$

For all sufficiently large n ,

$$\boxed{G_n^{\text{spl}}(y_r) = x_r + \Delta_r(4^{-n}).} \quad (\text{C.10})$$

The function Δ_r is algebraic over $\mathbb{Q}(Q)$, and its Taylor coefficients are rational.

Proof. At $(z, Q) = (0, 0)$, $\mathcal{A}_r = 0$ and

$$\partial_z \mathcal{A}_r(0, 0) = J'_r(0) = F'(x_r) > 0.$$

The analytic implicit-function theorem gives a unique analytic germ Δ_r . Because (C.5) is a polynomial with rational coefficients, Δ_r is algebraic and its recursive Taylor coefficients are rational. Moreover $\Delta_r(Q) = \mathcal{O}(Q)$, so $|\Delta_r(4^{-n})| < 2^{-n}$ for all sufficiently large n . On that interval (C.4) applies, and its derivative remains positive near zero. The unique local solution of $F_n^{\text{spl}}(x) = y_r$ is therefore exactly $x_r + \Delta_r(4^{-n})$. $\square$

If

$$\Delta_r(Q) = \sum_{m \geq 1} \delta_{r,m} Q^m, \quad (\text{C.11})$$

then the coefficients can be obtained without radicals:

$$\delta_{r,m} = -\frac{1}{F'(x_r)} [Q^m] \mathcal{A}_r \left(\sum_{k=1}^{m-1} \delta_{r,k} Q^k, Q \right). \quad (\text{C.12})$$

This is a finite implicit version of (B.3). Expanding the powers of the truncated series recovers partial Bell polynomials.

C.3 The first four algebraic germs

The first jet polynomials are

$$J_1(z) = 2z, \quad (\text{C.13})$$

$$J_2(z) = z + 4z^2, \quad (\text{C.14})$$

$$J_3(z) = \frac{5}{36}z + 2z^2 + \frac{32}{3}z^3, \quad (\text{C.15})$$

$$J_4(z) = \frac{1}{144}z + \frac{5}{18}z^2 + \frac{16}{3}z^3 + \frac{128}{3}z^4. \quad (\text{C.16})$$

Consequently,

$$\mathcal{A}_1(z, Q) = 2z, \quad (\text{C.17})$$

$$\mathcal{A}_2(z, Q) = z + 4z^2 - \frac{4}{9}Q, \quad (\text{C.18})$$

$$\mathcal{A}_3(z, Q) = \frac{5}{36}z + 2z^2 + \frac{32}{3}z^3 - \frac{2}{9}Q - \frac{32}{9}Qz, \quad (\text{C.19})$$

$$\begin{aligned} \mathcal{A}_4(z, Q) = & \frac{1}{144}z + \frac{5}{18}z^2 + \frac{16}{3}z^3 + \frac{128}{3}z^4 \\ & - \frac{5}{162}Q - \frac{16}{9}Qz - \frac{256}{9}Qz^2 + \frac{3968}{2025}Q^2. \end{aligned} \quad (\text{C.20})$$

The corresponding outputs are

$$y_1 = \frac{1}{2}, \quad y_2 = \frac{5}{72}, \quad y_3 = \frac{1}{288}, \quad y_4 = \frac{143}{2073600}. \quad (\text{C.21})$$

Thus $G_n^{\text{spl}}(1/288)$ is, for large n , the small real root of the cubic (C.19), while $G_n^{\text{spl}}(143/2073600)$ is governed by the quartic (C.20). Their first coefficients are

$$\Delta_3(Q) = \frac{8}{5}Q + \frac{512}{125}Q^2 - \frac{1245184}{3125}Q^3 + \mathcal{O}(Q^4), \quad (\text{C.22})$$

$$\Delta_4(Q) = \frac{40}{9}Q + \frac{132608}{2025}Q^2 + \frac{126943232}{18225}Q^3 + \mathcal{O}(Q^4). \quad (\text{C.23})$$

C.4 The quarter anchor: an exact square root

At $r = 2$, (C.18) is quadratic and can be solved without any asymptotic step.

Theorem C.4 (Exact quarter-quantile formula). *For every $n \geq 3$,*

$$G_n^{\text{spl}}\left(\frac{5}{72}\right) = \frac{1}{4} + \frac{\sqrt{1 + \frac{64}{9}4^{-n} - 1}}{8}. \quad (\text{C.24})$$

Proof. For $r = 2$ and $n \geq 3$, the positive root

$$z_n = \frac{\sqrt{1 + (64/9)4^{-n} - 1}}{8}$$

satisfies $0 < z_n < 2^{-n}$. The exact local identity

$$F_n^{\text{spl}}(1/4 + z) = \frac{5}{72} + z + 4z^2 - \frac{4}{9}4^{-n}$$

therefore applies at $z = z_n$, and the right side equals $5/72$. Positivity of the derivative $1 + 8z$ gives uniqueness on the local branch. $\square$

The singularity of the square root is at $Q = -9/64$, so its Taylor series converges for $|Q| < 9/64$. Let $C_m = \frac{1}{m+1} \binom{2m}{m}$ be the Catalan number. Then

$$\Delta_2(Q) = \sum_{m=1}^{\infty} (-1)^{m-1} C_{m-1} \frac{2^{4m-2}}{9^m} Q^m. \quad (\text{C.25})$$

The first terms are

$$\Delta_2(Q) = \frac{4}{9}Q - \frac{64}{81}Q^2 + \frac{2048}{729}Q^3 - \frac{81920}{6561}Q^4 + \frac{3670016}{59049}Q^5 - \dots \quad (\text{C.26})$$

This is a direct and previously absent bridge from the inverse Fabius problem to Catalan numbers.

C.5 Catalan–Gaussian–binomial Richardson constants

Apply the s -level filter (B.9) to the exact quarter-quantile sequence. Combining (C.25) and (B.8) gives the full convergent residual series

$$\mathcal{G}_{n,s}\left(\frac{5}{72}\right) - \frac{1}{4} = (-1)^{s-1} q^{\binom{s}{2}} \sum_{m=s}^{\infty} \delta_{2,m} \left[\begin{matrix} m-1 \\ s-1 \end{matrix} \right]_q q^{mn}, \quad q = \frac{1}{4}. \quad (\text{C.27})$$

In particular:

Corollary C.5 (Closed leading constant at every Richardson order). *For every fixed $s \geq 1$,*

$$\mathcal{G}_{n,s}\left(\frac{5}{72}\right) - \frac{1}{4} = \frac{C_{s-1} 2^{-s^2+5s-2}}{9^s} 4^{-sn} + \mathcal{O}(4^{-(s+1)n}). \quad (\text{C.28})$$

The leading constant is positive for every s .

Proof. Use the $m = s$ term of (C.27), for which the Gaussian binomial equals one, and insert $\delta_{2,s} = (-1)^{s-1} C_{s-1} 2^{4s-2}/9^s$ and $q^{\binom{s}{2}} = 2^{-s(s-1)}$. $\square$

The first three error laws are therefore

$$G_n^{\text{spl}}(5/72) - \frac{1}{4} \sim \frac{4}{9} 4^{-n}, \quad (\text{C.29})$$

$$\mathcal{G}_{n,2}(5/72) - \frac{1}{4} \sim \frac{16}{81} 16^{-n}, \quad (\text{C.30})$$

$$\mathcal{G}_{n,3}(5/72) - \frac{1}{4} \sim \frac{32}{729} 64^{-n}. \quad (\text{C.31})$$

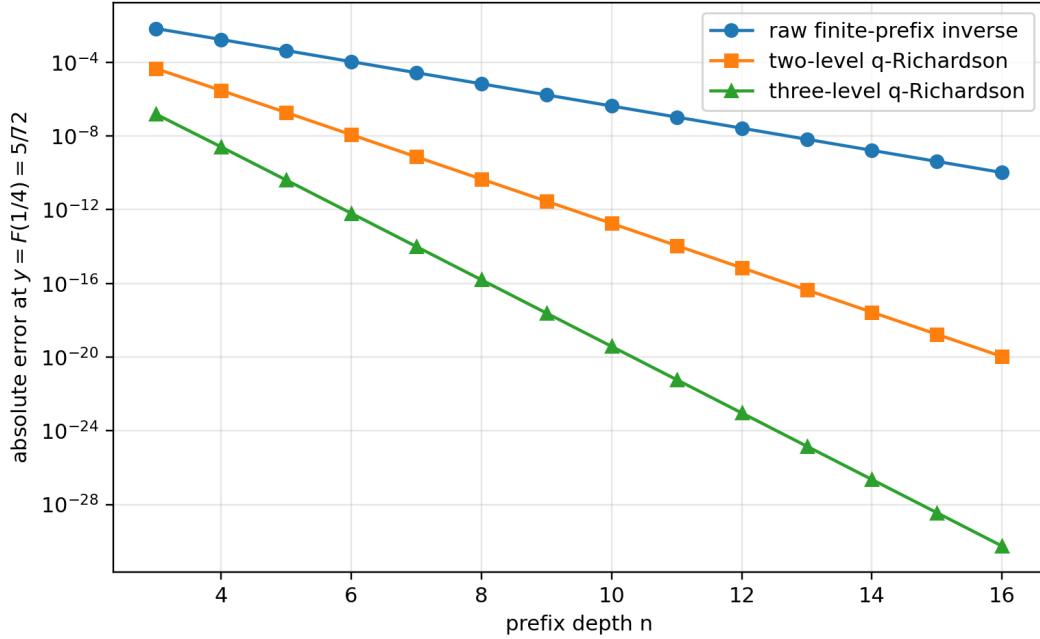


Figure 1: Raw and accelerated errors at the exact output $F(1/4) = 5/72$. The straight-line separation on the logarithmic scale reflects the exact geometric orders 4^{-n} , 16^{-n} , and 64^{-n} .

C.6 Exact sign and two-sided bounds for every accelerated quarter error

The preceding asymptotic constants can be strengthened to a finite- n interpolation statement. Put

$$f(Q) := \Delta_2(Q) = \frac{\sqrt{1 + (64/9)Q} - 1}{8}. \quad (\text{C.32})$$

For fixed n and s , the quantity $\mathcal{G}_{n,s}(5/72) - 1/4$ is the value at $Q = 0$ of the polynomial of degree at most $s - 1$ interpolating f at the nodes $4^{-n}, 4^{-(n+1)}, \dots, 4^{-(n+s-1)}$.

Theorem C.6 (Positive accelerated errors and sharp finite-depth bounds). *Let*

$$K_s := \frac{C_{s-1} 2^{-s^2+5s-2}}{9^s}. \quad (\text{C.33})$$

For every $s \geq 1$ and $n \geq 3$,

$$0 < K_s 4^{-sn} \left(1 + \frac{64}{9} 4^{-n}\right)^{\frac{1}{2}-s} < \mathcal{G}_{n,s}\left(\frac{5}{72}\right) - \frac{1}{4} < K_s 4^{-sn}. \quad (\text{C.34})$$

In particular, the ratio of the exact error to its leading term is squeezed to one inside an explicitly shrinking enclosure.

Proof. Let $p_{n,s-1}$ be the Lagrange interpolant just described. The interpolation remainder at zero gives a point $\xi_{n,s} \in (0, 4^{-n})$ such that

$$-p_{n,s-1}(0) = \frac{f^{(s)}(\xi_{n,s})}{s!} \prod_{j=0}^{s-1} (-4^{-(n+j)}). \quad (\text{C.35})$$

Now

$$f^{(s)}(Q) = \frac{1}{8} \left(\frac{1}{2}\right)^s \left(\frac{64}{9}\right)^s \left(1 + \frac{64}{9}Q\right)^{\frac{1}{2}-s}, \quad (\text{C.36})$$

where the falling factorial has sign $(-1)^{s-1}$. The signs in (C.35) therefore imply $p_{n,s-1}(0) > 0$. Using

$$\frac{|\left(\frac{1}{2}\right)^s|}{8s!} \left(\frac{64}{9}\right)^s 4^{-\binom{s}{2}} = K_s \quad (\text{C.37})$$

produces the middle expression in (C.34) with $\xi_{n,s}$ in place of either endpoint. Since the exponent $1/2 - s$ is negative, evaluation at 0 and 4^{-n} gives the two strict bounds. $\square$

Remark C.7. The positivity in [theorem C.6](#) is not visible term-by-term in (C.27), whose tail alternates. It is a consequence of the complete monotonicity pattern of the derivatives of the square-root germ and the sign of the Lagrange remainder. This is one concrete place where exact interpolation gives more information than formal Richardson cancellation.

Proposition C.8 (Large-order Catalan coefficients). *The Taylor coefficients of the quarter germ satisfy*

$$\delta_{2,m} \sim (-1)^{m-1} \frac{1}{16\sqrt{\pi}} \left(\frac{64}{9}\right)^m m^{-3/2}. \quad (\text{C.38})$$

Thus the branch point $Q = -9/64$ controls both the convergence radius and the $m^{-3/2}$ algebraic prefactor.

Proof. Insert the classical Catalan asymptotic $C_{m-1} \sim 4^{m-1}/(\sqrt{\pi} m^{3/2})$ into (C.25). $\square$

C.7 A universal first coefficient at every inverse-dyadic anchor

The leading displacement of the algebraic germ admits a particularly compact expression in the dyadic moment numbers d_r from (1.19).

Proposition C.9 (First algebraic-germ coefficient). *For every $r \geq 2$,*

$$\delta_{r,1} = \frac{1}{18} \frac{F''(2^{-r})}{F'(2^{-r})} = \frac{2^{r-1}(r-1)d_{r-2}}{9d_{r-1}} > 0. \quad (\text{C.39})$$

For $r = 1$, $\Delta_1 \equiv 0$ and $G_n^{\text{spl}}(1/2) = 1/2$ exactly.

Proof. The first equality is (B.4) evaluated at x_r . Formula (1.18) gives

$$\frac{F''(x_r)}{F'(x_r)} = 4 \frac{F(2^{2-r})}{F(2^{1-r})}.$$

Substitute (1.19) at indices $r-2$ and $r-1$ and simplify the powers of two. Positivity follows from $d_j > 0$. The case $r = 1$ follows from symmetry of every prefix density. $\square$

The values $4/9$, $8/5$, and $40/9$ in (C.26), (C.22), and (C.23) are therefore not isolated coincidences; they are the first members of a single exact rational sequence controlled by adjacent dyadic moments.

C.8 The two random series

Recall the common variables from [equation \(1.1\)](#): $(V_j)_{j \geq 1}$ are independent and uniform on $[0, 1]$, while $U_j = 2V_j - 1$ is uniform on $[-1, 1]$. Thus

$$X = \sum_{j \geq 1} 2^{-j} V_j, \quad Y = \sum_{j \geq 1} 2^{-j} U_j = 2X - 1. \quad (\text{C.40})$$

Then F is the distribution function of X and up is the density of Y . Their moment generating functions are entire:

$$L(t) = \mathbb{E}e^{tX} = \prod_{j \geq 1} \frac{e^{t/2^j} - 1}{t/2^j}, \quad (\text{C.41})$$

$$M(t) = \mathbb{E}e^{tY} = \prod_{j \geq 1} \frac{\sinh(t/2^j)}{t/2^j} = e^{-t} L(2t). \quad (\text{C.42})$$

Self-similarity gives

$$L(2t) = \frac{e^t - 1}{t} L(t), \quad M(2t) = \frac{\sinh t}{t} M(t). \quad (\text{C.43})$$

On the imaginary axis,

$$M(2\pi iz) = \prod_{k \geq 0} \text{sinc}_\pi(z/2^k) =: \Phi_{2\pi}(z), \quad \text{sinc}_\pi u = \frac{\sin(\pi u)}{\pi u}. \quad (\text{C.44})$$

Thus the moment problem and the infinite sinc product are two restrictions of the same entire function.

C.9 Moments and cumulants

Write $\kappa_n(X)$ and $\kappa_n(Y)$ for cumulants. The Bernoulli expansion of $\log((e^t - 1)/t)$ gives

$$\kappa_1(X) = \frac{1}{2}, \quad \kappa_{2r}(X) = \frac{B_{2r}}{2r(4^r - 1)}, \quad \kappa_{2r+1}(X) = 0 \quad (r \geq 1), \quad (\text{C.45})$$

$$\kappa_1(Y) = 0, \quad \kappa_{2r}(Y) = \frac{2^{2r-1} B_{2r}}{r(4^r - 1)}, \quad \kappa_{2r+1}(Y) = 0. \quad (\text{C.46})$$

The first centered moments are

$$m_0 = 1, \quad m_2 = \frac{1}{9}, \quad m_4 = \frac{19}{675}, \quad m_6 = \frac{583}{59535}, \quad m_8 = \frac{132809}{32531625}. \quad (\text{C.47})$$

These established rational sequences will become coefficients in the exact finite-prefix expansion.

D Algebraic Taylor shadows of the inverse Fabius function

D.1 The finite dyadic jet polynomial

Fix a reduced dyadic point

$$x_0 = \frac{a}{2^N} \in (0, 1), \quad a \text{ odd}, \quad y_0 = F(x_0). \quad (\text{D.1})$$

The derivative atlas [\(2.7\)](#) gives

$$f_k := F^{(k)}(x_0) = 2^{\binom{k+1}{2}} \Theta(2^k x_0), \quad Q_{x_0}(h) := \sum_{k=1}^N \frac{f_k}{k!} h^k. \quad (\text{D.2})$$

For $k > N$, the argument $2^k x_0$ is an even integer, so the block formula (2.4) gives $F^{(k)}(x_0) = 0$. Hence $y_0 + Q_{x_0}$ is the complete Taylor series of F at x_0 , not merely a finite-order truncation. The exact dyadic evaluator in [appendix J](#) shows that every f_k is rational. Moreover $f_1 = F'(x_0) > 0$.

Define the smooth remainder

$$\rho_{x_0}(h) := F(x_0 + h) - y_0 - Q_{x_0}(h). \quad (\text{D.3})$$

Every derivative of ρ_{x_0} at zero vanishes. Nevertheless ρ_{x_0} is not identically zero on any neighborhood: otherwise F would agree locally with a polynomial, contrary to [theorem 3.1](#).

Theorem D.1 (Dyadic algebraic inverse germ; FRONTIER). *There is a unique real-analytic germ S_{x_0} at zero satisfying*

$$Q_{x_0}(S_{x_0}(z)) = z, \quad S_{x_0}(0) = 0. \quad (\text{D.4})$$

It is algebraic over $\mathbb{Q}(z)$. On a neighborhood of zero,

$$\boxed{Q_F(y_0 + z) = x_0 + S_{x_0}(z) + R_{x_0}(z)}, \quad (\text{D.5})$$

where R_{x_0} is smooth, flat at zero, and nonzero on every neighborhood of zero. Consequently the complete Taylor series of Q_F at y_0 has positive radius and an algebraic sum, but that sum does not represent Q_F on any neighborhood.

Proof. Because $Q'_{x_0}(0) = f_1 > 0$, the real-analytic inverse-function theorem gives a unique analytic inverse germ S_{x_0} . It satisfies the polynomial equation $Q_{x_0}(w) - z = 0$ over $\mathbb{Q}(z)$, so it is algebraic.

Put

$$H(z) := Q_F(y_0 + z) - x_0.$$

Then $H(0) = 0$, $H'(0) = 1/f_1 > 0$, and

$$z = Q_{x_0}(H(z)) + \rho_{x_0}(H(z)). \quad (\text{D.6})$$

Faà di Bruno's formula shows directly that $\rho_{x_0} \circ H$ is flat: every term in every derivative at zero contains a derivative of ρ_{x_0} at $H(0) = 0$. Taking formal Taylor series in (D.6) therefore gives $Q_{x_0}(\mathbb{T}_H(z)) = z$. A formal series with zero constant term and nonzero linear coefficient has a unique compositional inverse; hence $\mathbb{T}_H = \mathbb{T}_{S_{x_0}}$. Thus $R_{x_0} := H - S_{x_0}$ is smooth and flat.

If R_{x_0} vanished throughout some neighborhood, then Q_F would agree there with the analytic germ $x_0 + S_{x_0}$. Its derivative at y_0 is positive, so the analytic inverse-function theorem would make F analytic at x_0 , again contradicting [theorem 3.1](#). Therefore the flat remainder is nonzero in every neighborhood. Finally, its vanishing Taylor series shows that the Taylor series of Q_F is precisely the convergent algebraic series of S_{x_0} . $\square$

Analytic shadow versus the actual inverse

At each dyadic output $F(a/2^N)$, every derivative of the inverse is encoded by a finite rational polynomial, yet the actual inverse differs from that algebraic germ by a nontrivial flat function. Positive Taylor radius and local analyticity are therefore sharply different properties here.

D.2 Lagrange–Bürmann and Bell formulas

Factor

$$Q_{x_0}(u) = u\varphi_{x_0}(u), \quad \varphi_{x_0}(0) = f_1 > 0. \quad (\text{D.7})$$

The coefficients of the algebraic inverse are completely explicit.

Theorem D.2 (All inverse derivatives at dyadic outputs; FRONTIER). *For every $m \geq 1$,*

$$\boxed{Q_F^{(m)}(y_0) = (m-1)! [u^{m-1}] \varphi_{x_0}(u)^{-m}.} \quad (\text{D.8})$$

Equivalently, if $g_m = Q_F^{(m)}(y_0)$ and $B_{m,k}$ denotes the partial exponential Bell polynomial, then

$$g_1 = \frac{1}{f_1}, \quad (\text{D.9})$$

$$g_m = -\frac{1}{f_1} \sum_{k=2}^{\min(m,N)} f_k B_{m,k}(g_1, g_2, \dots, g_{m-k+1}) \quad (m \geq 2). \quad (\text{D.10})$$

In particular, every $Q_F^{(m)}(y_0)$ is rational.

Proof. The Lagrange–Bürmann formula for the inverse of $z = u\varphi_{x_0}(u)$ gives

$$[z^m] S_{x_0}(z) = \frac{1}{m} [u^{m-1}] \varphi_{x_0}(u)^{-m}.$$

Multiplication by $m!$ proves (D.8). For the Bell recurrence, differentiate $F(Q_F(y)) = y$ m times at $y = y_0$. Faà di Bruno gives

$$\sum_{k=1}^m f_k B_{m,k}(g_1, \dots, g_{m-k+1}) = 0 \quad (m \geq 2).$$

Since $B_{m,1} = g_m$, solving for the newest derivative gives (D.10). All inputs are rational. $\square$

Remark D.3. The formulas use two different Bell-polynomial roles in the same theory. Partial Bell polynomials invert composition in (D.10); complete Bell polynomials will invert a moment generating function in [appendix E](#). The two appearances are not cosmetic variants: one organizes set partitions by the number of blocks, while the other sums over all block counts.

D.3 The midpoint and quarter-point shadows

At $x_0 = y_0 = 1/2$, reflection gives $F'(1/2) = 2$ and makes every higher derivative vanish. Thus

$$Q_{1/2}(h) = 2h, \quad S_{1/2}(z) = \frac{z}{2}, \quad Q_F(1/2 + z) = \frac{1}{2} + \frac{z}{2} + R_{1/2}(z). \quad (\text{D.11})$$

The reflection law for Q_F makes $R_{1/2}$ odd; [theorem D.1](#) makes it flat and nonzero in every neighborhood. Hence every derivative beyond the first vanishes even though the inverse is not affine on any interval.

The first nonlinear shadow has a classical coefficient sequence.

Theorem D.4 (Catalan inverse jet at $F(1/4)$; FRONTIER). *One has*

$$F(1/4) = \frac{5}{72}, \quad F'(1/4) = 1, \quad F''(1/4) = 8, \quad F^{(m)}(1/4) = 0 \quad (m \geq 3). \quad (\text{D.12})$$

Consequently

$$Q_{1/4}(h) = h + 4h^2, \quad S_{1/4}(z) = \frac{\sqrt{1+16z}-1}{8}, \quad (\text{D.13})$$

and, for every $m \geq 1$,

$$\boxed{Q_F^{(m)}(5/72) = m!(-1)^{m-1}4^{m-1}C_{m-1}, \quad C_r = \frac{1}{r+1} \binom{2r}{r}.} \quad (\text{D.14})$$

In particular,

$$\left| Q_F^{(m)}(5/72) \right| \sim \frac{m! 16^{m-1}}{\sqrt{\pi} (m-1)^{3/2}}. \quad (\text{D.15})$$

Proof. The exact dyadic formula of [appendix J](#) gives $F(1/4) = 5/72$. The derivative atlas and the values $\Theta(1/2) = F(1/2) = 1/2$ and $\Theta(1) = \text{up}(0) = 1$ give the first two derivatives in (D.12). For $m \geq 3$, the argument $2^m/4$ is an even integer, so (2.4) gives zero. Solving $z = h + 4h^2$ for the branch with $h(0) = 0$ proves (D.13). The Catalan generating function, after the substitution $w = -4z$, is

$$\frac{\sqrt{1+16z}-1}{8} = \sum_{m \geq 1} (-1)^{m-1} 4^{m-1} C_{m-1} z^m.$$

Because the flat remainder has zero derivatives, multiplying the coefficient of z^m by $m!$ proves (D.14). Finally use $C_r \sim 4^r / (\sqrt{\pi} r^{3/2})$. $\square$

For reference, the first values are

$$Q'_F(5/72) = 1, \quad Q''_F(5/72) = -8, \quad Q_F^{(3)}(5/72) = 192, \quad Q_F^{(4)}(5/72) = -7680.$$

These rapidly growing derivatives belong to the algebraic square-root shadow; they do not assert local analyticity of the actual inverse.

D.4 The eighth-point cubic and the singularity atlas

At $x_0 = 1/8$, exact dyadic evaluation and the derivative atlas give

$$F(1/8) = \frac{1}{288}, \quad F'(1/8) = \frac{5}{36}, \quad F''(1/8) = 4, \quad F^{(3)}(1/8) = 64. \quad (\text{D.16})$$

Thus the inverse Taylor series at $1/288$ is the origin branch of

$$z = \frac{5}{36}h + 2h^2 + \frac{32}{3}h^3, \quad (\text{D.17})$$

and [theorem D.2](#) specializes to

$$Q_F^{(m)}(1/288) = (m-1)! [u^{m-1}] \left(\frac{5}{36} + 2u + \frac{32}{3}u^2 \right)^{-m}. \quad (\text{D.18})$$

The finite polynomial Q_{x_0} also turns large-order derivative analysis into finite algebraic geometry. Singularities of its inverse branch occur at critical values $Q_{x_0}(u_c)$ with $Q'_{x_0}(u_c) = 0$ (or at infinity). A nearest simple critical value produces the universal square-root $m^{-3/2}$ coefficient factor seen above; higher critical multiplicity produces the corresponding Puiseux exponent.

Research problem D.5 (Dyadic inverse singularity atlas). *For every reduced dyadic $a/2^N$, classify the critical values of $Q_{a/2^N}$, determine those visible from the origin branch and hence its exact convergence radius, and describe the Galois and 2-adic structure of the associated Puiseux constants. The cases $N = 2$ and $N = 3$ are the quadratic Catalan and cubic examples above.*

E Kabaya–Iri polynomials and the Rvachev Appell inverse

E.1 The external precursor and the normalization used here

Lee’s scaling-polynomial construction associates to a compactly supported probability density ϕ the Appell sequence generated by $e^{xt}/\hat{\phi}_{\text{ang}}(it)$ and proves its biorthogonality to the distributional derivatives of ϕ . For the up-function, these are the Kabaya–Iri polynomials. In the Fabius normalization used in this report, it is convenient to distinguish the uncentered law $X \in [0, 1]$ from the centered law $Y = 2X - 1 \in [-1, 1]$.

Definition E.1 (Kabaya–Iri and centered Rvachev–Appell polynomials). Define $P_n, A_n \in \mathbb{Q}[x]$ by

$$\frac{e^{xt}}{L(t)} = \sum_{n \geq 0} P_n(x) \frac{t^n}{n!}, \quad (\text{E.1})$$

$$\frac{e^{xt}}{M(t)} = \sum_{n \geq 0} A_n(x) \frac{t^n}{n!}. \quad (\text{E.2})$$

The affine change $M(t) = e^{-t}L(2t)$ gives

$$A_n(x) = 2^n P_n\left(\frac{x+1}{2}\right), \quad P_n(x) = 2^{-n} A_n(2x-1). \quad (\text{E.3})$$

Thus the sequence P_n is the known Kabaya–Iri sequence, while A_n is its symmetric Rvachev normalization. The centering eliminates every odd cumulant and makes the pole analysis in [appendix I](#) transparent.

E.2 Appell, mean-value, and biorthogonality identities

Proposition E.2 (Classical Appell package; PRIOR). For $n \geq 1$ and, in the biorthogonality identities, every integer $r \geq 0$,

$$P'_n(x) = nP_{n-1}(x), \quad A'_n(x) = nA_{n-1}(x), \quad (\text{E.4})$$

$$P_n(x+y) = \sum_{k=0}^n \binom{n}{k} P_k(x) y^{n-k}, \quad A_n(x+y) = \sum_{k=0}^n \binom{n}{k} A_k(x) y^{n-k}. \quad (\text{E.5})$$

The polynomial averaging operators are inverted exactly:

$$\mathbb{E} P_n(x+X) = x^n, \quad \mathbb{E} A_n(x+Y) = x^n. \quad (\text{E.6})$$

If $f_X = F'$ is the density of X , then in the distributional sense

$$\left\langle (-1)^r f_X^{(r)}, \frac{P_n}{n!} \right\rangle = \delta_{rn}. \quad (\text{E.7})$$

Equivalently, for the centered density up ,

$$\int_{-1}^1 (-1)^r \text{up}^{(r)}(x) \frac{A_n(x)}{n!} dx = \delta_{rn}. \quad (\text{E.8})$$

Proof. Differentiate the generating functions with respect to x to obtain (E.4); multiply by e^{yt} to obtain (E.5). For the mean-value identities,

$$\mathbb{E} \sum_{n \geq 0} P_n(x+X) \frac{t^n}{n!} = \mathbb{E} \frac{e^{(x+X)t}}{L(t)} = e^{xt},$$

and similarly for A_n . Here is the distributional calculation, including the case in which the order of differentiation exceeds the polynomial degree. Regard f_X as a compactly supported smooth function on $\mathbb{R}$. By the definition of a distributional derivative and repeated integration by parts,

$$\left\langle (-1)^r f_X^{(r)}, \frac{P_n}{n!} \right\rangle = \left\langle f_X, \left(\frac{P_n}{n!} \right)^{(r)} \right\rangle. \quad (\text{E.9})$$

There are no boundary terms in this formulation. Iterating the Appell rule shows that the polynomial on the right is

$$\left(\frac{P_n}{n!}\right)^{(r)} = \begin{cases} P_{n-r}/(n-r)!, & 0 \leq r \leq n, \\ 0, & r > n. \end{cases}$$

For $r \leq n$, the right side of (E.9) is therefore

$$\frac{\mathbb{E}P_{n-r}(X)}{(n-r)!} = \delta_{rn},$$

where the second equality is (E.6) at $x = 0$ (the expectation is 1 for $n - r = 0$ and 0 for $n - r > 0$); for $r > n$ the pairing is zero because the test polynomial has vanished. This proves (E.7) for every $r \geq 0$.

For completeness, the centered identity is exactly the affine transport of this calculation. Since $Y = 2X - 1$,

$$\text{up}(y) = \frac{1}{2}f_X\left(\frac{y+1}{2}\right), \quad \text{up}^{(r)}(y) = 2^{-r-1}f_X^{(r)}\left(\frac{y+1}{2}\right),$$

while (E.3) gives $A_n(y) = 2^n P_n((y+1)/2)$. Thus the substitution $y = 2u - 1$ gives

$$\begin{aligned} \int_{-1}^1 (-1)^r \text{up}^{(r)}(y) \frac{A_n(y)}{n!} dy &= 2^{n-r} \left\langle (-1)^r f_X^{(r)}, \frac{P_n}{n!} \right\rangle \\ &= 2^{n-r} \delta_{rn} = \delta_{rn}, \end{aligned}$$

which is (E.8). □

Remark E.3. Equation (E.6) is an exact polynomial deconvolution formula. The operator $\mathcal{T}_Y : p \mapsto \mathbb{E}p(\cdot + Y)$ is triangular on $\mathbb{Q}[x]$ with ones on its diagonal; $A_n = \mathcal{T}_Y^{-1}x^n$. This is the linear inverse problem paired with quantile inversion in the common platform of [section 1](#).

E.3 Bell reconstruction from Bernoulli cumulants

Set

$$\alpha_n = A_n(0), \quad \frac{1}{M(t)} = \sum_{n \geq 0} \alpha_n \frac{t^n}{n!}. \quad (\text{E.10})$$

Symmetry gives $\alpha_{2r+1} = 0$. If $\mathcal{B}_n^{\text{exp}}$ denotes the complete exponential Bell polynomial, then (C.46) yields

$$\boxed{\alpha_n = \mathcal{B}_n^{\text{exp}}(-\kappa_1(Y), -\kappa_2(Y), \dots, -\kappa_n(Y))}. \quad (\text{E.11})$$

Equivalently,

$$\alpha_0 = 1, \quad \alpha_n = - \sum_{j=1}^n \binom{n-1}{j-1} \kappa_j(Y) \alpha_{n-j}. \quad (\text{E.12})$$

The first constants are

$$\alpha_0 = 1, \quad \alpha_2 = -\frac{1}{9}, \quad \alpha_4 = \frac{31}{675}, \quad \alpha_6 = -\frac{2347}{59535}, \quad \alpha_8 = \frac{1904369}{32531625}. \quad (\text{E.13})$$

The Appell translation law then gives

$$A_n(x) = \sum_{k=0}^n \binom{n}{k} \alpha_{n-k} x^k. \quad (\text{E.14})$$

For reference,

$$\begin{aligned}
A_0(x) &= 1, \\
A_1(x) &= x, \\
A_2(x) &= x^2 - \frac{1}{9}, \\
A_3(x) &= x^3 - \frac{1}{3}x, \\
A_4(x) &= x^4 - \frac{2}{3}x^2 + \frac{31}{675}, \\
A_5(x) &= x^5 - \frac{10}{9}x^3 + \frac{31}{135}x, \\
A_6(x) &= x^6 - \frac{5}{3}x^4 + \frac{31}{45}x^2 - \frac{2347}{59535}.
\end{aligned} \tag{E.15}$$

Using (E.3), the first uncentered polynomials are

$$\begin{aligned}
P_0(x) &= 1, \\
P_1(x) &= x - \frac{1}{2}, \\
P_2(x) &= x^2 - x + \frac{2}{9}, \\
P_3(x) &= x^3 - \frac{3}{2}x^2 + \frac{2}{3}x - \frac{1}{12}, \\
P_4(x) &= x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{16}{675}, \\
P_5(x) &= x^5 - \frac{5}{2}x^4 + \frac{20}{9}x^3 - \frac{5}{6}x^2 + \frac{16}{135}x - \frac{1}{270}.
\end{aligned} \tag{E.16}$$

These agree with Lee's displayed Kabaya–Iri list through degree five.

E.4 Bernoulli refinement recurrences

The self-similarity identities (C.43) become polynomial recurrences when the reciprocal generating functions are compared.

Proposition E.4 (Bernoulli refinement of the Appell sequences). *Let B_k be defined by $t/(e^t - 1) = \sum B_k t^k/k!$, and let $c_{2r} = 2(1 - 2^{2r-1})B_{2r}$ be the exponential coefficients of $t/\sinh t$. Then*

$$2^n P_n(x) = \sum_{k=0}^n \binom{n}{k} B_k P_{n-k}(2x), \tag{E.17}$$

$$2^n A_n(x) = \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} c_{2r} A_{n-2r}(2x). \tag{E.18}$$

Proof. Substitute (C.43) into the defining generating functions:

$$\begin{aligned}
\frac{e^{2xt}}{L(2t)} &= \frac{t}{e^t - 1} \frac{e^{2xt}}{L(t)}, \\
\frac{e^{2xt}}{M(2t)} &= \frac{t}{\sinh t} \frac{e^{2xt}}{M(t)}.
\end{aligned}$$

Coefficient comparison gives the two formulas. □

Remark E.5. The repository's Bernoulli recurrences usually govern moments, reciprocal sine products, or saddle coefficients. Here the same Bernoulli numbers govern a polynomial eigenbasis for the adjoint Rvachev averaging operator. The recurrence is a direct algebraic shadow of dyadic refinement.

E.5 A dyadic logistic dual

The complete Bell signs in (E.13) admit a probabilistic explanation that appears not to have been recorded for the Rvachev law.

Let Λ be a standard logistic random variable with density

$$\ell(x) = \frac{e^{-x}}{(1 + e^{-x})^2}. \quad (\text{E.19})$$

The beta integral gives

$$\mathbb{E}e^{it\Lambda} = \Gamma(1 + it)\Gamma(1 - it) = \frac{\pi t}{\sinh(\pi t)}. \quad (\text{E.20})$$

Let $(\Lambda_j)_{j \geq 1}$ be independent copies and define

$$Z = \sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j}. \quad (\text{E.21})$$

The series converges in L^2 and almost surely.

Theorem E.6 (Logistic dual of the Rvachev law; FRONTIER). *For real t ,*

$$\mathbb{E}e^{itZ} = \prod_{j \geq 1} \frac{t/2^j}{\sinh(t/2^j)} = \frac{1}{M(t)}. \quad (\text{E.22})$$

Consequently, as polynomial identities,

$$A_n(x) = \mathbb{E}(x + iZ)^n, \quad (\text{E.23})$$

$$P_n(x) = \mathbb{E}\left(x - \frac{1}{2} + \frac{iZ}{2}\right)^n. \quad (\text{E.24})$$

In particular,

$$\alpha_{2m} = (-1)^m \mathbb{E}Z^{2m}, \quad (-1)^m \alpha_{2m} > 0. \quad (\text{E.25})$$

Proof. We first record the integrability needed to turn the formal coefficient comparison into an analytic one. The logistic density satisfies $\ell(x) \leq e^{-|x|}$, so $\mathbb{E}e^{s|\Lambda|} < \infty$ for $0 \leq s < 1$. Fix $0 < a < 2\pi$ and put $s_j = a/(\pi 2^j)$. Since

$$\log \mathbb{E}e^{s|\Lambda|} = \mathcal{O}(s) \quad (s \downarrow 0),$$

independence and monotone convergence give

$$\mathbb{E} \exp\left(a \sum_{j \geq 1} \frac{|\Lambda_j|}{\pi 2^j}\right) = \prod_{j \geq 1} \mathbb{E}e^{s_j|\Lambda|} < \infty. \quad (\text{E.26})$$

In particular, the defining series for Z converges absolutely almost surely and $\mathbb{E}e^{a|Z|} < \infty$ for every $0 < a < 2\pi$.

Let $Z_N = \sum_{j=1}^N \Lambda_j/(\pi 2^j)$. Independence and (E.20) give, for real t ,

$$\mathbb{E}e^{itZ_N} = \prod_{j=1}^N \frac{t/2^j}{\sinh(t/2^j)}.$$

The almost-sure convergence $Z_N \rightarrow Z$ and bounded convergence pass the left side to $\mathbb{E}e^{itZ}$. On the right,

$$\log\left(\frac{w}{\sinh w}\right) = \mathcal{O}(w^2) \quad (w \rightarrow 0),$$

so the product converges locally uniformly for complex t in $|t| < 2\pi$; its factors are nonzero there, and its product is the reciprocal of the locally uniformly convergent product defining $M(t)$. The exponential-moment bound makes $t \mapsto \mathbb{E}e^{itZ}$ holomorphic on every strip $|\operatorname{Im} t| < a < 2\pi$. Thus the identity theorem extends the equality from real t throughout the disk $|t| < 2\pi$, and in particular proves (E.22).

It remains to justify coefficient comparison. The exponential-moment bound (E.26) makes

$$t \mapsto \mathbb{E}e^{t(x+iZ)}$$

analytic near zero. Indeed, on any smaller closed disk and for every n , the n th derivative of the integrand is dominated by

$$(|x| + |Z|)^n e^{r(|x| + |Z|)},$$

which is integrable after choosing $r < a < 2\pi$. Differentiation under the expectation is therefore valid to every order. Consequently

$$\frac{e^{xt}}{M(t)} = e^{xt} \mathbb{E}e^{itZ} = \mathbb{E}e^{t(x+iZ)} = \sum_{n \geq 0} \mathbb{E}(x + iZ)^n \frac{t^n}{n!}$$

near zero, and comparison with (E.2) proves (E.23) for every real x , hence as a polynomial identity. Substituting $2x - 1$ and using (E.3) gives (E.24). Finally Z is symmetric and nondegenerate; indeed, independence and $\operatorname{Var}(\Lambda) = \pi^2/3$ give $\operatorname{Var}(Z) = \sum_{j \geq 1} (3 \cdot 4^j)^{-1} = 1/9$. Therefore setting $x = 0$ gives $\alpha_{2m} = (-1)^m \mathbb{E}Z^{2m}$ and $\mathbb{E}Z^{2m} > 0$ (with the value 1 when $m = 0$), proving (E.25). $\square$

A duality of compact and noncompact laws

The Rvachev variable Y is compactly supported and has moment generating function M . Its polynomial deconvolution sequence is represented by imaginary moments of the noncompact variable Z , an infinite dyadic sum of logistic variables, whose characteristic function is M^{-1} . Thus $M(t)^{-1} = \Phi_{\text{ang}}(-it)^{-1}$ is not merely a formal series: on the real t -axis it is the positive-definite characteristic function of Z . This is the reciprocal *hyperbolic*-sinc product, or equivalently the analytic continuation of the reciprocal trigonometric-sinc product to the imaginary Fourier axis. The function $1/\Phi_{\text{ang}}(\xi)$ on the real Fourier axis has poles and is not a characteristic function.

E.6 Operator interpretation

Let $D = d/dx$. On polynomials,

$$\mathcal{T}_Y = \mathbb{E}e^{YD} = M(D), \quad \mathcal{T}_Y^{-1} = M(D)^{-1} = \mathbb{E}e^{iZD}. \quad (\text{E.27})$$

Therefore

$$A_n = M(D)^{-1} x^n. \quad (\text{E.28})$$

This identity packages the mean-value theorem, the logistic dual, and the Appell derivative rule in one line. It also prepares the finite Barnes and Thue–Morse difference calculus: replacing M by a finite product replaces the infinite-order differential operator by a finite product of Bernoulli operators.

F Exact distribution laws for all Rvachev derivatives

The latest formal development yields a distributional theorem stronger than any individual derivative norm or moment identity. It is logically distinct from Appell biorthogonality: the test function below is applied to the *value* of a derivative, not multiplied by an Appell polynomial in the spatial variable.

For $n \geq 0$, $0 \leq k < 2^n$, and $-1 \leq y \leq 1$, put

$$c_{n,k}(y) := -1 + \frac{2k + 1 + y}{2^n}, \quad \mathcal{D}_n := 2^{\binom{n+1}{2}},$$

and let $\text{wt}_2(k)$ be the binary digit sum.

Theorem F.1 (Cell tiling and derivative pushforward laws). *The n th derivative of the Rvachev function u satisfies*

$$u^{(n)}(c_{n,k}(y)) = \mathcal{D}_n (-1)^{\text{wt}_2(k)} u(y). \quad (\text{F.1})$$

Consequently, for every continuous real- or Banach-valued test function H ,

$$\int_{-1}^1 H(\mathcal{D}_n^{-1} |u^{(n)}(x)|) dx = \int_{-1}^1 H(u(y)) dy. \quad (\text{F.2})$$

For $n \geq 1$ the signed normalized law is the symmetric half-mixture

$$\int_{-1}^1 H(\mathcal{D}_n^{-1} u^{(n)}(x)) dx = \frac{1}{2} \int_{-1}^1 (H(u(y)) + H(-u(y))) dy. \quad (\text{F.3})$$

Equivalently, these are equalities of Borel pushforward measures under Lebesgue measure restricted to $[-1, 1]$. In particular, for $m \in \mathbb{N}_0$,

$$\int_{-1}^1 (u^{(n)}(x))^m dx = 2^{-n} (1 + (-1)^m)^n \mathcal{D}_n^m \int_{-1}^1 u(y)^m dy, \quad (\text{F.4})$$

and, for every real $p \geq 0$,

$$\int_{-1}^1 |u^{(n)}(x)|^p dx = \mathcal{D}_n^p \int_{-1}^1 u(y)^p dy. \quad (\text{F.5})$$

Proof. The derivative atlas gives exactly (F.1); the affine map $c_{n,k}$ carries $[-1, 1]$ onto the k th of the 2^n adjacent level- n cells and has Jacobian 2^{-n} . Partitioning $[-1, 1]$ into these cells and substituting $x = c_{n,k}(y)$ therefore gives

$$\int_{-1}^1 H(u^{(n)}(x)) dx = 2^{-n} \sum_{k=0}^{2^n-1} \int_{-1}^1 H(\mathcal{D}_n (-1)^{\text{wt}_2(k)} u(y)) dy.$$

Taking absolute values erases every sign, and the 2^n equal summands cancel the Jacobian, proving (F.2). For $n \geq 1$, toggling the lowest binary digit pairs the indices with opposite Thue–Morse signs; hence exactly 2^{n-1} signs are positive and 2^{n-1} are negative. This proves (F.3). Continuous test functions determine finite Borel measures on the compact range, so the integral identities imply the stated pushforward equalities. Finally take $H(t) = t^m$ and use the Boolean-cube identity

$$\sum_{k=0}^{2^n-1} ((-1)^m)^{\text{wt}_2(k)} = (1 + (-1)^m)^n$$

to obtain (F.4); take $H(t) = t^p$ in the nonnegative absolute law to obtain (F.5). $\square$

The cell identity, Banach-valued test-function formulas, both pushforward equalities, and all nonnegative real absolute moments are machine checked in `FabiusFunction.RvachevDerivativeDistribution`; the principal declarations are `iteratedDeriv_rvachev_cell`, `map_normalized_iteratedDeriv_rvachev_restrict_Icc`, and `map_normalized_abs_iteratedDeriv_rvachev_restrict_Icc`.

G Finite dyadic prefixes as Barnes–Bernoulli polynomials

G.1 Finite random sums and finite Appell inverses

For $N \geq 0$, define the convolution prefixes

$$X_N = \sum_{j=1}^N 2^{-j} V_j, \quad Y_N = \sum_{j=1}^N 2^{-j} U_j, \quad (\text{G.1})$$

with the empty-sum convention $X_0 = Y_0 = 0$. Empty products below are one, so $L_0 = M_0 = 1$ and $P_n^{[0]}(x) = A_n^{[0]}(x) = x^n$. This convention makes depth zero the identity averaging operator and will be used in the exact interpolation statements. Their moment generating functions are

$$L_N(t) = \prod_{j=1}^N \frac{e^{t/2^j} - 1}{t/2^j}, \quad (\text{G.2})$$

$$M_N(t) = \prod_{j=1}^N \frac{\sinh(t/2^j)}{t/2^j}. \quad (\text{G.3})$$

Define finite-prefix Appell polynomials by

$$\frac{e^{xt}}{L_N(t)} = \sum_{n \geq 0} P_n^{[N]}(x) \frac{t^n}{n!}, \quad (\text{G.4})$$

$$\frac{e^{xt}}{M_N(t)} = \sum_{n \geq 0} A_n^{[N]}(x) \frac{t^n}{n!}. \quad (\text{G.5})$$

They satisfy the exact finite mean-value identities

$$\mathbb{E} P_n^{[N]}(x + X_N) = x^n, \quad \mathbb{E} A_n^{[N]}(x + Y_N) = x^n. \quad (\text{G.6})$$

Because X_N and Y_N are finite convolutions of boxes, these polynomials are the exact polynomial duals of the spline approximants developed elsewhere in the repository.

G.2 Normalized Barnes multiple Bernoulli polynomials

For nonzero periods $b_1, \dots, b_N$, define the normalized Barnes multiple Bernoulli polynomials $\mathfrak{B}_n^{(N)}$ by

$$e^{xt} \prod_{j=1}^N \frac{b_j t}{e^{b_j t} - 1} = \sum_{n \geq 0} \mathfrak{B}_n^{(N)}(x \mid b_1, \dots, b_N) \frac{t^n}{n!}. \quad (\text{G.7})$$

This normalization has leading coefficient one and reduces to the ordinary Bernoulli polynomial when $N = 1$ and $b_1 = 1$.

Theorem G.1 (Dyadic Barnes identification; FRONTIER). *For every $N, n \geq 0$,*

$$\boxed{P_n^{[N]}(x) = \mathfrak{B}_n^{(N)}(x \mid 2^{-1}, 2^{-2}, \dots, 2^{-N})}. \quad (\text{G.8})$$

Let

$$s_N = 1 - 2^{-N}, \quad (a_1, \dots, a_N) = (1, 1/2, \dots, 2^{1-N}). \quad (\text{G.9})$$

Then

$$\boxed{A_n^{[N]}(x) = \mathfrak{B}_n^{(N)}\left(x + s_N \mid 1, \frac{1}{2}, \dots, 2^{1-N}\right)}. \quad (\text{G.10})$$

Proof. Equation (G.8) is immediate by comparing (G.4) with (G.7). For the centered product, write

$$\frac{\sinh(a_j t/2)}{a_j t/2} = e^{-a_j t/2} \frac{e^{a_j t} - 1}{a_j t}.$$

Since $\frac{1}{2} \sum_{j=1}^N a_j = s_N$, one has

$$\frac{e^{xt}}{M_N(t)} = e^{(x+s_N)t} \prod_{j=1}^N \frac{a_j t}{e^{a_j t} - 1},$$

which is (G.10). $\square$

Remark G.2. The period vectors are geometric rather than repeated. This is a Barnes analogue of the repository's ubiquitous $q = 1/2$ geometry: each new convolution scale appends one period to the multiple Bernoulli polynomial. The infinite Kabaya–Iri sequence is therefore a geometric-period limit of a Barnes tower.

G.3 A complete Thue–Morse difference operator

For a step h , write $E_h = e^{hD}$ for translation: $(E_h p)(x) = p(x + h)$. The complete binary signed difference operator is

$$\Delta_N^{\text{TM}}(b_1, \dots, b_N) := \prod_{j=1}^N (1 - E_{b_j}). \quad (\text{G.11})$$

Expanding the product assigns a sign $(-1)^{|J|}$ to each subset J . With geometric steps, subset sums are binary integers and the signs are precisely Thue–Morse.

Theorem G.3 (Uncentered Thue–Morse–Barnes collapse; FRONTIER). *Let $\varepsilon_k = (-1)^{\text{wt}_2(k)}$. For every $N \geq 1$ and $n \geq 0$,*

$$\sum_{k=0}^{2^N-1} \varepsilon_k P_n^{[N]} \left(x + \frac{k}{2^N} \right) = (-1)^N 2^{-N(N+1)/2} (n)_N^N x^{n-N}, \quad (\text{G.12})$$

where the right side is interpreted as zero when $n < N$.

Proof. Multiply the generating function (G.4) by the complete signed exponential sum:

$$\begin{aligned} \sum_{k=0}^{2^N-1} \varepsilon_k \frac{e^{(x+k/2^N)t}}{L_N(t)} &= \frac{e^{xt}}{L_N(t)} \prod_{j=1}^N (1 - e^{t/2^j}) \\ &= (-1)^N e^{xt} t^N \prod_{j=1}^N 2^{-j} \\ &= (-1)^N 2^{-N(N+1)/2} t^N e^{xt}. \end{aligned}$$

The coefficient of $t^n/n!$ is the stated monomial. $\square$

Corollary G.4 (Prouhet cancellation with a canonical correction). *For $n < N$, the complete Thue–Morse sum in (G.12) vanishes. For $n = N$, it is the constant*

$$(-1)^N N! 2^{-N(N+1)/2}. \quad (\text{G.13})$$

For $n > N$, it is exactly the monomial response of an N th finite difference. Thus the finite Kabaya–Iri polynomial is a canonical polynomial on which the Prouhet cancellation defect is completely diagonalized.

Proof. Specialize (G.12). The falling factorial $(n)^{\underline{N}}$ is zero for $n < N$. At $n = N$ it equals $N!$ and $x^{n-N} = 1$, which gives (G.13). When $n > N$, the same formula is a constant multiple of x^{n-N} ; this is precisely the response of the N th finite-difference operator to a degree- n Appell polynomial. $\square$

The centered form is sign-cleaner because the support is arranged symmetrically.

Theorem G.5 (Centered Thue–Morse–Rvachev collapse; FRONTIER). *With $s_N = 1 - 2^{-N}$, for every $N \geq 1$ and $n \geq 0$,*

$$\sum_{k=0}^{2^N-1} \varepsilon_k A_n^{[N]} \left(x + s_N - \frac{k}{2^{N-1}} \right) = 2^{-N(N-1)/2} (n)^{\underline{N}} x^{n-N}. \quad (\text{G.14})$$

Proof. The binary subset sums of the period vector $(1, 1/2, \dots, 2^{1-N})$ are $k/2^{N-1}$. Hence

$$\begin{aligned} \sum_{k=0}^{2^N-1} \varepsilon_k \frac{e^{(x+s_N-k/2^{N-1})t}}{M_N(t)} \\ = \frac{e^{(x+s_N)t}}{M_N(t)} \prod_{j=1}^N (1 - e^{-a_j t}). \end{aligned}$$

Now

$$\prod_{j=1}^N (1 - e^{-a_j t}) = e^{-2s_N t} \prod_{j=1}^N (e^{a_j t} - 1),$$

while

$$M_N(t)^{-1} = e^{s_N t} \prod_{j=1}^N \frac{a_j t}{e^{a_j t} - 1}.$$

All exponentials cancel, leaving $e^{xt} t^N \prod_j a_j = 2^{-N(N-1)/2} t^N e^{xt}$. Coefficient comparison proves the theorem. $\square$

Example G.6 (The first genuinely centered identity). For $N = 2$,

$$A_4^{[2]}(x) = x^4 - \frac{5}{8}x^2 + \frac{53}{1280}. \quad (\text{G.15})$$

The four-point Thue–Morse difference is

$$\begin{aligned} A_4^{[2]}(x + 3/4) - A_4^{[2]}(x + 1/4) - A_4^{[2]}(x - 1/4) + A_4^{[2]}(x - 3/4) \\ = 6x^2. \end{aligned} \quad (\text{G.16})$$

The constants and lower even powers disappear exactly, not asymptotically.

G.4 A q-binomial form of the difference operator

The Thue–Morse block polynomial

$$T_N(z) = \sum_{k=0}^{2^N-1} \varepsilon_k z^k = \prod_{r=0}^{N-1} (1 - z^{2^r}) \quad (\text{G.17})$$

encodes the same operation as (G.11). The collapse theorems can therefore be read in three interchangeable languages:

binary subset signs $\longleftrightarrow$ Thue–Morse block polynomial $\longleftrightarrow$ product of Barnes difference factors. $\quad (\text{G.18})$

This is a new bridge between the formula atlas and the finite convolution/Barnes hierarchy. It also suggests a higher-rank extension: replace the simple product of differences by the Pascal-weighted multiplicities of the repository's rank- r Rvachev hierarchy, which should produce multiple Thue–Morse–Barnes operators with repeated geometric periods.

H Exact finite-prefix expansions and q -Richardson recovery

H.1 Tail self-similarity is polynomial in the truncation scale

Let X', Y' be independent copies of X, Y , independent of the first N summands. Splitting the random series after level N gives the exact distributional identities

$$X \stackrel{d}{=} X_N + 2^{-N} X', \quad Y \stackrel{d}{=} Y_N + 2^{-N} Y'. \quad (\text{H.1})$$

Hence

$$L(t) = L_N(t)L(2^{-N}t), \quad M(t) = M_N(t)M(2^{-N}t). \quad (\text{H.2})$$

Write $\mu_r = \mathbb{E}X^r$ and $m_{2r} = \mathbb{E}Y^{2r}$.

Theorem H.1 (Exact finite-prefix expansion; FRONTIER). *For every $N, n \geq 0$,*

$$P_n^{[N]}(x) = \sum_{r=0}^n \binom{n}{r} \mu_r 2^{-Nr} P_{n-r}(x). \quad (\text{H.3})$$

For the centered sequence,

$$A_n^{[N]}(x) = \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} m_{2r} 4^{-Nr} A_{n-2r}(x). \quad (\text{H.4})$$

Equivalently, before specializing x , $P_n^{[N]}$ is exactly a $\mathbb{Q}[x]$ -valued polynomial of degree n in 2^{-N} , while $A_n^{[N]}$ is exactly a $\mathbb{Q}[x]$ -valued polynomial of degree $\lfloor n/2 \rfloor$ in 4^{-N} . The uncentered degree remains exact after a fixed- x specialization, but the centered degree can drop: for odd n , the leading coefficient contains $A_1(x) = x$ and vanishes at $x = 0$.

Proof. From (H.2),

$$\begin{aligned} \frac{e^{xt}}{L_N(t)} &= L(2^{-N}t) \frac{e^{xt}}{L(t)}, \\ \frac{e^{xt}}{M_N(t)} &= M(2^{-N}t) \frac{e^{xt}}{M(t)}. \end{aligned}$$

Expand the first extra factor in moments and compare exponential generating coefficients. In the centered case all odd moments vanish, so only powers $2^{-2Nr} = 4^{-Nr}$ survive. $\square$

Lean correspondence. The eleven-definition, seventeen-theorem module `FabiusFunction.FinitePrefixAppellRecovery` proves both displayed identities for every $N, n \in \mathbb{N}$, including $N = 0$, in exact rational polynomial arithmetic. The direct declarations are `Fabius.uncenteredDyadicPrefixAppellPolynomialRat_eq_sum` and `Fabius.centeredDyadicPrefixAppellPolynomialRat_eq_sum_even`; their coefficients identify the manuscript's μ_r with `halfMomentr` and m_{2r} with `momentr`. The free-scale definitions `Fabius.uncenteredDyadicPrefixAppellScalePolynomialRat` and `Fabius.centeredDyadicPrefixAppellScalePolynomialRat`,

the bridges `Fabius.uncenteredDyadicPrefixAppellPolynomialRat_eq_eval_scale` and `Fabius.centeredDyadicPrefixAppellPolynomialRat_eq_eval_scale`, and `Fabius.natDegree_uncenteredDyadicPrefixAppellScalePolynomialRat` and `Fabius.natDegree_centeredDyadicPrefixAppellScalePolynomialRat` formalize exact outer degrees n and $\lfloor n/2 \rfloor$ over the coefficient ring $\mathbb{Q}[x]$. This is the symbolic degree statement above, not the false assertion that every fixed- x centered specialization retains its leading coefficient. The prefix moment sequences are constructed noncircularly by finite binomial convolution. The module does not additionally realize them as random variables or supply a HasLaw or analytic-MGF theorem; those probabilistic identities motivate, but are not hypotheses of, the polynomial conclusion certified here.

Remark H.2. Earlier extrapolation results in the repository accelerate an infinite sinc product or extract periodic endpoint coefficients from asymptotic series. Here there is no asymptotic remainder at all. For each fixed polynomial degree, the dependence on the prefix depth is finite and algebraic. Richardson extrapolation becomes exact interpolation at the missing node 0.

H.2 Geometric Lagrange weights

Fix $0 < q < 1$ and an order $p \geq 0$. The Lagrange weights that evaluate a degree- p polynomial at zero from its values at $1, q, \dots, q^p$ are

$$w_{p,j}(q) = \prod_{\substack{0 \leq \ell \leq p \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell} \quad (0 \leq j \leq p). \quad (\text{H.5})$$

They have the Gaussian-binomial form

$$w_{p,j}(q) = \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_j (q; q)_{p-j}} = \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_p} \begin{bmatrix} p \\ j \end{bmatrix}_q. \quad (\text{H.6})$$

The interpolation moments are

$$\sum_{j=0}^p w_{p,j}(q) q^{jr} = \begin{cases} 1, & r = 0, \\ 0, & 1 \leq r \leq p. \end{cases} \quad (\text{H.7})$$

Lean boundary for the weights. The source-level individual quotient is already represented by `geometricLagrangeWeight_eq_gaussianBinomial_div` in `GeometricQBinomialLagrange.lean`, over a field and under injectivity of $j \mapsto q^j$ on $\{0, \dots, p\}$. Its numerator deliberately uses `gaussianBinomialqp(p-j)`; the symmetric-index notation in (H.6) is the equivalent paper convention, not a newly added generic symmetry theorem. This closed-weight API predates the principal-specialization and generic-residual tranche described below. In the present real range $0 < q < 1$, its finite-node hypothesis is automatic.

Theorem H.3 (Exact q -Richardson recovery; FRONTIER). *For every $n, N \geq 0$,*

$$P_n(x) = \sum_{j=0}^n w_{n,j}(1/2) P_n^{[N+j]}(x). \quad (\text{H.8})$$

Let $d = \lfloor n/2 \rfloor$. Then

$$A_n(x) = \sum_{j=0}^d w_{d,j}(1/4) A_n^{[N+j]}(x). \quad (\text{H.9})$$

Both identities hold for every starting depth N , not merely in a limit.

Proof. In (H.3), set $z = 2^{-N}$. The right side is a polynomial in z of degree at most n , and its value at $z = 0$ is $P_n(x)$. The values at depths $N + j$ are the values at zq^j with $q = 1/2$. Lagrange interpolation at zero is scale-invariant, so the weights are $w_{n,j}(1/2)$. The centered proof is identical with $z = 4^{-N}$, degree d , and $q = 1/4$. $\square$

Lean correspondence. The direct declarations `Fabius.kabayaIriAppellPolynomialRat_eq_sum_prefix` and `Fabius.rvachevAppellPolynomialRat_eq_sum_prefix` prove these identities for all $N, n \in \mathbb{N}$, including the empty starting prefix $N = 0$. They use exactly $n + 1$ consecutive prefixes at geometric base $1/2$ in the uncentered case and $\lfloor n/2 \rfloor + 1$ consecutive prefixes at base $1/4$ in the centered case. The conclusions are finite polynomial identities, not limits. As above, their finite-prefix inputs are the algebraic binomial-convolution moment model; no random-variable realization is inferred.

Richardson without an error term

For fixed degree n , the infinite Rvachev-Appell polynomial is a finite linear combination of the Barnes polynomials associated with any $d + 1$ consecutive dyadic convolution prefixes. The outer weights are Gaussian q -binomial coefficients. The construction therefore joins four structures that the repository previously treated in separate contexts:

finite splines $\longleftrightarrow$ Barnes periods $\longleftrightarrow$ q -Lagrange interpolation $\longleftrightarrow$ the infinite Rvachev law.

H.3 Residual moments beyond the exact degree

For later perturbations it is useful to know every nonvanishing geometric moment of the weights. The following is the common engine behind all quarter-scale, half-scale, and scaled Richardson formulas in this volume.

Theorem H.4 (Complete geometric Lagrange residual moments). *Let h_r be the complete homogeneous symmetric polynomial. For $r \geq p + 1$,*

$$\sum_{j=0}^p w_{p,j}(q) q^{jr} = (-1)^p q^{\binom{p+1}{2}} h_{r-p-1}(1, q, \dots, q^p) = (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} r-1 \\ p \end{bmatrix}_q. \quad (\text{H.10})$$

Together with (H.7), this determines every geometric moment of the weights.

Proof. Set $S(z) = \sum_{j=0}^p w_{p,j}(q)/(1 - q^j z)$ and $D(z) = \prod_{j=0}^p (1 - q^j z)$. The numerator of $S(z) - 1$ over the denominator $D(z)$ has degree at most $p + 1$. By Lagrange exactness, the first $p + 1$ Taylor coefficients of $S(z) - 1$ vanish. Hence that numerator is a constant multiple of z^{p+1} . Since the numerator of S has degree at most p , comparison with the leading coefficient of $-D$ gives the constant $(-1)^p q^{\binom{p+1}{2}}$. Therefore

$$S(z) = 1 + (-1)^p q^{\binom{p+1}{2}} \frac{z^{p+1}}{\prod_{j=0}^p (1 - q^j z)}.$$

Finally $D(z)^{-1} = \sum_{s \geq 0} h_s(1, q, \dots, q^p) z^s$, and the finite principal specialization of the complete homogeneous polynomial is $h_s(1, q, \dots, q^p) = \begin{bmatrix} s+p \\ p \end{bmatrix}_q$. Coefficient comparison with $s = r - p - 1$ proves (H.10). $\square$

This identity is the exact analogue of the transformed-moment formulas used in classical Richardson error analysis. Here it also quantifies what happens when a numerical or symbolic approximation replaces the exact prefix polynomial.

Post-snapshot Lean status. At source checkpoint 1e761ed77583e9870ceeaeb2c74ac824094f0f3f, `completeHomogeneousEval_geometric` proves

$$h_s(1, q, \dots, q^p) = \left[\begin{matrix} s+p \\ p \end{matrix} \right]_q$$

over every commutative semiring, without division or a distinct-node hypothesis. Its literal right-hand side is the recursive term `gaussianBinomialq(s+p)p`, which agrees with the displayed quotient in the present field setting. The Lagrange formula (H.10) is the surviving range of `sum_geometricLagrangeWeight_mul_pow_of_pos`; its field hypothesis and finite-node injectivity are exactly those needed to construct the weights. The generic theorem `sum_weight_mul_geometric_pow_of_pos` assumes only the low moments themselves and therefore permits repeated nodes over an arbitrary commutative ring.

There is also a denominator-free scaled exact-row form. If weights $a_0, \dots, a_p$ over a commutative ring satisfy

$$\sum_{j=0}^p a_j (cq^j)^d = 0^d \quad (0 \leq d \leq p),$$

then `sum_weight_mul_scaled_geometric_pow_succ_add` gives, for $s \geq 0$,

$$\sum_{j=0}^p a_j (cq^j)^{p+1+s} = c^{p+1+s} (-1)^p q^{\binom{p+1}{2}} \left[\begin{matrix} p+s \\ p \end{matrix} \right]_q.$$

Exactness is assumed directly after scaling, so c need not be a unit or even nonzero. These residual declarations alone discharge the finite principal-specialization and residual-moment algebra; they do not imply the prefix-polynomial identifications or [theorem H.3](#). Those identities are now formalized separately, with their algebraic-model boundary, in `FabiusFunction.FinitePrefixAppellRecovery`.

H.4 Worked centered recovery in degree six

For $A_6(0) = \alpha_6$, the exact prefix constants at depths 1, 2, 3, 4 are

$$-\frac{31}{1344}, \quad -\frac{2995}{86016}, \quad -\frac{210499}{5505024}, \quad -\frac{13784195}{352321536}. \quad (\text{H.11})$$

Since $d = 3$, the $q = 1/4$ weights are

$$\left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835} \right). \quad (\text{H.12})$$

Their exact combination is

$$\sum_{j=0}^3 w_{3,j}(1/4) A_6^{[1+j]}(0) = -\frac{2347}{59535} = A_6(0). \quad (\text{H.13})$$

No limiting argument or numerical approximation enters.

H.5 Worked uncentered recovery in degree three

For $P_3(0) = -1/12$, the finite constants at depths 1, 2, 3, 4 are

$$0, \quad -\frac{3}{128}, \quad -\frac{49}{1024}, \quad -\frac{525}{8192}. \quad (\text{H.14})$$

The $q = 1/2$ weights

$$\left(-\frac{1}{21}, \frac{2}{3}, -\frac{8}{3}, \frac{64}{21}\right) \quad (\text{H.15})$$

recover the limit exactly:

$$\sum_{j=0}^3 w_{3,j}(1/2) P_3^{[1+j]}(0) = -\frac{1}{12}. \quad (\text{H.16})$$

H.6 A finite algorithm for any fixed degree

Algorithm H.5 (Exact Appell polynomial recovery). To compute $A_n(x)$ exactly:

1. Set $d = \lfloor n/2 \rfloor$ and choose any starting depth $N \geq 1$.
2. For $j = 0, \dots, d$, form the finite Barnes polynomial

$$A_n^{[N+j]}(x) = \mathfrak{B}_n^{(N+j)} \left(x + 1 - 2^{-(N+j)} \mid 1, \frac{1}{2}, \dots, 2^{1-N-j} \right).$$

3. Combine them with $w_{d,j}(1/4)$ from (H.6).

The result is exactly $A_n(x)$. The analogous uncentered algorithm uses $n + 1$ levels and $q = 1/2$.

Remark H.6 (Why centering halves the interpolation order). The raw expansion contains every power 2^{-Nr} through $r = n$. Symmetry removes odd moments, so the centered expansion contains only 4^{-Nr} through $r = \lfloor n/2 \rfloor$. Centering thus halves both the polynomial degree in the tail parameter and the number of prefix levels required for exact recovery. This is the exact-polynomial counterpart of the conditioning improvement obtained by centering logarithmic sinc-product tails in the repository's acceleration reports.

I Reciprocal-sinc poles and the asymptotics of Appell constants

I.1 The reciprocal moment generating function

Put

$$R(t) = \frac{1}{M(t)} = \sum_{n \geq 0} \alpha_n \frac{t^n}{n!}. \quad (\text{I.1})$$

The function M is entire, but R is meromorphic. By (C.44), the zeros of M are the imaginary lifts of the integer zeros of the Rvachev Fourier product. For $n \neq 0$, set

$$T_n = 2\pi i n, \quad \mu(n) = \nu_2(n) + 1. \quad (\text{I.2})$$

Then T_n is a zero of M and a pole of R of order $\mu(n)$.

I.2 Exact leading Laurent coefficient

Write $n = 2^v u$ with u odd and define

$$C_u = M(\pi i u) \neq 0. \quad (\text{I.3})$$

The number C_u is a real sinc product:

$$C_u = \prod_{j \geq 1} \frac{\sin(\pi u / 2^j)}{\pi u / 2^j}. \quad (\text{I.4})$$

Theorem I.1 (Laurent coefficient from the odd core; FRONTIER). *For every nonzero integer $n = 2^v u$ with u odd,*

$$\lim_{t \rightarrow T_n} (t - T_n)^{\mu(n)} R(t) = -\frac{T_n^{\mu(n)}}{C_u}. \quad (\text{I.5})$$

Thus the pole order is controlled by the 2-adic valuation of n , while the leading amplitude is controlled only by the odd core u .

Proof. Write $t = 2^{v+1}s$ and $s_0 = \pi i u$. Iterating $M(2s) = (\sinh s/s)M(s)$ gives

$$M(2^{v+1}s) = M(s) \prod_{r=0}^v \frac{\sinh(2^r s)}{2^r s}. \quad (\text{I.6})$$

At $s = s_0$, the factor $M(s_0) = C_u$ is nonzero. Every displayed hyperbolic-sine factor has a simple zero. Since u is odd,

$$\cos(\pi u) = -1, \quad \cos(2^r \pi u) = 1 \quad (r \geq 1).$$

Consequently

$$M(2^{v+1}s) \sim -C_u \left(\frac{s - s_0}{s_0} \right)^{v+1} = -C_u \left(\frac{t - T_n}{T_n} \right)^{\mu(n)}.$$

Taking reciprocals proves (I.5). □

Exact Lean status. The compiled module `FabiusFunction.RvachevLaurentLeading` formalizes this theorem with the manuscript's normalization. Its definition `Fabius.rvachevCenteredMGF` is connected to the sinc product by `Fabius.rvachevCenteredMGF_eq_rvachevFourierProduct`; the odd-core value and its nonvanishing are `Fabius.rvachevCenteredMGF_pi_mul_I_int` and `Fabius.rvachevCenteredMGF_pi_mul_I_int_ne_zero_of_odd`. The reusable product-coordinate limit is `Fabius.tendsto_sub_pow_mul_inv_rvachevFourierProduct_int`, the arbitrary nonzero-integer centered-MGF form is `Fabius.tendsto_rvachevCenteredMGF_laurent_int`, and `Fabius.tendsto_rvachevCenteredMGF_laurent_two_pow_mul_odd` is exactly (I.5) for $n = 2^v u$, u odd. These limit theorems use a punctured neighborhood of the pole. This is the precise meaning of the displayed meromorphic limit: Lean's inverse is totalized to zero at a zero of M , so the value at the pole itself must be excluded. No lower Laurent coefficients or subsequent coefficient asymptotics are claimed by this formalization.

Remark I.2. The repository now identifies both the zero multiplicity $\nu_2(n) + 1$ and the exact leading Laurent amplitude. Propagating the full divisor into coefficient asymptotics for the polynomial deconvolution sequence remains a separate step.

I.3 A 2-adic hierarchy of exponentially small corrections

Let

$$r_N = [t^N]R(t) = \frac{\alpha_N}{N!}. \quad (\text{I.7})$$

For a fixed positive integer n , the conjugate pair of poles $\pm T_n$ contributes to r_{2m} . The highest Laurent term in (I.5) gives

$$\frac{2(-1)^{m+\mu(n)+1}}{C_u} \binom{2m+\mu(n)-1}{\mu(n)-1} \frac{1}{(2\pi n)^{2m}}, \quad (\text{I.8})$$

while the lower Laurent coefficients contribute the same exponential scale multiplied by polynomials in m of degree at most $\mu(n) - 2$.

Corollary I.3 (Pole multiplicity becomes polynomial degree; FRONTIER). *For each fixed $n = 2^v u$, the complete contribution of the pair $\pm 2\pi i n$ to $\alpha_{2m}/(2m)!$ has the form*

$$\frac{(-1)^m}{(2\pi n)^{2m}} \Pi_n(m), \quad (\text{I.9})$$

where Π_n is a real polynomial of degree exactly $v = \nu_2(n)$. Its leading term is

$$\Pi_n(m) = \frac{2(-1)^v}{C_u v!} (2m)^v + \mathcal{O}(m^{v-1}); \quad (\text{I.10})$$

equivalently, its coefficient of m^v is $2^{v+1}(-1)^v/(C_u v!)$. Hence the binary valuation of the Fourier zero is visible as the polynomial degree of the corresponding exponentially small correction to the Appell constants.

Proof. By “the contribution of the pole pair” we mean the coefficient contribution of the two finite principal parts in the Laurent expansions at $\pm T_n$. Write the principal part at T_n as $\sum_{r=1}^{v+1} a_{-r}(t-T_n)^{-r}$. For fixed r ,

$$[t^{2m}](t-T_n)^{-r} = (-T_n)^{-r} T_n^{-2m} \binom{2m+r-1}{r-1},$$

so, after adding the conjugate pole, it is $(-1)^m(2\pi n)^{-2m}$ times a real polynomial in m of degree at most $r-1$. Summing over r proves (I.9) with degree at most v . The top Laurent coefficient from (I.5) gives exactly (I.8); since $\binom{2m+v}{v} = (2m)^v/v! + \mathcal{O}(m^{v-1})$ and $C_u \neq 0$, its leading term is the nonzero quantity in (I.10). Thus the degree is exactly v . $\square$

I.4 The first three pole shells explicitly

Define

$$C = C_1 = M(\pi i) = 0.55377127588881009534421741634454888128 \dots, \quad (\text{I.11})$$

$$C_3 = M(3\pi i) = -0.04620229913430671299257486221028792308 \dots, \quad (\text{I.12})$$

$$\begin{aligned} \eta &= \frac{d}{dx} \log M(ix) \Big|_{x=\pi} = \sum_{j \geq 1} \left(2^{-j} \cot \frac{\pi}{2^j} - \frac{1}{\pi} \right) \\ &= -0.40861837118802483147907598511740221545 \dots, \end{aligned} \quad (\text{I.13})$$

so

$$\begin{aligned} \pi\eta &= -1.2837124730461260445945438959726042474 \dots, \\ \frac{C}{C_3} &= -11.985794782182535885188878444711216907 \dots \end{aligned} \quad (\text{I.14})$$

The nearest poles $\pm 2\pi i$ are simple. The poles $\pm 4\pi i$ are double. At $T = 4\pi i$, the principal part is

$$R(t) = \frac{16\pi^2/C}{(t-T)^2} + \frac{4\pi i(\pi\eta - 2)/C}{t-T} + \mathcal{O}(1). \quad (\text{I.15})$$

The poles $\pm 6\pi i$ are simple with odd-core constant C_3 .

Theorem I.4 (Three-shell asymptotic; FRONTIER). *As $m \rightarrow \infty$,*

$$\boxed{\frac{\alpha_{2m}}{(2m)!} = \frac{2(-1)^m}{C(2\pi)^{2m}} \left[1 + (1 - 2m - \pi\eta)4^{-m} + \frac{C}{C_3}9^{-m} + \mathcal{O}(m^2 16^{-m}) \right]}. \quad (\text{I.16})$$

In particular,

$$\alpha_{2m} \sim \frac{2(-1)^m(2m)!}{C(2\pi)^{2m}}. \quad (\text{I.17})$$

Proof. A simple pole at T with residue a contributes $-aT^{-2m-1}$ to r_{2m} . Applying (I.5) at $\pm 2\pi i$ gives the leading term in (I.16).

For $T = 4\pi i$, use (I.15). The coefficient of t^{2m} in $D(t - T)^{-2} + E(t - T)^{-1}$ is

$$D(2m + 1)T^{-2m-2} - ET^{-2m-1}.$$

Adding the conjugate pole and simplifying gives

$$\frac{2(-1)^m}{C(4\pi)^{2m}}(1 - 2m - \pi\eta),$$

which is the relative 4^{-m} correction. The simple poles at $\pm 6\pi i$ contribute $2(-1)^m/(C_3(6\pi)^{2m})$, giving the relative term $(C/C_3)9^{-m}$.

For completeness, fix a radius ρ with $8\pi < \rho < 10\pi$. In $|t| < \rho$, the only unremoved singularities are the triple poles at $\pm 8\pi i$. Subtract their complete principal parts as well. Every term $(t - T)^{-j}$ with $1 \leq j \leq 3$ has coefficient $\mathcal{O}(m^{j-1}|T|^{-2m})$, so the two triple principal parts contribute $\mathcal{O}(m^2(8\pi)^{-2m})$. The function remaining after all eight principal parts have been removed is holomorphic on a neighbourhood of $\{|t| \leq \rho\}$; Cauchy's coefficient estimate bounds its $2m$ th coefficient by $\mathcal{O}(\rho^{-2m})$, which is smaller than the preceding bound. Thus the absolute remainder is $\mathcal{O}(m^2(8\pi)^{-2m})$. Division by the leading scale $(2\pi)^{-2m}$ yields $\mathcal{O}(m^2 16^{-m})$. $\square$

m	normalized exact value $\mathcal{R}_m$	three-shell approximation	difference
4	0.976481902429846	0.975843927464026	6.38×10^{-4}
6	0.997609767253799	0.997605306103035	4.46×10^{-6}
8	0.999790455959967	0.999790427625023	2.83×10^{-8}
10	0.999983101160970	0.999983100994111	1.67×10^{-10}
12	0.999998705566884	0.999998705565958	9.26×10^{-13}
20	0.99999999965697235338431	0.99999999965697235337828	6.03×10^{-22}

Table 1: Verification of [theorem I.4](#). Here $\mathcal{R}_m = \frac{C(2\pi)^{2m}}{2(-1)^m} \frac{\alpha_{2m}}{(2m)!}$.

I.5 Why the correction polynomial is genuinely dyadic

The shell $n = 2$ is closer than the odd shell $n = 3$, but its pole is double. Its correction is not a constant multiple of 4^{-m} ; it carries the linear factor $1 - 2m - \pi\eta$. The next power-of-two shell $n = 4$ is triple and produces a quadratic polynomial times 16^{-m} . In general, the shell $n = 2^v$ produces a degree- v polynomial times 4^{-vm} . This is the coefficient-side analogue of the zero-counting identities in the spectral reports: 2-adic multiplicity does not disappear under inversion; it changes from a divisor multiplicity into a hierarchy of polynomially weighted exponentials.

I.6 A Mittag-Leffler frontier

The local theorem suggests a global expansion

$$R(t) = H(t) + \sum_{n \neq 0} \mathcal{P}_n(t), \quad (\text{I.18})$$

where $\mathcal{P}_n$ is a suitably regularized principal part at $2\pi in$ and H is entire. A convergent, symmetry-preserving Mittag-Leffler decomposition would give exact rapidly convergent series for all α_{2m} , with each summand labeled by the odd core and 2-adic valuation of n . It would also expose whether the constants C_u satisfy additional cyclotomic or Thue-Morse product relations along arithmetic rays.

J A Lambert- W threshold for the Appell coefficients

J.1 From pole asymptotics to factorial growth

By (I.17) and Stirling's formula,

$$\begin{aligned}\log |\alpha_{2m}| &= \log \left(\frac{2(2m)!}{C(2\pi)^{2m}} \right) + o(1) \\ &= 2m \log \frac{m}{\pi e} + \frac{1}{2} \log(4\pi m) + \log \frac{2}{C} + o(1).\end{aligned}\quad (\text{J.1})$$

Thus the constants grow faster than any fixed exponential in m , but slower than $(2m)!$ by the geometric factor $(2\pi)^{-2m}$.

For a continuous inversion, let $\mathfrak{m}(X)$ be the large positive real solution of

$$\frac{2\Gamma(2\mathfrak{m} + 1)}{C(2\pi)^{2\mathfrak{m}}} = X. \quad (\text{J.2})$$

Let $H = \log X$ and

$$w = W\left(\frac{H}{2\pi e}\right), \quad m_0 = \frac{H}{2w}. \quad (\text{J.3})$$

The identity $we^w = H/(2\pi e)$ implies

$$\log \frac{m_0}{\pi e} = w, \quad 2m_0 \log \frac{m_0}{\pi e} = H. \quad (\text{J.4})$$

Theorem J.1 (Lambert coefficient threshold; FRONTIER). *As $X \rightarrow \infty$,*

$$\boxed{\mathfrak{m}(X) = m_0 - \frac{\frac{1}{2} \log(4\pi m_0) + \log(2/C)}{2 \log(m_0/\pi)} + o(1), \quad m_0 = \frac{\log X}{2W(\log X/(2\pi e))}.} \quad (\text{J.5})$$

If

$$m_{\min}(X) = \min\{m \in \mathbb{N}_0 : |\alpha_{2m}| \geq X\}, \quad (\text{J.6})$$

then

$$\boxed{m_{\min}(X) = \frac{\log X}{2W(\log X/(2\pi e))} + \mathcal{O}(1).} \quad (\text{J.7})$$

Proof. The leading equation obtained from (J.1) is $2m \log(m/(\pi e)) = H$, whose exact solution is m_0 by (J.4). Write the right side of (J.1) as $f(m) + g(m) + o(1)$, where

$$f(m) = 2m \log \frac{m}{\pi e}, \quad g(m) = \frac{1}{2} \log(4\pi m) + \log \frac{2}{C}.$$

Since

$$f'(m) = 2 \log \frac{m}{\pi},$$

one Newton correction gives

$$\mathfrak{m} = m_0 - \frac{g(m_0)}{f'(m_0)} + o(1),$$

which is (J.5). To justify the least-index statement, note from the coefficient asymptotic that

$$\frac{|\alpha_{2m+2}|}{|\alpha_{2m}|} \sim \frac{(2m+2)(2m+1)}{(2\pi)^2} \rightarrow \infty.$$

Thus $|\alpha_{2m}|$ is strictly increasing for all sufficiently large m . The exact coefficients differ from their leading envelope by a relative quantity tending to zero; because the logarithmic envelope has derivative $2 \log(m/\pi) \rightarrow \infty$, its threshold crossing and the exact crossing differ by a bounded number of integer indices. Rounding the continuous root changes the answer by at most one more index, proving (J.7). $\square$

J.2 A second Lambert geometry in the same theory

The repository's endpoint asymptotics use the lower branch W_{-1} to invert $x = \lambda 2^{-\lambda}$ as $x \downarrow 0$. The present threshold uses the principal branch W to invert $H = 2m \log(m/(\pi e))$ as coefficient size grows. The two appearances have different origins:

$$\begin{aligned} \text{endpoint geometry} & : \text{ dyadic scale count balanced against a small spatial argument,} \\ \text{coefficient geometry} & : \text{ factorial growth balanced against a reciprocal-sinc pole radius.} \end{aligned} \quad (\text{J.8})$$

The common feature is an unknown occurring both linearly and inside a logarithm. The branch is determined by the direction of inversion: a vanishing endpoint requires W_{-1} , while an increasing factorial threshold uses the principal branch.

What the Lambert law does and does not say

Equation (J.7) predicts where the absolute Appell constants first cross a large threshold. It is not an endpoint asymptotic for F , and it does not approximate F^{-1} . Its input is the nearest reciprocal-sinc pole, not the saddle point of a Laplace inversion. The connection is structural rather than an identification of the two asymptotic problems.

Part V

All-Orders Endpoint Asymptotics

K Imported inputs and the two-scale inverse problem

K.1 Forward input

With $X = \sum_{j \geq 1} 2^{-j} V_j$ (V_j i.i.d. uniform on $[0, 1]$) and $F(x) = \mathbb{P}(X \leq x)$, the Laplace transform and exponent are

$$M(t) = \mathbb{E} e^{-tX} = \prod_{j \geq 1} \frac{1 - e^{-t/2^j}}{t/2^j}, \quad K(t) = \log M(t), \quad (\text{K.1})$$

and the Bromwich integral $F(x) = \frac{1}{2\pi i} \int_{(\sigma)} t^{-1} \exp(xt + K(t)) dt$ is the analytic source of the saddle expansion. For small $x > 0$ the exact lower-Lambert phase is $\lambda(x) = -W_{-1}(-Lx)/L$, i.e. $x = \lambda e^{-L\lambda}$ exactly, and the repository's forward theorem states: for every fixed $N \geq 1$,

$$\log F(x) = -\frac{L}{2} \lambda^2 + L\beta\lambda - \frac{1}{2} \log \lambda + C_{\sharp} + \Psi(\lambda) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}_N(\lambda^{-N}), \quad (\text{K.2})$$

uniformly in the phase. More precisely, after the displayed terms are subtracted the remainder is bounded in absolute value by $C_N \lambda^{-N}$, with a constant independent of the real phase. Every A_j is a bounded real-analytic one-periodic differential polynomial in the Gamma-zeta fluctuation

$$\Psi(u) = \sum_{k \neq 0} c_{\Psi}(k) e^{2\pi i k u}, \quad c_{\Psi}(k) = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}, \quad (\text{K.3})$$

$c_{\Psi}(0) = 0$. The explicit first correction and the forward top-jet law are

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}, \quad [\Psi^{(2j)}]A_j = \frac{(-1)^j}{2^j j! L^{2j}}. \quad (\text{K.4})$$

The second correction, needed for the third inverse order, is

$$A_2 = \frac{P_1}{24L} + \frac{2P_1 + 1}{4L^2} - \frac{P_1^3 + 3P_1^2 + 3P_1P_2 + 3P_2 + P_3}{6L^3} + \frac{4P_1^2P_2 + 4P_1P_3 + 2P_2^2 + P_4}{8L^4}, \quad (\text{K.5})$$

with $P_r = \Psi^{(r)}(\theta)$, and $A'_1 = -(P_3 + 2P_1P_2)/(2L^2)$.

For reference, with Euler's constant γ and the Stieltjes convention $\zeta(1+s) = s^{-1} + \gamma - \gamma_1 s + O(s^2)$, the two constants used below are

$$C_\# = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi, \quad C_B = \frac{L}{12} + \frac{\pi^2 - 6\gamma^2 - 12\gamma_1}{12L}. \quad (\text{K.6})$$

In particular $C_\# + C_B = -\frac{1}{2} \log(2\pi)$ by direct simplification.

K.2 Inverse prior art

The corpus's inverse volume establishes, and this volume imports: the leading factor and first two phase corrections

$$d_1 = \frac{\beta^2}{2} + \frac{C_\# + \Psi(\theta) - \frac{1}{2}\ell}{L}, \quad d_2 = \frac{\Psi'(\theta) d_1 + A_1(\theta) - \frac{1}{2}\beta}{L}; \quad (\text{K.7})$$

the logarithmic corrections $h_1 = \frac{1}{2}\ell + \kappa_0 - \Psi(\theta)$ with $\kappa_0 = \beta - \frac{L}{2}\beta^2 - C_\#$, and h_2 ; the nondegenerate first normalized cluster interval; the two-term elasticity; and a triangular all-orders recursion. The consolidated results below convert that recursion into closed formulas and extend every structural layer.

Until [appendix R](#), the unadorned symbols D, d_n, h_n, g_n always refer to this ρ -carrier. The Wright-omega carrier introduced there has a different correction and different coefficients; they will be written D_W, d_n^W, h_n^W, g_n^W . This superscript is essential: for example $d_1 = \beta^2/2 + (C_\# + \Psi - \ell/2)/L$, whereas $d_1^W = \Psi/L$.

K.3 The two-scale problem

For $y \downarrow 0$ set T, ρ, z, ℓ, θ as in the notation table and

$$G_0(y) = \rho e^{-L\rho - L\beta} = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}, \quad \lambda(G(y)) = \rho + \beta + D, \quad D = \sum_{n \geq 1} d_n z^n. \quad (\text{K.8})$$

Substituting $\lambda = \rho + \beta + D$ into (K.2) and using $-\frac{L}{2}\lambda^2 + L\beta\lambda = -T - L\rho D + \frac{L}{2}\beta^2 - \frac{L}{2}D^2$ (the quadratic collapses exactly because $T = L\rho^2/2$) reduces $\log F(G(y)) = -T$ to the fixed-point equation

$$\boxed{D = z \mathcal{K}(z, D)}, \quad \mathcal{K}(z, w) = -\frac{1}{L} \mathcal{R}(z, w), \quad (\text{K.9})$$

with the formal residual

$$\mathcal{R}(z, w) = -\frac{L}{2}\beta^2 + \frac{L}{2}w^2 + \frac{1}{2}\ell + \frac{1}{2} \log(1 + \beta z + zw) - C_\# - \Psi(\theta + w) - \sum_{j \geq 1} \frac{A_j(\theta + w) z^j}{(1 + \beta z + zw)^j}. \quad (\text{K.10})$$

Coefficients are computed in the two-scale ring $\mathcal{A}[\ell][[z]]$, where $\mathcal{A}$ is the differential algebra of one-periodic functions generated by Ψ and the A_j : ℓ and θ are treated as independent coefficient variables and only afterwards evaluated at $\log \rho$ and $\rho + \beta$. This convention is forced, not cosmetic: θ winds around the circle and ℓ is unbounded, so expanding either as a slow series would destroy terms at their own order — there is no phase-free complete expansion ([appendix N](#)), and the correct coefficient space is $\mathbb{R}[\ell] \otimes C^\infty(\mathbb{R}/\mathbb{Z})$.

Theorem K.1 (All-orders inverse endpoint expansion). *For every fixed $N \geq 1$ there are unique coefficients $d_n, h_n, g_n \in \mathcal{A}[\ell]$ ($1 \leq n \leq N$), polynomials in ℓ with real-analytic one-periodic coefficients, such that as $y \downarrow 0$, uniformly in the phase,*

$$\lambda(G(y)) = \rho + \beta + \sum_{n=1}^N \frac{d_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1+\ell)^N}{\rho^{N+1}}\right), \quad (\text{K.11})$$

$$\begin{aligned} \log \frac{G(y)}{G_0(y)} &= \sum_{n=1}^N \frac{h_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1+\ell)^N}{\rho^{N+1}}\right), \\ \frac{G(y)}{G_0(y)} &= 1 + \sum_{n=1}^N \frac{g_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1+\ell)^{N+1}}{\rho^{N+1}}\right). \end{aligned} \quad (\text{K.12})$$

d_n depends only on $\Psi, A_1, \dots, A_{n-1}$ and finitely many of their derivatives. By reflection, $G(1-y) = 1 - G(y)$ transfers every formula to the right endpoint.

Proof. Fix N and truncate the forward expansion after A_{N-1} . Denote by $\mathcal{R}_N$ the corresponding truncation of (K.10) and put

$$\mathcal{H}_N(w) := \frac{L}{z}w + \mathcal{R}_N(z, w).$$

The exact phase correction D satisfies $\mathcal{H}_N(D) = -E_N(D)$, where the value remainder from the forward theorem gives $|E_N(D)| \leq C_N z^N$ uniformly in the phase. No derivative of E_N is used below.

Substituting $D^{[N]} = \sum_{n \leq N} d_n z^n$ into the formal equation and cancelling successive powers determines $d_1, \dots, d_N$ uniquely: at the stage that determines d_n , the new coefficient occurs only in $(L/z)D^{[N]}$ and has nonzero multiplier L . Since the only explicit ℓ in $\mathcal{R}_N$ is affine, induction gives polynomial dependence on ℓ ; since Ψ and the A_j are finite differential polynomials with normally convergent Fourier series, their Taylor coefficients are real analytic and so are the periodic coefficients of every d_n . Formal cancellation gives

$$|\mathcal{H}_N(D^{[N]})| \leq C_N(1+\ell)^N z^N.$$

Both D and $D^{[N]}$ are $\mathcal{O}((1+\ell)z)$, and throughout the segment joining them the explicitly known residual map $\mathcal{H}_N$ satisfies

$$\partial_w \mathcal{H}_N(w) = \frac{L}{z} + \mathcal{O}_N(1), \quad |\partial_w \mathcal{H}_N(w)| \geq \frac{L}{2z}$$

for all sufficiently small z . The mean-value theorem, applied only to $\mathcal{H}_N$, now yields

$$|D - D^{[N]}| \leq \frac{2z}{L}(|E_N(D)| + |\mathcal{H}_N(D^{[N]})|) = \mathcal{O}_N((1+\ell)^N z^{N+1}),$$

which is (K.11). The exact Lambert identity gives

$$\frac{G}{G_0} = (1 + \beta z + zD) e^{-LD}, \quad H := \log \frac{G}{G_0} = \log(1 + \beta z + zD) - LD, \quad (\text{K.13})$$

The logarithm and exponential are analytic at the origin. Taylor's theorem applied to these two finite compositions therefore gives the expansion for H with remainder $\mathcal{O}_N((1+\ell)^N z^{N+1})$ and the expansion for $\exp H$ with remainder $\mathcal{O}_N((1+\ell)^{N+1} z^{N+1})$. These are precisely (K.12). Finally, reflection of the strictly increasing CDF gives $G(1-y) = 1 - G(y)$ and transfers the formulas to the right endpoint. $\square$

L The diagonal Lagrange–Bürmann calculus

The triangular recursion computes order n only after all lower orders. A catalytic variable turns the same fixed point into a closed finite extraction at any single prescribed order — the central shared theorem of the two absorbed sources.

Theorem L.1 (Diagonal coefficient formulas). *For every $n \geq 1$,*

$$d_n = \sum_{k=1}^n \frac{1}{k} [z^{n-k} w^{k-1}] \mathcal{K}(z, w)^k = \sum_{k=1}^n \frac{(-1)^k}{k L^k} [z^{n-k} w^{k-1}] \mathcal{R}(z, w)^k, \quad (\text{L.1})$$

a finite formula: only the z^{n-1} -truncation of $\mathcal{K}$, the jets $\Psi, \dots, \Psi^{(n-1)}$, and $A_1, \dots, A_{n-1}$ enter. More generally, for any formal $K(z, w)$,

$$[z^n] K(z, D(z)) = [z^n] K(z, 0) + \sum_{k=1}^n \frac{1}{k} [z^{n-k} w^{k-1}] \partial_w K(z, w) \mathcal{K}(z, w)^k, \quad (\text{L.2})$$

which with $K = \log(1 + \beta z + zw) - Lw$ and $K = (1 + \beta z + zw)e^{-Lw}$ gives closed diagonal formulas for h_n and g_n directly.

Proof. Let $W(t, z)$ solve $W = t\mathcal{K}(z, W)$. For fixed z , ordinary Lagrange–Bürmann inversion in t gives $[t^k]W = \frac{1}{k} [w^{k-1}] \mathcal{K}(z, w)^k$ and the composition form $K(z, W) = K(z, 0) + \sum_k \frac{t^k}{k} [w^{k-1}] \partial_w K \cdot \mathcal{K}^k$. Since $D(z) = W(z, z)$, set $t = z$ and extract total degree n . Finiteness: a term with Lagrange index k asks for $z^{n-k} w^{k-1}$, so no $A_j z^j$ with $j > n - k$ and at most $k - 1$ phase derivatives occur. $\square$

Expanding $\mathcal{K} = \sum \phi_{r,s} z^r w^s$ gives the fully unrolled partition form

$$d_n = \sum_k \frac{1}{k} \sum_{r_1 + \dots + r_k = n-k, s_1 + \dots + s_k = k-1} \prod_j \phi_{r_j, s_j}$$

— convenient for proofs, while (L.1) is convenient for implementation.

Theorem L.2 (Direct partial-Bell formula for g_n). *With $x_j = j! d_j$ and the partial Bell normalization $(\sum_j d_j z^j)^k = \sum_{n \geq k} \frac{k!}{n!} \mathcal{B}_{n,k}^{\text{exp}}(x_1, x_2, \dots) z^n$,*

$$g_n = \frac{1}{n!} \sum_{k=1}^n (-L)^k \mathcal{B}_{n,k}^{\text{exp}} + \frac{\beta}{(n-1)!} \sum_{k=0}^{n-1} (-L)^k \mathcal{B}_{n-1,k}^{\text{exp}} + \frac{1}{(n-1)!} \sum_{k=1}^{n-1} k (-L)^{k-1} \mathcal{B}_{n-1,k}^{\text{exp}}, \quad (\text{L.3})$$

all $\mathcal{B}_{m,k}^{\text{exp}} = \mathcal{B}_{m,k}^{\text{exp}}(x_1, x_2, \dots)$; combined with (L.1) this is a nonrecursive finite formula for the arbitrary-order multiplicative coefficient. The logarithmic-to-multiplicative conversion is the complete-Bell identity $g_n = \frac{1}{n!} \mathcal{B}_n^{\text{exp}}(1!h_1, \dots, n!h_n)$, i.e. $ng_n = \sum_{j \leq n} j h_j g_{n-j}$.

Proof. Expand the exact factorization (K.13): the three lines are $[z^n]e^{-LD}$, $[z^{n-1}]\beta e^{-LD}$, and $[z^{n-1}]De^{-LD}$, each a standard partial-Bell expansion. $\square$

Algorithm L.3 (Arbitrary-order symbolic generation). Fix an output order N : obtain $A_1, \dots, A_{N-1}$ (from (K.4)–(K.5) or the saddle pipeline of appendix S); Taylor-expand $\Psi(\theta + w)$, $A_j(\theta + w)$ only as far as the bidegrees require; truncate $\mathcal{K}$ at z^{N-1} ; evaluate (L.1) for each needed n (no lower d_j is mathematically required); and produce g_n by (L.3) or the h -route. The triangular and diagonal algorithms are complementary: the former reuses lower-order work, the latter is an independent certificate — the absorbed sources’ implementations verify $d_j^{\text{diag}} \equiv d_j^{\text{tri}}$ symbolically for $j \leq 3$ over algebraically independent jet symbols.

M The explicit third correction

Theorem M.1 (Closed third-order coefficients). *With $P_r = \Psi^{(r)}(\theta)$ and d_1, d_2 from (K.7),*

$$d_3 = \frac{1}{L} \left(P_1 d_2 + \frac{1}{2} P_2 d_1^2 + A'_1 d_1 - \beta A_1 + A_2 - \frac{L}{2} d_1^2 - \frac{1}{2} d_1 + \frac{\beta^2}{4} \right), \quad (\text{M.1})$$

and

$$h_1 = \beta - L d_1, \quad h_2 = d_1 - \frac{\beta^2}{2} - L d_2, \quad h_3 = d_2 - \beta d_1 + \frac{\beta^3}{3} - L d_3, \quad (\text{M.2})$$

$$g_1 = h_1, \quad g_2 = h_2 + \frac{1}{2} h_1^2 = d_1 - L d_2 - \beta L d_1 + \frac{L^2}{2} d_1^2, \quad g_3 = h_3 + h_1 h_2 + \frac{1}{6} h_1^3, \quad (\text{M.3})$$

with the direct d -form $g_3 = -L d_3 + (1 - \beta L) d_2 + L^2 d_1 d_2 + (\frac{\beta L^2}{2} - L) d_1^2 - \frac{L^3}{6} d_1^3$.

Proof. Extract $[z^2]$ from (K.9): the contributions are $\frac{L}{2} d_1^2, \frac{1}{2} d_1 - \frac{\beta^2}{4}$ (from the logarithm), $-P_1 d_2 - \frac{1}{2} P_2 d_1^2$ (Taylor transport of Ψ), $-A'_1 d_1 + \beta A_1$ (transport of A_1), and $-A_2$; their sum plus $L d_3$ vanishes. The h - and g -chains are the expansions of (K.13). $\square$

Remark M.2 (Cross-validation of the sources). The two absorbed reports derived (M.1) independently and grouped it differently (one as $-\frac{1}{L} \{ \frac{L}{2} d_1^2 + \frac{1}{2} d_1 - \frac{\beta^2}{4} - P_1 d_2 - \frac{P_2}{2} d_1^2 - A'_1 d_1 + \beta A_1 - A_2 \}$); the forms are algebraically identical, and both were checked symbolically against the triangular recursion in their independent implementations. The compact grouping displays the five mechanisms separately — quadratic saddle correction, logarithmic Jacobian, transport of Ψ , transport of A_1 , and the genuinely new A_2 — and is the stabler form for evaluation.

N Structure laws

The diagonal formula does more than compute: its bidegree geometry yields laws invisible in a triangular expansion.

N.1 Logarithmic degrees and top logarithms

Write $\mathcal{R} = \frac{1}{2} \ell + \mathcal{R}_0$ with $\mathcal{R}_0$ independent of ℓ , so the z^0 slice of $\mathcal{K}$ is $\mathcal{K}(0, w) = a\ell + \phi(w)$ with

$$a = -\frac{1}{2L}, \quad \phi(w) = \frac{\beta^2}{2} + \frac{C_\# + \Psi(\theta + w)}{L} - \frac{w^2}{2}. \quad (\text{N.1})$$

Theorem N.1 (Degree and top-log laws). $\deg_\ell d_1 = 1$ and $\deg_\ell d_n \leq n - 1$ for $n \geq 2$, with

$$[\ell^{n-1}] d_n = a^{n-1} [w^{n-1}] \phi(w) \quad (n \geq 2); \quad (\text{N.2})$$

explicitly $[\ell] d_2 = -\Psi'(\theta)/(2L^2)$, $[\ell^2] d_3 = (\Psi''(\theta) - L)/(8L^3)$, and

$$[\ell^{n-1}] d_n = (-1)^{n-1} \frac{\Psi^{(n-1)}(\theta)}{2^{n-1} L^n (n-1)!} \quad (n \geq 4).$$

For the logarithmic coefficients, $\deg_\ell h_1 = \deg_\ell h_2 = 1$, $\deg_\ell h_n \leq n - 1$ ($n \geq 3$), with $[\ell] h_1 = \frac{1}{2}$, $[\ell] h_2 = (\Psi'(\theta) - 1)/(2L)$, $[\ell^2] h_3 = (L - \Psi''(\theta))/(8L^2)$, and $[\ell^{n-1}] h_n = (-1)^n \Psi^{(n-1)}(\theta)/(2^{n-1} L^{n-1} (n-1)!)$ for $n \geq 4$. Finally $\deg_\ell g_n \leq n$ and the multiplicative top logarithm is universal:

$$[\ell^n] g_n = \frac{1}{2^n n!}, \quad g_n = \frac{\ell^n}{2^n n!} + \mathcal{O}_n((1 + |\ell|)^{n-1}) \text{ uniformly in } \theta. \quad (\text{N.3})$$

Proof. In (L.1) a product $\mathcal{K}^k$ has ℓ -degree at most $k \leq n$, and for $k = n$ the coefficient requested is $[w^{n-1}]\mathcal{K}(0, w)^n/n$, whose nominal ℓ^n term carries no w and dies for $n \geq 2$; the $\ell^{n-1}w^{n-1}$ term takes $a\ell$ from $n-1$ factors and ϕ from one, the multiplicity n cancelling $1/n$ — this is (N.2), and expanding ϕ gives the special cases. In $H = \log(1 + \beta z + zD) - LD$ the logarithm at order z^n involves only $d_{<n}$ (degree $\leq n-2$ for $n \geq 3$), so the top logarithm of h_n is $-L[\ell^{n-1}]d_n$; orders 1, 2 are read off (M.2). For g_n , the only way to reach ℓ -degree n at z -weight n in $\exp(H)$ is $(h_1 z)^n/n!$ with the term $\ell/2$ chosen in every factor: $[\ell^n]g_n = (1/2)^n/n!$. (Equivalently, in the direct d -route: only $(-LD)^n/n!$ with $d_1 = -\ell/(2L) + \mathcal{O}(1)$ from every copy attains degree n , giving $(-L)^n(-1/2L)^n/n!$ — the two absorbed proofs of the same law.) $\square$

Corollary N.2 (First subleading multiplicative logarithm). Write $h_1 = \frac{1}{2}\ell + q(\theta)$, $q = \kappa_0 - \Psi(\theta)$, and $\alpha_j(\theta) = [\ell^{j-1}]h_j$ ($j \geq 2$). Then for $n \geq 2$

$$[\ell^{n-1}]g_n = \frac{q(\theta)}{2^{n-1}(n-1)!} + \sum_{j=2}^n \frac{\alpha_j(\theta)}{2^{n-j}(n-j)!} : \quad (\text{N.4})$$

the first nonuniversal logarithmic layer is closed-form in the top-log sequence of the h_j .

Proof. Write $H(z) = \sum_{j \geq 1} h_j z^j$, so that $1 + \sum_{n \geq 1} g_n z^n = \exp H(z)$. To obtain $z^n \ell^{n-1}$, first take n copies of $h_1 z$ in the exponential. Exactly one copy must contribute $q(\theta)$ and the other $n-1$ copies contribute $\ell/2$; the resulting coefficient is $q(\theta)/(2^{n-1}(n-1)!)$.

The other possibility is to take one copy of $h_j z^j$, with $2 \leq j \leq n$, and $n-j$ copies of $h_1 z$. The largest possible ℓ -degree is then $(j-1) + (n-j) = n-1$, and choosing the top term $\alpha_j \ell^{j-1}$ from h_j gives $\alpha_j/(2^{n-j}(n-j)!)$. Two or more factors h_j with $j \geq 2$ have maximal total ℓ -degree at most $n-2$, because each such factor loses one degree relative to its z -weight. The listed cases are therefore exhaustive, and summing them proves (N.4). $\square$

The universal law explains why elementary Lambert approximations look deceptively good — the largest slow logarithms exponentiate as if the periodic term were absent — while the phase data, entering one degree lower, are of the same size as everything else at that degree and can never be dropped.

N.2 Transfer of the newest periodic derivative

Theorem N.3 (Inverse top-jet law). Viewing each coefficient as a differential polynomial in Ψ , for every $n \geq 1$ the monomial linear in the newest jet $\Psi^{(2n-2)}$ has

$$[\Psi^{(2n-2)}]d_n = \frac{(-1)^{n-1}}{2^{n-1}(n-1)!L^{2n-1}}, \quad [\Psi^{(2n-2)}]h_n = [\Psi^{(2n-2)}]g_n = \frac{(-1)^n}{2^{n-1}(n-1)!L^{2n-2}}. \quad (\text{N.5})$$

Proof. For $n \geq 2$ the unique source of jet order $2n-2$ in (L.1) is the $k=1$ term through $A_{n-1}(\theta)z^{n-1}/L$: a factor $A_j(\theta + w)z^j$ in a term of Lagrange index k has $j \leq n-k$ and gains at most $k-1$ derivatives from the w -extraction, so its maximal jet order $2j+k-1 \leq 2n-k-1$ reaches $2n-2$ only at $(k, j) = (1, n-1)$; pure Ψ -factors rank lower. The forward top-jet law (K.4) then gives the d -statement. At order n the logarithm in H uses only $d_{<n}$ (jets $\leq 2n-4$), so h_n inherits $-L$ times the d_n coefficient, and Bell products of lower h_j leave g_n with the same newest-jet coefficient. $\square$

At each order a genuinely new derivative $\Psi^{(2n-2)}$ enters linearly with a nonzero universal coefficient: the periodic hierarchy never closes at finite jet order, and an omission or sign error in A_{n-1} is immediately visible in h_n, g_n — a practical computer-algebra diagnostic.

O Fourier-mode algebra and the analyticity strip

Proposition O.1 (Finite Gamma–zeta convolutions). *For a finite jet monomial $\mathcal{M} = \prod_r (\Psi^{(r)})^{m_r}$,*

$$\widehat{\mathcal{M}}_{\text{FS}}[k] = \sum_{\substack{k_{r,s} \in \mathbb{Z} \setminus \{0\} \\ \sum k_{r,s} = k}} \prod_r \prod_{s=1}^{m_r} (2\pi i k_{r,s})^r c_{\Psi}(k_{r,s}), \quad (\text{O.1})$$

and every Fourier coefficient of every d_n, h_n, g_n is a finite linear combination of such convolutions with coefficients polynomial in ℓ and belonging to the real constant algebra generated by $L^{-1}, \beta, C_{\sharp}$ and the nonoscillatory saddle coefficients.

Proof. Termwise differentiation of (K.3) gives

$$\Psi^{(r)}(u) = \sum_{k \neq 0} (2\pi i k)^r c_{\Psi}(k) e^{2\pi i k u}.$$

The Gamma factor gives exponential decay in $|k|$, so this series and all its differentiated series converge absolutely on the real axis. The Fourier coefficient of a finite product is consequently its ordinary Cauchy convolution. Indexing the m_r copies of $\Psi^{(r)}$ by $s = 1, \dots, m_r$ gives exactly (O.1); absolute convergence justifies every rearrangement.

By theorems L.1 and L.2, each fixed coefficient d_n, h_n, g_n is a finite polynomial in finitely many derivatives of Ψ and of the A_j . Each A_j is itself a finite differential polynomial in Ψ . Expanding these finite polynomials reduces every nonconstant Fourier coefficient to a finite linear combination of the displayed convolutions. The scalar multipliers belong to the real algebra generated by $L^{-1}, \beta, C_{\sharp}$ and the nonoscillatory saddle constants, and their dependence on ℓ is polynomial. This is the claimed form. $\square$

Proposition O.2 (Common strip). *For every $0 \leq \sigma < \pi/(2L)$, all periodic coefficient functions of all d_n, h_n, g_n extend holomorphically and boundedly to $|\Im \theta| \leq \sigma$, with normally convergent Fourier series: Stirling on vertical lines gives $|c_{\Psi}(k)| \leq C_{\varepsilon} \exp[-(\pi^2/L - \varepsilon)|k|]$, and finite differentiation, multiplication, and coefficient extraction preserve the exponential type. Consequently, truncating all Fourier sums at $|k| \leq K$ costs at most $C(1 + \ell)^N K^{\mathcal{O}(1)} e^{-(\pi^2/L - 2\pi\sigma)K}$ at fixed inverse order $N - a$ few modes suffice for high-precision real-phase evaluation, since already $|c_{\Psi}(1)| = 1.06271 \dots \times 10^{-6}$.*

Proof. Put $\tau_k = 2\pi k/L$. Stirling's estimate on vertical lines and a polynomial vertical-line bound for $\zeta(1 - i\tau)$ give, for a fixed exponent A ,

$$|c_{\Psi}(k)| \leq C(1 + |k|)^A \exp\left(-\frac{\pi^2}{L}|k|\right).$$

Every phase derivative only inserts another power of k . A finite convolution of sequences bounded by a polynomial times $e^{-a|k|}$ is again bounded by a (possibly larger) polynomial times $e^{-a|k|}$: use $|k_1| + \dots + |k_r| \geq |k|$ and sum the remaining polynomially weighted geometric tails. The convolution formula of proposition O.1 therefore yields, for every periodic coefficient c occurring through a fixed inverse order N ,

$$|\widehat{c}_{\text{FS}}[k]| \leq C_N(1 + \ell)^N (1 + |k|)^{A_N} e^{-\pi^2|k|/L}.$$

On $|\Im \theta| \leq \sigma$, the modulus of $e^{2\pi i k \theta}$ is at most $e^{2\pi\sigma|k|}$. Since $\pi^2/L - 2\pi\sigma > 0$, the Weierstrass test gives normal convergence and hence a bounded holomorphic extension to the closed substrip. Summing the same majorant for $|k| > K$ gives

$$C_N(1 + \ell)^N K^{A'_N} e^{-(\pi^2/L - 2\pi\sigma)K},$$

after increasing the polynomial exponent to absorb the geometric tail. This is the stated truncation estimate. $\square$

P Phase locking and coefficient tomography

The fast phase can be frozen exactly while the endpoint is approached, turning the formal coefficient functions into observable limits.

Theorem P.1 (Phase-locked endpoint sequences). *Fix $u \in [0, 1)$ and set $\rho_{m,u} = m + u - \beta$, $y_{m,u} = \exp(-\frac{\ell}{2}\rho_{m,u}^2)$ for large integers m . Then $\theta \equiv u \pmod{1}$ along the sequence, and (K.12) holds with $(\ell, \theta) = (\log \rho_{m,u}, u)$, uniformly in u . In particular, the limit set of the first normalized correction $\rho(G/G_0 - 1) - \frac{1}{2}\ell$ is exactly the interval $[\kappa_0 - \max \Psi, \kappa_0 - \min \Psi]$, every point being realized by a phase-locked sequence.*

Proof. By definition, the phase in the ρ -scale is $\theta = \rho + \beta$. Hence at $\rho = \rho_{m,u}$ it equals $m + u$ and is congruent to u modulo one. Moreover $\inf_{u \in [0,1]} \rho_{m,u} \rightarrow \infty$, so the uniform remainder in theorem K.1 remains uniform after this substitution.

At first order, $g_1 = h_1 = \ell/2 + \kappa_0 - \Psi(\theta)$, and therefore

$$\rho \left(\frac{G}{G_0} - 1 \right) - \frac{\ell}{2} = \kappa_0 - \Psi(\theta) + O\left(\frac{(1+\ell)^2}{\rho}\right).$$

Along the phase-locked sequence the error tends to zero uniformly in u , so every value of the continuous function $u \mapsto \kappa_0 - \Psi(u)$ occurs as a subsequential limit. Conversely, given any sequence $y_j \downarrow 0$, compactness of the phase circle provides a subsequence on which $\theta_j \pmod{1} \rightarrow u$; the same formula then forces the normalized correction to converge to $\kappa_0 - \Psi(u)$. The image of a continuous real-valued function on the circle is the interval from its minimum to its maximum, namely $[\kappa_0 - \max \Psi, \kappa_0 - \min \Psi]$. $\square$

Writing $g_n(\ell, u) = \sum_{r \leq n} g_{n,r}(u)\ell^r$, the renormalized statistic

$$\begin{aligned} \mathcal{T}_{N,m}(u) &= \rho_{m,u}^N \left\{ \frac{G(y_{m,u})}{G_0(y_{m,u})} - 1 - \sum_{n < N} \frac{g_n(\log \rho_{m,u}, u)}{\rho_{m,u}^n} \right\} \\ &= g_N(\log \rho_{m,u}, u) + \mathcal{O}_N\left(\frac{(1 + \log \rho_{m,u})^{N+1}}{\rho_{m,u}}\right). \end{aligned} \quad (\text{P.1})$$

reconstructs, phase by phase, every periodic layer $g_{N,r}(u)$ from a polynomial fit in $\log \rho$; the universal $g_{N,N} = 1/(2^N N!)$ supplies an exact normalization. The endpoint has no single asymptotic constant beyond the elementary factor: it has a hierarchy of periodic coefficient *profiles*, whose Fourier spectra (O.1) are predicted explicitly.

Q Elasticity and every inverse derivative

The value remainder in (K.2) cannot be differentiated merely because it is $O(\lambda^{-N})$. The results in this section therefore use the following explicit derivative-stability hypothesis on the forward saddle expansion: for every fixed N and $r \geq 0$, its remainder $R_N(\lambda)$ is C^r for large λ and

$$\frac{d^j}{d\lambda^j} R_N(\lambda) = O_{N,r}(\lambda^{-N}) \quad (0 \leq j \leq r), \quad (\text{Q.1})$$

uniformly in the real phase. For any fixed derivative order, the usual parameter-uniform saddle proof supplies this strengthened input by differentiating the localized integral and repeating the same Gaussian-tail majorant. It is stated here because the value-only theorem used in theorem K.1 is insufficient by itself.

Since $T = L\rho^2/2$ and $y = e^{-T}$, the endpoint Euler operator acts through the two-scale derivation

$$y \frac{d}{dy} = -\frac{1}{L\rho} \mathfrak{d}, \quad \mathfrak{d} = \partial_\theta + z \partial_\ell - z^2 \partial_z, \quad \mathfrak{d}(fz^n) = (\partial_\theta f)z^n + (\partial_\ell f - nf)z^{n+1}. \quad (\text{Q.2})$$

Theorem Q.1 (All-orders elasticity). *Assume (Q.1) to the finite order required by the requested truncation. The elasticity $\mathcal{E}(y) = yG'(y)/G(y)$ has $\mathcal{E} = \sum_{m \geq 1} e_m z^m$ with $e_1 = 1$ and, for $m \geq 2$ (with $h_0 := 0$),*

$$\boxed{e_m = -\frac{\delta_{m,2}}{L} - \frac{1}{L} \left(\partial_\theta h_{m-1} + \partial_\ell h_{m-2} - (m-2)h_{m-2} \right);} \quad \mathcal{E} = \frac{1}{\rho} + \frac{\Psi'(\theta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{1+\ell}{\rho^3}\right). \quad (\text{Q.3})$$

Proof. Implicit differentiation of the truncated phase equation, together with (Q.1) and the lower bound $|\partial_w \mathcal{H}_N| \geq L/(2z)$ from the proof of theorem K.1, shows inductively that applying any fixed number of two-scale derivatives to the omitted inverse remainder preserves its stated symbol order. Thus the finite expansion may be differentiated to any fixed order covered by the hypothesis.

Apply $-(L\rho)^{-1}\mathfrak{d}$ to $\log G = \ell - L\rho - L\beta + \sum h_n \rho^{-n}$: the elementary part contributes $z - z^2/L$, and (Q.2) termwise gives the recurrence; $\partial_\theta h_1 = -\Psi'$ gives the two-term form, recovering the corpus's theorem. $\square$

Two complementary routes organize *all* higher derivatives — one recursive in the derivative order, one closed via Stirling–Bell conversion.

Theorem Q.2 (Derivative hierarchy and the universal two-term law). *Assume (Q.1) through the fixed derivative order under consideration. The ratios $Q_m = y^m G^{(m)}/G$ obey the exact recurrence $Q_{m+1} = y \frac{d}{dy} Q_m + (\mathcal{E} - m)Q_m$ (linear at fixed $\mathcal{E}$), whose coefficient form via (Q.2) computes every fixed $G^{(m)}$ to arbitrary order. Its two-term consequence is universal: for each fixed $m \geq 1$,*

$$\boxed{G^{(m)}(y) = (-1)^{m-1} (m-1)! \frac{G(y)}{y^m \rho} \left[1 + \frac{(\Psi'(\theta) - 1)/L - H_{m-1}}{\rho} + \mathcal{O}_m\left(\frac{1+\ell}{\rho^2}\right) \right],} \quad (\text{Q.4})$$

so the quantile is eventually increasing, concave, with alternating higher derivatives near the endpoint (each fixed m on its own interval $0 < y < y_m$; no uniformity in m is asserted). Alternatively, with $E = y \, d/dy$, $f = \log G$, and signed Stirling numbers $s(k, j)$,

$$H_k := y^k f^{(k)}(y) = \sum_{j=1}^k s(k, j) E^j f, \quad \frac{y^r G^{(r)}(y)}{G(y)} = \mathcal{B}_r^{\text{exp}}(H_1, H_2, \dots, H_r), \quad (\text{Q.5})$$

a closed finite algorithm assembling every ordinary derivative from repeated applications of (Q.2) to $\log G$.

Proof. The recurrence is the product rule on $G^{(m+1)} = (G^{(m)})'$. Writing $Q_m = a_m z(1 + b_m z + \dots)$, the operator $y \, d/dy$ raises the displayed order, so $a_{m+1} = -ma_m$ and $b_{m+1} = b_m - 1/m$; from $Q_1 = \mathcal{E}$, $a_m = (-1)^{m-1} (m-1)!$ and $b_m = e_2 - H_{m-1}$ with $e_2 = (\Psi'(\theta) - 1)/L$. The Stirling–Bell route is the classical Euler-to-ordinary conversion followed by the complete-Bell exponential formula for $G = e^f$. $\square$

R A log-free resummation: the Wright-omega carrier

Every coefficient of the ρ -scale expansion carries polynomials in $\ell = \log \rho$. Those logarithms are artifacts, not intrinsic features: they enter because completing the square leaves $\frac{1}{2} \log \lambda$ (and the constants) outside the carrier. The third absorbed source resums them completely.

R.1 The carrier

Let ω denote the Wright omega function, $\omega(s) + \log \omega(s) = s$ on the real line (equivalently $\omega(s) = W_0(e^s)$), and set

$$H := T + C_\sharp + \frac{L\beta^2}{2}, \quad w := \sqrt{\frac{\omega(4H + \log(2L))}{2L}}, \quad \vartheta := w + \beta. \quad (\text{R.1})$$

Proposition R.1 (Carrier equation). *w is the unique positive solution of*

$$T = \frac{L}{2}w^2 - \frac{L\beta^2}{2} + \frac{1}{2}\log w - C_\sharp, \quad (\text{R.2})$$

i.e. $w^2 = W_0(2Le^{4H})/(2L)$ — the omega form being the numerically stable one, since it never forms the astronomically large argument. (Newton iteration on $\Omega + \log \Omega = S$ from the seed $\Omega_0 = S - \log S + (\log S)/S$ converges immediately for large S .)

Proof. Multiply (R.2) by 4 and substitute $\Omega = 2Lw^2$: the equation becomes $\Omega + \log \Omega = 4H + \log(2L)$; strict monotonicity gives uniqueness. Note W_0 solves the *positive* quadratic-logarithmic equation in T , while W_{-1} parametrizes the exact map $x \mapsto \lambda$ — no contradiction. $\square$

Definition R.2 (W-resummed carrier). $G_W(y) := \vartheta e^{-L\vartheta} = (w + \beta) 2^{-(w+\beta)}$.

Writing the exact phase as $\lambda = \vartheta + D_W$ with $z = 1/w$ and negating the forward expansion against (R.2) leaves the residual

$$\begin{aligned} 0 &= \frac{L}{z}D_W + \mathcal{R}_W(z, D_W), \\ \mathcal{R}_W(z, u) &= \frac{L}{2}u^2 + \frac{1}{2}\log(1 + \beta z + zu) - \Psi(\vartheta + u) \\ &\quad - \sum_{j \geq 1} \frac{A_j(\vartheta + u) z^j}{(1 + \beta z + zu)^j}. \end{aligned} \quad (\text{R.3})$$

every constant and every logarithm has disappeared. The entire diagonal calculus of [appendix L](#) applies verbatim to $\mathcal{K}_W = -\mathcal{R}_W/L$, and the ratio identity becomes $G/G_W = (1 + \frac{zD_W}{1+\beta z})e^{-LD_W}$.

Theorem R.3 (Log-free all-orders expansion, with an extra order). *With d_m^W, h_m^W, g_m^W the diagonal coefficients of (R.3) (via [theorem L.1](#) and its composition form), each is a bounded real-analytic one-periodic differential polynomial in $\Psi, A_1, \dots, A_{m-1}$ — after inserting the closed forward formulas, a polynomial over $\mathbb{Q}(L^{-1})$ in $\Psi, \Psi', \dots, \Psi^{(2m-2)}$ evaluated at ϑ , with no occurrence of $\log w$ — and, uniformly in the phase,*

$$G(y) = G_W(y) \left(1 + \sum_{m=1}^N \frac{g_m^W(\vartheta)}{w^m} + \mathcal{O}_N(w^{-N-1}) \right). \quad (\text{R.4})$$

Knowledge of the forward phase through A_{N-1} thus determines the inverse through order N with a clean w^{-N-1} remainder (no logarithmic grading), one power better than the forward input — the steep-phase Newton gain of [appendix V](#) with no $(1 + \ell)$ factors left to track.

Proof. Fix N , truncate the forward expansion after A_{N-1} , and let $\mathcal{R}_{W,N}$ be the corresponding truncation of (R.3). The exact correction satisfies

$$\mathcal{H}_{W,N}(D_W) := \frac{L}{z}D_W + \mathcal{R}_{W,N}(z, D_W) = -E_{W,N}(D_W), \quad |E_{W,N}(D_W)| \leq C_N z^N.$$

The constant and logarithmic terms were absorbed by the carrier equation, so all coefficients of $\mathcal{R}_{W,N}$ are bounded uniformly in the phase; there is no ℓ variable. The formal fixed-point equation $D_W = z\mathcal{K}_W(z, D_W)$ therefore has unique coefficients $D_W^{[N]} = \sum_{m \leq N} d_m^W z^m$, given by the diagonal formula of [theorem L.1](#). Each coefficient uses only finitely many Taylor coefficients of $\Psi, A_1, \dots, A_{m-1}$, and is consequently a bounded real-analytic one-periodic differential polynomial with no $\log w$.

Formal cancellation gives $|\mathcal{H}_{W,N}(D_W^{[N]})| \leq C_N z^N$. Both the exact and truncated corrections are $O(z)$, and on the segment joining them

$$|\partial_u \mathcal{H}_{W,N}(u)| = \left| \frac{L}{z} + O_N(1) \right| \geq \frac{L}{2z}$$

for small z . Applying the mean-value theorem to the explicit truncated function, not to the uncontrolled value remainder, yields

$$D_W - D_W^{[N]} = O_N(z^{N+1}).$$

Finally the exact identity

$$\frac{G}{G_W} = \left(1 + \frac{zD_W}{1 + \beta z} \right) e^{-LD_W}$$

is analytic in (z, D_W) at the origin. Substitution of the phase expansion and Taylor's theorem give unique coefficients h_m^W, g_m^W and the $O_N(z^{N+1})$ remainder in [\(R.4\)](#). Since $z = w^{-1}$, this is the asserted result. $\square$

R.2 First coefficients and the universal 5/24

With $P_r = \Psi^{(r)}(\vartheta)$ (and $P = P_0$), the diagonal extraction gives

$$\begin{aligned} d_1^W &= \frac{P}{L}, & d_2^W &= \frac{A_1 - \beta/2}{L} + \frac{PP_1}{L^2}, \\ d_3^W &= \frac{A_2 - \beta A_1 + \beta^2/4}{L} + \frac{A_1 P_1 + A_1' P - \frac{1}{2}(P^2 + P + \beta P_1)}{L^2} + \frac{P(PP_2 + 2P_1^2)}{2L^3}, \end{aligned} \quad (\text{R.5})$$

The algebraic shape matches the ρ -scale recursion after its carrier term is changed, but the coefficients are not identified: d_m^W and d_m belong to different asymptotic coordinates. The Wright-omega logarithmic coefficients are $h_1^W = -P$, $h_3^W = \beta A_1 - A_2 - \frac{\beta^2}{4} + [-A_1 P_1 + A_1 - A_1' P + \frac{1}{2}(P^2 - 2\beta P + P + \beta P_1 - \beta)]/L + P(-PP_2 - 2P_1^2 + 2P_1)/(2L^2)$, and, after substituting the explicit A_1 , every L^{-1} constant in h_2^W cancels:

$$\boxed{h_2^W = \frac{5}{24} + \frac{P(1 - P_1)}{L} + \frac{P_2 + P_1^2}{2L^2}, \quad g_1^W = -P, \quad g_2^W = h_2^W + \frac{P^2}{2},} \quad (\text{R.6})$$

with $g_3^W = h_3^W + h_1^W h_2^W + (h_1^W)^3/6$. The constant 5/24 is a robust invariant of the carrier normalization. Suppressing all periodic data while keeping the nonoscillatory parts $A_1^{(0)} = \frac{1}{24} + \frac{1}{2L}$, $A_2^{(0)} = \frac{1}{4L^2}$, $A_3^{(0)} = -\frac{7}{2880} - \frac{1}{24L} - \frac{1}{4L^2} + \frac{1}{6L^3}$ gives the regression targets

$$\begin{aligned} (h_1^W)^{(0)} &= 0, & (h_2^W)^{(0)} &= \frac{5}{24}, & (h_3^W)^{(0)} &= -\frac{L+4}{24L}, \\ (h_4^W)^{(0)} &= \frac{37L^2 + 120L + 1080}{2880L^2}, & (g_4^W)^{(0)} &= \frac{199L^2 + 240L + 2160}{5760L^2}. \end{aligned} \quad (\text{R.7})$$

The highest-derivative law is *carrier-invariant*: the W-scale coefficients obey exactly the right-hand coefficients in [\(N.5\)](#), with (d_m, h_m, g_m) replaced by (d_m^W, h_m^W, g_m^W) and evaluated at ϑ (the

unique source is still A_{m-1} , whose top jet is normalization-independent), so the first-harmonic contribution of this single linear top-jet monomial has the precisely defined amplitude

$$\text{amp}_{\{\pm 1\}}((g_m^W)_{\text{top,lin}}) = 2 |c_\Psi(1)| \frac{(2\pi^2/L^2)^{m-1}}{(m-1)!} \quad (\text{R.8})$$

The same identity holds for the corresponding top-jet contribution in the ρ -normalization. It describes one explicitly isolated summand, not the amplitude or large-order growth of the complete coefficient: other differential-polynomial terms can reinforce or cancel its first Fourier mode.

R.3 The inversion is not the source of divergence

Theorem R.4 (Convergent bare skeleton). *Set $\Psi \equiv 0$, $A_j \equiv 0$ in (R.3). The bare reversion $D_{\text{bare}}(z)$ of $0 = \frac{L}{z}D + \frac{L}{2}D^2 + \frac{1}{2}\log(1 + \beta z + zD)$ is a convergent analytic germ at $z = 0$,*

$$D_{\text{bare}}(z) = -\frac{\beta}{2L}z^2 + \frac{\beta^2}{4L}z^3 + \left(-\frac{\beta^3}{6L} + \frac{\beta}{4L^2}\right)z^4 + \mathcal{O}(z^5). \quad (\text{R.9})$$

Hence any divergence of the complete inverse series enters through the large-order saddle data A_j and the differentiated Fourier modes — not through formal inversion of the quadratic–logarithmic carrier.

Proof. The equation forces $D = z^2V$; dividing by z , $0 = LV + \frac{L}{2}z^3V^2 + \frac{1}{2z}\log(1 + \beta z + z^3V)$, whose logarithmic quotient is analytic at $z = 0$ with unique root $V(0) = -\beta/(2L)$ and V -derivative $L \neq 0$; the analytic implicit function theorem gives the germ, and coefficient comparison the series. $\square$

Theorem R.5 (Conditional Gevrey-one transfer). *For $0 < \sigma' < \sigma$, write $\|f\|_\sigma = \sup_{|\Im u| \leq \sigma} |f(u)|$. Suppose Ψ and every A_j extend holomorphically to $|\Im u| < \sigma$, continuously to the closed strip, and for some $C, K \geq 1$ satisfy*

$$\|\Psi\|_\sigma \leq C, \quad \|A_j\|_\sigma \leq CK^j j! \quad (j \geq 1).$$

Then there are constants C', K' , depending only on $C, K, L, \beta, \sigma - \sigma'$, such that

$$\|d_m^W\|_{\sigma'} + \|h_m^W\|_{\sigma'} + \|g_m^W\|_{\sigma'} \leq C'(K')^m m! \quad (m \geq 1).$$

Thus W -resummed inversion preserves Gevrey order one; the theorem makes no claim about a sharp exponential constant or about Borel summability.

Proof. Choose an intermediate strip $\sigma' < \sigma_1 < \sigma$ and put $\delta = \sigma - \sigma_1$. Cauchy's estimate gives

$$\|\Psi^{(r)}\|_{\sigma_1} \leq C r! \delta^{-r}, \quad \|A_j^{(r)}\|_{\sigma_1} \leq CK^j j! r! \delta^{-r}.$$

The Taylor coefficients in the auxiliary variable u of $\Psi(\vartheta + u)$ and $A_j(\vartheta + u)$ therefore lose the factor $r!$ and are bounded by geometric powers of δ^{-1} . The expansions of $(1 + \beta z + zu)^{-j}$ and of the logarithm in (R.3) also have coefficients bounded by geometric powers, after enlarging a constant depending on L and β . Consequently there are C_0, K_0 such that every coefficient of total weight $r + s$ in $\mathcal{K}_W(z, u)$ is bounded on the σ_1 strip by

$$C_0 K_0^{r+s} (r + s)!.$$

In the diagonal formula for d_m^W , a term of Lagrange index k is a product whose total weight is $m - 1$. For every decomposition $n_1 + \dots + n_k = m - 1$ one has $n_1! \dots n_k! \leq (m - 1)!$, and the number of ordered decompositions and choices of factors is at most exponential in m . Summing over $1 \leq k \leq m$ and absorbing these exponential counts into K_1^m gives

$$\|d_m^W\|_{\sigma_1} \leq C_1 K_1^m m!.$$

Equivalently, one may close the same majorant induction with $\sum_{j=0}^m j!(m - j)! \leq C_2 m!$.

The exact logarithmic ratio and multiplicative ratio are analytic compositions of z and D_W at the origin. Their coefficient formulas are finite convolution/Bell sums of total weight m ; the same factorial-product bound and exponential count give the asserted estimates for h_m^W and g_m^W . Restricting from σ_1 to σ' completes the proof. $\square$

R.4 Derivatives, numerics, and the pre-asymptotic hierarchy

Under the derivative-stability hypothesis (Q.1), the W-scale remainder inherits the same fixed-order differentiability by the implicit argument used in [theorem Q.1](#). Differentiating (R.2) then gives $dT/dw = Lw + (2w)^{-1}$, so with $\mathcal{H}_W(w) = \sum_m h_m^W(w + \beta)w^{-m}$,

$$\frac{yG'(y)}{G(y)} = \frac{L - (w + \beta)^{-1} - \mathcal{H}'_W(w)}{Lw + (2w)^{-1}} = \frac{1}{w} + \frac{\Psi'(\vartheta) - 1}{Lw^2} + \frac{\beta + h_1^W - (h_2^W)' - \frac{1}{2}}{Lw^3} + \mathcal{O}(w^{-4}), \quad (\text{R.10})$$

and repeated $y \, d/dy$ derivatives are powers of the single operator $-(Lw + (2w)^{-1})^{-1} d/dw$ – no second inversion, consistent with [appendix Q](#).

On the exact dyadic quantiles the improvement over the ρ -carrier is dramatic – the bare W-carrier already beats the ρ -scale’s two-correction approximant at small n :

n	w	carrier only	order 1	order 2	order 3
5	6.066	-4.79×10^{-3}	-4.79×10^{-3}	8.58×10^{-4}	-4.13×10^{-4}
10	11.845	-1.35×10^{-3}	-1.35×10^{-3}	1.36×10^{-4}	-3.32×10^{-5}
20	22.680	-3.84×10^{-4}	-3.84×10^{-4}	2.11×10^{-5}	-2.95×10^{-6}
40	43.566	-1.07×10^{-4}	-1.07×10^{-4}	3.16×10^{-6}	-2.30×10^{-7}
80	84.491	-2.87×10^{-5}	-2.87×10^{-5}	4.50×10^{-7}	-1.66×10^{-8}
160	165.444	-7.55×10^{-6}	-7.55×10^{-6}	6.11×10^{-8}	-1.14×10^{-9}

Table 2: Relative errors of the W-resummed approximants $G_W \exp(\sum_{j \leq r} h_j^W/w^j)$ at $y_n = F(2^{-n})$. Compare [table 3](#): order for order the W-scale wins throughout (at $n = 160$, third order -1.14×10^{-9} versus 4.92×10^{-8} ; second order 6.1×10^{-8} versus -3.6×10^{-6}), and the *bare* carrier alone outperforms the ρ -scale’s full h_2 -approximant through $n \approx 40$ (-1.07×10^{-4} versus -1.43×10^{-4} at $n = 40$; the comparison reverses beyond $n \approx 80$ as the ρ -corrections catch up) by absorbing the logarithmic drift.

Remark R.6 (Pre-asymptotic hierarchy). Since $\|\Psi\|_\infty \approx 2.13 \times 10^{-6}$ while $(h_2^W)^{(0)} = 5/24$, the nominally first correction $h_1^W/w = -\Psi/w$ is numerically *invisible* against h_2^W/w^2 until $w \gtrsim (5/24)/\|\Psi\|_\infty \approx 10^5$, i.e. $T \sim 3 \times 10^9$. Practical approximations must therefore include h_2^W whenever they include h_1^W (cf. the table’s near-identical first two columns). This does not violate asymptotic ordering; one coefficient simply happens to be extraordinarily small – the same smallness that makes the phase oscillation resolvable only at high precision.

Remark R.7 (Candidate beyond-all-orders scales; heuristic only). The exact dyadic tail suggests an inverse perturbation on the scale $w^{-1}e^{-2^{q_*}}$, where q_* is the exact saddle coordinate; replacing it by ϑ is not uniform, as explained after (T.7). A second candidate comes from the isolated linear top-jet contribution (R.8). Combining its k th Fourier factor with $|c_\Psi(k)|$ and optimizing in k places the largest term of that *single contribution* near $k_* \sim 2(m-1)L/\pi^2$ and suggests the factorial scale $m!(8/\pi^2)^m$, least term near $m \sim \pi^2 w/8$, and exponential size $e^{-\pi^2 w/8}$.

None of these optimizations controls the remaining saddle diagrams. They may reinforce or cancel the isolated Fourier mode, and no analytic continuation of the Borel transform is proved here. In particular this calculation does *not* establish an exact limsup, Borel summability, a Borel singularity at $\pi^2/8$, or a Stokes multiplier. It records only the candidate constants used to formulate [conjectures W.1, W.2 and W.5](#). Locating the nearest complex singularity of the bare implicit germ (R.9) is a separate elimination problem and would measure the carrier’s geometric, rather than factorial, coefficient growth.

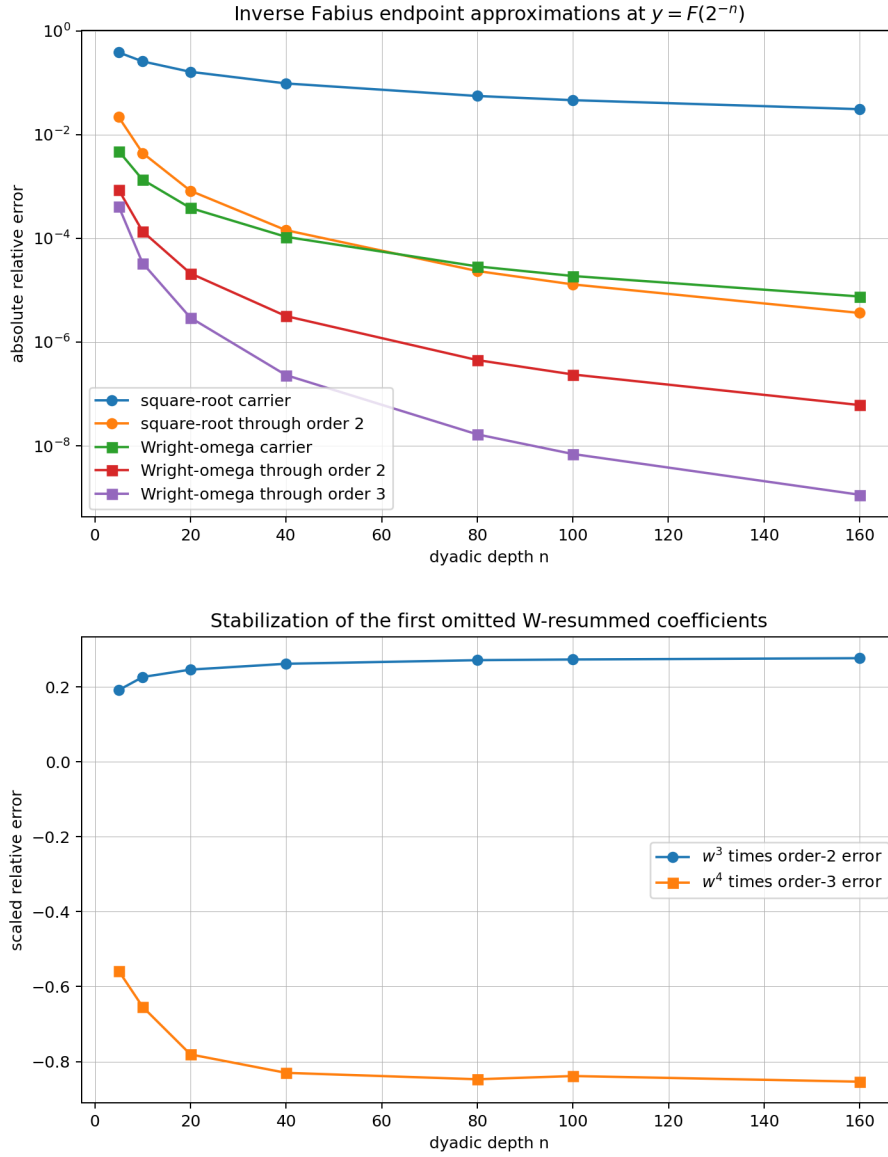


Figure 2: Top: absolute relative errors of the ρ - and W-carrier approximants on the exact dyadic quantiles. Bottom: scaled residuals after the second and third W-corrections, consistent with the predicted next powers w^{-3} , w^{-4} .

S The forward saddle engine: constrained partition sums

The inverse formulas consume the periodic forward coefficients A_j . This chapter closes their construction with a finite partition formula for the universal Gaussian layer of the saddle analysis of the Bromwich integral.

Let $s > 0$ solve the saddle equation $x = -K'(s)$ and normalize the saddle jets and their ratios

$$\varkappa_r = s^r K^{(r)}(s), \quad \varrho_r = \frac{\varkappa_r}{\varkappa_2} \quad (r \geq 3). \quad (\text{S.1})$$

Scaling the contour by $t = s(1 + iZ/\sqrt{\varkappa_2})$, $Z \sim N(0, 1)$, produces the formal correction factor

$$\mathcal{S} = \mathbb{E} \left[\left(1 + \frac{iZ}{\sqrt{\varkappa_2}} \right)^{-1} \exp \left(\sum_{r \geq 3} \frac{\varrho_r}{r!} \frac{(iZ)^r}{\varkappa_2^{(r-2)/2}} \right) \right] \sim 1 + \sum_{n \geq 1} \frac{C_n}{\varkappa_2^n}, \quad (\text{S.2})$$

the reciprocal factor being the CDF's Bromwich $1/t$ – deleting it would produce the *density* coefficients instead.

Theorem S.1 (Constrained-partition formula). *For $n \geq 1$,*

$$C_n = \sum_{\substack{m_0, m_3, m_4, \dots \geq 0 \\ m_0 + \sum_{r \geq 3} (r-2)m_r = 2n}} (-1)^{m_0 + N/2} (N-1)!! \prod_{r \geq 3} \frac{\varrho_r^{m_r}}{m_r! (r!)^{m_r}}, \quad N := m_0 + \sum_{r \geq 3} r m_r, \quad (\text{S.3})$$

a finite sum (the constraint forces N even and $r \leq 2n+2$). When all ϱ_r vanish it reduces to the Gaussian Mills-ratio coefficients $C_n = (-1)^n (2n-1)!!$ – a sensitive sign and normalization check, verified symbolically $(-1, 3, -15)$. The logarithmic coefficients $\log \mathcal{S} \sim \sum \Lambda_n \varkappa_2^{-n}$ follow nonrecursively from

$$\Lambda_n = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \sum_{n_1 + \dots + n_k = n, n_j \geq 1} C_{n_1} \cdots C_{n_k}, \quad \Lambda_1 = C_1, \quad \Lambda_2 = C_2 - \frac{1}{2} C_1^2. \quad (\text{S.4})$$

Proof. Expand the reciprocal geometrically (m_0 factors of $-iZ/\sqrt{\varkappa_2}$) and the exponential by multiplicities m_r . The total inverse half-power of $\varkappa_2$ is $\frac{1}{2}(m_0 + \sum (r-2)m_r)$, giving the constraint; the Gaussian degree is N with $N - 2n = 2 \sum m_r$ even, so $\mathbb{E} Z^N = (N-1)!!$; the powers of i and the reciprocal's sign give $(-1)^{m_0 + N/2}$; and the exponential series supplies the multiplicity factorials. The Λ formula is $\log(1 + \cdot)$ composed with the C -series. $\square$

The first coefficients are $C_1 = -1 - \frac{1}{2}\varrho_3 + \frac{1}{8}\varrho_4 - \frac{5}{24}\varrho_3^2$ and

$$C_2 = 3 + \frac{5}{2}\varrho_3 - \frac{5}{8}\varrho_4 + \frac{1}{8}\varrho_5 - \frac{1}{48}\varrho_6 + \frac{35}{24}\varrho_3^2 - \frac{35}{48}\varrho_3\varrho_4 + \frac{7}{48}\varrho_3\varrho_5 + \frac{35}{384}\varrho_4^2 + \frac{35}{48}\varrho_3^3 - \frac{35}{64}\varrho_3^2\varrho_4 + \frac{385}{1152}\varrho_3^4. \quad (\text{S.5})$$

With $q = \log_2 s$ and $J(q) = K(2^q)$, the normalized jets are generated exactly by the operator identity $\varkappa_r(q) = L^{-r} \prod_{j=0}^{r-1} (D_q - jL) J(q)$; inserting the exact decomposition of K from [appendix T](#) and converting from the saddle coordinate q to the exact Lambert phase completes the pipeline

$$\Psi \longrightarrow \{\varkappa_r\} \xrightarrow{(\text{S.3})} \{C_n\} \xrightarrow{(\text{S.4})} \{\Lambda_n\} \longrightarrow \{A_j\} \xrightarrow{(\text{L.1})} \{d_n\} \xrightarrow{(\text{L.3})} \{g_n\}, \quad (\text{S.6})$$

every arrow a finite differential-polynomial or coefficient-extraction operation at fixed order. The raw term count of (S.3) grows like an integer-partition function – subexponentially – rather than like all set partitions, which removes the practical bottleneck to generating $A_3, A_4, \dots$ in compact Fourier-differential form.

For implementers, the corpus also supplies a second, fully unrolled closed route to the same A_j (recorded in the third absorbed source from the archived small-argument study): with the signed

Stirling coefficients $\prod_{r=1}^n (z - r) = \sum_m c(n, m) z^m$ and harmonic numbers H_n , the bounded saddle jets are

$$q_n(u) = (-1)^n n! \left(\frac{H_n}{L} - \frac{1}{2} \right) + \sum_{m=0}^n c(n, m) \frac{\Psi^{(m+1)}(u)}{L^{m+1}}, \quad (\text{S.7})$$

the normalized mass coefficients a_j are finite sums over compositions of $2j$ and subsets weighted by double factorials $(2j + 2|S| - 1)!!$, and $A_j = \sum_{r \leq j} \frac{(-1)^{r-1}}{r} \sum_{\mathcal{C}(j,r)} a_{s_1} \cdots a_{s_r}$ — an explicit finite differential polynomial in $\Psi', \dots, \Psi^{(2j)}$ whose top jet reproduces (K.4). The partition engine (S.3) and this composition route validate each other symbolically.

T The exact 2-adic completion of the Laplace exponent

Everything so far is asymptotic in ρ . This chapter is *exact*: it completes the quadratic–periodic normal form of K by an explicitly 2-adic, everywhere-convergent tail, and shows that the periodic phase and the beyond-all-orders sectors are two faces of one Mellin kernel.

T.1 The decomposition and the dyadic Bose tail

The product (K.1) gives the dyadic difference equation $K(2t) - K(t) = \log(1 - e^{-t}) - \log t$. Define the *dyadic Bose tail*

$$B(t) = - \sum_{k \geq 0} \log(1 - e^{-2^k t}), \quad B(t) = - \log(1 - e^{-t}) + B(2t). \quad (\text{T.1})$$

Theorem T.1 (Exact quadratic–periodic–dyadic decomposition). *For every $t > 0$,*

$$K(t) = - \frac{\log^2 t}{2L} + \frac{1}{2} \log t - C_B + \Psi(\log_2 t) + B(t). \quad (\text{T.2})$$

Proof. With $Q(q) = -\frac{L}{2}q^2 + \frac{L}{2}q$ one checks $Q(q+1) - Q(q) = -Lq$, so $\Omega(q) := K(2^q) - Q(q) - B(2^q)$ is 1-periodic by the two difference equations. The small- t Mellin expansion of B (theorem T.3 below) is $B(t) = \log^2(1/t)/(2L) + \frac{1}{2} \log(1/t) + C_B - \Psi(\log_2 t) + o(1)$, and $K(t) \rightarrow 0$ as $t \downarrow 0$; letting $q = m + u \rightarrow -\infty$ at fixed u identifies $\Omega(u) = \Psi(u) - C_B$, and periodicity upgrades the identity to all q . $\square$

Theorem T.2 (Exact 2-adic coefficients). *For every $t > 0$,*

$$B(t) = \sum_{n \geq 1} b_n e^{-nt}, \quad b_n = \frac{2^{\nu_2(n)+1} - 1}{n}, \quad 0 < b_n < 2, \quad (\text{T.3})$$

with the elementary tail enclosure $0 < B(t) - \sum_{n \leq N} b_n e^{-nt} \leq 2e^{-(N+1)t}/(1 - e^{-t})$.

Proof. Expand each Bose logarithm and regroup $B(t) = \sum_{k \geq 0} \sum_{m \geq 1} m^{-1} e^{-m2^k t}$: the representations $n = m2^k$ are indexed by $0 \leq k \leq \nu_2(n)$, with total weight $\frac{1}{n} \sum_{k \leq \nu_2(n)} 2^k = (2^{\nu_2(n)+1} - 1)/n$. $\square$

The first terms, $B(t) = e^{-t} + \frac{3}{2}e^{-2t} + \frac{1}{3}e^{-3t} + \frac{7}{4}e^{-4t} + \frac{1}{5}e^{-5t} + \frac{1}{2}e^{-6t} + \dots$, spike at powers of two: the exact arithmetic trace of the dyadic scale hierarchy. The integer weight $f(n) = nb_n = 2^{\nu_2(n)+1} - 1$ satisfies $f(2n) = 2f(n) + 1$, $f(2n+1) = 1$, and its generating function obeys the Mahler equation $C(z) = 2C(z^2) + z/(1 - z)$, placing the sectors in the same digital universe as the Thue–Morse and ruler sequences — but with geometric accumulation over the full 2-adic valuation in place of digit-sum parity. Note the crosswalk: $\nu_2(n) + 1$ is precisely the integer zero order of $\widehat{\text{up}}_{2\pi}$ that drives the exactness theorems of the companion volume *Dyadic-Comb Frontiers*; here the same statistic returns as the arithmetic seed of the beyond-all-orders sectors.

T.2 One Mellin kernel, two asymptotic ends

Theorem T.3 (Mellin bridge). *For $\Re s > 0$,*

$$\sum_{n \geq 1} \frac{b_n}{n^s} = \frac{\zeta(s+1)}{1-2^{-s}}, \quad \int_0^\infty B(t) t^{s-1} dt = \Gamma(s) \frac{\zeta(s+1)}{1-2^{-s}}, \quad (\text{T.4})$$

and contour shifting in the Mellin inversion gives, for every fixed integer $M \geq 1$, with $R = \log(1/t)$,

$$B(t) = \frac{R^2}{2L} + \frac{R}{2} + C_B - \Psi(\log_2 t) + \sum_{m=1}^{M-1} \frac{(-1)^m \zeta(1-m)}{m!(1-2^{-m})} t^m + \mathcal{O}_M(t^M) \quad (t \downarrow 0). \quad (\text{T.5})$$

Proof. For $\Re s > 0$, Tonelli's theorem applied to absolute values gives

$$\sum_{n \geq 1} \frac{b_n}{n^s} = \sum_{k \geq 0} \sum_{m \geq 1} \frac{1}{m(m2^k)^s} = \zeta(s+1) \sum_{k \geq 0} 2^{-ks} = \frac{\zeta(s+1)}{1-2^{-s}}.$$

The exact series (T.3) is locally uniformly convergent for $t > 0$. A second application of Tonelli, followed by $\int_0^\infty e^{-nt} t^{s-1} dt = n^{-s} \Gamma(s)$, proves the Mellin transform identity. To make the endpoint bounds explicit, put $J = \lceil \log_2(1/t) \rceil$ when $0 < t \leq 1$. For $k \leq J$, $-\log(1 - e^{-2^k t}) \leq C + |\log(2^k t)|$, whose sum is $O((1 + |\log t|)^2)$; for $k > J$, the inequality $-\log(1 - e^{-u}) \leq C e^{-u}$ for $u \geq 1$ and dyadic growth give a bounded tail. Thus $B(t)$ has at most quadratic logarithmic growth at zero. Its defining series also gives exponential decay at infinity, so Mellin inversion is valid on every line $\Re s = c > 0$:

$$B(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \frac{\zeta(s+1)}{1-2^{-s}} t^{-s} ds.$$

Shift a rectangular contour to $\Re s = -M - \frac{1}{2}$, choosing its horizontal edges between the poles on $\Re s = 0$. Stirling's bound gives exponential decay $e^{-\pi|\Im s|/2}$ for $\Gamma(s)$, whereas $\zeta(s+1)$ has only polynomial vertical growth. Thus the horizontal integrals vanish and the residue series on $\Re s = 0$ converges normally. The pole at $s = 0$ has order three. The Laurent expansions

$$\Gamma(s) = \frac{1}{s} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12} \right) s + O(s^2), \quad \zeta(1+s) = \frac{1}{s} + \gamma - \gamma_1 s + O(s^2),$$

$$\frac{1}{1-e^{-Ls}} = \frac{1}{Ls} + \frac{1}{2} + \frac{Ls}{12} + O(s^3), \quad t^{-s} = e^{Rs} = 1 + Rs + \frac{R^2 s^2}{2} + O(s^3)$$

show that its residue contribution is

$$\frac{R^2}{2L} + \frac{R}{2} + \frac{L}{12} + \frac{\pi^2 - 6\gamma^2 - 12\gamma_1}{12L} = \frac{R^2}{2L} + \frac{R}{2} + C_B.$$

The other zeros of $1 - 2^{-s}$ are the simple points $s = \chi_k$, $k \neq 0$. Their residues sum, after replacing k by $-k$, to

$$\sum_{k \neq 0} \frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L} t^{\chi_k} = -\Psi(\log_2 t).$$

At $s = -m$, $1 \leq m \leq M$, the residue of Γ is $(-1)^m/m!$, so the corresponding term is

$$\frac{(-1)^m \zeta(1-m)}{m!(1-2^{-m})} t^m.$$

On the new vertical line the denominator is bounded away from zero, and the same Gamma estimate bounds the remaining integral by $\mathcal{O}_M(t^{M+1/2})$. Retain the residues with $m < M$ and absorb the $m = M$ term and the new-line integral into $\mathcal{O}_M(t^M)$. This proves (T.5). $\square$

Two faces of one kernel

The Gamma–zeta phase Ψ (small- t residues at the complex dimensions $2\pi i k/L$) and the exact 2-adic exponential sectors (large- t Dirichlet regrouping) are generated by the *same* kernel $\Gamma(s)\zeta(s+1)/(1-2^{-s})$. No new frequencies ever appear downstream: the entire hierarchy of inverse oscillations consists of integer convolutions of the modes χ_k ([proposition O.1](#)), and the beyond-all-orders remainder is not an opaque error but the explicit arithmetic object ([T.3](#)). Comparing the two constant terms independently recovers the relation $C_{\sharp} + C_B = -\frac{1}{2} \log(2\pi)$ already visible in ([K.6](#)).

T.3 Exact saddle map and ultraflat jets

Corollary T.4 (Exact 2-adic saddle map). *With $t = 2^q$, the saddle abscissa $x = -K'(t)$ is exactly*

$$x(q) = 2^{-q} \left(q - \frac{1}{2} - \frac{\Psi'(q)}{L} \right) + \sum_{n \geq 1} (2^{\nu_2(n)+1} - 1) e^{-n2^q}, \quad (\text{T.6})$$

the first term the complete algebraic–periodic saddle map, the second an exact convergent ultraflat correction. Derivative bookkeeping is closed: with Stirling numbers of the second kind,

$$\frac{d^r}{dq^r} B(2^q) = L^r \sum_{n \geq 1} b_n \sum_{j=0}^r \left\{ \begin{matrix} r \\ j \end{matrix} \right\} (-n2^q)^j e^{-n2^q}, \quad t^r B^{(r)}(t) = (-1)^r \sum_{n \geq 1} b_n (nt)^r e^{-nt}. \quad (\text{T.7})$$

Proof. Differentiate the exact decomposition ([T.2](#)). Since $q = \log_2 t = (\log t)/L$,

$$-K'(t) = \frac{1}{t} \left(q - \frac{1}{2} - \frac{\Psi'(q)}{L} \right) - B'(t).$$

The series in ([T.3](#)) and all of its termwise derivatives converge uniformly on each half-line $t \geq t_0 > 0$, so

$$-B'(t) = \sum_{n \geq 1} n b_n e^{-nt} = \sum_{n \geq 1} (2^{\nu_2(n)+1} - 1) e^{-nt}.$$

Putting $t = 2^q$ gives ([T.6](#)).

Ordinary termwise differentiation gives

$$t^r B^{(r)}(t) = (-1)^r \sum_{n \geq 1} b_n (nt)^r e^{-nt}.$$

For derivatives in q , use $d/dq = L, t d/dt$ and the operator identity

$$(t d_t)^r = \sum_{j=0}^r \left\{ \begin{matrix} r \\ j \end{matrix} \right\} t^j d_t^j.$$

Applying it termwise to e^{-nt} proves the first formula in ([T.7](#)). Uniform convergence on compact q -intervals justifies all differentiations and interchanges. $\square$

The jet formulas expose the technical obstacle to naive hyperasymptotics: each q -derivative of an e^{-n2^q} sector produces powers of 2^q , so the ordinary inverse-power saddle hierarchy is *not uniform* inside a fixed dyadic sector; and although the algebraic inversion gives the saddle coordinate $q_{\star}(y) = \theta + \mathcal{O}(\ell/\rho)$, the induced change in the exponent $n2^{q_{\star}}$ is of order $2^{\theta} \ell/\rho \rightarrow \infty$ – the exact saddle coordinate is part of the beyond-all-orders datum ([conjecture W.5](#)).

U Verification

The two absorbed packages verify overlapping claims by independent implementations; both suites live under `assets/`.

Symbolic. Over algebraically independent symbols $L, \beta, C_\sharp, \ell, P_0, \dots, P_4, A_1, A'_1, A_2$, both codes confirm $d_j^{\text{diagonal}} - d_j^{\text{triangular}} \equiv 0$ ($j \leq 3$), the third correction (M.1), the universal top-log (N.3), and the top-jet transfer (N.5); the saddle generator reproduces $C_n = (-1)^n(2n-1)!!$ at vanishing ϱ and archives C_1, C_2, C_3, Λ_2 (`generated_coefficients.tex`).

Exact identities. At 70–100 digits, the decomposition (T.2) agrees with the defining product for K to $\sim 10^{-16}$ at $t \in \{0.1, 1, 3, 10\}$ (twelve Fourier modes); the 2-adic series (T.3) agrees with the product to 7.5×10^{-39} at $t = 0.7$ and 3.8×10^{-55} at $t = 1.5$; the small- t expansion (T.5) to $\sim 4 \times 10^{-16}$ on $0.02 \leq t \leq 0.2$; and the Dirichlet identity (T.4) to the expected algebraic truncation tail with 5×10^4 unaccelerated terms.

Wright-omega layer. The third source’s suite independently reconstructs the W-scale d_m^W, h_m^W, g_m^W symbolically from the residual equation (SymPy; the identities of appendix R including the $5/24$ cancellation and the zero-periodic values (R.7)), verifies the carrier equation and Newton evaluation of ω , and generates the carrier-comparison tables of table 2 on the same exact dyadic quantiles at 100 digits with nine Fourier modes.

Exact dyadic quantiles. With $a_0 = 1$ and $a_n = (2^n - 1)^{-1} \sum_{k < n} a_k / (n - k + 1)!$, the repository’s exact values give $y_n = F(2^{-n}) = 2^{-n(n-1)/2} a_n$ with $G(y_n) = 2^{-n}$ *exactly* – no inversion error, only the asymptotic formula is evaluated (at 100 digits, twelve modes). Signed relative errors of $G^{[r]} = G_0 \exp(\sum_{j \leq r} h_j \rho^{-j})$:

n	ρ	G_0 only	through h_1	through h_2	through h_3
5	6.44	-3.83×10^{-1}	7.89×10^{-2}	-2.17×10^{-2}	6.41×10^{-3}
10	12.08	-2.57×10^{-1}	2.69×10^{-2}	-4.40×10^{-3}	7.19×10^{-4}
20	22.82	-1.62×10^{-1}	8.76×10^{-3}	-8.24×10^{-4}	7.40×10^{-5}
40	43.65	-9.66×10^{-2}	2.72×10^{-3}	-1.43×10^{-4}	6.96×10^{-6}
80	84.54	-5.53×10^{-2}	8.05×10^{-4}	-2.34×10^{-5}	6.05×10^{-7}
120	125.08	-3.94×10^{-2}	3.88×10^{-4}	-7.92×10^{-6}	1.40×10^{-7}
160	165.47	-3.08×10^{-2}	2.30×10^{-4}	-3.64×10^{-6}	4.92×10^{-8}

Table 3: Exact-quantile tests: the new third correction improves the two-correction error by factors 11–74 across the tested range.

V Proof details

Finiteness bookkeeping. In (L.1) the index- k summand requests $z^{n-k} w^{k-1}$: no $\mathcal{K}$ -power beyond z^{n-k} , no Taylor jet beyond order $k-1$, and A_j only for $j \leq n-k$; hence A_{n-1} occurs only at $k=1$ – the key to the top-jet proof.

Formal-to-asymptotic transfer. Separate the explicit truncated residual from the value-only error:

$$\mathcal{F}_N(w) = \frac{L}{z} w + \mathcal{R}_N(w), \quad \mathcal{F}_N(D) = -E_N(D), \quad |E_N(D)| = O_N(z^N).$$

Formal cancellation gives $\mathcal{F}_N(D^{[N]}) = O_N((1+\ell)^N z^N)$, while $\inf |\partial_w \mathcal{F}_N| \geq L/(2z)$ throughout the relevant $O(\ell z)$ neighborhood. Applying the mean-value theorem to $\mathcal{F}_N$ alone gives

$$|D - D^{[N]}| \leq \frac{2z}{L} (|E_N(D)| + |\mathcal{F}_N(D^{[N]})|) = O_N((1+\ell)^N z^{N+1}).$$

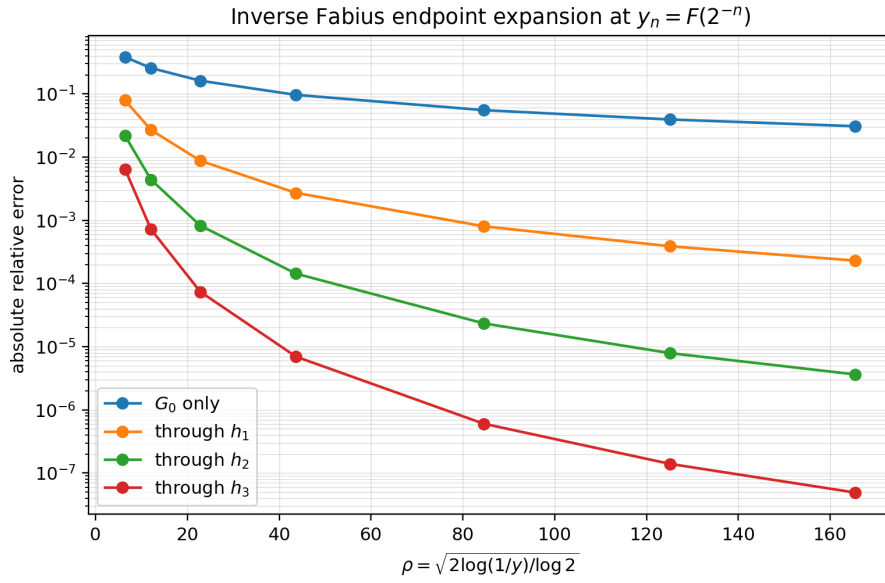


Figure 3: Absolute relative errors on the exact dyadic quantiles; each added logarithmic correction produces the expected strong reduction once ρ is moderately large.

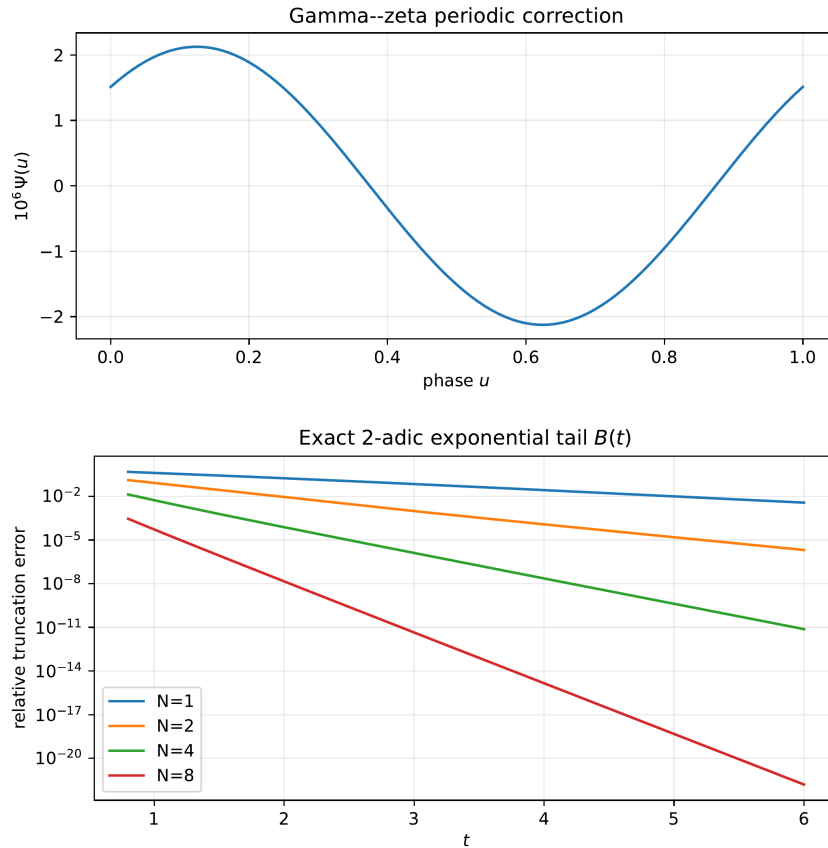


Figure 4: Top: the centered Gamma-zeta fluctuation Ψ – tiny amplitude, same asymptotic order as the constants. Bottom: relative error of $N = 1, 2, 4, 8$ terms of the exact 2-adic series for $B(t)$, with the elementary enclosure of [theorem T.2](#).

No derivative of E_N is assumed. This is the gain-of-one-power argument that lets a forward expansion through A_{N-1} certify an inverse expansion through d_N . The map $\lambda \mapsto \lambda e^{-L\lambda}$ then transfers phase error to relative inverse error of the same order.

W Conjectures and frontier directions

Conjecture W.1 (Gevrey-one growth). *For every $0 < \sigma' < \pi/(2L)$ there are $C, A > 0$ such that*

$$\|d_n^W\|_{\sigma'} + \|h_n^W\|_{\sigma'} + \|g_n^W\|_{\sigma'} \leq CA^n n! \quad (n \geq 1).$$

Writing a ρ -coefficient as $p_n(\ell, \theta) = \sum_r p_{n,r}(\theta) \ell^r$, the same estimate holds for the coefficient norm $\max_r \|p_{n,r}\|_{\sigma'}$, for each of $p_n = d_n, h_n, g_n$. The calculation in the preceding heuristic suggests $8/\pi^2$ as a candidate exponential constant for one linear contribution, but this conjecture asserts only Gevrey order one, not that constant and not Borel summability.

Conjecture W.2 (Optimal truncation). *Fix $u \in \mathbb{R}/\mathbb{Z}$ and approach the endpoint along a W -phase-locked sequence $\vartheta \equiv u \pmod{1}$. If the leading saddle-resurgent multiplier at that phase is nonzero, then the least term of the W -series occurs at $N_{\text{opt}} = (\pi^2/8 + o(1))w$, and truncation there has relative remainder $R_u(w)$ satisfying*

$$\log |R_u(w)| = - \left(\frac{\pi^2}{8} + o(1) \right) w.$$

The qualification on the multiplier allows exceptional phases at which the first sector cancels. No summability assertion is included.

Conjecture W.3 (Sharp strip inheritance). *Every nonconstant periodic coefficient occurring in $d_n, h_n, g_n, d_n^W, h_n^W, g_n^W$ has maximal open strip $|\Im \theta| < \pi/(2L)$, the maximal strip of Ψ , unless its boundary singularities cancel by an algebraic identity. In particular the generic coefficient at each inverse order has Fourier exponential type exactly π^2/L rather than merely at least that type as proved in [proposition O.2](#).*

Conjecture W.4 (Nonconstant phase at every order). *For every $n \geq 1$, at least one nonzero Fourier coefficient of each $d_n(\ell, \cdot), h_n(\ell, \cdot), g_n(\ell, \cdot)$ is a nonzero polynomial in ℓ ; equivalently, none of the three coefficients is phase-blind for all ℓ . Likewise each of d_n^W, h_n^W, g_n^W is nonconstant in its phase. The formal top-jet law proves that these differential polynomials are not formally constant; the remaining content is to exclude an identity after substitution of the particular Gamma–zeta function Ψ .*

The geometry of the joint phase profiles—immersion, self-intersection, and possible embeddings after several coefficients are recorded—is a separate problem and is not part of the preceding nonconstancy conjecture.

Conjecture W.5 (Two-level exponentially improved transseries). *After truncating the algebraic W -series at its least term, the relative remainder decomposes into two asymptotically distinct families. The first is a saddle family on scales $e^{-c_j w}$ determined by large-order asymptotics of the forward coefficients A_j . The second is an ultraflat family on the exact scales*

$$\exp(-n2^{q_*(y)}) \quad (n \geq 1),$$

where $q_(y)$ is the exact saddle coordinate of [\(T.6\)](#), not the leading phase θ . At each fixed ultraflat product order, its arithmetic coefficient sequence belongs to the finite Dirichlet-convolution algebra generated by $b_n = (2^{\nu_2(n)+1} - 1)/n$; equivalently its Dirichlet series is a finite linear combination of products of shifted copies of $\zeta(s+1)/(1-2^{-s})$. The conjecture asserts this scale and coefficient algebra, but neither Borel summability nor particular Stokes constants.*

Conjecture W.6 (Universal diagonal inversion class). *Let $a > 0$. Suppose an endpoint CDF has a strictly monotone exact phase $\lambda \rightarrow \infty$ and a forward expansion*

$$-a\lambda^2 + b\lambda - c \log \lambda + C + P(\lambda) + \sum_{j < N} A_j(\lambda) \lambda^{-j} + R_N(\lambda),$$

where P, A_j are periodic and holomorphic on a common strip, the remainders and every fixed finite set of their phase derivatives are uniform, and the phase equation has derivative $2a\lambda + O(1)$ bounded away from zero. Then its inverse has a unique diagonal expansion with polynomial logarithms, phase-locked limits, and Bell conversion; the highest logarithms depend only on a and c . Any newest-jet conclusion additionally requires a stated top-jet law for the forward A_j .

For the separate base- $b > 1$ product of uniform smoothing factors, the exact digital tail has coefficients $(b^{\nu_b(n)+1} - 1)/((b-1)n)$, Mellin kernel $\zeta(s+1)/(1-b^{-s})$, and period one in $\log_b$. The conjectural part is that the preceding analytic hypotheses and inverse package hold uniformly in this base- b family; the coefficient regrouping itself is an elementary finite-valuation calculation.

Concrete projects: generate A_3, A_4 compactly through the pipeline (S.6) and make g_4, g_5 explicit in Ψ alone; prove weighted Wiener-algebra bounds uniform in phase; run optimal truncation experiments to see whether the first visible beyond-all-orders scale is saddle-resurgent or the far smaller dyadic one; quantify the uniform-in- m breakdown of the derivative law (Q.4); locate the nearest complex critical values of F (the analytic continuation radius of G and the Borel singularities); and build certified few-mode Fourier-polynomial coefficient tables $g_n \approx \sum_{r \leq n} \sum_{|k| \leq K} c_{n,r,k} \ell^r e^{2\pi i k \theta}$ that, with one Newton step on the exact functional equation, give a fast globally certified quantile algorithm.

Lean formalization roadmap

Four layers, the first three purely formal: (1) a two-scale coefficient ring (truncated series over $\mathbb{R}[\ell] \otimes C^\infty(\mathbb{R}/\mathbb{Z})$, closure under composition, log, exp); (2) parameterized Lagrange inversion $[t^k]W = \frac{1}{k}[w^{k-1}]\mathcal{K}^k$ and the diagonal specialization (L.1); (3) the Gaussian contraction (S.3) over an algebraic moment functional, and the 2-adic regrouping and Dirichlet identities of appendix T (a bijection and two Tonelli arguments); (4) the analytic transfer (appendix V) from formal cancellation to the fixed-order remainder. The degree/top-jet filtrations of appendix N are finite combinatorics on top of (2).

Part VI

Dyadic Self-Sampling and Alias Superconvergence

A Normalization and established structure

A.1 The up-law, Fabius function, and Fourier convention

Let $u = \text{up}$ be Rvachev's even nonnegative C^∞ function supported on $[-1, 1]$, normalized by

$$\int_{-1}^1 u(x) dx = 1, \quad u(0) = 1.$$

We use

$$\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad \Phi_{2\pi}(\xi) = \widehat{u}_{2\pi}(\xi) = \prod_{j=0}^{\infty} \text{sinc}_{\pi}\left(\frac{\xi}{2^j}\right), \quad (\text{A.1})$$

where

$$\operatorname{sinc}_\pi z = \begin{cases} \frac{\sin(\pi z)}{\pi z}, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

The product is entire. If $m \in \mathbb{Z} \setminus \{0\}$, then

$$\operatorname{ord}_{\xi=m} \Phi_{2\pi} = 1 + \nu_2(m). \quad (\text{A.2})$$

Indeed, the factors indexed by $0 \leq j \leq \nu_2(m)$ vanish simply and all later factors are nonzero.

This foundation is now formal rather than merely inherited from the human corpus. The standalone product is entire by `Fabius.rvachevFourierProduct_differentiable`; for every nonzero integer m , Lean proves

$$\operatorname{ord}_{\xi=m} \Phi_{2\pi} = 1 + \nu_2(|m|)$$

as `Fabius.analyticOrderAt_rvachevFourierProduct_int`, together with all lower derivative vanishings and the exact nonzero leading derivative in `IntegerZeroAnalyticOrder.lean`.

The standard Fabius distribution function $F : [0, 1] \rightarrow [0, 1]$ is related to u by

$$u(x) = F(1 - |x|) \quad (|x| \leq 1), \quad F(t) = u(t - 1) \quad (0 \leq t \leq 1). \quad (\text{A.3})$$

If Y has density u , then

$$\mathbb{P}(Y \leq x) = F\left(\frac{x+1}{2}\right), \quad -1 \leq x \leq 1. \quad (\text{A.4})$$

Thus the centered quantile is

$$q(t) = 2F^{-1}(t) - 1. \quad (\text{A.5})$$

A.2 Moments, moment generating function, and cumulants

Write

$$\mu_n = \int_{-1}^1 x^n u(x) \, dx, \quad c_n = \mu_{2n}. \quad (\text{A.6})$$

Odd moments vanish. The even moments are rational and satisfy

$$(2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k, \quad c_0 = 1. \quad (\text{A.7})$$

The moment generating function is

$$\begin{aligned} M(z) &= \int_{-1}^1 e^{zx} u(x) \, dx = \Phi_{2\pi}\left(\frac{iz}{2\pi}\right) \\ &= \prod_{j=0}^{\infty} \frac{\sinh(z/2^{j+1})}{z/2^{j+1}}. \end{aligned} \quad (\text{A.8})$$

Its logarithm has even cumulants

$$\log M(z) = \sum_{r=1}^{\infty} \kappa_{2r} \frac{z^{2r}}{(2r)!}, \quad \boxed{\kappa_{2r} = \frac{B_{2r}}{2r} \frac{2^{2r}}{2^{2r}-1}}, \quad (\text{A.9})$$

where B_n are Bernoulli numbers with $B_2 = 1/6$. Consequently

$$\mu_n = Y_n(0, \kappa_2, 0, \kappa_4, \dots, \kappa_n), \quad (\text{A.10})$$

with Y_n the complete exponential Bell polynomial.

A.3 Exact dyadic arithmetic

For $0 \leq a \leq 2^n$, the corpus gives the finite moment–Thue–Morse formula

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-n(n+1)/2}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} (2a - 2h - 1)^{n-2k} c_k, \quad (\text{A.11})$$

where $\varepsilon_h = (-1)^{\text{wt}_2(h)}$. In particular every dyadic sample of F and u is rational. A second exact formula, recalled in [appendix J](#), expresses the same value through Gaussian binomial coefficients at $q = 1/2$ and finite signed Thue–Morse power sums.

B The shifted dyadic self-sampling theorem

B.1 Definition of the rules

For $N \geq 0$, $h_N = 2^{-N}$, and phase $\theta \in \mathbb{R}$, define

$$Q_{N,\theta}[f] := h_N \sum_{k \in \mathbb{Z}} f(h_N(k + \theta)) u(h_N(k + \theta)). \quad (\text{B.1})$$

Because u is supported on $[-1, 1]$, this is a finite sum. The nodes are equally spaced, but the weights

$$w_{N,\theta,k} = h_N u(h_N(k + \theta)) \quad (\text{B.2})$$

are samples of the target density itself. They are nonnegative and strictly positive at every interior node.

Theorem B.1 (Shifted self-sampling quadrature). *For every $N \geq 0$, every $\theta \in \mathbb{R}$, and every polynomial p with $\deg p \leq N$,*

$$\boxed{\int_{-1}^1 p(x) u(x) \, dx = 2^{-N} \sum_{k \in \mathbb{Z}} p\left(\frac{k + \theta}{2^N}\right) u\left(\frac{k + \theta}{2^N}\right)}. \quad (\text{B.3})$$

In particular, $\sum_k w_{N,\theta,k} = 1$, so the rule is a positive probability measure matching the first N moments of u .

Proof. Let $0 \leq r \leq N$ and set $f_r(x) = x^r u(x)$. Since $u \in C_c^\infty(\mathbb{R})$, Poisson summation applies without regularization. For the shifted lattice,

$$h_N \sum_{k \in \mathbb{Z}} f_r(h_N(k + \theta)) = \sum_{\ell \in \mathbb{Z}} e^{2\pi i \ell \theta} \widehat{f_r}_{2\pi}(2^N \ell). \quad (\text{B.4})$$

Differentiating (A.1),

$$\widehat{f_r}_{2\pi}(\xi) = \frac{\Phi_{2\pi}^{(r)}(\xi)}{(-2\pi i)^r}. \quad (\text{B.5})$$

The term $\ell = 0$ is μ_r . If $\ell \neq 0$, then (A.2) gives

$$\text{ord}_{\xi=2^N \ell} \Phi_{2\pi} = N + 1 + \nu_2(\ell) > r,$$

so $\Phi_{2\pi}^{(r)}(2^N \ell) = 0$. Thus every nonzero alias vanishes, proving the identity for monomials and hence for all polynomials of degree at most N . $\square$

Remark B.2 (A dyadic Strang–Fix phenomenon). Classical Strang–Fix conditions turn zeros of a generator’s Fourier transform at the dual lattice into polynomial reproduction by shifts. Here the generator also supplies the quadrature weights. The result is therefore a self-weighted, positive, shifted Strang–Fix rule rather than a cardinal interpolation formula. The exact order is read directly from the 2-adic multiplicities of the sinc product.

B.2 Rationality and denominator control

Corollary B.3 (Positive rational rules). *If $\theta = a/2^s$ is dyadic, every node and every weight in (B.3) is rational. More explicitly, with*

$$D = N + s, \quad n_k = 2^s k + a, \quad b_k = 2^D - |n_k|,$$

one has, whenever $|n_k| < 2^D$,

$$x_k = \frac{n_k}{2^D}, \quad w_k = 2^{-N} F\left(\frac{b_k}{2^D}\right) \in \mathbb{Q}_{>0}. \quad (\text{B.6})$$

A common denominator for all such weights is obtained by multiplying the repository's level- D dyadic denominator bound by 2^N .

Proof. Use (A.3) at the dyadic node x_k . The denominator assertion follows from the established bound

$$D! 2^{\binom{D+1}{2}} (2 \lfloor D/2 \rfloor + 1)!! \prod_{j=1}^{\lfloor D/2 \rfloor} (2^{2j} - 1) u(n/2^D) \in \mathbb{N}_0$$

for $|n| < 2^D$. □

B.3 Finite moment identities in Fabius values

For the centered phase $\theta = 0$, symmetry yields particularly transparent formulas. For $r \geq 1$ with $2r \leq N$,

$$c_r = 2^{1-N-2rN} \sum_{j=1}^{2^N-1} (2^N - j)^{2r} F\left(\frac{j}{2^N}\right). \quad (\text{B.7})$$

For $r = 0$, normalization reads

$$1 = 2^{-N} \left(1 + 2 \sum_{j=1}^{2^N-1} F\left(\frac{j}{2^N}\right) \right). \quad (\text{B.8})$$

These are positive finite identities: the moment on the left is reconstructed from values of the Fabius function with no alternating coefficients.

Proof. Apply theorem B.1 to $p(x) = x^{2r}$. Pair the nodes $\pm k/2^N$, use $u(k/2^N) = F(1 - k/2^N)$, and then put $j = 2^N - k$. □

First frontier bridge

The exact dyadic values in the corpus were previously used primarily as point evaluations, denominator data, or inputs to signed extrapolation. Equations (B.3)–(B.7) instead organize them as a hierarchy of positive moment designs. Fourier zero multiplicities, dyadic rationality, and probability positivity become three faces of one formula.

C The master alias function and the 2-adic error filtration

C.1 An entire generating function for all moment defects

Define

$$G_{N,\theta}(z) := Q_{N,\theta}[e^{zx}] - M(z). \quad (\text{C.1})$$

The coefficient of $z^r/r!$ is exactly the r th moment defect

$$E_{N,r}(\theta) := Q_{N,\theta}[x^r] - \mu_r. \quad (\text{C.2})$$

Theorem C.1 (Master shifted-alias identity). *For every $N \geq 0$, $\theta \in \mathbb{R}$, and $z \in \mathbb{C}$,*

$$G_{N,\theta}(z) = \sum_{\ell \in \mathbb{Z} \setminus \{0\}} e^{2\pi i \ell \theta} \Phi_{2\pi} \left(2^N \ell + \frac{iz}{2\pi} \right). \quad (\text{C.3})$$

The series is locally normally convergent. Moreover $G_{N,\theta}$ has a zero of order at least $N + 1$ at $z = 0$.

Proof. Apply shifted Poisson summation to $f_z(x) = e^{zx}u(x)$. Its Fourier transform is

$$\widehat{f_z}_{2\pi}(\xi) = \Phi_{2\pi} \left(\xi + \frac{iz}{2\pi} \right).$$

The zero-frequency term is $M(z)$, giving (C.3). Compact support and smoothness imply the local normal convergence needed to differentiate termwise. At $z = 0$, the summand indexed by $\ell \neq 0$ has order $N + 1 + \nu_2(\ell)$, so the full sum begins at order $N + 1$. $\square$

Corollary C.2 (Valuation filtration of polynomial errors). *For $r > N$,*

$$E_{N,r}(\theta) = \frac{1}{(-2\pi i)^r} \sum_{a=0}^{r-N-1} \sum_{\substack{q \in \mathbb{Z} \setminus \{0\} \\ q \text{ odd}}} e^{2\pi i 2^a q \theta} \Phi_{2\pi}^{(r)}(2^{N+a} q). \quad (\text{C.4})$$

Thus the excess degree $r - N$ exposes exactly the first $r - N$ layers of the 2-adic alias tree.

Proof. In the coefficient formula obtained from (C.3), an alias ℓ can contribute only if $N + 1 + \nu_2(\ell) \leq r$. Write $\ell = 2^a q$ with q odd. $\square$

Remark C.3. The first failure, $r = N + 1$, sees only odd aliases. The next failure, $r = N + 2$, sees odd aliases and twice-odd aliases. Higher defects are therefore not arbitrary: they are a triangular sequence of 2-adic spectral layers. This observation will be the basis of the next-defect conjectures in [appendix L](#).

C.2 An exact local factorization at every integer zero

The integer zero jets can be organized in a form that makes Bell polynomials appear automatically.

Proposition C.4 (Exact local factorization). *Let $m = 2^{d-1}q \neq 0$, where q is odd and $d = 1 + \nu_2(m)$. Then, for w near 0,*

$$\Phi_{2\pi}(m + w) = - \left(\frac{w}{m + w} \right)^d \left(\prod_{j=0}^{d-1} \text{sinc}_\pi \left(\frac{w}{2^j} \right) \right) \Phi_{2\pi} \left(\frac{q}{2} + \frac{w}{2^d} \right). \quad (\text{C.5})$$

In particular,

$$\Phi_{2\pi}(m+w) = C_m w^d \exp\left(\sum_{j \geq 1} \gamma_j(m) \frac{w^j}{j!}\right), \quad C_m = -\frac{\Phi_{2\pi}(q/2)}{m^d}, \quad (\text{C.6})$$

where the logarithmic jet $\gamma_j(m)$ is explicit.

Proof. For $0 \leq j \leq d-1$, $m/2^j$ is an integer and

$$\text{sinc}_\pi\left(\frac{m+w}{2^j}\right) = (-1)^{m/2^j} \frac{w}{m+w} \text{sinc}_\pi\left(\frac{w}{2^j}\right).$$

The product of the signs is -1 because

$$\sum_{j=0}^{d-1} \frac{m}{2^j} = q(2^d - 1)$$

is odd. The remaining tail is

$$\prod_{j=d}^{\infty} \text{sinc}_\pi\left(\frac{m+w}{2^j}\right) = \Phi_{2\pi}\left(\frac{q}{2} + \frac{w}{2^d}\right).$$

This proves (C.5); taking the analytic logarithm after removing $C_m w^d$ gives (C.6). $\square$

Lean status of the local factorization

Equation (C.5) is now formal in a stronger total, denominator-cleared form valid for every complex w , including the removable points. The all-nonzero-integer statement is `Fabius.rvachevFourierProduct_int_add_factorization`; its divided germ form is `Fabius.rvachevFourierProduct_int_add_eventuallyEq_pow_mul_cofactor`. The cofactor is analytic and nonzero at the local origin, so Lean proves the exact order, every lower-jet vanishing, and the leading identity (D.5). More generally, if that cofactor is denoted by U_m and $d = 1 + \nu_2(|m|)$, Lean proves every higher jet as

$$\Phi_{2\pi}^{(d+r)}(m) = \binom{d+r}{d} d! U_m^{(r)}(0) \quad (r \in \mathbb{N}_0)$$

in `Fabius.iteratedDeriv_rvachevFourierProduct_int_add_order`. The generic engine in `LeadingJet.lean` is vector-valued and computes every derivative index over an arbitrary nontrivially normed field. The explicit normalization in (C.6), the logarithmic coefficients (C.7)–(C.8) and the Bell-polynomial formula (C.10) for $s > 0$ remain human-readable frontier results, not named Lean theorems.

The first logarithmic coefficient is especially simple:

$$\gamma_1(m) = -\frac{d}{m} + 2^{-d} \frac{\Phi'_{2\pi}(q/2)}{\Phi_{2\pi}(q/2)}. \quad (\text{C.7})$$

For general j ,

$$\gamma_j(m) = d(-1)^j \frac{(j-1)!}{m^j} + \sum_{h=0}^{d-1} 2^{-hj} \ell_j + 2^{-dj} L_j(q/2), \quad (\text{C.8})$$

where

$$\ell_j = \frac{d^j}{dz^j} \log \text{sinc}_\pi z \Big|_{z=0}, \quad L_j(t) = \frac{d^j}{dt^j} \log \Phi_{2\pi}(t).$$

The universal local coefficients satisfy

$$\ell_{2r} = -\frac{(2r)!}{r} \zeta(2r), \quad \ell_{2r+1} = 0, \quad (\text{C.9})$$

so Bernoulli numbers may be substituted through Euler’s formula for $\zeta(2r)$.

Corollary C.5 (Bell-polynomial zero jets). *With the notation of [proposition C.4](#), for every $s \geq 0$,*

$$\Phi_{2\pi}^{(d+s)}(m) = C_m \frac{(d+s)!}{s!} Y_s(\gamma_1(m), \dots, \gamma_s(m)). \quad (\text{C.10})$$

Here $Y_0 = 1$. Consequently every layer of (C.4) has a finite Bell-polynomial description in half-integer logarithmic derivatives, zeta values, and elementary 2-adic data.

Proof. Use the defining exponential generating function

$$\exp\left(\sum_{j \geq 1} x_j \frac{w^j}{j!}\right) = \sum_{s \geq 0} Y_s(x_1, \dots, x_s) \frac{w^s}{s!}$$

in (C.6) and compare coefficients. □

Second frontier bridge

The corpus already contains the integer zero multiplicities and several zero-jet formulas. The new point here is their use as an *error calculus*: the excess polynomial degree selects a finite set of 2-adic alias layers, and every derivative on those layers is generated by a complete Bell polynomial. Bernoulli/zeta data enter locally through $\log \operatorname{sinc}_\pi$, while Thue–Morse data enter globally through the half-integer tail.

D The first defect: half-integer spectrum and Thue–Morse signs

Put

$$r = N + 1, \quad E_N(\theta) := E_{N,N+1}(\theta). \quad (\text{D.1})$$

Only odd aliases survive.

Theorem D.1 (Exact first-defect series). *For every $N \geq 0$ and $\theta \in \mathbb{R}$,*

$$E_N(\theta) = -\frac{r!}{(-2\pi i)^r 2^{Nr}} \sum_{\substack{\ell \in \mathbb{Z} \setminus \{0\} \\ \ell \text{ odd}}} \frac{\Phi_{2\pi}(\ell/2)}{\ell^r} e^{2\pi i \ell \theta}. \quad (\text{D.2})$$

Equivalently, if $r = 2s$,

$$E_N(\theta) = \frac{2(-1)^{s+1} r!}{(2\pi)^r 2^{Nr}} \sum_{\substack{m \geq 1 \\ m \text{ odd}}} \frac{\Phi_{2\pi}(m/2)}{m^r} \cos(2\pi m \theta), \quad (\text{D.3})$$

and if $r = 2s + 1$,

$$E_N(\theta) = \frac{2(-1)^s r!}{(2\pi)^r 2^{Nr}} \sum_{\substack{m \geq 1 \\ m \text{ odd}}} \frac{\Phi_{2\pi}(m/2)}{m^r} \sin(2\pi m \theta). \quad (\text{D.4})$$

Proof. For $m = 2^N \ell$ with ℓ odd, [proposition C.4](#) with $d = r$ gives

$$\Phi_{2\pi}^{(r)}(2^N \ell) = -\frac{r! \Phi_{2\pi}(\ell/2)}{(2^N \ell)^r}. \quad (\text{D.5})$$

Insert this into (C.4). Pair positive and negative odd frequencies to obtain the real forms. □

D.1 The spectral sign is exactly Thue–Morse

Proposition D.2 (Half-integer sign law). *For $j \geq 0$,*

$$\operatorname{sgn} \Phi_{2\pi} \left(\frac{2j+1}{2} \right) = (-1)^{\operatorname{wt}_2(j)} = \varepsilon_j. \quad (\text{D.6})$$

Hence the first quadrature defect is a Thue–Morse-signed half-integer spectral Dirichlet series.

Proof. On the positive real axis, $\Phi_{2\pi}$ changes sign whenever an integer zero has odd multiplicity. At the point $(2j+1)/2 = j + 1/2$, the total number of zero crossings, counted with multiplicity, is

$$\sum_{m=1}^j (1 + \nu_2(m)) = j + \nu_2(j!) = 2j - \operatorname{wt}_2(j),$$

where the last equality is Legendre’s formula $\nu_2(j!) = j - \operatorname{wt}_2(j)$. Its parity is therefore $\operatorname{wt}_2(j)$. Since $\Phi_{2\pi}(1/2) > 0$, the claimed sign follows. $\square$

Writing

$$a_j = |\Phi_{2\pi}((2j+1)/2)|, \quad (\text{D.7})$$

we may rewrite the spectral coefficient as $\Phi_{2\pi}((2j+1)/2) = \varepsilon_j a_j$. Thus the signs are purely digital, while the amplitudes record the sinc-product decay.

D.2 Symmetries

Corollary D.3 (Phase symmetries). *The first defect is 1-periodic and satisfies*

$$E_N(-\theta) = (-1)^r E_N(\theta), \quad E_N(\theta + 1/2) = -E_N(\theta). \quad (\text{D.8})$$

Proof. The first identity follows from the sine/cosine parity. The second follows because every frequency in (D.2) is odd. $\square$

E A Bernoulli convolution formula for the first defect

E.1 The periodic up-kernel

Define a 1-periodic continuous function $\mathcal{K}$ by

$$\mathcal{K}(t) = 2u(2t) - 1, \quad -\frac{1}{2} \leq t \leq \frac{1}{2}, \quad (\text{E.1})$$

with periodic extension. Its mean is zero. For $n \neq 0$, its Fourier coefficient is

$$\widehat{\mathcal{K}}_{\text{FS}}[n] = \int_{-1/2}^{1/2} \mathcal{K}(t) e^{-2\pi i n t} dt = \Phi_{2\pi}(n/2). \quad (\text{E.2})$$

In particular the even nonzero coefficients vanish, and the odd coefficients carry the Thue–Morse signs of [proposition D.2](#).

Let $\widetilde{B}_r(t) = B_r(\{t\})$ be the periodic Bernoulli polynomial, understood in the standard Fourier/distributional sense when $r = 1$. For $n \neq 0$,

$$\widehat{\widetilde{B}_r}_{\text{FS}}[n] = -\frac{r!}{(2\pi i n)^r}. \quad (\text{E.3})$$

Theorem E.1 (Bernoulli–Rvachev error kernel). *For $r = N + 1$,*

$$E_N(\theta) = (-1)^r 2^{-Nr} (\tilde{B}_r * \mathcal{K})(\theta), \quad (\text{E.4})$$

where convolution is on $\mathbb{R}/\mathbb{Z}$.

Proof. Multiply the nonzero Fourier coefficients in (E.2) and (E.3). The resulting Fourier series is exactly (D.2); the even frequencies vanish automatically. $\square$

Remark E.2. This formula is not an Euler–Maclaurin expansion with an unspecified remainder. It is an exact identity. The universal periodic Bernoulli kernel describes polynomial aliasing, while $\mathcal{K}$ inserts the complete half-integer spectrum of the up-law. It therefore provides a direct analytical bridge from Bernoulli polynomials to the Rvachev sinc product.

E.2 A rigorous first-harmonic bound

Lemma E.3 (Finite product loss bound). *If $0 \leq a_j \leq 1$ for $0 \leq j \leq J$, then*

$$\prod_{j=0}^J (1 - a_j) \geq 1 - \sum_{j=0}^J a_j. \quad (\text{E.5})$$

Proof. The assertion is equality for $J = 0$. If it holds through $J - 1$, put $S = \sum_{j=0}^{J-1} a_j$. Multiplication by the nonnegative number $1 - a_J$ and the induction hypothesis give

$$\prod_{j=0}^J (1 - a_j) \geq (1 - S)(1 - a_J) = 1 - S - a_J + S a_J \geq 1 - S - a_J.$$

This proves the induction step. $\square$

Lemma E.4 (A uniform lower bound at the first half-integer).

$$\Phi_{2\pi}(1/2) > 1 - \frac{\pi^2}{18} > \frac{4}{9}. \quad (\text{E.6})$$

Proof. For $0 \leq x \leq 1/2$, $\text{sinc}_\pi x \geq 1 - \pi^2 x^2/6$. Set $a_j = (\pi^2/6)4^{-(j+1)}$. All a_j lie in $(0, 1)$, and lemma E.3, followed by passage to the limit, gives

$$\Phi_{2\pi}(1/2) \geq \prod_{j=0}^{\infty} (1 - a_j).$$

This product is strictly larger than $1 - \sum_j a_j$. Indeed, if $S = \sum_{j=2}^{\infty} a_j$, another application of the finite product bound to the tail and passage to the limit yield

$$\begin{aligned} \prod_{j=0}^{\infty} (1 - a_j) &\geq (1 - a_0)(1 - a_1)(1 - S) \\ &= 1 - a_0 - a_1 - S + a_0 a_1 + S(a_0 + a_1 - a_0 a_1) \\ &> 1 - a_0 - a_1 - S. \end{aligned}$$

Consequently

$$\Phi_{2\pi}(1/2) > 1 - \frac{\pi^2}{6} \sum_{j=0}^{\infty} 4^{-(j+1)} = 1 - \frac{\pi^2}{18}.$$

The final inequality follows from $\pi^2 < 10$. $\square$

For $r \geq 2$,

$$\sum_{\substack{m \geq 3 \\ m \text{ odd}}} \frac{|\Phi_{2\pi}(m/2)|}{m^r} \leq \sum_{\substack{m \geq 3 \\ m \text{ odd}}} \frac{1}{m^r} \leq \frac{\pi^2}{8} - 1 < \frac{1}{4}. \quad (\text{E.7})$$

Thus the first coefficient is larger than the sum of all remaining absolute coefficients for every $r \geq 2$.

Theorem E.5 (Uniform sinusoidal asymptotics). *Let $N \geq 1$, put $r = N + 1$, and define*

$$A_{N,r} = \frac{2r! \Phi_{2\pi}(1/2)}{(2\pi)^r 2^{Nr}}. \quad (\text{E.8})$$

Then, uniformly in θ ,

$$E_N(\theta) = (-1)^{r/2+1} A_{N,r} \cos(2\pi\theta) + \mathcal{O}(A_{N,r} 3^{-r}), \quad r \text{ even}, \quad (\text{E.9})$$

$$E_N(\theta) = (-1)^{(r-1)/2} A_{N,r} \sin(2\pi\theta) + \mathcal{O}(A_{N,r} 3^{-r}), \quad r \text{ odd}. \quad (\text{E.10})$$

The implied constant is absolute.

Proof. Separate the frequency $m = 1$ in (D.3) or (D.4). Since $|\Phi_{2\pi}(m/2)| \leq 1$ and $\sum_{m \geq 3, m \text{ odd}} m^{-r} = \mathcal{O}(3^{-r})$, the remaining series has the stated uniform bound. $\square$

The exact numerical ratios are far smaller than the elementary majorant:

Table 4: Absolute spectral tail divided by the first harmonic.

$r = N + 1$	N	absolute tail / first harmonic
2	1	0.0099476026
3	2	0.0032220116
4	3	0.001055965
5	4	0.00034847136
6	5	0.00011546539
7	6	$3.835166e - 5$
8	7	$1.2756733e - 5$
9	8	$4.2468422e - 6$
10	9	$1.4145376e - 6$

F Parity-selected positive superconvergence

F.1 Forced phases

The real forms immediately show two parity-dependent phase families.

Corollary F.1 (One extra exact degree). *Let $N \geq 0$.*

- If N is even, then $Q_{N,0}$ and $Q_{N,1/2}$ are exact for every polynomial of degree at most $N + 1$.*
- If N is odd, then $Q_{N,1/4}$ and $Q_{N,3/4}$ are exact for every polynomial of degree at most $N + 1$.*

All four phase families are positive; at these dyadic phases they are rational.

Proof. If N is even, $r = N + 1$ is odd, and every sine in (D.4) vanishes at $\theta = 0, 1/2$. If N is odd, r is even, and every cosine in (D.3) vanishes at $\theta = 1/4, 3/4$ because m is odd. $\square$

Lean status of parity-selected superconvergence

Corollary F.1 now has an exact compiled counterpart. The predicate `Fabius.isRvachevSuperconvergentPhase` records the four selected phases, and `Fabius.isRvachevSuperconvergentPhase_two_pow_iff` specializes it at the mesh $M = 2^N$ to precisely the even- N endpoint/midpoint pair and the odd- N quarter-phase pair above. The physical-coordinate statement `Fabius.integral_polynomial_mul_rvachevUp_eq_normalized_tsum_superconvergent` then gives the displayed 2^{-N} -normalized rule for every real polynomial of degree at most $N + 1$.

The positivity and rationality clauses use already compiled kernels rather than a new classification theorem: `Fabius.rvachevUp_nonneg` gives nonnegative weights, `Fabius.rvachev_pos_iff_mem_Ioo` gives strict positivity at exactly the interior nodes, and `Fabius.rvachevDyadic_cast`, together with the zero-off-support theorem `Fabius.rvachevUp_eq_zero_iff_not_mem_Ioo`, identifies every dyadic sample with a rational value. This finite family matches every clause of the corollary. It does not classify all possible superconvergent phases or assert sharpness beyond the selected families.

F.2 Classification of the superconvergent phases

The first harmonic does more than suggest the correct phases: from level two onward it proves that there are no others.

Lemma F.2 (Odd trigonometric factor bounds). *For odd $m \geq 1$ and real x ,*

$$|\sin(mx)| \leq m |\sin x|, \quad |\cos(mx)| \leq m |\cos x|. \quad (\text{F.1})$$

Proof. The sine inequality is standard. For the cosine inequality put $x = t + \pi/2$; because m is odd, $\cos(mx) = \pm \sin(mt)$ and $\cos x = -\sin t$. $\square$

Theorem F.3 (Exact phase classification). *For $N \geq 2$, the first defect $E_N(\theta)$ vanishes modulo 1 exactly at*

$$\begin{cases} \theta \equiv 0, 1/2 \pmod{1}, & N \text{ even}, \\ \theta \equiv 1/4, 3/4 \pmod{1}, & N \text{ odd}. \end{cases} \quad (\text{F.2})$$

Consequently these are precisely the shifted self-sampling rules with degree of exactness at least $N + 1$.

Proof. Here $r = N + 1 \geq 3$. Suppose first that r is odd. Factor $\sin(2\pi\theta)$ from (D.4). By lemma F.2, the absolute contribution of the remaining frequencies to the bracket is at most

$$\sum_{\substack{m \geq 3 \\ m \text{ odd}}} m^{1-r} \leq \sum_{\substack{m \geq 3 \\ m \text{ odd}}} m^{-2} = \frac{\pi^2}{8} - 1 < \frac{1}{4}.$$

The first bracket coefficient is $\Phi_{2\pi}(1/2) > 4/9$, so the bracket never vanishes. Hence the zero set is exactly $\sin(2\pi\theta) = 0$.

If r is even, factor $\cos(2\pi\theta)$ and use the cosine bound in the same way. The bracket again has a fixed nonzero sign, so its only zeros are those of $\cos(2\pi\theta)$. $\square$

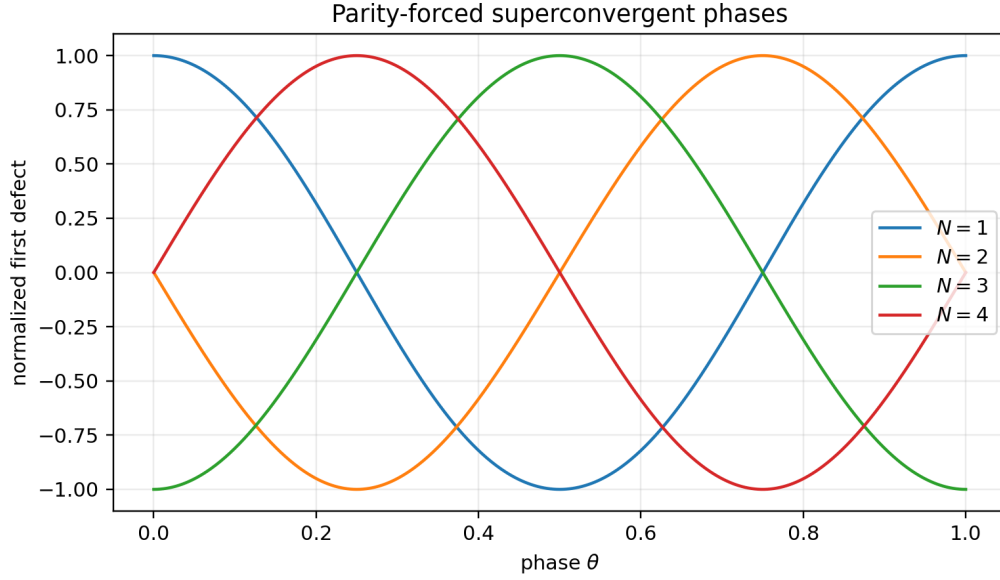


Figure 5: Normalized exact first-defect profiles. The parity-forced phases are common zeros of every odd harmonic in the corresponding sine or cosine series.

F.3 A canonical phase schedule and exact experiments

Choose

$$\theta_N = \begin{cases} 0, & N \text{ even,} \\ 1/4, & N \text{ odd.} \end{cases} \quad (\text{F.3})$$

Then Q_{N,θ_N} is positive, rational, and exact through degree $N + 1$. Its number of positive nodes is $2^{N+1} - 1$ at centered levels and 2^{N+1} at quarter-shifted levels.

Table 5: Exact rational checks of the canonical phase schedule. “Proved degree” is the theorem above. Each next defect is computed as an exact fraction; a decimal display is used here for legibility, while the full fractions are retained in the CSV and exact TeX table in the archive.

N	phase θ_N	positive nodes	proved degree	next defect (decimal display)
0	0	1	1	−0.11111111
1	1/4	4	2	0.011067708
2	0	7	3	0.0002806713
3	1/4	16	4	−2.189698e − 6
4	0	31	5	−4.8411566e − 9
5	1/4	64	6	3.0609753e − 12
6	0	127	7	5.4077218e − 16
7	1/4	256	8	−2.6505694e − 20
8	0	511	9	−3.5647819e − 25

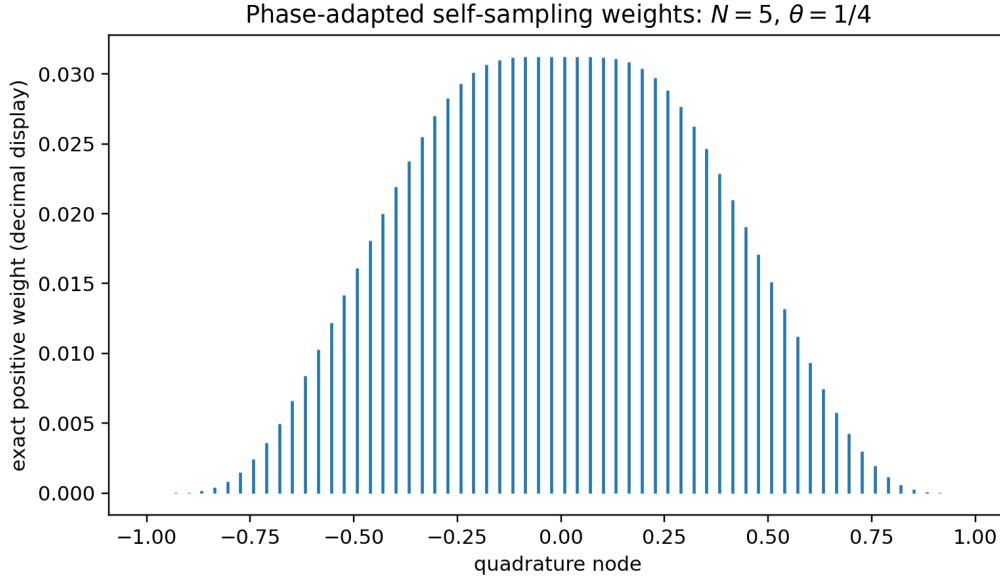


Figure 6: The positive rational weights at $N = 5$, $\theta = 1/4$. The rule integrates every polynomial through degree six exactly.

The table proves by exact rational arithmetic that the canonical rule has degree exactly $N + 1$ for $0 \leq N \leq 8$. The corresponding assertion for all N , together with its experimentally observed sign refinement, is stated once as [conjecture L.1](#); its exact two-layer formula is derived in [appendix N](#).

G Positive alias filters as Richardson extrapolation

G.1 Coset rules

For $s \geq 0$ and $0 \leq a < 2^s$, define the phase-coset rule

$$Q_{N;s,a}[f] := 2^{-N} \sum_{k \in \mathbb{Z}} f\left(\frac{2^s k + a}{2^{N+s}}\right) u\left(\frac{2^s k + a}{2^{N+s}}\right). \quad (\text{G.1})$$

This is simply $Q_{N,a/2^s}$, hence is positive and exact through degree N .

Theorem G.1 (Positive phase refinement). *For every $s \geq 0$,*

$$Q_{N+s,0}[f] = 2^{-s} \sum_{a=0}^{2^s-1} Q_{N;s,a}[f]. \quad (\text{G.2})$$

Thus averaging all 2^s positive phase classes gains exactly s guaranteed degrees of polynomial exactness.

Proof. The integers $2^s k + a$, with $k \in \mathbb{Z}$ and $0 \leq a < 2^s$, partition $\mathbb{Z}$. Multiplying the outer factor 2^{-N} by 2^{-s} gives $2^{-(N+s)}$, exactly the fine-grid rule. $\square$

For $s = 1$, let

$$\nu_N = Q_{N,0}, \quad \rho_N = Q_{N,1/2}.$$

Then

$$\nu_{N+1} = \frac{1}{2}(\nu_N + \rho_N), \quad \boxed{\rho_N = 2\nu_{N+1} - \nu_N}. \quad (\text{G.3})$$

The right side is the familiar signed Richardson difference, yet the left side is manifestly a positive quadrature measure. The cancellation that is hidden in scale space becomes positivity after resolution into even and odd lattice cosets.

G.2 Uniqueness in alias space

Suppose coefficients $c_0, \dots, c_{2^s-1}$ combine the phase classes. At Fourier alias ℓ , the multiplier is

$$C(\ell) = \sum_{a=0}^{2^s-1} c_a e^{2\pi i a \ell / 2^s}. \quad (\text{G.4})$$

To gain all s degrees, every residue class $\ell \not\equiv 0 \pmod{2^s}$ must be annihilated.

Theorem G.2 (Unique full residue-class filter). *Assume*

$$C(0) = 1, \quad C(\ell) = 0 \quad (1 \leq \ell < 2^s). \quad (\text{G.5})$$

Then

$$c_0 = c_1 = \dots = c_{2^s-1} = 2^{-s}. \quad (\text{G.6})$$

Hence the positive average in (G.2) is the unique filter on all dyadic phases that cancels every nonzero residue class modulo 2^s .

Proof. The vector $(C(0), \dots, C(2^s-1))$ is the discrete Fourier transform of $(c_0, \dots, c_{2^s-1})$. Inverting the DFT of $(1, 0, \dots, 0)$ gives the uniform vector. $\square$

Remark G.3 (Comparison with q -Richardson). The repository’s geometric Lagrange/Richardson filters cancel powers of a scale parameter and generally use signed Gaussian- q -binomial weights. The filter above cancels Fourier residue classes and uses equal positive weights. The two mechanisms act in complementary variables: scale depth versus lattice phase. Their tensor product is a natural candidate for a stable two-parameter accelerator; see [research problem L.3](#).

Third frontier bridge

Subdividing a self-sampling rule does not merely refine a Riemann sum. In Fourier space it applies an exact DFT projector to the alias tree. The usual Richardson difference $2\nu_{N+1} - \nu_N$ is positive because it isolates the odd lattice coset. This gives a rare instance in which an extrapolation operator has a hidden positive-measure realization.

H Inverse-moment Rvachev–Appell polynomials

H.1 The umbral inverse of the moment family

The frontier volume already uses the forward moment Appell sequence

$$\mathcal{P}_n(x) = \int_{-1}^1 (x+y)^n u(y) dy, \quad \sum_{n \geq 0} \mathcal{P}_n(x) \frac{z^n}{n!} = e^{xz} M(z). \quad (\text{H.1})$$

We now reverse the moment multiplier.

Definition H.1 (Sampling notation for the canonical Rvachev–Appell sequence). Set

$$\mathcal{A}_n(x) := A_n(x),$$

where A_n is the centered Rvachev–Appell sequence already defined in [definition E.1](#). Equivalently, its canonical generating function reads

$$\frac{e^{xz}}{M(z)} = \sum_{n=0}^{\infty} \mathcal{A}_n(x) \frac{z^n}{n!}. \quad (\text{H.2})$$

Thus $\mathcal{A}_n$ is notation, not a second inverse-moment family. Because $M(0) = 1$, the displayed series is well-defined formally and analytically near $z = 0$. Its logarithm is

$$xz - \sum_{r \geq 1} \kappa_{2r} \frac{z^{2r}}{(2r)!}. \quad (\text{H.3})$$

Theorem H.2 (Canonical Appell package in sampling notation). *Under the identification $\mathcal{A}_n = A_n$, the established Appell, Bell, and operator identities take the following forms.*

1. *It is Appell and parity adapted:*

$$\mathcal{A}'_n(x) = n\mathcal{A}_{n-1}(x), \quad \mathcal{A}_n(-x) = (-1)^n \mathcal{A}_n(x). \quad (\text{H.4})$$

2. *It is a complete Bell polynomial in the Bernoulli cumulants:*

$$\boxed{\mathcal{A}_n(x) = Y_n(x, -\kappa_2, 0, -\kappa_4, 0, \dots, -\kappa_n)}. \quad (\text{H.5})$$

3. *It satisfies the triangular recurrence*

$$\mathcal{A}_{n+1}(x) = x\mathcal{A}_n(x) - \sum_{j=1}^{\lfloor (n+1)/2 \rfloor} \binom{n}{2j-1} \kappa_{2j} \mathcal{A}_{n-2j+1}(x). \quad (\text{H.6})$$

4. *The forward and inverse families are umbral inverses:*

$$M(D)\mathcal{A}_n(x) = x^n, \quad M(D)^{-1}\mathcal{P}_n(x) = x^n, \quad D = \frac{d}{dx}. \quad (\text{H.7})$$

Proof. The Appell law is (E.4), and parity follows at once from the evenness of M in the canonical generating function (E.2). Combining the canonical Bell reconstruction (E.11) with the translation formula (E.14) gives (H.5); equivalently, it is the defining complete-Bell expansion of (H.3). The only additional displayed formula is the convenient triangular form. Differentiating the generating function with respect to z gives

$$\frac{\partial}{\partial z} \frac{e^{xz}}{M(z)} = \left(x - \frac{M'(z)}{M(z)} \right) \frac{e^{xz}}{M(z)},$$

and coefficient comparison gives (H.6). Finally, the first identity in (H.7) is the canonical operator formula (E.28). The generating function (H.1) says $\mathcal{P}_n = M(D)x^n$; applying $M(D)^{-1}$ proves the second identity. $\square$

The first polynomials are reproduced in [appendix M](#). In compact notation,

$$\begin{aligned} \mathcal{A}_2(x) &= x^2 - \frac{1}{9}, \\ \mathcal{A}_4(x) &= x^4 - \frac{2}{3}x^2 + \frac{31}{675}, \\ \mathcal{A}_6(x) &= x^6 - \frac{5}{3}x^4 + \frac{31}{45}x^2 - \frac{2347}{59535}. \end{aligned}$$

The constants are the signed complete Bell transforms of the rational cumulants; no transcendental constants remain.

H.2 Continuous deconvolution

Theorem H.3 (Exact deconvolution identity). *For every $n \geq 0$ and $x \in \mathbb{R}$,*

$$\boxed{\int_{-1}^1 \mathcal{A}_n(x-y)u(y) \, dy = x^n.} \quad (\text{H.8})$$

Equivalently, the convolution operator by u maps the inverse-moment Appell basis to the monomial basis.

Proof. This is the centered mean-value inversion (E.6), with symmetry replacing Y by $-Y$. For completeness, the generating-function proof that will be discretized below is as follows. Multiply generating functions:

$$\begin{aligned} \int_{-1}^1 \frac{e^{(x-y)z}}{M(z)} u(y) \, dy &= \frac{e^{xz}}{M(z)} M(-z) \\ &= e^{xz}, \end{aligned}$$

using the evenness $M(-z) = M(z)$. Compare coefficients of $z^n/n!$. □

Corollary H.4 (Mean-zero Appell cancellations). *For $n \geq 1$,*

$$\int_{-1}^1 \mathcal{A}_n(x)u(x) \, dx = 0. \quad (\text{H.9})$$

Proof. Set $x = 0$ in (H.8) and use parity. □

H.3 Finite lattice reproduction

The self-sampling theorem discretizes the deconvolution identity without loss.

Theorem H.5 (Finite Rvachev–Appell translation formula). *Let $N \geq 0$, $\theta \in \mathbb{R}$, and $0 \leq n \leq N$. Then, for every $x \in \mathbb{R}$,*

$$\boxed{x^n = 2^{-N} \sum_{k \in \mathbb{Z}} \mathcal{A}_n\left(x - \frac{k + \theta}{2^N}\right) u\left(\frac{k + \theta}{2^N}\right).} \quad (\text{H.10})$$

The sum is finite. At a superconvergent phase from corollary F.1, the identity remains valid for $n = N + 1$.

Proof. For fixed x , the map $y \mapsto \mathcal{A}_n(x-y)$ is a polynomial of degree n . Apply theorem B.1 to the integral in (H.8). □

More generally, for a polynomial p define its Rvachev deconvolution

$$\mathcal{D}_{\text{up}} p := M(D)^{-1} p = \exp\left(-\sum_{r \geq 1} \kappa_{2r} \frac{D^{2r}}{(2r)!}\right) p. \quad (\text{H.11})$$

The exponential truncates because p is a polynomial.

Corollary H.6 (Polynomial reproduction by deconvolved translates). *If $\deg p \leq N$, then*

$$p(x) = 2^{-N} \sum_{k \in \mathbb{Z}} (\mathcal{D}_{\text{up}} p)\left(x - \frac{k + \theta}{2^N}\right) u\left(\frac{k + \theta}{2^N}\right). \quad (\text{H.12})$$

Proof. Write $p(x) = \sum_{n=0}^N c_n x^n$. The defining generating function gives $M(D)^{-1} x^n = \mathcal{A}_n(x)$, hence $\mathcal{D}_{\text{up}} p = \sum_{n=0}^N c_n \mathcal{A}_n$. Apply Theorem H.5 to each monomial and sum with coefficients c_n . Every sum is finite because u is compactly supported, so the interchange is purely finite and yields (H.12). □

Lean status of Appell deconvolution and lattice reproduction

Three source rows in this section now have exact compiled counterparts. The centered identity [theorem H.3](#) is `Fabius.integral_eval_rvachevAppellPolynomial_sub_mul_rvachev`; its general polynomial companion is `Fabius.integral_eval_rvachevDeconvolvedPolynomial_sub_mul_rvachev`. The mean-zero corollary [corollary H.4](#) is `Fabius.integral_eval_rvachevAppellPolynomial_mul_rvachev_eq_zero`, and [corollary H.6](#) is `Fabius.normalized_tsum_shifted_rvachevDeconvolvedPolynomial_mul_rvachevUp`. Compact support reduces the first two whole-line integrals to the displayed interval $[-1, 1]$. The last declaration has the same nonzero natural mesh, degree bound, arbitrary real phase, normalized physical-coordinate nodes, and conclusion as the paper.

The full finite-lattice theorem is now formalized by an explicit two-part family. For arbitrary phase and $0 \leq n \leq N$, specialize `Fabius.normalized_tsum_shifted_rvachevDeconvolvedPolynomial_mul_rvachevUp` to $M = 2^N$ and $P = X^n$. At a selected phase, including the endpoint and midpoint pair for even N and the quarter pair for odd N , `Fabius.normalized_tsum_shifted_rvachevAppellPolynomial_mul_rvachevUp_superconvergent` proves the same formula through $n = N + 1$; the exact phase dictionary is `Fabius.isRvachevSuperconvergentPhase_two_pow_iff`. Finally, `Fabius.finite_support_comb` supplies the asserted finiteness. Thus [theorem H.5](#) is Lean-proved. No uniqueness or sharpness assertion about the selected phases is included in this status.

Fourth frontier bridge

The forward moment Appell family packages convolution by the up-law; the reciprocal family $\mathcal{A}_n$ packages deconvolution. The same positive dyadic samples that integrate ordinary polynomials now reproduce ordinary monomials from translates of $\mathcal{A}_n$. This simultaneously connects the Fourier zeros, Bernoulli cumulants, complete Bell polynomials, and finite rational Fabius values.

H.4 Root geometry: an exact counterexample to real-rootedness

Numerically, the first seven polynomials have only real zeros. Degree eight already supplies an exact counterexample to global real-rootedness; without exact certificates in degrees one through seven, we do not claim that it is the first such degree.

Proposition H.7 (Exact degree-eight nonreal-zero certificate). *The exact polynomial*

$$\mathcal{A}_8(x) = x^8 - \frac{28}{9}x^6 + \frac{434}{135}x^4 - \frac{9388}{8505}x^2 + \frac{1904369}{32531625} \quad (\text{H.13})$$

has exactly four real zeros. Hence its other four zeros are nonreal and form two conjugate pairs.

Computer-assisted exact proof. Let $S_0 = \mathcal{A}_8$, $S_1 = S'_0$, and $S_{j+1} = -\text{rem}(S_{j-1}, S_j)$ in $\mathbb{Q}[x]$. Exact Euclidean division produces nine nonzero terms, ending in the positive constant $1904369/32531625$. After clearing only positive rational factors, their degrees and limiting signs are

j	0	1	2	3	4	5	6	7	8	V
$\deg S_j$	8	7	6	5	4	3	2	1	0	
$\text{sgn } S_j(-\infty)$	+	−	+	−	+	+	+	−	+	6
$\text{sgn } S_j(+\infty)$	+	+	+	+	+	−	+	+	+	2

The complete primitive integer sequence and its positive scale factors are recorded in `assets/self-sampling/appell_a8_sturm_certificate.txt`; each row may be checked by one exact polynomial division. Sturm's theorem therefore gives $V(-\infty) - V(+\infty) = 6 - 2 = 4$ distinct real

roots. Because the sequence ends in a nonzero constant, S_0 is square-free. Its remaining four roots are nonreal; real coefficients pair them into two conjugate pairs. No floating-point root classification enters the proof. $\square$

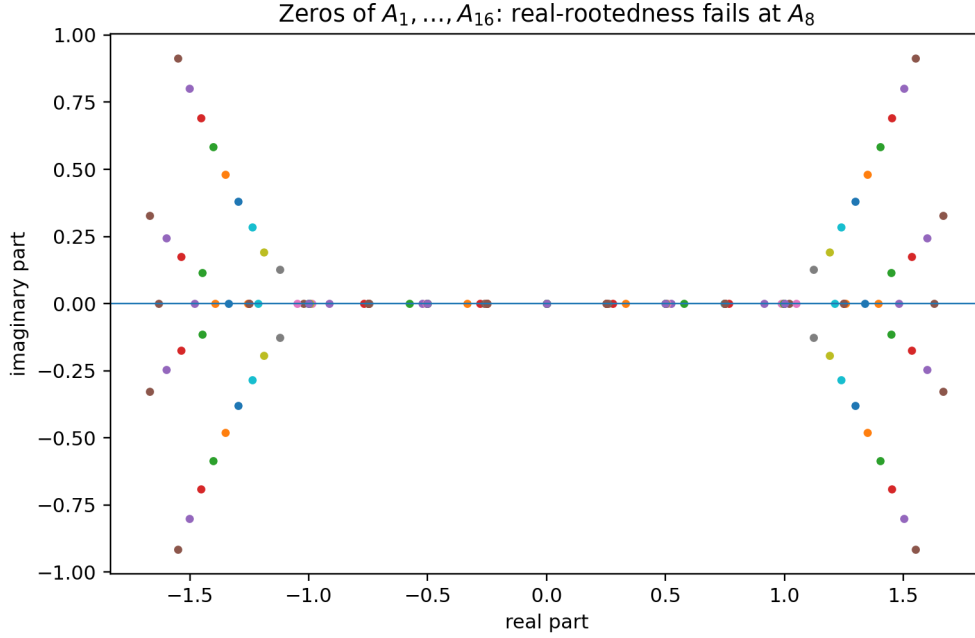


Figure 7: Numerical locations of the zeros of $\mathcal{A}_1, \dots, \mathcal{A}_{16}$. The Sturm calculation certifies the degree-eight departure from the real axis; the apparent absence of earlier departures and the later pattern are exploratory.

The computed nonreal-root counts through degree 30 are

Degree range	1–7	8–12	13–17	18–22	23–28	29–30
Nonreal roots	0	4	8	12	16	20

This staircase is not asserted as a theorem; it motivates [conjecture L.6](#).

I Transport through the inverse Fabius function

The centered quantile $q(t) = 2F^{-1}(t) - 1$ pushes Lebesgue measure on $[0, 1]$ to the up-law. Therefore

$$\int_0^1 g(q(t)) \, dt = \int_{-1}^1 g(x) u(x) \, dx \quad (\text{I.1})$$

for every bounded Borel function g .

Theorem I.1 (Rational inverse-Fabius quadrature). *Let $\theta = a/2^s$ and retain the dyadic nodes $x_k = n_k/2^D$ and weights w_k from [corollary B.3](#). Define*

$$t_k = F\left(\frac{1+x_k}{2}\right) \in \mathbb{Q}. \quad (\text{I.2})$$

Then $q(t_k) = x_k$, and for every polynomial p with $\deg p \leq N$,

$$\int_0^1 p(2F^{-1}(t) - 1) \, dt = \sum_k w_k p(2F^{-1}(t_k) - 1) = \sum_k w_k p(x_k). \quad (\text{I.3})$$

At a phase-superconvergent level the exactness extends to $\deg p \leq N + 1$.

Proof. Equation (A.4) implies $q(t_k) = x_k$. Apply (I.1) and then [theorem B.1](#). Rationality follows because both $(1 + x_k)/2$ and the weight arguments are dyadic. $\square$

Remark I.2 (A nonlinear exactness space). The rule in (I.3) is not claimed to integrate all ordinary polynomials in t . It integrates exactly the non-elementary space

$$\mathcal{V}_N = \text{span}\{1, q(t), q(t)^2, \dots, q(t)^N\}, \quad q(t) = 2F^{-1}(t) - 1, \quad (\text{I.4})$$

using rational nodes t_k , rational weights w_k , and no numerical inversion. This is an exact finite interface between the inverse Fabius function and dyadic arithmetic.

I.1 Endpoint nodes and the Lambert coordinate

For centered rules, the smallest nonzero endpoint distance is 2^{-N} and the corresponding extreme weights are

$$2^{-N} F(2^{-N}). \quad (\text{I.5})$$

For quarter-shifted rules, the leftmost distance is 2^{-N-2} , so one extreme weight is

$$2^{-N} F(2^{-N-2}). \quad (\text{I.6})$$

The corpus's sharp endpoint theory expresses $F(x)$ through the exact lower-Lambert coordinate

$$\lambda(x) = -\frac{1}{\log 2} W_{-1}(-x \log 2), \quad x = \lambda 2^{-\lambda}, \quad (\text{I.7})$$

with a nonconstant periodic correction in the dyadic phase of λ . Thus the new quadrature rules supply exact rational global moment constraints whose smallest weights lie precisely in the Lambert boundary layer. Turning these constraints into identities for the endpoint periodic factor is an open program formulated in [research problem L.8](#).

J Finite Gaussian- q -binomial identities forced by quadrature

J.1 The exact dyadic evaluator

Let

$$(q; q)_n = \prod_{j=1}^n (1 - q^j), \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}. \quad (\text{J.1})$$

For integers $m, n \geq 0$ and $c \in \mathbb{Q}$, define

$$\begin{aligned} \mathfrak{Q}_n(m; c) &:= \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{j=0}^n \frac{\begin{bmatrix} n \\ j \end{bmatrix}_{1/2}}{2^{j(j-1)} (n+j)!} \\ &\quad \times \sum_{r=0}^{m2^j-1} \varepsilon_r (r - m2^j + c)^{n+j}. \end{aligned} \quad (\text{J.2})$$

The established arbitrary-numerator theorem states

$$\mathfrak{Q}_n(m; c) = F(m/2^n) \quad (0 \leq m \leq 2^n), \quad (\text{J.3})$$

and the right side of (J.2) is independent of c . Its Gaussian coefficients arise from Lagrange interpolation on a geometric grid.

J.2 A q -binomial–Thue–Morse–Bell identity

Theorem J.1 (Finite arithmetic moment identity). *Let $\theta = a/2^s$, $D = N + s$, $n_k = 2^s k + a$, and $b_k = 2^D - |n_k|$. For every $0 \leq r \leq N$ and every $c \in \mathbb{Q}$,*

$$\mu_r = 2^{-N-Dr} \sum_{|n_k| < 2^D} n_k^r \mathfrak{Q}_D(b_k; c). \quad (\text{J.4})$$

At the superconvergent phases of [corollary F.1](#), the identity also holds for $r = N + 1$. Therefore the finite Gaussian- q -binomial–Thue–Morse sum on the right equals

$$Y_r(0, \kappa_2, 0, \kappa_4, \dots, \kappa_r), \quad (\text{J.5})$$

where the cumulants are the Bernoulli rationals in [\(A.9\)](#).

Proof. At node $x_k = n_k/2^D$, [\(A.3\)](#) and [\(J.3\)](#) give

$$u(x_k) = F(b_k/2^D) = \mathfrak{Q}_D(b_k; c).$$

Substitute this into [\(B.3\)](#), then use [\(A.10\)](#). □

Remark J.2 (Why this identity is structurally new). Each ingredient existed separately: the q -binomial evaluator, the Bell moment formula, and the zero multiplicities. The self-sampling theorem forces them into a single positive finite identity. The common translation c disappears after a double finite summation, giving a family of nontrivial translation-invariance identities among Thue–Morse power sums.

Corollary J.3 (Finite Appell cancellation in q -arithmetic). *For $1 \leq n \leq N$,*

$$\sum_{|n_k| < 2^D} \mathcal{A}_n\left(-\frac{n_k}{2^D}\right) \mathfrak{Q}_D(b_k; c) = 0. \quad (\text{J.6})$$

After inserting [\(J.2\)](#), this is a finite identity containing Gaussian q -binomial coefficients, Thue–Morse signs, Bernoulli cumulants, and complete Bell polynomials.

Proof. Set $x = 0$ in [\(H.10\)](#) and clear the common factor 2^{-N} . □

K Base- b and multidimensional extensions

K.1 Integer dilation bases

Let $b \geq 2$ be an integer and define

$$\Phi_b(\xi) = \prod_{j=0}^{\infty} \text{sinc}_{\pi}\left(\frac{\xi}{b^j}\right). \quad (\text{K.1})$$

Proposition K.1 (Construction and regularity of the base- b density). *Let $(U_j)_{j \geq 0}$ be independent, with U_j uniform on*

$$I_j = \left[-\frac{1}{2b^j}, \frac{1}{2b^j}\right].$$

Then the series $S_b = \sum_{j \geq 0} U_j$ converges absolutely. Its law has a nonnegative density $u_b \in C_c^\infty(\mathbb{R})$ satisfying

$$\widehat{u}_{b \cdot 2\pi}(\xi) = \Phi_b(\xi), \quad \int_{\mathbb{R}} u_b(x) \, dx = 1, \quad \text{supp}(u_b) = \left[-\frac{b}{2(b-1)}, \frac{b}{2(b-1)}\right]. \quad (\text{K.2})$$

Thus u_b is precisely the infinite convolution of the centered uniform density factors.

Proof. The deterministic estimate

$$\sum_{j \geq 0} |U_j| \leq \frac{1}{2} \sum_{j \geq 0} b^{-j} = \frac{b}{2(b-1)}$$

proves absolute convergence. The j th uniform law has Fourier transform $\text{sinc}_\pi(\xi/b^j)$, so independence and convergence of the partial sums give $\mathbb{E}e^{-2\pi i \xi S_b} = \Phi_b(\xi)$.

It remains to justify that this probability law is smooth rather than merely a compactly supported measure. The elementary estimate

$$|\text{sinc}_\pi t| \leq \min\left(1, \frac{1}{\pi |t|}\right)$$

shows, by retaining any fixed number K of factors in (K.1), that

$$|\Phi_b(\xi)| \leq C_K (1 + |\xi|)^{-K} \quad (\xi \in \mathbb{R})$$

for some constant C_K . Hence $\xi^r \Phi_b(\xi)$ is integrable for every $r \geq 0$. Fourier inversion therefore supplies a C^r density for every r , and differentiating under the inverse-transform integral identifies all of its derivatives. Thus $u_b \in C^\infty$. Since it represents a probability law, it is nonnegative and has integral one.

For the exact support, equip $\prod_{j \geq 0} I_j$ with the product measure. It has full support, and the summation map is continuous because the defining series converges uniformly. Its image is the interval

$$\sum_{j \geq 0} I_j = \left[-\frac{1}{2} \sum_{j \geq 0} b^{-j}, \frac{1}{2} \sum_{j \geq 0} b^{-j} \right] = \left[-\frac{b}{2(b-1)}, \frac{b}{2(b-1)} \right].$$

(For example, every point in this interval is obtained by choosing all coordinates in the same fixed proportion of their respective endpoints.) The continuous image of this full-support product measure consequently has exactly that interval as its support. This proves (K.2). $\square$

If $m \neq 0$ is an integer, then

$$\text{ord}_{\xi=m} \Phi_b = 1 + \nu_b(m), \quad (\text{K.3})$$

where $\nu_b(m)$ is the largest a such that $b^a \mid m$. Indeed, precisely the factors with $0 \leq j \leq \nu_b(m)$ vanish at m ; each has a simple zero there, and every later factor is nonzero.

Theorem K.2 (Base- b self-sampling). *For every $N \geq 0$, phase θ , and polynomial p of degree at most N ,*

$$\int p(x) u_b(x) dx = b^{-N} \sum_{k \in \mathbb{Z}} p\left(\frac{k + \theta}{b^N}\right) u_b\left(\frac{k + \theta}{b^N}\right). \quad (\text{K.4})$$

Moreover, averaging the b^s phases a/b^s gives the level- $N + s$ rule and is the unique normalized full DFT filter: among coefficient vectors (c_a) with $\sum_a c_a = 1$, it is the only one annihilating every nonzero residue modulo b^s .

Proof. By [proposition K.1](#), the functions $x^r u_b(x)$ are smooth and compactly supported, so Poisson summation applies. For $0 \leq r \leq N$, its nonzero alias at $b^N \ell$ is a constant multiple of $\Phi_b^{(r)}(b^N \ell)$. Equation (K.3) gives

$$\text{ord}_{\xi=b^N \ell} \Phi_b = N + 1 + \nu_b(\ell) > r,$$

so every such alias vanishes. The zero alias is the r th moment. This proves (K.4) for monomials and hence for all stated polynomials.

For the phase refinement, average the rules with phases a/b^s . Writing $r = b^s k + a$ identifies their nodes bijectively with r/b^{N+s} , and the averaged coefficient is $b^{-(N+s)}$; the result is exactly the

level- $N + s$ rule. Finally, normalized weights c_a with $\sum_a c_a = 1$ annihilate every nonzero residue precisely when their b^s -point discrete Fourier transform is $(1, 0, \dots, 0)$. Fourier inversion gives $c_a = b^{-s}$ for every a , proving uniqueness. $\square$

The binary case is distinguished arithmetically because half-integer spectral signs reduce to the classical Thue–Morse sequence and dyadic Fabius values are rational. For general b , the same analytic framework suggests generalized digital sign sequences and $q = b^{-1}$ arithmetic.

K.2 Tensor-product cubature

Let

$$U_d(x_1, \dots, x_d) = \prod_{j=1}^d u(x_j). \quad (\text{K.5})$$

Tensoring [theorem B.1](#) gives a positive cubature rule.

Corollary K.3 (Positive tensor cubature). *For a polynomial p whose degree in each coordinate is at most N ,*

$$\int_{[-1,1]^d} p(\mathbf{x}) U_d(\mathbf{x}) \, d\mathbf{x} = 2^{-Nd} \sum_{\mathbf{k} \in \mathbb{Z}^d} p\left(\frac{\mathbf{k} + \boldsymbol{\theta}}{2^N}\right) U_d\left(\frac{\mathbf{k} + \boldsymbol{\theta}}{2^N}\right). \quad (\text{K.6})$$

The sum is finite and positive. Coordinatewise phase superconvergence raises the exactness by one in every selected coordinate.

Proof. It is enough by linearity to take $p(\mathbf{x}) = \prod_{j=1}^d x_j^{r_j}$ with $0 \leq r_j \leq N$. Fubini's theorem factors the integral into d one-dimensional moments. Apply [Theorem B.1](#) to each factor and multiply the resulting finite sums; their Cartesian product is exactly (K.6). Compact support makes the sum finite, and every weight is nonnegative because $u \geq 0$. If a chosen coordinate phase satisfies the cancellation condition of [Corollary F.1](#), the same argument admits $r_j = N + 1$ in that coordinate; applying this independently proves the final coordinatewise assertion. $\square$

The node count grows exponentially with d , so sparse-grid combinations and hyperbolic-cross alias filters are a natural next step; see [research problem L.11](#).

L Conjectures and frontier research program

L.1 The next alias layer

Conjecture L.1 (Exact phase-adapted precision and sign). *Let θ_N be the canonical phase in (F.3). Then*

$$E_{N,N+2}(\theta_N) \neq 0 \quad (N \geq 0). \quad (\text{L.1})$$

Moreover, for $N \geq 1$, the experimental sign law is

$$\text{sgn } E_{N,N+2}(\theta_N) = (-1)^{\lfloor (N-1)/2 \rfloor}. \quad (\text{L.2})$$

The exact two-layer formula is given in [appendix N](#). A proof should combine:

1. the odd-alias contribution with one Bell correction γ_1 ;
2. the twice-odd contribution with its leading zero jet;
3. the parity-specific phase factors;
4. a first-frequency dominance estimate sharp enough to control the logarithmic derivative $\Phi'_{2\pi}/\Phi_{2\pi}$ at half-integers.

A successful argument would determine the exact degree of every positive phase-adapted rule.

Research problem L.2 (All higher phase gains). *For fixed excess $s \geq 2$, classify all phases θ for which*

$$E_{N,N+1}(\theta) = \cdots = E_{N,N+s}(\theta) = 0.$$

The valuation filtration makes this a finite hierarchy of trigonometric equations built from alias layers $0, \dots, s-1$. Determine whether any single positive phase can gain more than one degree for infinitely many N .

L.2 Optimal phase and scale filters

Research problem L.3 (Hybrid DFT–Gaussian- q filter). *Combine the positive DFT projector in (G.2) with the repository’s geometric Gaussian- q -binomial Richardson filters in the scale variable. For a prescribed exactness order, minimize:*

1. *total variation of the combined coefficients;*
2. *number of distinct dyadic Fabius evaluations;*
3. *largest denominator of an exact rational coefficient;*
4. *sensitivity near the flat endpoints.*

A two-stage construction may first remove low 2-adic alias classes by positive phase averaging and then remove the remaining scale modes by a short signed q -Richardson filter.

Research problem L.4 (Sparse positive phase design). *Fix $s \geq 2$ and $1 \leq q < 2^s$. Among probability vectors $c = (c_0, \dots, c_{2^s-1})$ with at most q nonzero entries, determine*

$$\min_c \max_{1 \leq r < 2^s} \left| \sum_{a=0}^{2^s-1} c_a e^{2\pi i ar/2^s} \right|,$$

classify the minimizers up to cyclic symmetry, and, among tied minimizers, minimize $\sum_a c_a^2$. Determine how the optimal leakage controls the worst first surviving alias of the associated positive sampling rule.

The former “positive extremality” conjecture imposed all exact DFT annihilation equations. By [Theorem G.2](#) those equations admit only the uniform vector, so optimization over that singleton was vacuous. The support constraint above creates a genuine tradeoff between positivity, sparsity, and alias suppression.

L.3 Denominators and arithmetic geometry

Research problem L.5 (Exact valuations of self-sampling weights). *For dyadic $\theta = a/2^s$, determine the exact prime valuations of*

$$w_{N,\theta,k} = 2^{-N} F\left(1 - \frac{|2^s k + a|}{2^{N+s}}\right)$$

in lowest terms. Natural targets are:

1. *exact 2-adic valuations of the extreme weights;*
2. *odd-prime divisibility controlled by Mersenne factors $2^{2^j} - 1$;*
3. *the least common denominator of the complete rule;*
4. *cancellation in the rational next defect $E_{N,N+2}(\theta_N)$.*

The q -binomial identity (J.4) supplies a second route to the same valuations and may reveal cancellations invisible in the moment formula.

The former “primitive normalized phase rules” conjecture is also retired. If L is defined to be the least common denominator of a rational weight vector, then the integers Lw_k automatically have greatest common divisor one: their gcd divides L because $\sum_k Lw_k = L$, and a larger gcd would produce a smaller common denominator. The nontrivial arithmetic question is therefore the exact value and prime factorization of L , already posed in [Research problem L.5](#).

L.4 Rvachev–Appell zero geometry

Conjecture L.6 (Unbounded nonreal root count). *The number of nonreal zeros of $\mathcal{A}_n$ tends to infinity with n .*

This statement deliberately does not extrapolate the short numerical staircase. A stronger, separately testable target is weak convergence of the empirical measures

$$\nu_n := \frac{1}{n} \sum_{\mathcal{A}_n(z)=0} \delta_{2\pi z/n}$$

(zeros counted with multiplicity) to a probability measure with compact support. Identifying such a measure and its support belongs to the spectral problem below; measures converge to measures, not to compact sets.

Research problem L.7 (A spectral asymptotic for $\mathcal{A}_n$). *Use residues of $e^{xz}/M(z)$ to derive an expansion for $\mathcal{A}_n(x)/n!$ that is uniform in the zero-scaling regime. Determine:*

1. *the first curve on which the contributions from $\pm 2\pi i$ balance;*
2. *how the next dyadic poles perturb that curve;*
3. *whether the observed root-count plateaus have an exact spectral explanation;*
4. *whether a Jensen-polynomial or multiplier-sequence normalization restores real-rootedness.*

L.5 Lambert endpoint sum rules

Research problem L.8 (Exact quadrature constraints on the periodic endpoint factor). *Insert the repository’s all-orders endpoint expansion for $F(x)$, written in the lower-Lambert coordinate (I.7), into the exact identities (B.7) and (J.4). After separating the bulk from a shrinking endpoint boundary layer, derive explicit linear or nonlinear sum rules for the periodic coefficient functions.*

The principal difficulty is uniformity: the number of dyadic samples grows with N , while the Lambert expansion is strongest only near the endpoints. A viable strategy is to choose a cutoff $j \leq J(N)$, use the endpoint transseries on that range, and control the remaining samples by smooth quadrature estimates. Exactness of the full moment identity should force cancellation between the two endpoint layers and the bulk.

Research problem L.9 (Uniform endpoint/alias matching for extreme quantile nodes). *For the inverse-Fabius quadrature in (I.3), choose an explicit cutoff $J(N) \rightarrow \infty$ with $J(N) = o(2^N)$. Derive a lower-Lambert expansion, uniform for the extreme indices $0 \leq k \leq J(N)$ and their reflected partners, with a stated remainder after each retained inverse-phase order. Then define an explicit two-level combination of the phases in (F.3) and determine whether its normalized endpoint contribution cancels through one additional inverse-phase order; bound the complementary bulk uniformly so that the resulting assertion is a theorem about the complete quadrature error.*

The earlier phase-locking wording did not specify the extreme-index schedule, the two-level coefficients, the normalization, or the meaning of one inverse-Lambert order. Those data are made explicit requirements here rather than hidden inside an unfalsifiable conjecture.

L.6 Digital generalizations and higher dimensions

Research problem L.10 (Base- b spectral signs). *For the product (K.1), determine the sign and phase of $\Phi_b((bk + r)/b)$ on each nonzero residue class r . Identify the resulting automatic sequence and its analogues of Prouhet cancellation, Mahler equations, and q -binomial arithmetic.*

Research problem L.11 (Sparse-grid Rvachev cubature). *Construct Smolyak or hyperbolic-cross combinations of the tensor rules (K.6) that preserve as much positivity as possible. The alias description suggests indexing components by 2-adic valuation vectors rather than by total scale alone. Determine optimal node counts for coordinatewise or total-degree exactness.*

Research problem L.12 (Nonproduct Rvachev laws). *Replace the tensor product by compactly supported refinable densities associated with a dilation matrix. Characterize when Fourier zero multiplicities on the dual lattice make the density's own samples into a positive polynomial cubature rule.*

M First inverse-moment Appell polynomials

For compactness the generated table writes A_n in place of $\mathcal{A}_n$.

$$\begin{aligned}
A_0(x) &= 1 \\
A_1(x) &= x \\
A_2(x) &= x^2 - \frac{1}{9} \\
A_3(x) &= x^3 - \frac{x}{3} \\
A_4(x) &= x^4 - \frac{2x^2}{3} + \frac{31}{675} \\
A_5(x) &= x^5 - \frac{10x^3}{9} + \frac{31x}{135} \\
A_6(x) &= x^6 - \frac{5x^4}{3} + \frac{31x^2}{45} - \frac{2347}{59535} \\
A_7(x) &= x^7 - \frac{7x^5}{3} + \frac{217x^3}{135} - \frac{2347x}{8505} \\
A_8(x) &= x^8 - \frac{28x^6}{9} + \frac{434x^4}{135} - \frac{9388x^2}{8505} + \frac{1904369}{32531625} \\
A_9(x) &= x^9 - 4x^7 + \frac{434x^5}{75} - \frac{9388x^3}{2835} + \frac{1904369x}{3614625} \\
A_{10}(x) &= x^{10} - 5x^8 + \frac{434x^6}{45} - \frac{4694x^4}{567} + \frac{1904369x^2}{722925} - \frac{3309760579}{24405225075}
\end{aligned}$$

N Exact two-layer formula for the next defect

Put $R = N + 2$. The valuation filtration (C.4) contains only $a = 0$ and $a = 1$. For odd q , define

$$m_0 = 2^N q, \quad d_0 = N + 1, \quad (\text{N.1})$$

$$m_1 = 2^{N+1} q, \quad d_1 = N + 2 = R. \quad (\text{N.2})$$

Then corollary C.5 gives

$$\Phi_{2\pi}^{(R)}(m_0) = -\frac{R! \Phi_{2\pi}(q/2)}{m_0^{N+1}} \gamma_1(m_0), \quad (\text{N.3})$$

$$\Phi_{2\pi}^{(R)}(m_1) = -\frac{R! \Phi_{2\pi}(q/2)}{m_1^R}. \quad (\text{N.4})$$

Consequently

$$E_{N,N+2}(\theta) = -\frac{R!}{(-2\pi i)^R} \sum_{\substack{q \in \mathbb{Z} \setminus \{0\} \\ q \text{ odd}}} \Phi_{2\pi}(q/2) \left[\frac{\gamma_1(2^N q)}{(2^N q)^{N+1}} e^{2\pi i q \theta} + \frac{1}{(2^{N+1} q)^R} e^{4\pi i q \theta} \right]. \quad (\text{N.5})$$

Here

$$\gamma_1(2^N q) = -\frac{N+1}{2^N q} + 2^{-N-1} \frac{\Phi'_{2\pi}(q/2)}{\Phi_{2\pi}(q/2)}. \quad (\text{N.6})$$

This formula is the proposed starting point for [conjecture L.1](#). It isolates exactly what is new beyond the first defect: a half-integer logarithmic derivative and one newly activated 2-adic alias layer.

Part VII

Exact Inverse Moduli and Computability

O Main result and proof architecture

O.1 The theorem in one statement

Throughout the report, $F : \mathbb{R} \rightarrow [0, 1]$ denotes the bounded Fabius function: it is 0 on $(-\infty, 0]$, 1 on $[1, \infty)$, and on $[0, 1]$ it is the strictly increasing C^∞ function satisfying

$$F(x) + F(1-x) = 1, \quad F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}). \quad (\text{O.1})$$

Let $G : [0, 1] \rightarrow [0, 1]$ be its inverse, and define the totalized inverse

$$\overline{G}(y) := G(\text{clamp}(y)), \quad \text{clamp}(y) := \min(1, \max(0, y)). \quad (\text{O.2})$$

Thus $\overline{G}(y) = 0$ for $y \leq 0$ and $\overline{G}(y) = 1$ for $y \geq 1$.

Theorem O.1 (Computability of the inverse Fabius function). *The inverse $G : [0, 1] \rightarrow [0, 1]$, in the subset-domain version of the Grzegorzczuk definition, and its totalization $\overline{G} : \mathbb{R} \rightarrow \mathbb{R}$ are computable real functions. More explicitly:*

1. G and $\overline{G}$ are sequentially computable. From one algorithm that names a computable sequence (y_i) , one can uniformly compute dyadic approximations to $G(y_i)$ (or $\overline{G}(y_i)$) of any requested precision.
2. For $r \geq 0$, put

$$\Delta_r = \frac{1}{2^{r(r+1)/2} (r+1)^{r+1}}. \quad (\text{O.3})$$

Then, for all $u, v \in \mathbb{R}$,

$$|u - v| < \Delta_r \implies |\overline{G}(u) - \overline{G}(v)| < 2^{-r}. \quad (\text{O.4})$$

3. For $n \geq 1$, let

$$r(n) := \min \{r \geq 1 : 2^r > n\}, \quad (\text{O.5})$$

and define

$$d(n) := 2^{r(n)(r(n)+1)/2} (r(n) + 1)^{r(n)+1}, \quad d(0) := 1. \quad (\text{O.6})$$

The integer function d is recursive (indeed primitive recursive after a standard bounded-search coding), and

$$|u - v| < \frac{1}{d(n)} \implies |\overline{G}(u) - \overline{G}(v)| < \frac{1}{n} \quad (n \geq 1). \quad (\text{O.7})$$

Proof. The claims are assembled below from independently certified ingredients; we record the dependency chain here so that the headline theorem has a proof at the point where it is stated. The tolerant-bisection realizer proved in [theorem U.5](#) uniformly transforms a fast name of a sequence (y_i) in $[0, 1]$ into a fast name of $(G(y_i))$. This is [corollary U.6](#). Applying the computable, 1-Lipschitz clamp first gives the same uniform transformation for $\overline{G}$ by [lemma U.7](#) and [corollary U.8](#), proving item 1.

For item 2, [lemma R.1](#) gives $\Delta_r \leq F(2^{-r})$. The exact inverse-modulus theorem, in the effective-injectivity form [corollary S.5](#), therefore gives

$$|u - v| < \Delta_r \implies |\overline{G}(u) - \overline{G}(v)| < 2^{-r};$$

this is [theorem T.1](#). For $n \geq 1$, the least integer $r(n) \geq 1$ with $2^{r(n)} > n$ is computable by bounded search, and the displayed formula for $d(n)$ is a computable composition of integer operations. Moreover $d(n)^{-1} = \Delta_{r(n)}$ and $2^{-r(n)} < 1/n$. Chaining these two strict inequalities with item 2 proves item 3, as detailed in [theorem T.2](#). Thus G (on its subset domain) and $\overline{G}$ satisfy both clauses of [definition P.3](#). $\square$

The theorem is stronger than a nonconstructive assertion that a continuous inverse on a compact interval is uniformly continuous. It gives a concrete integer modulus, a finite algorithm on names, and a sharp structural identity for the best possible modulus.

O.2 Four proof layers

The argument is organized so that every hidden effectiveness issue is exposed.

1. **Forward oracle.** A centered finite convolution has an exact Thue–Morse truncated-power formula. At rational inputs it is rational, and its error from F is at most 2^{-N} .
2. **Effective injectivity.** Symmetric unimodality shows that an interval of geometric length h has probability mass at least $F(h)$; strict radial unimodality shows that equality occurs only for a degenerate interval or an endpoint interval. A direct event gives the computable lower bound $F(2^{-r}) \geq \Delta_r$.
3. **Uniform continuity.** Effective injectivity of F becomes an effective modulus for G . In fact the ordinary modulus of G is exactly G itself.
4. **Name transformation.** Tolerant bisection combines the forward oracle and the inverse modulus. A midpoint is used for a left/right update only when the sign is certified; otherwise it is already a certified output.

The fourth layer matters. Naive bisection that waits until it decides whether $F(c) < y$, $F(c) = y$, or $F(c) > y$ is not an algorithm on arbitrary computable reals: equality of computable reals is undecidable in general. The tolerance branch is what removes that obstruction.

P Computable-analysis definitions

P.1 Fast names and computable sequences

Following standard computable analysis [\[8, 9, 10\]](#), a convenient representation of a real number x is a fast dyadic Cauchy name: an algorithm returns a dyadic rational q_p such that

$$|x - q_p| \leq 2^{-p}. \quad (\text{P.1})$$

A real sequence $(x_i)_{i \in \mathbb{N}_0}$ is computable when one algorithm returns $q_{i,p}$ satisfying (P.1) uniformly in both i and p . The repository uses an equivalent signed-natural code for the dyadic numerator, avoiding an opaque integer encoding.

Definition P.1 (Sequential computability). A function f is *sequentially computable* if there is one effective realizer transformer with the following property. Given as a subroutine any algorithm $Q(i, p)$ returning a dyadic rational with $|Q(i, p) - x_i| \leq 2^{-p}$ for a sequence (x_i) in the domain, the transformer produces an algorithm $R(i, p)$ returning a dyadic rational with $|R(i, p) - f(x_i)| \leq 2^{-p}$. The same transformer works for every input name, uniformly in the sequence index i and requested precision p .

Thus this is a realizability assertion, not merely the extensional statement that each computable input sequence happens to have a computable image. The construction in [appendix U](#) supplies the single uniform transformer explicitly.

P.2 The reciprocal uniform-continuity convention

Definition P.2 (Effective uniform continuity). A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *effectively uniformly continuous* if there is a recursive $d : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ with $d(n) > 0$ for $n > 0$ such that

$$|x - y| < \frac{1}{d(n)} \implies |f(x) - f(y)| < \frac{1}{n} \quad (n > 0). \quad (\text{P.2})$$

This is the reciprocal convention used in the cited Wikipedia article [11]. The repository implements it in `FabiusComputability.lean`. It is equivalent to the more common binary modulus convention

$$|x - y| < 2^{-m(p)} \implies |f(x) - f(y)| < 2^{-p}, \quad (\text{P.3})$$

provided m is recursive. We give both forms because the binary version reveals the endpoint conditioning much more clearly.

Definition P.3 (Computable real function). In the Grzegorzczuk formulation used here, a real function is computable if it is sequentially computable and effectively uniformly continuous.

For G defined only on $[0, 1]$, the same definitions are restricted to sequences and pairs of points in that computable compact domain. The totalized $\overline{G}$ lets one use the literal type $\mathbb{R} \rightarrow \mathbb{R}$ without changing the mathematics.

Q Probability, Fourier structure, and the forward function

Q.1 The random-series model

The classical probability model may be written as follows [1, 3, 7]. Recall that $V_1, V_2, \dots$ are independent uniform random variables on $[0, 1]$ and put

$$X := \sum_{k=1}^{\infty} 2^{-k} V_k. \quad (\text{Q.1})$$

The series converges absolutely, $0 \leq X \leq 1$, and

$$F(x) = \mathbb{P}[X \leq x]. \quad (\text{Q.2})$$

Reflection $V_k \mapsto 1 - V_k$ sends X to $1 - X$ and proves the symmetry in (O.1).

The centered random variable

$$Y := 2X - 1 = \sum_{k=1}^{\infty} 2^{-k} (2V_k - 1) \quad (\text{Q.3})$$

has density up. With $\text{sinc}_{\text{rad}} z = \sin z / z$,

$$\mathbb{E} e^{itY} = \prod_{k=1}^{\infty} \text{sinc}_{\text{rad}} \left(\frac{t}{2^k} \right). \quad (\text{Q.4})$$

Thus the probability model, the Rvachev bump, and the infinite sinc product are three representations of the same object. The computability proof uses the probability representation in physical space; the Fourier and polynomial alternatives are developed in the shared platform and the deconvolution and self-sampling parts of this volume.

Q.2 Finite sums and their exact CDFs

For $N \geq 1$, define

$$X_N := \sum_{k=1}^N 2^{-k} V_k, \quad R_N := X - X_N. \quad (\text{Q.5})$$

Then

$$0 \leq R_N \leq 2^{-N}. \quad (\text{Q.6})$$

Let $F_N(x) = \mathbb{P}[X_N \leq x]$. The general inclusion–exclusion formula for a sum of independent, nonidentically scaled uniforms becomes especially rigid for dyadic scales.

Proposition Q.1 (Finite Thue–Morse CDF). *For $N \geq 1$ and $x \in \mathbb{R}$,*

$$F_N(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{j=0}^{2^N-1} (-1)^{\text{wt}_2(j)} \left(x - \frac{j}{2^N} \right)_+^N, \quad (\text{Q.7})$$

where $\text{wt}_2(j)$ is the sum of the binary digits of j .

Proof. For positive lengths $a_1, \dots, a_N$, the CDF of $\sum a_k V_k$ is

$$\frac{1}{N! \prod_{k=1}^N a_k} \sum_{A \subseteq \{1, \dots, N\}} (-1)^{|A|} \left(x - \sum_{k \in A} a_k \right)_+^N.$$

This follows by integrating the box indicator over the half-space $\sum a_k u_k \leq x$, or inductively by finite differences of truncated powers. Here $a_k = 2^{-k}$ and $\prod a_k = 2^{-N(N+1)/2}$. The map

$$A \mapsto j(A) := \sum_{k \in A} 2^{N-k}$$

is a bijection from subsets to $\{0, \dots, 2^N - 1\}$, with $\sum_{k \in A} 2^{-k} = j(A)/2^N$ and $|A| \equiv \text{wt}_2(j(A)) \pmod{2}$. Substitution gives (Q.7). $\square$

The alternating signs are exactly the Thue–Morse signs. No limiting argument is hidden in (Q.7): at rational x it is an exact rational obtained by finitely many integer operations.

Q.3 Centered finite splines and a uniform certificate

The deterministic midpoint of the omitted tail is 2^{-N-1} . Define

$$Q_N := X_N + 2^{-N-1}, \quad S_N(x) := \mathbb{P}[Q_N \leq x] = F_N(x - 2^{-N-1}). \quad (\text{Q.8})$$

Then [proposition Q.1](#) gives

$$S_N(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{j=0}^{2^N-1} (-1)^{\text{wt}_2(j)} \left(x - \frac{2j+1}{2^{N+1}} \right)_+^N. \quad (\text{Q.9})$$

This is the centered uniform spline used by the repository's primitive- recursive evaluator.

Lemma Q.2 (Uniform density bound). *The finite CDFs F_N and S_N are globally 2-Lipschitz.*

Proof. The density of $2^{-1}U_1$ is $2\mathbf{1}_{[0,1/2]}$. Convolution with any probability measure cannot increase the L^∞ norm of a density. Hence the density of X_N is bounded by 2. Translating by 2^{-N-1} does not change this bound. Integrating the density over an interval proves the Lipschitz estimate. $\square$

Theorem Q.3 (Certified centered approximation). *For every $N \geq 1$ and every real x ,*

$$|S_N(x) - F(x)| \leq 2^{-N}. \quad (\text{Q.10})$$

Proof. Write

$$X = Q_N + E_N, \quad E_N = R_N - 2^{-N-1}.$$

By [\(Q.6\)](#), $|E_N| \leq 2^{-N-1}$. Therefore

$$S_N(x - 2^{-N-1}) \leq F(x) \leq S_N(x + 2^{-N-1}).$$

By [lemma Q.2](#), moving the argument by 2^{-N-1} changes S_N by at most $2 \cdot 2^{-N-1} = 2^{-N}$. This gives both sides of [\(Q.10\)](#). $\square$

Corollary Q.4 (Self-contained computability of F). *There is an explicit uniform algorithm which, given a rational x and a precision p , returns a rational q with $|q - F(x)| \leq 2^{-p}$. One may take $q = S_N(x)$ with $N \geq p$. Consequently F is sequentially computable; together with its global 2-Lipschitz bound, it is a computable real function.*

Proof. Formula [\(Q.9\)](#) is a finite rational computation, and [theorem Q.3](#) supplies its error certificate. At an arbitrary computable input, first replace the input by a sufficiently accurate dyadic; the 2-Lipschitz bound controls this replacement. The construction is uniform in a sequence index. Effective uniform continuity follows from $|F(x) - F(y)| \leq 2|x - y|$; for example the reciprocal modulus $d(n) = 2n$ works. $\square$

Remark Q.5 (Efficiency versus computability). The direct sum has 2^N terms and is not intended as an efficient evaluator. The repository's exact dyadic recurrence, highest-bit reduction, and primitive-recursive natural-number implementation are substantially better. For the theorem here, only termination and a proved error bound are required.

R Effective positivity, strict increase, and the density shape

R.1 A closed positive lower bound at dyadic points

The existence of an inverse requires strict increase; effective inversion requires a quantitative version. Both begin with an elementary positive probability event.

Lemma R.1 (Dyadic endpoint box event). *For every $r \geq 0$,*

$$F(2^{-r}) \geq \Delta_r := 2^{-r(r+1)/2} (r+1)^{-(r+1)}. \quad (\text{R.1})$$

In particular, $F(x) > 0$ for every $x > 0$.

Proof. Put $m = r + 1$. For $1 \leq k \leq m$, impose the independent restrictions

$$V_k \leq a_k := \frac{2^{k-r-1}}{m}. \quad (\text{R.2})$$

The largest a_k is $a_m = 1/m \leq 1$, so these are valid events. On their intersection,

$$2^{-k} V_k \leq \frac{2^{-r-1}}{m} \quad (1 \leq k \leq m).$$

After summation the first m terms contribute at most 2^{-r-1} . The unrestricted tail contributes at most

$$\sum_{k=m+1}^{\infty} 2^{-k} = 2^{-m} = 2^{-r-1}.$$

Thus the event in (R.2) implies $X \leq 2^{-r}$. Its probability is

$$\begin{aligned} \prod_{k=1}^m a_k &= m^{-m} 2^{\sum_{k=1}^m (k-r-1)} \\ &= m^{-m} 2^{m(m+1)/2 - m(r+1)} \\ &= 2^{-r(r+1)/2} (r+1)^{-(r+1)} = \Delta_r. \end{aligned}$$

This proves (R.1). Given $x > 0$, choose r with $2^{-r} \leq x$ and use monotonicity of a CDF. $\square$

The bound is deliberately elementary: it uses only independence, the support of the tail, and integer arithmetic. It is not numerically sharp, but [appendix T.3](#) shows that its logarithm has the correct two largest orders for computability purposes.

The numerical conclusion of [lemma R.1](#) is now Lean-formalized, but by a different and stronger route. Here is the complete recurrence argument. Let $a_0 = 1$ and use the repository's exact inverse-dyadic identities

$$a_r = \frac{\sum_{0 \leq k < r} a_k / (r - k + 1)!}{2^r - 1} \quad (r \geq 1), \quad F(2^{-r}) = 2^{-\binom{r}{2}} a_r.$$

Every a_k is positive. Retaining only the $k = 0$ summand therefore gives

$$a_r \geq \frac{1}{(r+1)!(2^r - 1)}, \quad F(2^{-r}) \geq \frac{1}{2^{\binom{r}{2}} (r+1)!(2^r - 1)} \quad (r \geq 1).$$

Since $2^r - 1 \leq 2^r$ and $\binom{r}{2} + r = \binom{r+1}{2}$, taking the larger denominator yields the first inequality below; the exceptional index $r = 0$ is simply $F(1) = 1$. Thus for every $r \geq 0$,

$$F(2^{-r}) \geq \frac{1}{2^{\binom{r+1}{2}} (r+1)!} \geq \frac{1}{2^{\binom{r+1}{2}} (r+1)^{r+1}} = \Delta_r. \quad (\text{R.3})$$

Indeed, $(r+1)! \leq (r+1)^{r+1}$ proves the second inequality after multiplication by the positive power of two and reversal under reciprocation.

For the executable arithmetic, write

$$d_{\text{fac}}(r) := 2^{\binom{r+1}{2}}(r+1)!, \quad d_{\Delta}(r) := 2^{\binom{r+1}{2}}(r+1)^{r+1} = \Delta_r^{-1}.$$

Lean defines the first denominator by primitive recursion,

$$d_{\text{fac}}(0) = 1, \quad d_{\text{fac}}(r+1) = d_{\text{fac}}(r) 2^{r+1}(r+2).$$

Induction, using $\binom{r+2}{2} = \binom{r+1}{2} + r+1$, proves the displayed closed form. The recursive clause uses only successor, multiplication, and exponentiation, so d_{fac} is primitive recursive. Likewise $\binom{r+1}{2} = r(r+1)/2$, and primitive-recursive closure under successor, multiplication, natural-number division, and exponentiation proves that d_{Δ} is primitive recursive. Finally $d_{\text{fac}}(r) \leq d_{\Delta}(r)$ is exactly the factorial–power inequality above, and positivity reverses this comparison between their reciprocals.

The declarations `Fabius.fabiusReal_inverse_two_pow_one_term_lower_bound`, `Fabius.inv_inverseFabiusFactorialDenominator_le_fabiusReal`, and `Fabius.inv_inverseFabiusDeltaDenominator_le_fabiusReal` certify this chain. They do not formalize the event in (R.2), its containment, or its product probability; that probabilistic proof remains report-level. The argument in this paragraph uses only the already formalized dyadic recurrence and therefore preserves the distinction between the two proof routes.

R.2 Strict monotonicity from the random series

Proposition R.2 (Strict increase). *For $0 \leq a < b \leq 1$,*

$$F(a) < F(b). \tag{R.4}$$

Hence $F : [0, 1] \rightarrow [0, 1]$ is a homeomorphism and $G = F^{-1}$ exists.

Proof. The density of X_N is strictly positive on the interior of its support $(0, 1 - 2^{-N})$. This follows inductively: a convolution of two densities that are positive on the interiors of compact intervals is positive on the interior of their Minkowski-sum interval, because the convolution integral contains a nonempty open region on which both factors are positive.

Choose N so large that $2^{-N} < b - a$. Then the open interval $(a, b - 2^{-N})$ is nonempty and lies below the right endpoint of the support of X_N . Choose a nonempty open subinterval J of it. The preceding positivity gives $\mathbb{P}[X_N \in J] > 0$. For $X_N \in J$ and every possible $0 \leq R_N \leq 2^{-N}$, one has $a < X_N + R_N < b$. Consequently

$$F(b) - F(a) = \mathbb{P}[a < X \leq b] \geq \mathbb{P}[X_N \in J] > 0.$$

Continuity and the endpoint values $F(0) = 0$, $F(1) = 1$ then give the homeomorphism statement. $\square$

R.3 Symmetric strict unimodality

Let

$$p(x) := F'(x), \quad 0 \leq x \leq 1. \tag{R.5}$$

The repository proves that F is C^∞ , so p is continuous. The functional and reflection equations completely determine its qualitative shape.

Proposition R.3 (Shape of the Fabius density). *The density p satisfies*

$$p(x) = p(1 - x), \quad (\text{R.6})$$

is strictly increasing on $[0, 1/2]$, and is strictly decreasing on $[1/2, 1]$. Equivalently, $p(x)$ is a strictly decreasing function of $|x - 1/2|$.

Proof. On $[0, 1/2]$, (O.1) gives

$$p(x) = 2F(2x).$$

By [proposition R.2](#), this is strictly increasing. Differentiating $F(x) + F(1 - x) = 1$ gives $p(x) = p(1 - x)$, which reflects the first-half behavior to the second half. $\square$

This simple observation is the geometric heart of the inverse modulus. It is also a point where Rvachev's function enters directly: by (2.1),

$$p(x) = 2 \operatorname{up}(2x - 1). \quad (\text{R.7})$$

Thus [proposition R.3](#) is the standard single-peaked shape of the compactly supported atomic bump.

S The least-mass interval and the exact inverse modulus

S.1 Intervals of a prescribed geometric length

For $0 \leq h \leq 1$ and $0 \leq x \leq 1 - h$, define the fixed-length CDF increment

$$M_h(x) := F(x + h) - F(x). \quad (\text{S.1})$$

If μ is the Fabius probability law with CDF F , then the canonical measure-theoretic interpretation is

$$M_h(x) = \mu((x, x + h]).$$

The law is atomless—indeed it has the continuous density $p = F'$ —so the value is unchanged if either or both endpoints are included. In particular,

$$\mu((x, x + h]) = \mu([x, x + h]) = \mu((x, x + h)) = \mu([x, x + h)).$$

We therefore continue to call $M_h(x)$ an interval mass, while (S.1) fixes the canonical CDF convention.

Theorem S.1 (Strict interval-mass shape and exact minimizers). *For every $0 \leq h \leq 1$,*

$$\min_{0 \leq x \leq 1-h} (F(x + h) - F(x)) = F(h). \quad (\text{S.2})$$

If $0 < h \leq 1$ and $m = (1 - h)/2$, then the globally extended increment M_h is strictly increasing on $[-h, m]$, symmetric about m , and strictly decreasing on $[m, 1]$; outside $[-h, 1]$ it is identically zero. On the admissible-start interval $0 \leq x \leq 1 - h$ one has the boundary-safe equality law

$$M_h(x) = F(h) \iff h = 0 \text{ or } x = 0 \text{ or } x = 1 - h. \quad (\text{S.3})$$

Equivalently, whenever $0 \leq a \leq b \leq 1$,

$$F(b - a) = F(b) - F(a) \iff a = b \text{ or } a = 0 \text{ or } b = 1. \quad (\text{S.4})$$

Thus every genuinely interior interval $0 < a < b < 1$ has strictly more Fabius mass than an endpoint interval of the same length. At $h = 1$ the two endpoint formulas coincide at the single admissible start $x = 0$.

Proof. The case $h = 0$ gives $M_0 \equiv 0$, so suppose $0 < h \leq 1$ and put $m = (1 - h)/2$. Reflection gives

$$\begin{aligned} M_h(1 - h - x) &= F(1 - x) - F(1 - h - x) \\ &= (1 - F(x)) - (1 - F(x + h)) = M_h(x), \end{aligned}$$

so M_h is symmetric about m .

Consider $-h < x < m$. Then $0 < x + h < 1$. If $x \leq 0$, the constant left tail gives $p(x) = 0$, whereas strict positivity on the open support gives $p(x + h) > 0$. If $x > 0$, compute

$$\left(x + h - \frac{1}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2 = h(2x + h - 1) < 0. \quad (\text{S.5})$$

Thus $x + h$ is strictly closer to $1/2$ than x is. The strict radial unimodality in [proposition R.3](#) again gives $p(x + h) > p(x)$. Hence

$$M'_h(x) = p(x + h) - p(x) > 0 \quad (-h < x < m). \quad (\text{S.6})$$

The mean-value theorem makes M_h strictly increasing on the closed interval $[-h, m]$. Reflection makes it strictly decreasing on $[m, 1]$. The clamped tails of F give $M_h = 0$ to the left of $-h$ and to the right of 1 ; this is why the global half-line shape is only non-strict, whereas the two finite branches above are maximally strict.

Finally,

$$M_h(0) = F(h) - F(0) = F(h) \quad \text{and} \quad M_h(1 - h) = F(1) - F(1 - h) = F(h).$$

If $0 < x < 1 - h$, compare x strictly with 0 on the left branch when $x \leq m$, and compare it strictly with $1 - h$ on the right branch when $m < x$. In either case $M_h(x) > F(h)$. This proves the minimum and the complete equality locus ([S.3](#)). Substituting $h = b - a$ and $x = a$ gives ([S.4](#)); its diagonal alternative $a = b$ is exactly the degenerate case $h = 0$. $\square$

Corollary S.2 (Global increment shape and constrained superadditivity). *For every $h \geq 0$, the globally extended increment M_h is nondecreasing on the half-line*

$$x \leq \frac{1 - h}{2}$$

and nonincreasing on the complementary half-line. Consequently, whenever $a, b \geq 0$ and $a + b \leq 1$,

$$F(a) + F(b) \leq F(a + b). \quad (\text{S.7})$$

Its complete equality locus is

$$F(a) + F(b) = F(a + b) \iff a = 0 \text{ or } b = 0 \text{ or } a + b = 1. \quad (\text{S.8})$$

Proof. The derivative argument above never used $x \geq 0$: from $x \leq (1 - h)/2$ and $h \geq 0$, either $x + h \leq 1/2$ and the monotonicity of p on the left half gives $p(x) \leq p(x + h)$, or $x + h > 1/2$ and reflection replaces $p(x + h)$ by $p(1 - x - h)$, where

$$x \leq 1 - x - h \leq \frac{1}{2}.$$

Thus $M'_h \geq 0$ on the entire left half-line. Reflection $M_h(1 - h - x) = M_h(x)$ gives the right-half statement. For ([S.7](#)), apply [theorem S.1](#) to the interval from a to $a + b$: its length is b , so

$$F(b) \leq F(a + b) - F(a).$$

Rearranging proves the inequality. Equality is exactly the equality case of [theorem S.1](#) for the interval $[a, a + b]$: its three alternatives are $a = a + b$, $a = 0$, and $a + b = 1$, and the first is equivalent to $b = 0$. $\square$

Remark S.3 (A general principle). The lower bound used only that a probability density on $[0, 1]$ is symmetric and unimodal. Hence the minimum-value part of [theorem S.1](#) holds for every continuous symmetric unimodal distribution. The endpoint-only equality locus uses the genuinely stronger hypothesis that the density decreases strictly with distance from the midpoint; plateaus can create additional minimizers. The Fabius system is special not because the geometric lemma is exotic, but because $F(h)$ has unusually rich exact arithmetic and endpoint asymptotics.

S.2 The inverse's modulus is the inverse itself

For a bounded function H define its ordinary modulus of continuity by

$$\omega_H(\eta) := \sup \{|H(u) - H(v)| : |u - v| \leq \eta\}. \quad (\text{S.9})$$

For G the variables are restricted to $[0, 1]$; for $\overline{G}$ they range over $\mathbb{R}$.

Theorem S.4 (Exact modulus of the inverse). *For every $\eta \geq 0$, with the inputs in the definition of ω_G restricted to $[0, 1]$,*

$$\omega_G(\eta) = G(\min(\eta, 1)). \quad (\text{S.10})$$

In particular this is $G(\eta)$ for $0 \leq \eta \leq 1$ and is 1 thereafter. For the totalized inverse and every $\eta \geq 0$,

$$\omega_{\overline{G}}(\eta) = G(\min(\eta, 1)). \quad (\text{S.11})$$

More strongly, for all real u, v ,

$$|\overline{G}(u) - \overline{G}(v)| \leq \overline{G}(|u - v|) = G(\min(|u - v|, 1)). \quad (\text{S.12})$$

The associated ordered-gap and subadditivity laws are

$$\overline{G}(v) - \overline{G}(u) \leq \overline{G}(v - u) \quad (u, v \in \mathbb{R}) \quad (\text{S.13})$$

and

$$\overline{G}(x + y) \leq \overline{G}(x) + \overline{G}(y) \quad (x, y \in \mathbb{R}). \quad (\text{S.14})$$

On the unit interval the ordered-gap equality cases are exactly

$$G(v) - G(u) = G(v - u) \iff u = v \text{ or } u = 0 \text{ or } v = 1 \quad (0 \leq u \leq v \leq 1). \quad (\text{S.15})$$

Equivalently, without ordering the inputs,

$$|G(u) - G(v)| = G(|u - v|) \iff u = v \text{ or } u \in \{0, 1\} \text{ or } v \in \{0, 1\} \quad (u, v \in [0, 1]). \quad (\text{S.16})$$

These two equality classifications are intentionally unit-domain statements: the constant tails of $\overline{G}$ create additional configurations on $\mathbb{R}$.

Proof. First let $c, d \in [0, 1]$. Interchanging them if necessary, assume $c \leq d$, and put

$$a = G(c), \quad b = G(d).$$

Monotonicity of G gives $a \leq b$. The inverse identities and [theorem S.1](#) give

$$F(b - a) \leq F(b) - F(a) = d - c. \quad (\text{S.17})$$

Both $b - a$ and $d - c$ lie in $[0, 1]$. Applying the increasing inverse and using $G(F(b - a)) = b - a$ yields

$$b - a \leq G(d - c).$$

Removing the temporary ordering assumption gives

$$|G(c) - G(d)| \leq G(|c - d|) \quad (c, d \in [0, 1]). \quad (\text{S.18})$$

The same argument records equality. Under $c \leq d$, equality in the inverse gap is equivalent, after applying the strictly increasing F , to equality in (S.17). By (S.4), this says

$$a = b, \quad a = 0, \quad \text{or} \quad b = 1.$$

Because $a = G(c)$ and $b = G(d)$ and F, G are inverse order isomorphisms, these alternatives are respectively $c = d$, $c = 0$, and $d = 1$. This proves (S.15). Applying it after ordering c, d proves (S.16); reversing the order exchanges the left and right endpoint alternatives.

Now put $c = \text{clamp}(u)$ and $d = \text{clamp}(v)$. Projection onto $[0, 1]$ is nonexpansive and has diameter one, so

$$|c - d| \leq \min(|u - v|, 1).$$

Combining this with (S.18) proves (S.12). If $|u - v| \leq \eta$, monotonicity then gives the upper bound $G(\min(\eta, 1))$ for both moduli. Let $\theta = \min(\eta, 1)$. The pair $(0, \theta)$ belongs to $[0, 1]^2$, has input separation $\theta \leq \eta$, and has output separation

$$|G(\theta) - G(0)| = G(\theta).$$

Thus the displayed upper bound is attained, not merely approached, proving both exact supremum identities.

For (S.13), if $u \leq v$, remove the absolute value from (S.12); then $v - u = |v - u|$. If $v < u$, the left side is nonpositive by monotonicity, whereas $\overline{G}(v - u) \geq 0$. Finally apply (S.13) with $u = x$ and $v = x + y$:

$$\overline{G}(x + y) - \overline{G}(x) \leq \overline{G}(y),$$

which rearranges to (S.14). □

Corollary S.5 (Effective-injectivity form). *For $0 \leq h \leq 1$ and $u, v \in \mathbb{R}$,*

$$|u - v| < F(h) \implies |\overline{G}(u) - \overline{G}(v)| < h. \quad (\text{S.19})$$

The closed-threshold and contrapositive forms are

$$|u - v| \leq F(h) \implies |\overline{G}(u) - \overline{G}(v)| \leq h, \quad (\text{S.20})$$

$$h \leq |\overline{G}(u) - \overline{G}(v)| \implies F(h) \leq |u - v|. \quad (\text{S.21})$$

The same statements hold for G on $[0, 1]$.

Proof. The strict hypothesis gives $|u - v| < F(h) \leq 1$, so the saturation in the exact modulus disappears. Strict monotonicity of G on $[0, 1]$ therefore gives

$$|\overline{G}(u) - \overline{G}(v)| \leq G(|u - v|) < G(F(h)) = h.$$

The same chain with non-strict inequalities proves (S.20); its contrapositive is (S.21). □

Theorem S.6 (Exact strict threshold). *For $0 \leq h \leq 1$ and every real δ ,*

$$[\forall u, v \in \mathbb{R}, |u - v| < \delta \implies |\overline{G}(u) - \overline{G}(v)| < h] \iff \delta \leq F(h). \quad (\text{S.22})$$

Thus $F(h)$ is exactly the largest strict universal input threshold for output error below h .

Proof. If $\delta \leq F(h)$, then $|u - v| < \delta$ implies $|u - v| < F(h)$, and [corollary S.5](#) applies. Conversely, suppose the universal implication holds but $F(h) < \delta$. Take

$$u = 0, \quad v = F(h).$$

Their input separation is $F(h) < \delta$, whereas the inverse identities give

$$|\overline{G}(0) - \overline{G}(F(h))| = |0 - h| = h,$$

contradicting the asserted strict output bound. Notice that no sign hypothesis on δ is needed: for $\delta \leq 0$ the left implication is vacuous and the right inequality follows from $F(h) \geq 0$. $\square$

T Effective uniform continuity

T.1 A closed primitive-recursive modulus

Combining the exact inverse modulus with the box event gives the promised fully explicit implication.

Theorem T.1 (Closed dyadic modulus). *For every $r \geq 0$ and all $u, v \in \mathbb{R}$,*

$$|u - v| < \Delta_r \implies |\overline{G}(u) - \overline{G}(v)| < 2^{-r}, \quad (\text{T.1})$$

where Δ_r is given by [\(R.1\)](#).

Proof. By [lemma R.1](#), $\Delta_r \leq F(2^{-r})$. Apply [corollary S.5](#) with $h = 2^{-r}$. $\square$

The exact statement [\(T.1\)](#) is also formalized as `Fabius.abs_fabiusInv_sub_lt_inverse_two_pow_of_lt_deltaDenominator`. The stronger factorial-denominator version is `Fabius.abs_fabiusInv_sub_lt_inverse_two_pow_of_lt_factorialDenominator`. The formal API also closes both endpoints:

$$|u - v| \leq \Delta_r \implies |\overline{G}(u) - \overline{G}(v)| \leq 2^{-r},$$

with declaration `Fabius.abs_fabiusInv_sub_le_inverse_two_pow_of_le_deltaDenominator`; `Fabius.abs_fabiusInv_sub_le_inverse_two_pow_of_le_factorialDenominator` is its stronger factorial-denominator companion. Thus the strict and closed input/output boundaries are separately available rather than recovered by an unstated limiting argument. Their common proof is direct. Let d be either $d_{\text{fac}}(r)$ or $d_{\Delta}(r)$. The recurrence argument above gives $d^{-1} \leq F(2^{-r})$. Hence

$$|u - v| < d^{-1} \implies |u - v| < F(2^{-r}) \implies |\overline{G}(u) - \overline{G}(v)| < 2^{-r}$$

by strict effective injectivity. Replacing both strict inequalities by weak ones and using its closed-threshold companion proves the two closed versions. This proves all four formal dyadic implications without a limiting argument; only the choice of denominator distinguishes the factorial and Delta forms.

Theorem T.2 (Reciprocal modulus required by the definition). *For $n \geq 1$, let $r(n)$ be the least positive integer with $2^{r(n)} > n$ and put*

$$d(n) = 2^{r(n)(r(n)+1)/2} (r(n) + 1)^{r(n)+1}, \quad d(0) = 1. \quad (\text{T.2})$$

Then d is recursive and satisfies [\(P.2\)](#) for $\overline{G}$. Thus $\overline{G}$ and the restricted inverse G are effectively uniformly continuous.

Proof. The function $r(n)$ can be found by bounded integer search, or equivalently by computing the binary length of n ; hence it is recursive. Exponentiation, multiplication, and composition preserve recursiveness, so d is recursive. By construction,

$$\frac{1}{d(n)} = \Delta_{r(n)} \quad \text{and} \quad 2^{-r(n)} < \frac{1}{n}.$$

If $|u - v| < 1/d(n)$, [theorem T.1](#) gives $|\overline{G}(u) - \overline{G}(v)| < 2^{-r(n)} < 1/n$. $\square$

Remark T.3 (An even simpler, much larger modulus). Taking $r = n$ gives the one-line primitive-recursive choice

$$d_0(n) = 2^{n(n+1)/2} (n+1)^{n+1} \quad (n \geq 1),$$

because $2^{-n} < 1/n$. The logarithmic choice $r(n)$ in [theorem T.2](#) is far smaller and has the correct leading complexity scale.

The Lean development uses the stronger one-line witness

$$d_{\text{fac}}(n) = 2^{\binom{n+1}{2}} (n+1)!,$$

which is no larger than $d_0(n)$. The theorem `Fabius.fabiusInv_effectivelyUniformContinuous` proves `EffectivelyUniformContinuous` for the totalized inverse as follows. The recurrence above makes d_{fac} primitive recursive and positive. For $n \geq 1$, the strict factorial implication gives

$$|u - v| < \frac{1}{d_{\text{fac}}(n)} \implies |\overline{G}(u) - \overline{G}(v)| < 2^{-n}.$$

The elementary inequality $n < 2^n$ between positive integers reverses under reciprocation, so $2^{-n} < 1/n$. Chaining these inequalities is precisely the reciprocal convention in [definition P.2](#); $d_{\text{fac}}(0) = 1$ supplies the harmless zeroth value. Thus effective uniform continuity itself is formalized. The downstream `FabiusInverseLogarithmicModulus.lean` now also formalizes the smaller choice $r(n)$ in [theorem T.2](#). Its declaration `Fabius.inverseFabiusLogarithmicOrder_isLeast` identifies the least order, `Fabius.inverseFabiusLogarithmicOrder_primrec` proves primitive recursiveness, and `Fabius.fabiusInv_effectivelyUniformContinuous_logarithmicDelta` packages the report-exact denominator $d(n)$. The same module supplies the stronger logarithmic factorial witness.

T.2 The exact-dyadic repository modulus

Put $h_n := 2^{-r(n)}$. The corpus proves that every $F(h_n)$ is an exactly computable positive rational. Therefore one can replace the elementary lower bound $\Delta_{r(n)}$ by the exact endpoint mass while retaining an exact dyadic output proxy.

Proposition T.4 (Repository-native exact-dyadic modulus). *Let*

$$d_*(n) := \left\lceil \frac{1}{F(h_n)} \right\rceil \quad (n \geq 1), \quad h_n = 2^{-r(n)}, \quad d_*(0) := 1. \quad (\text{T.3})$$

Then d_ is recursive and*

$$|u - v| < \frac{1}{d_*(n)} \implies |\overline{G}(u) - \overline{G}(v)| < h_n < \frac{1}{n}. \quad (\text{T.4})$$

Moreover, $d_(n)$ is the least positive integer d satisfying*

$$\frac{1}{d} \leq F(h_n).$$

Equivalently, it is the least reciprocal denominator certified by the exact inverse-modulus law for the fixed dyadic output target h_n . It is not claimed to be the least integer modulus for the weaker target $1/n$.

Proof. The exact dyadic evaluator computes the positive rational $F(h_n)$, and rational reciprocal and ceiling are computable integer operations. By the defining property of the ceiling,

$$\frac{1}{d_*(n)} \leq F(h_n).$$

The strict input hypothesis and [corollary S.5](#) therefore give output separation strictly below h_n , while the definition of $r(n)$ gives $h_n < 1/n$.

For the qualified minimality statement, let $1 \leq d < d_*(n)$. Then

$$d < \frac{1}{F(h_n)}, \quad \text{hence} \quad \frac{1}{d} > F(h_n).$$

The endpoint pair $u = 0, v = F(h_n)$ has input separation strictly below $1/d$ and output separation exactly h_n . Thus d cannot certify the stricter conclusion $|\overline{G}(u) - \overline{G}(v)| < h_n$. $\square$

This proposition now has an exact Lean counterpart in `FabiusInverseExactDyadicModulus.lean`. At a fixed order r , `Fabius.inverseFabiusExactDyadicDenominator` defines $\lceil 1/F(2^{-r}) \rceil$ through the exact rational evaluator; `Fabius.inverseFabiusExactDyadicDenominator_primrec` certifies this fixed-order denominator as primitive recursive; `Fabius.inverseFabiusExactDyadicDenominator_pos`, `Fabius.inv_inverseFabiusExactDyadicDenominator_le_fabiusAtInverseTwoPow`, and `Fabius.inverseFabiusExactDyadicDenominator_isLeast` prove its positivity, endpoint-mass bound, and arithmetic leastness. The strict modulus, the endpoint counterexample for every smaller positive denominator, and leastness among strict integer moduli are respectively `Fabius.abs_fabiusInv_sub_lt_inverse_two_pow_of_lt_exactDyadicDenominator`, `Fabius.exists_fabiusInv_gap_of_lt_exactDyadicDenominator`, and `Fabius.inverseFabiusExactDyadicDenominator_isLeast_strictModulus`.

The definition `Fabius.inverseFabiusExactLogarithmicDenominator` then sets the value at zero to 1 and composes the fixed-target denominator with $r(n)$ at positive inputs. The theorem `Fabius.inverseFabiusExactLogarithmicDenominator_primrec` proves that this totalized function is primitive recursive. Its positive-input equation is `Fabius.inverseFabiusExactLogarithmicDenominator_of_pos`, and `Fabius.abs_fabiusInv_sub_lt_inv_nat_of_lt_exactLogarithmicDenominator` is exactly the implication to the weaker $1/n$ target. That last theorem is only a witness: it does not assert leastness for $1/n$, and its zero branch is only the convention $d_*(0) = 1$ with no modulus conclusion there.

The two recursion certificates use a natural code rather than a computability instance for normalized rationals. If $F(2^{-r}) = H_r/D_r$ is the formalized exact quotient, then the ceiling is computed as $(D_r + H_r - 1)/H_r$. The development proves H_r and D_r primitive recursive, and then composes the resulting fixed-order denominator with the already primitive-recursive $r(n)$. Consequently every clause of [proposition T.4](#) now has a compiled Lean counterpart, and its concordance row is classified as *Lean-proved*.

Remark T.5 (The genuine integer optimum for the $1/n$ target). The exact inverse-modulus law shows that a positive integer d has the universal property

$$|u - v| < \frac{1}{d} \implies |\overline{G}(u) - \overline{G}(v)| < \frac{1}{n}$$

if and only if

$$\frac{1}{d} \leq F(1/n).$$

Consequently the mathematically least such denominator is

$$d_{\text{opt}}(n) = \left\lceil \frac{1}{F(1/n)} \right\rceil.$$

The present report does not prove that $n \mapsto d_{\text{opt}}(n)$ is recursive: computability of the real number $F(1/n)$ alone does not make the discontinuous ceiling operation effective at an unknown integer boundary. The dyadic quantity d_* avoids that issue because $F(h_n)$ is an exact rational.

The distinction is already visible at $n = 1$. Here $r(1) = 1$, $F(1/2) = 1/2$, and hence $d_*(1) = 2$, whereas $d = 1$ already satisfies the required implication with output tolerance 1.

The exact-ceiling construction d_* , including its recursion and qualified fixed-dyadic denominator minimality, is therefore translated into Lean. The Delta and stronger factorial denominators remain useful closed-form witnesses that avoid invoking the exact dyadic evaluator at the formalized least order $r(n)$.

T.3 Binary conditioning is quadratic

The exact modulus permits a sharp, representation-independent measurement of how many input bits are intrinsically needed near the flat endpoint.

Definition T.6 (Optimal binary modulus index). For $p \geq 1$, define

$$\mu(p) := \min \{m \in \mathbb{N}_0 : 2^{-m} \leq F(2^{-p})\} = \left\lceil \log_2 \frac{1}{F(2^{-p})} \right\rceil. \quad (\text{T.5})$$

Then

$$|u - v| < 2^{-\mu(p)} \implies |\overline{G}(u) - \overline{G}(v)| < 2^{-p}, \quad (\text{T.6})$$

and no smaller m has this universal property.

The lower event bound already gives an upper bound for $\mu(p)$. A matching lower bound follows from endpoint flatness.

Lemma T.7 (Elementary flatness upper bound). For every integer $q \geq 1$ and $0 \leq x \leq 2^{-q}$,

$$F(x) \leq \frac{2^{q(q+1)/2}}{q!} x^q. \quad (\text{T.7})$$

In particular,

$$F(2^{-p}) \leq \frac{2^{-p(p-1)/2}}{p!}. \quad (\text{T.8})$$

Proof. Repeated differentiation of $F'(x) = 2F(2x)$ gives

$$F^{(q)}(x) = 2^{q(q+1)/2} F(2^q x) \quad (0 \leq x \leq 2^{-q}). \quad (\text{T.9})$$

All derivatives of F vanish at 0, and $0 \leq F \leq 1$. Taylor's formula in integral form therefore gives

$$\begin{aligned} F(x) &= \frac{1}{(q-1)!} \int_0^x (x-t)^{q-1} F^{(q)}(t) dt \\ &\leq \frac{2^{q(q+1)/2}}{(q-1)!} \int_0^x (x-t)^{q-1} dt = \frac{2^{q(q+1)/2}}{q!} x^q. \end{aligned}$$

Set $q = p$ and $x = 2^{-p}$ to obtain (T.8). □

Theorem T.8 (Asymptotically sharp input-bit requirement). For every $p \geq 1$,

$$\left\lceil \frac{p(p-1)}{2} + \log_2(p!) \right\rceil \leq \mu(p) \quad (\text{T.10})$$

$$\leq \left\lceil \frac{p(p+1)}{2} + (p+1) \log_2(p+1) \right\rceil. \quad (\text{T.11})$$

Consequently

$$\boxed{\mu(p) = \frac{p^2}{2} + p \log_2 p + \mathcal{O}(p).} \quad (\text{T.12})$$

Proof. The flatness bound (T.8) yields

$$\log_2 \frac{1}{F(2^{-p})} \geq \frac{p(p-1)}{2} + \log_2(p!),$$

while lemma R.1 yields

$$\log_2 \frac{1}{F(2^{-p})} \leq \frac{p(p+1)}{2} + (p+1) \log_2(p+1).$$

Take ceilings and use (T.5). Stirling's formula $\log_2(p!) = p \log_2 p - (\log_2 e)p + \mathcal{O}(\log p)$ makes both bounds equal to $p^2/2 + p \log_2 p + \mathcal{O}(p)$. $\square$

Remark T.9 (Intrinsic endpoint cost). The quadratic precision loss is not an artifact of the spline evaluator or of bisection. The two admissible inputs 0 and $F(2^{-p})$ have outputs separated by 2^{-p} . Any uniform oracle algorithm that promises p output bits must, in the worst endpoint case, obtain enough input information to distinguish those values. The exact modulus quantifies that information barrier.

Corollary T.10 (Growth of exact and explicit reciprocal moduli). *Let*

$$d_{\text{opt}}(n) = \lceil 1/F(1/n) \rceil, \quad d_*(n) = \left\lceil 1/F(2^{-r(n)}) \right\rceil,$$

and let $d_\Delta(n)$ denote the explicit denominator $d(n)$ of theorem T.2. Then each of d_{opt} , d_* , and d_Δ has

$$\log_2 d_\bullet(n) = \frac{1}{2}(\log_2 n)^2 + (\log_2 n) \log_2 \log_2 n + \mathcal{O}(\log n). \quad (\text{T.13})$$

Equivalently,

$$d_\bullet(n) = n^{\frac{1}{2} \log_2 n + \log_2 \log_2 n + \mathcal{O}(1)}. \quad (\text{T.14})$$

Here d_{opt} is the mathematically least denominator for the $1/n$ target, d_* is recursive and exact-dyadic, and d_Δ is the closed elementary witness; no computability claim is made here for d_{opt} .

Proof. Put $p = r(n)$. Minimality of p gives

$$2^{-p} < \frac{1}{n} \leq 2^{-(p-1)}, \quad p = \log_2 n + \mathcal{O}(1).$$

Since F is increasing, with $\Lambda(q) = \log_2(1/F(2^{-q}))$ this implies

$$\Lambda(p-1) \leq \log_2 \frac{1}{F(1/n)} < \Lambda(p).$$

By theorem T.8, and because $\mu(q) = \lceil \Lambda(q) \rceil$,

$$\Lambda(q) = \frac{q^2}{2} + q \log_2 q + \mathcal{O}(q).$$

The preceding sandwich therefore gives the same formula with $q = p$ for $\log_2(1/F(1/n))$. For every $x \geq 1$, $x \leq \lceil x \rceil < 2x$, so applying the outer ceiling changes its base-2 logarithm by at most one. This proves (T.13) for d_{opt} .

The same ceiling observation applied directly to $F(2^{-p})$ proves it for d_* . Finally the closed formula for d_Δ gives

$$\log_2 d_\Delta(n) = \frac{p(p+1)}{2} + (p+1) \log_2(p+1) = \frac{p^2}{2} + p \log_2 p + \mathcal{O}(p).$$

Substitute $p = \log_2 n + \mathcal{O}(1)$ in all three cases. Exponentiating the resulting identity proves (T.14). $\square$

U Sequential computability by tolerant bisection

U.1 Why ordinary bisection is insufficient

Suppose y is given only by a computable name. At a dyadic midpoint c , one can approximate both $F(c)$ and y , but in general no algorithm can decide whether the two represented reals are exactly equal. An implementation that loops until it proves either $F(c) < y$ or $F(c) > y$ can therefore fail to terminate when $y = F(c)$.

The inverse modulus provides the missing third branch. If the approximations are too close to certify a sign, then $F(c)$ is close to y ; effective injectivity says that c is close to $G(y)$, so the algorithm may return it.

U.2 The algorithm on a computable sequence

Let (y_i) be a computable sequence in $[0, 1]$. Fix a naming algorithm $q(i, s) \in \mathbb{Q}$ with

$$|q(i, s) - y_i| \leq 2^{-s}. \quad (\text{U.1})$$

We describe an algorithm which, from (i, p) , returns a dyadic $z(i, p)$ with

$$|z(i, p) - G(y_i)| < 2^{-p}. \quad (\text{U.2})$$

Algorithm U.1 (Tolerant certified bisection). Given an index i and output precision p :

1. Set

$$r = p + 2, \quad h = 2^{-r}, \quad \delta = \Delta_r.$$

Choose integers s, N effectively so that

$$2^{-s} \leq \delta/8, \quad 2^{-N} \leq \delta/8. \quad (\text{U.3})$$

Compute the rational $\tilde{y} = q(i, s)$.

2. Initialize the certified bracket $[a, b] = [0, 1]$.

3. Repeat at most $p + 2$ times:

- (a) Put $c = (a + b)/2$ and evaluate the exact rational $S_N(c)$ by (Q.9).
- (b) Form the rational approximate difference

$$\tilde{D} = S_N(c) - \tilde{y}. \quad (\text{U.4})$$

- (c) If $\tilde{D} > \delta/2$, replace b by c .
- (d) If $\tilde{D} < -\delta/2$, replace a by c .
- (e) Otherwise return c .

4. If no midpoint was returned, return $(a + b)/2$ after the final update.

Every comparison in [algorithm U.1](#) is between exact rationals. The order N can be extremely large, but it is obtained by a finite integer calculation and the spline sum is finite. Hence the procedure is an algorithm in the computability-theoretic sense.

U.3 Correctness of every branch

Lemma U.2 (Difference certificate). *At every midpoint c in [algorithm U.1](#),*

$$\left| \tilde{D} - (F(c) - y_i) \right| \leq \delta/4. \quad (\text{U.5})$$

Proof. By [theorem Q.3](#) and (U.3), $|S_N(c) - F(c)| \leq \delta/8$. By (U.1) and the same choice of s , $|\tilde{y} - y_i| \leq \delta/8$. The triangle inequality gives (U.5). $\square$

Lemma U.3 (Safe left/right updates). *If $\tilde{D} > \delta/2$, then $G(y_i) < c$. If $\tilde{D} < -\delta/2$, then $c < G(y_i)$.*

Proof. In the first case, [lemma U.2](#) gives

$$F(c) - y_i \geq \tilde{D} - \delta/4 > \delta/4 > 0.$$

Since F is strictly increasing, $c > G(y_i)$. The second case is symmetric. Thus every update preserves the invariant

$$G(y_i) \in [a, b]. \quad (\text{U.6})$$

$\square$

Lemma U.4 (The inconclusive branch is a success branch). *If $|\tilde{D}| \leq \delta/2$, then*

$$|c - G(y_i)| < h = 2^{-r} < 2^{-p}. \quad (\text{U.7})$$

Proof. By [lemma U.2](#),

$$|F(c) - y_i| \leq |\tilde{D}| + \delta/4 \leq 3\delta/4 < \delta.$$

The choice $\delta = \Delta_r$ and [theorem T.1](#), applied to the input pair $F(c), y_i$, give

$$|G(F(c)) - G(y_i)| < 2^{-r}.$$

Because $c \in [0, 1]$, $G(F(c)) = c$. Finally, $r = p + 2$. $\square$

Theorem U.5 (Correctness and uniformity of tolerant bisection). *Algorithm [U.1](#) terminates and returns a dyadic $z(i, p)$ satisfying (U.2). The map $(i, p) \mapsto z(i, p)$ is computable whenever the name $(i, s) \mapsto q(i, s)$ is computable.*

Proof. The loop has a fixed finite bound. If it returns through the inconclusive branch, [lemma U.4](#) proves the error estimate. Otherwise [lemma U.3](#) preserves the bracket invariant through $p + 2$ bisections. Its final width is $2^{-(p+2)}$, so the final midpoint is at distance at most $2^{-(p+3)} < 2^{-p}$ from $G(y_i)$.

All quantities r, h, δ, s, N are computable from p . Midpoints are dyadic; $S_N(c)$ is an exact finite rational expression; and every branch is a rational comparison. The only access to the sequence is the single call to its naming algorithm. Therefore the construction is uniform in i and p . $\square$

Corollary U.6 (Sequential computability on the unit interval). *The inverse $G : [0, 1] \rightarrow [0, 1]$ is sequentially computable.*

Proof. Given any computable sequence (y_i) in $[0, 1]$, the output of [algorithm U.1](#) is a computable fast dyadic name for the sequence $(G(y_i))$. $\square$

U.4 Clamping and the totalized inverse

Lemma U.7 (Computability of clamping). *If (y_i) is a computable real sequence, then $(\text{clamp}(y_i))$ is a computable sequence.*

Proof. Clamping is 1-Lipschitz. If $q(i, p)$ approximates y_i within 2^{-p} , then the exactly rational quantity $\text{clamp}(q(i, p))$ approximates $\text{clamp}(y_i)$ within the same error. The rational min/max operations are decidable and computable. $\square$

Corollary U.8 (Sequential computability of the totalized inverse). *The totalized inverse $\overline{G} : \mathbb{R} \rightarrow \mathbb{R}$ is sequentially computable.*

Proof. Given a computable sequence (y_i) , first compute the clamped sequence by [lemma U.7](#), then apply [corollary U.6](#). By [\(O.2\)](#), the result is $(\overline{G}(y_i))$. $\square$

U.5 Main-theorem dependency checkpoint

At this point every result cited in the proof of [theorem O.1](#) has been established: tolerant bisection supplies one uniform realizer transformer, clamping totalizes it, and the closed dyadic threshold supplies the recursive reciprocal modulus. Thus the early proof is now entirely discharged rather than relying on a compactness or choice argument.

U.6 An abstract inverse-computability principle

The Fabius proof illustrates a reusable theorem. Stating it clarifies which properties are genuinely needed and which belong only to this example.

Theorem U.9 (Effective inversion on a computable compact interval). *Let $f : [0, 1] \rightarrow [0, 1]$ be a strictly increasing homeomorphism. Suppose:*

1. *there is one algorithm which, given a dyadic rational $c \in [0, 1]$ and $s \in \mathbb{N}_0$, returns a rational within 2^{-s} of $f(c)$;*
2. *there is a computable positive rational sequence (α_p) such that for all $0 \leq x \leq 1 - 2^{-p}$,*

$$f(x + 2^{-p}) - f(x) \geq \alpha_p. \quad (\text{U.8})$$

Then f^{-1} is sequentially computable and effectively uniformly continuous. A tolerant-bisection algorithm is obtained by replacing Δ_r with α_r in [algorithm U.1](#).

Proof. Condition [\(U.8\)](#) implies $|u - v| < \alpha_p \Rightarrow |f^{-1}(u) - f^{-1}(v)| < 2^{-p}$ by the same contrapositive argument as [corollary S.5](#): if two inverse values were at least 2^{-p} apart, monotonicity and [\(U.8\)](#) would force their inputs to be at least α_p apart.

This implication gives the uniform realizer. On input precision p , run the tolerant bisection of [algorithm U.1](#) for $p + 2$ steps, replacing Δ_{p+2} by the positive rational α_{p+2} . Use hypothesis 1 and the input name to approximate both sides within $\alpha_{p+2}/8$. A certified sign preserves the inverse bracket; in the inconclusive branch their true difference is below $3\alpha_{p+2}/4$, so the preceding implication certifies output error below $2^{-(p+2)}$. If that branch never occurs, the final bracket has width $2^{-(p+2)}$. All loop bounds, precisions, and comparisons are effective, so this is one uniform name transformer and proves sequential computability.

For the reciprocal convention, set $d(0) = 1$. For $n \geq 1$, compute any $p(n) \geq 1$ with $2^{-p(n)} < 1/n$ and define

$$d(n) := \left\lfloor \frac{1}{\alpha_{p(n)}} \right\rfloor + 1.$$

Because $\alpha_{p(n)}$ is a positive computable rational, d is a positive recursive integer function and $d(n) > 1/\alpha_{p(n)}$, hence $1/d(n) < \alpha_{p(n)}$. Therefore

$$|u - v| < \frac{1}{d(n)} \implies |f^{-1}(u) - f^{-1}(v)| < 2^{-p(n)} < \frac{1}{n}.$$

This is effective uniform continuity in the subset-domain sense. $\square$

For the Fabius function, [theorem S.1](#) identifies the best possible choice $\alpha_p = F(2^{-p})$; its minimizing intervals are exactly the left and right endpoint intervals. The box event in [lemma R.1](#) supplies a closed lower substitute requiring no invocation of the exact dyadic evaluator.

V Analytic consequences of the exact modulus

V.1 Uniform continuity without any Hölder exponent

Effective uniform continuity is sometimes mentally associated with a power-law modulus. The inverse Fabius function is a clean counterexample to that intuition.

Theorem V.1 (No positive Hölder exponent at the endpoints). *For every $\alpha > 0$ and every $C > 0$, there are arbitrarily small $y > 0$ such that*

$$G(y) > Cy^\alpha. \quad (\text{V.1})$$

By reflection, there are arbitrarily small $1 - y > 0$ such that

$$1 - G(y) > C(1 - y)^\alpha. \quad (\text{V.2})$$

Thus neither G nor $\overline{G}$ is Hölder continuous of any positive order at 0 or 1, despite being effectively uniformly continuous.

Proof. Choose an integer q with $q\alpha > 1$. By [lemma T.7](#), for all sufficiently small $x > 0$, $F(x) \leq C_q x^q$ with a fixed C_q . If $G(y) \leq Cy^\alpha$ held near zero, substitution $y = F(x)$ would give

$$x = G(F(x)) \leq CF(x)^\alpha \leq CC_q^\alpha x^{q\alpha}.$$

After division by x , the right side tends to zero, a contradiction. The right endpoint follows from $G(1 - y) = 1 - G(y)$. $\square$

Corollary V.2 (The exact modulus is sub-Hölder). *For every $\alpha > 0$,*

$$\frac{\omega_G(\eta)}{\eta^\alpha} = \frac{G(\eta)}{\eta^\alpha} \longrightarrow +\infty \quad (\eta \downarrow 0). \quad (\text{V.3})$$

Proof. Choose an integer q with $q\alpha > 1$ and write $x = G(\eta)$, so that $\eta = F(x)$ and $x \downarrow 0$ with $\eta \downarrow 0$. By [lemma T.7](#), for all sufficiently small x , $F(x) \leq C_q x^q$. Hence

$$\frac{G(\eta)}{\eta^\alpha} = \frac{x}{F(x)^\alpha} \geq C_q^{-\alpha} x^{1-q\alpha} \longrightarrow +\infty.$$

The equality with ω_G is [theorem S.4](#). $\square$

The endpoint growth used here is also formalized, independently of the new exact-modulus module, in `FabiusInverseAsymptotic.lean`. The declarations `Fabius.rpow_isLittleO_fabiusInv_at_zero_right` and `Fabius.tendsto_fabiusInv_div_rpow_atTop_at_zero_right` give the sub-Hölder divergence, while `Fabius.fabiusInv_not_isBigO_rpow_at_zero_right` and `Fabius.not_exists_fabiusInv_le_const_mul_rpow_near_zero` give the corresponding impossibility of a positive power-law upper bound. These are analytic inputs already present in the repository, not consequences claimed to have been newly proved by `InverseModulus.lean`.

V.2 Endpoint asymptotics become modulus asymptotics

The live formal corpus proves the leading inverse equivalent through `fabiusInv_isEquivalent_fabiusInverseAsymptoticMain` in `FabiusInverseAsymptotic.lean`. The companion frontier reports give stronger paper-level inverse expansions [7]. Let

$$T = \log(1/\eta), \quad L = \log 2, \quad \rho = \sqrt{\frac{2T}{L}}. \quad (\text{V.4})$$

Suppressing the paper-level first log-periodic correction, the formalized leading equivalent is

$$G(\eta) \sim G_0(\eta) := \frac{\sqrt{T}}{e\sqrt{L}} \exp(-\sqrt{2LT}) \quad (\eta \downarrow 0). \quad (\text{V.5})$$

More precisely, the all-orders inversion theorem [theorem K.1](#) gives the relative expansion

$$\frac{G(\eta)}{G_0(\eta)} = 1 + \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta)}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right), \quad (\text{V.6})$$

where Ψ is a nonconstant smooth one-periodic Gamma-zeta fluctuation and $\theta := \rho + \beta$ is the leading saddle phase. It is not the exact lower-Lambert phase: the latter is

$$\lambda(G(\eta)) = -\frac{1}{L} W_{-1}(-LG(\eta)) = \theta + \mathcal{O}\left(\frac{\log \rho}{\rho}\right).$$

Thus replacing the exact phase by θ is justified at the displayed algebraic order, but the distinction matters for exponentially small terms.

Corollary V.3 (Paper-level transfer of the frontier inverse expansion). *At the paper level, the expressions (V.5) and (V.6) transfer, without any further inversion, to the leading and first-correction asymptotics of the exact continuity modulus of G and $\bar{G}$:*

$$\omega_G(\eta) = \omega_{\bar{G}}(\eta) = G(\eta) \quad (0 \leq \eta \leq 1). \quad (\text{V.7})$$

Likewise, every valid all-orders endpoint expansion for the inverse transfers verbatim to an all-orders modulus-of-continuity expansion. Lean currently proves the leading equivalent and an implicit expansion pulled back to exact inverse coordinates, but not the displayed explicit periodic correction or an explicit all-orders reversion.

Proof. This is exactly [theorem S.4](#). □

This observation closes a conceptual loop in the corpus. The lower-Lambert phase and Bell-polynomial inverse transfer were developed to approximate quantiles. The geometric theorem shows that the same quantiles are the optimal global sensitivity profile of the inverse map.

V.3 Derivative blow-up versus computability

On $0 < y < 1$, the repository proves

$$G'(y) = \frac{1}{F'(G(y))} = \frac{1}{2 \operatorname{up}(2G(y) - 1)} > 0. \quad (\text{V.8})$$

The derivative tends to $+\infty$ at both endpoints. There is therefore no global Lipschitz constant and no endpoint condition number bounded in the ordinary differential sense. Nonetheless the exact nonlinear modulus $G(\eta)$ is computable, and tolerant bisection uses precisely that nonlinear conditioning. This is why derivative-based numerical intuition alone is not a substitute for a computable-analysis proof.

W Further deductions, conjectures, and research directions

W.1 A certified practical inverse evaluator

The proof algorithm is intentionally elementary rather than efficient. A practical certified implementation should retain its logical shell while replacing the forward oracle by a portfolio of enclosures:

- exact dyadic evaluation and terminating local Taylor formulae near dyadic anchors;
- centered Thue–Morse splines at moderate depth;
- quarter-base q -Richardson acceleration of finite-prefix errors;
- Fourier/sinc evaluation in regions where frequency-space decay is favorable;
- lower-Lambert endpoint enclosures when the requested target is extremely close to 0 or 1.

Each backend must return a rational interval containing $F(c)$, not merely a floating-point approximation. The bisection controller only asks whether that interval lies wholly above or below the target interval; overlapping intervals trigger refinement or the modulus-based acceptance branch. Thus backend selection is a performance concern, not part of the correctness argument.

Research problem W.1 (Near-optimal certified evaluation). *Construct and formally verify an inverse evaluator whose oracle-bit consumption is $\mu(p) + \mathcal{O}(\log p)$ for p output bits at the worst endpoint, and whose arithmetic running time is quasi-linear or softly quadratic in the size of the exact integers produced. Separate information-theoretic oracle precision from the cost of evaluating F to that precision.*

W.2 Sharp integer moduli and the log-periodic phase

The real-valued optimal input-bit threshold is

$$\Lambda(p) := \log_2 \frac{1}{F(2^{-p})}, \quad \mu(p) = \lceil \Lambda(p) \rceil.$$

The all-orders endpoint theory expresses $\Lambda(p)$ in a lower-Lambert phase with a nonconstant one-periodic Gamma–zeta contribution and a hierarchy of inverse powers. Passing to the integer ceiling introduces a second, sawtooth nonlinearity. It is therefore natural to study the arithmetic interaction between the smooth periodic saddle phase and fractional parts of its growing nonperiodic carrier.

For $p \geq 1$ define the exact lower-Lambert phase

$$\lambda_p := -\frac{1}{L} W_{-1}(-L2^{-p}), \quad 2^{-p} = \lambda_p e^{-L\lambda_p}.$$

The first valid forward remainder gives the precise reduction

$$\Lambda(p) = \frac{\lambda_p^2}{2} - \beta\lambda_p + \frac{\log \lambda_p}{2L} - \frac{C_\sharp}{L} - \frac{\Psi(\lambda_p)}{L} + \mathcal{O}(\lambda_p^{-1}). \quad (\text{W.1})$$

Indeed, substitute $x = 2^{-p}$ and its exact phase λ_p into (K.2) with $N = 1$, negate, and divide by $L = \log 2$; this is exactly the definition of $\Lambda(p)$. Put

$$A(p) := \frac{\lambda_p^2}{2} - \beta\lambda_p + \frac{\log \lambda_p}{2L} - \frac{C_\sharp}{L}, \quad R(p) := \mu(p) - A(p).$$

Equation (W.1) and $0 \leq \lceil \Lambda(p) \rceil - \Lambda(p) < 1$ show that $R(p)$ is bounded.

Conjecture W.2 (Nontrivial fluctuation of the optimal integer modulus). *The bounded sequence $(R(p))_{p \geq 1}$ has at least two accumulation points and is not eventually periodic.*

The conjecture isolates a definite sequence and makes no unsupported claim that its accumulation set is a simple sum of two independent sets. A stronger description or equidistribution theorem would require joint information about $\lambda_p \bmod 1$ and the fractional parts of $\Lambda(p)$, which the present corpus does not supply. Exact values $F(2^{-p}) \in \mathbb{Q}$ make the sequence amenable to arbitrary-precision experimentation without numerical ambiguity.

W.3 Base- b atomic distributions

Replace the dyadic random series by

$$X_b = \sum_{k \geq 1} b^{-k} V_k, \quad b \geq 2,$$

with independent uniform coordinates. The finite convolution is still computable, but binary Thue–Morse signs are replaced by base- b subset or digit combinatorics, and the support is $[0, 1/(b-1)]$ before normalization. The least-interval-mass theorem continues to follow whenever the normalized density is symmetric and unimodal. The same proof would then identify the optimal inverse modulus with the quantile itself.

Research problem W.3 (Uniform computability across the base). *Develop a theorem uniform in the integer parameter b : construct a single algorithm that, given b , a name of y , and a precision p , computes the normalized quantile of X_b . Determine the leading dependence of the optimal input-bit modulus on both b and p , and identify the corresponding base- b log-periodic correction.*

W.4 Certified spectral and polynomial representations

The Legendre, Lagrange, and shifted-up representations in the corpus are mathematically exact, but a representation becomes an algorithm only after a recursive uniform tail bound is attached. This suggests three concrete projects:

1. derive explicit Legendre coefficient envelopes strong enough to certify pointwise and uniform evaluation of up;
2. combine the finite shifted-up representation of polynomials with interval arithmetic to obtain certified local surrogates for F ;
3. compare physical-space spline, Legendre, and sinc-product evaluators by bit complexity, not merely by observed decimal convergence.

A cross-certified implementation could evaluate the same point by two independent representations and intersect the resulting enclosures. This is particularly attractive near transition regions where one representation’s natural error bound is pessimistic.

W.5 Higher-dimensional monotone transport

The one-dimensional theorem

$$\min_x \mathbb{P}\{X \in (x, x+h]\} = \mathbb{P}\{X \in (0, h]\}$$

is a sharp anti-concentration lower bound tailored to a symmetric unimodal law. Because the law is atomless, the same identity holds with any choice of open or closed endpoints. In higher dimensions, inverse CDFs are replaced by triangular Rosenblatt maps or optimal-transport maps. Effective invertibility would require computable lower bounds on masses of suitable boxes or sections.

Research problem W.4 (Atomic transport computability). *For products or coupled variants of Fabius-type laws, determine conditions under which the triangular transport from Lebesgue measure has a computable inverse with an explicit multivariate modulus. Investigate whether total positivity or variation-diminishing properties supply the needed lower section-mass estimates.*

W.6 Branchwise inversion of the signed global extension

The signed global Fabius extension is not globally injective, so it has no single inverse. On every maximal monotonicity interval, however, it has a branch inverse. Local finiteness of its Thue–Morse translate expansion makes forward evaluation computable. The remaining issue is effective branch separation near critical values.

Research problem W.5 (Computable branch atlas). *Construct an effective enumeration of monotonicity intervals of the signed global extension, together with certified branch ranges and recursive inverse moduli. Determine whether the self-similar translation law permits one finite branch template plus a Thue–Morse address.*

Part VIII

Certification, Formalization, and Provenance

X Status discipline

This synthesis uses three noninterchangeable kinds of mathematical evidence.

1. A result is *Lean-proved* only when every clause of the human-readable statement has the same hypotheses and conclusion as a named Lean declaration. A paper theorem assembled from several Lean declarations qualifies only when the finite family is explicitly identified; the presence of all the right ingredients is not enough.
2. A *human-proved frontier result* has a complete proof in this volume but no exact Lean counterpart yet.
3. A *conjecture* or *research problem* retains an explicit, mathematically substantive unresolved obligation.

Executable symbolic or numerical checks are useful regression evidence, but they do not upgrade a manuscript claim into a theorem. Conversely, a theorem whose special case is formalized is not labelled Lean-proved unless the specialization and all of its hypotheses match exactly.

In the current reviewed concordance, the resulting classification is 54 Lean-proved rows, 91 human-proved frontier rows, 10 conjectures, 15 open problems, and 24 nonassertoric rows, for 194 rows in all. These are source-result counts, not counts of distinct canonical theorem environments: deduplication may send several source rows to one stronger surviving result.

The normalization in the shifted self-sampling theorem is matched exactly by the Lean declaration `Fabius.integral_polynomial_mul_rvachevUp_eq_dyadic_tsum` in `FabiusFunction.PolynomialCombExactness`. It has the same whole-line integral, dyadic mesh factor, arbitrary real phase, polynomial degree bound, sampling nodes, and Rvachev-density weights; compact support reduces its integral to $[-1, 1]$. Thus no informal rescaling bridge is being hidden in the Lean-proved classification.

Five further source rows now have exact compiled counterparts. The centered Appell convolution is `Fabius.integral_eval_rvachevAppellPolynomial_sub_mul_rvachev`, the positive-degree mean-zero corollary is `Fabius.integral_eval_rvachevAppellPolynomial_mul_rvachev_eq_zero`, and the arbitrarily phased polynomial formula is `Fabius.normalized_tsum_shifted_rvachevDeconvolvedPolynomial_mul_rvachevUp`. Compact support reduces the first two whole-line integrals to $[-1, 1]$; after that normalization, their hypotheses and conclusions match the corresponding canonical rows. The two newest exact rows are the parity-selected one-extra-degree rule [corollary F.1](#), whose principal physical-coordinate declaration is `Fabius.integral_polynomial_mul_rvachevUp_eq_normalized_tsum_superconvergent`, and the finite Appell

lattice formula [theorem H.5](#), whose selected-phase clause is `Fabius.normalized_tsum_shifted_rvachevAppellPolynomial_mul_rvachevUp_superconvergent`. The exact dyadic phase dictionary is `Fabius.isRvachevSuperconvergentPhase_two_pow_iff`. Existing positivity, strict-interior positivity, rational dyadic evaluation, and finite-support declarations complete the finite families recorded in the Lean crosswalk.

The leading Laurent coefficient theorem [theorem I.1](#) is also Lean-proved by the one-definition, six-theorem module `FabiusFunction.RvachevLaurentLeading`. Its principal manuscript-normalized declaration is `Fabius.tendsto_rvachevCenteredMGF_laurent_two_pow_mul_odd`; the crosswalk records the normalization, odd-core nonvanishing, generic cofactor limit, and general integer-pole companions. The limits are deliberately punctured, because the inverse operation is totalized at the zero itself. This exact row promotion does not extend to lower Laurent coefficients or the later pole-shell and Appell-coefficient asymptotics.

Two further source rows are now Lean-proved by the eleven-definition, seventeen-theorem module `FabiusFunction.FinitePrefixAppellRecovery`. Its direct uncentered and centered sum theorems give the finite-prefix expansions for every starting depth, including zero; the free-scale definitions, evaluation bridges, and degree theorems certify exact symbolic outer degrees n at base $1/2$ and $\lfloor n/2 \rfloor$ at base $1/4$. The latter is an equality in $\text{Polynomial}(\mathbb{Q}[x])$: specializing the inner variable can lower the centered degree. Its two direct recovery theorems use exactly $n + 1$ and $\lfloor n/2 \rfloor + 1$ consecutive prefixes at those respective bases, for all N , with no limit. The formal prefix is an algebraic finite-convolution moment model; this classification does not assert a random-variable, HasLaw, or analytic-MGF realization theorem.

Five canonical results were added after the immutable source snapshot and therefore do not alter the 194-row totals. Three are Lean-proved: the derivative-distribution theorem, the complete geometric-Lagrange residual theorem (by `Fabius.sum_geometricLagrangeWeight_mul_pow_of_pos` and `Fabius.completeHomogeneousEval_geometric`), and the finite-product loss lemma (by `Fabius.one_sub_sum_range_le_prod_range_one_sub`). The base- b density proposition is a new human-proved strengthening, and the dyadic singularity atlas is explicitly an open problem. All five dispositions and their exact formalization boundaries are recorded in the Lean crosswalk.

The source theorem [theorem D.4](#) illustrates why the count uses whole-statement rather than partial-clause certification. The declaration `Fabius.iteratedDeriv_fabiusInv_fifty_seventy_two_succ` formalizes its all-order Catalan derivative formula, but not the packaged quarter jet, quadratic algebraic shadow, and large-order asymptotic as one finite exact family. The complete canonical theorem therefore remains human-proved.

X.1 Partial formalizations are recorded as partial

Two results in the self-sampling chapter have substantial formal kernels but nonidentical or stronger canonical conclusions. The whole labelled results are therefore human-proved frontier results, with the precise Lean boundary as follows.

proposition C.4. Lean proves the denominator-cleared integer-zero factorization and the analytic-cofactor derivative engine. The proposition also asserts the normalized local exponential form and its explicit logarithmic jet; those extra clauses, and hence the whole proposition, are not a single exact Lean result.

theorem D.1. Lean proves the odd-alias representation, and a companion theorem evaluates the surviving derivative coefficient. No named declaration assembles these pieces into the displayed complex series together with both trigonometric forms, so the canonical theorem remains human-proved.

Y Deduplication and result provenance

The machine-readable theorem concordance records every theorem-style source environment at the immutable pre-retirement revision. Several source rows may map to one canonical theorem when one statement is a specialization or a later duplicate of a stronger result. Such a many-to-one mapping is deduplication, not loss: the source statement, source line, surviving label, logical status, and reason for the disposition remain explicit.

The companion asset ledger performs the analogous job for scripts, exact output, figures, generated fragments, checksum ledgers, and historical PDFs. Only unique reproducibility material is migrated into the canonical `assets/` tree. Superseded layouts remain recoverable at the revision recorded in `audit/SOURCE_REVISION` and throughout Git history.

Z Reproducibility boundary

The publication gate has four independent layers: a structural source audit, replay of retained exact computations, three serial $\text{\LaTeX}$ passes, and visual inspection of the rendered PDF. Lean validation, when required by a source change, is serialized separately; no PDF or source-level claim is treated as compiler evidence.

For the new arbitrary-phase polynomial self-sampling result, the focused command `lakebuild+FabiusFunction.PolynomialCombExactness` completed successfully after a serialized replay of its (3474)-job dependency graph. The terminal exit code was zero, and the target module contains no placeholder proof. This compiler evidence is narrower than an umbrella-library build but exact for the declaration used in the crosswalk.

The focused command `lakebuild+FabiusFunction.FinitePrefixAppellRecovery` likewise completed its (2947)-job target successfully before this documentation pass. No Lean or Lake command was run while updating the package-local ledgers.

At the same editorial checkpoint, `audit/validate_canonical.py` verified all nine chapter inputs, (748) distinct labels, (596) references, and all (194) immutable source rows with no incomplete disposition. The independent asset check verified all (88) source-file dispositions, all (63) deduplicated live payloads, and no duplicate canonical payload. These checks establish structural and provenance completeness; they do not replace mathematical proof or visual inspection.

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